

GE Healthcare

BrightSpeed Elite, Edge, Excel Installation Manual

Book 1 of 2: Mechanical Installation

OPERATING DOCUMENTATION



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Rev 10

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Book 1 of 2: Mechanical Installation

Pages 1 - 174

IMPORTANT PRECAUTIONS

LANGUAGE

警告 (ZH-CN)	• 本维修手册仅提供英文版本。 • 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。 • 未详细阅读和完全理解本维修手册之前，不得进行维修。 • 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。
ПРЕДУПРЕЖДЕНИЕ (BG)	• Това упътване за работа е налично само на английски език. • Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод. • Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа. • Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациент в резултат на токов удар или механична или друга опасност.
VÝSTRAHA (CS)	• Tento provozní návod existuje pouze v anglickém jazyce. • V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. • Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. • V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	• Denne servicemanual findes kun på engelsk. • Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. • Forsøg ikke at servicere udstyret medmindre denne servicemanual har været konsulteret og er forstået. • Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk, mekanisk eller anden fare for teknikeren, operatøren eller patienten.
WAARSCHUWING (NL)	• Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. • Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. • Probeer de apparatuur niet te onderhouden voordat deze onderhoudshandleiding werd geraadpleegd en begrepen is. • Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING (EN)	• This Service Manual is available in English only. • If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services. • Do not attempt to service the equipment unless this service manual has been consulted and is understood. • Failure to heed this warning may result in injury to the service provider, operator, or patient, from electric shock or from mechanical or other hazards.
HOIATUS (ET)	• Käesolev teenindusjuhend on saadaval ainult inglise keeles. • Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. • Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. • Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VAROITUS (FI)	• Tämä huolto-ohje on saatavilla vain englanniksi. • Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. • Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. • Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
ATTENTION (FR)	• Ce manuel de service n'est disponible qu'en anglais. • Si le technicien du client a besoin de ce manuel dans une autre langue que l'anglais, c'est au client qu'il incombe de le faire traduire. • Ne pas tenter d'intervenir sur les équipements tant que le manuel service n'a pas été consulté et compris • Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.
WARNUNG (DE)	• Diese Serviceanleitung existiert nur in Englischer Sprache. • Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es aufgabe des Kunden für eine Entsprechende Übersetzung zu sorgen. • Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben. • Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, Mechanische oder Sonstige gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	• Το παρόν εγχειρίδιο σέρβις διατίθεται στα αγγλικά μόνο. • Εάν το άτομο παροχής σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει υπηρεσίες μετάφρασης. • Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό εκτός εάν έχετε συμβουλευτεί και έχετε κατανοήσει το παρόν εγχειρίδιο σέρβις. • Εάν δε λάβετε υπόψη την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στο άτομο παροχής σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.
FIGYELMEZTETÉS (HU)	• Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. • Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése. • Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. • Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.
AÐVÖRUN (IS)	• Þessi þjónustuhandbók er eingöngu áanleg á ensku. • Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálapjónustu. • Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin. • Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnaða eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.
AVVERTENZA (IT)	• Il presente manuale di manutenzione è disponibile soltanto in inglese. • Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. • Si proceda alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. • Il non rispetto della presente avvertenza potrebbe far compiere operazioni da cui derivino lesioni all'addetto, alla manutenzione, all'utilizzatore ed al paziente per folgorazione elettrica, per urti meccanici od altri rischi.
警告 (JA)	• このサービスマニュアルには英語版しかありません。 • サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。 • このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。 • この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。
경고 (KO)	• 본 서비스 지침서는 영어로만 이용하실 수 있습니다 . • 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우 , 번역 서비스를 제공하는 것은 고객의 책임입니다 . • 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오 . • 이 경고에 유의하지 않으면 전기 쇼크 , 기계상의 혹은 다른 위험으로부터 서비스 제공자 , 운영자 혹은 환자에게 위해를 가할 수 있습니다 .

BRDINJUMS (LV)	• Šī apkalpes rokasgrāmata ir pieejama tikai angļu valodā. • Ja klienta apkalpes sniedzējam nepieciešama informācija citā valodā, nevis angļu, klienta pienākums ir nodrošināt tulkošanu. • Neveiciet aprīkojuma apkalpi bez apkalpes rokasgrāmatas izlasīšanas un saprašanas. • Šī brīdinājuma neievērošana var radīt elektriskās strāvas trieciena, mehānisku vai citu risku izraisītu traumu apkalpes sniedzējam, operatoram vai pacientam.
ĮSPĖJIMAS (LT)	• Šis eksploatavimo vadovas yra prieinamas tik anglų kalba. • Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, numatyti vertimo paslaugas yra kliento atsakomybė. • Nemėginkite atlikti įrangos techninės priežiūros, nebent atsižvelgėte į šį eksploatavimo vadovą ir jį supratote. • Jei neatkreipsite dėmesio į šį perspėjimą, galimi sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų paslaugų tiekėjui, operatoriui ar pacientui.
ADVARSEL (NO)	• Denne servicehåndboken finnes bare på engelsk. • Hvis kundens serviceleverandør trenger et annet språk, er det kundens ansvar å sørge for oversettelse. • Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. • Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	• Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. • Jeśli dostawca usług klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. • Nie próbować serwisować wyposażenia bez zapoznania się i zrozumienia niniejszego podręcznika serwisowego. • Niezastosowanie się do tego ostrzeżenia może spowodować urazy dostawcy usług, operatora lub pacjenta w wyniku porażenia elektrycznego, zagrożenia mechanicznego bądź innego.
ATENÇÃO (PT)	• Este manual de assistência técnica só se encontra disponível em inglês. • Se qualquer outro serviço de assistência técnica solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. • Não tente consertar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • O não cumprimento deste aviso pode pôr em perigo a segurança do técnico, do operador ou do paciente devido a choques elétricos, mecânicos ou outros.

ATENȚIE (RO)	• Acest manual de service este disponibil numai în limba engleză. • Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. • Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. • Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	• Данное руководство по обслуживанию предлагается только на английском языке. • Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. • Перед обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. • Несоблюдение требований данного предупреждения может привести к тому, что специалист по обслуживанию, оператор или пациент получат удар электрическим током, механическую травму или другое повреждение.
UPOZORNENIE (SK)	• Tento návod na obsluhu je k dispozícii len v angličtine. • Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. • Nepokúšajte sa o obsluhu zariadenia skôr, ako si neprečítate návod na obsluhu a neporozumiete mu. • Zanedbanie tohto upozornenia môže vyústiť do zranenia poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, do mechanického alebo iného nebezpečenstva.
ATENCION (ES)	• Este manual de servicio sólo existe en inglés. • Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. • No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. • La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	• Den här servicehandboken finns bara tillgänglig på engelska. • Om en kunds servicetekniker har behov av ett annat språk än engelska ansvarar kunden för att tillhandahålla översättningstjänster. • Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. • Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
DIKKAT (TR)	• Bu servis kilavuzunun sadece ingilizcesi mevcuttur. • Eğer müşteri teknisyeni bu kilavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. • Servis kilavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. • Bu uyariya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "damage in shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on <http://egems.med.ge.com/edq/home.jsp>
- Contact your local service coordinator for more information on this process.

Rev. June 13, 2006

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS



CAUTION
Risk of
Explosion

Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.



ATTENTION
Danger
d'Explosion

Il y a danger d'explosion s'il y a remplacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

Revision History

Revision	Date	Reason for change
10	11/04/08	Chapter 1: Section 4.0 updated Table/Gantry Alignment Procedure. Chapter 2: Section 2.0 added system interconnected diagram with AC Outlet Box Section 4.4 added figure of power connection with AC Outlet Box updated Section 4.6 Modem Option for AIO console updated Section 4.8, added AIO Interconnect added Section 5.0 Install Options updated Section 5.3, 5.4, 5.5 for Install Options Section 9.0 updated figure of CT System Ground Connections Appendix B: added new FWS (part# 5168666) installation procedure. Chapter 5: updated Section 2.10, added AIO Console Boot-up Flow Chart Section 3.4 updated CT System X-Ray ON Indicators, Cautions & Warning Labels Section 3.11.3 Deleted Alignment Light Characterization, it is duplicated with 3.9.1. Chapter 6: Section 3.0 updated the procedure with Laser Alignment Tool. Chapter 7: Section 2.0 Deleted install Injector and UPS, they are duplicated with Chapter 2, Section 5. added Section 11 for Modem setup
9	04/21/ 08	Chapter 5: updated Section 3.9, 3.11 for Table/Gantry interference test specifications based on changes with software version 07MW11.10. Customer Configuration Preference for Language requirements: Updated Chapter 5, Section 2.6.2, Step 7c Procedure, and Section 3.5 Warning Labels. Also updated GE form e4879 to record information.

Revision	Date	Reason for change
8	12/03/07	Important Precautions: Updated content for damage in transportation Chapter 1: added Section 1.0 installer/FE notices updated Section 2.1 Training Process updated Section 2.5 common tools updated Section 4.3 procedure, added Cable entrance width added Section 4.4 Gantry bearing gap inspection procedures Chapter 2: updated Section 4.4 injector connection port updated Section 4.7 connection cable added Section 5.0 Install Options Chapter 3: updated Section 4.0 Mechanical Installation Completion Section Appendix B: added Section 2.4 install Seismic kit Appendix E: updated Regulatory Clearance Quick Reference Guide Chapter 4: updated Section 4.3 phantom added Section 6.0 FE Workflow added Section 7.0 Checklist for Completed Installation added Section 8.0 GE and Regulatory Forms Chapter 5: updated Section 2.3 system state DVD location updated Section 2.7 Check/Set Date and Time added Section 2.10 GRE console boot-up flow chart updated Section 3.7 Verify Service Cabinet Installation Chapter 7: updated Section 2.5 UPS installation Chapter 9: removed this chapter ("Forms" chapter). Forms now reside on the Service CD.
7	8/29/07	Important Precautions: Added Bulgarian warnings Chapter 8 - Updated Patient Touch Leakage Test. Added the CT System Chassis Leakage Test. Chapter 9 - Added CT System Chassis Leakage Test Completion Form.
6	6/28/07	Important Precautions: updated multi-language warnings Chapter 5 - Added Section 3.10 for Short Footprint setting procedure.
5	12/12/06	Chapter 2 - Added note: use dry cleaning for electro components. Chapter 3 - Updated Mechanical Installation completion section. Chapter 5 - Added warning and notice about line transformer settings. Chapter 6 - Updated anchors tighten torque in section 3 table/gantry alignment procedure. Chapter 9 - Updated electrical calibration installation checklist. Updated GE Form 4879 installation data verification.
4	9/30/06	Updated for Small Room: Chapter 1: Section 3.2 Position Gantry Appendix A: Added Section 9.0 - Gantry Auxiliary (Mini) Dolly Installation w/Wood Block Appendix E: New appendix (Regulatory Clearance Quick Reference Guide) for Small Room

Revision	Date	Reason for change
3	8/28/06	Chapter 2 - Power, Ground & Interconnect Cables: Updated gantry cable routing: PQR 13078793.
2	6/22/06	Chapter 1 - Position Subsystems: Updated Section 11.0 - Seismic Mounting. Chapter 2 - Power, Ground & Interconnect Cables: Updated Figure 2-13 GOC4/LCGOC Interconnect. Chapter 3 - System Continuity & Ground Checks: Added Section 3.0 - Axial Head Holder Shim Installation. Appendix B - FWS Assembly and Adjustment: Update whole chapter. Chapter 5 - Electrical Integration and Safety Verifications: Updated Hardware Reconfig illustration. Chapter 8 - System-Level Safety Tests: Updated procedure.
1	2/20/06	Initial Release.

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Preface

Publication Conventions

Please become familiar with the conventions used within this publication before proceeding.

Section 1.0 Safety & Hazard Information

1.1 Text and Character Representation

Within this publication, different paragraph and character styles have been used to indicate potential hazards. Paragraph prefixes, such as hazard, caution, danger and warning, are used to identify important safety information. Text (Hazard) styles are applied to the paragraph contents that are applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information will normally include:

- Type of potential hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS USED

A few examples are provided that have been adapted from GEMS' global document standard (2119696-100). They include paragraph prefixes and modified text styles.



CAUTION
Pinch Points
Loss of Data
Sharp Objects

Caution is used when a hazard exists that can or could cause minor injury to self or others if instructions are ignored. They include for example:

- Loss of critical patient data
- Crush or pinch points
- Sharp objects



DANGER
EXCESSIVE
VOLTAGE
CRUSH
POINT

DANGER IS USED WHEN A HAZARD EXISTS THAT WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- ELECTROCUTION
- CRUSHING
- RADIATION



WARNING
ROTATING
EQUIPMENT
BARE WIRES

WARNING IS USED WHEN A HAZARD EXISTS WHICH COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- Potential for shock
- Exposed wires
- Failure to Tag and lockout system power could allow for un-command motion.



NOTICE
Equipment
Damage
Possible

Notice is used when a hazard is present that can cause property damage but has absolutely no personal injury risk. They can include:

- Disk drive will crash
- Internal mechanical damage, such as to the x-ray tube
- Coasting the rotor through resonance.

It's important that the reader not ignore hazard statements in this document.

1.2 Graphical Representation

Important information will always be preceded by the exclamation point  contained within a triangle, as seen throughout this chapter. In addition to text, several different graphical icons (symbols) may be used to make you aware of specific types of hazards that could possibly cause harm.

ELECTRICAL



LASER



MECHANICAL



HEAT



RADIATION



PINCH



Some others make you aware of specific procedures that should be followed.

AVOID STATIC ELECTRICITY



TAG AND LOCK OUT



WEAR EYE PROTECTION



Section 2.0 Publication Conventions

2.1 General Paragraph and Character Styles

Prefixes are used to highlight important non-safety related information. Paragraph prefixes (such as Purpose, Example, Comment and Note) are used to identify important but non-safety related information. Text styles are also applied to text within each paragraph modified by the specific prefix.

EXAMPLES OF PREFIXES USED FOR GENERAL INFORMATION

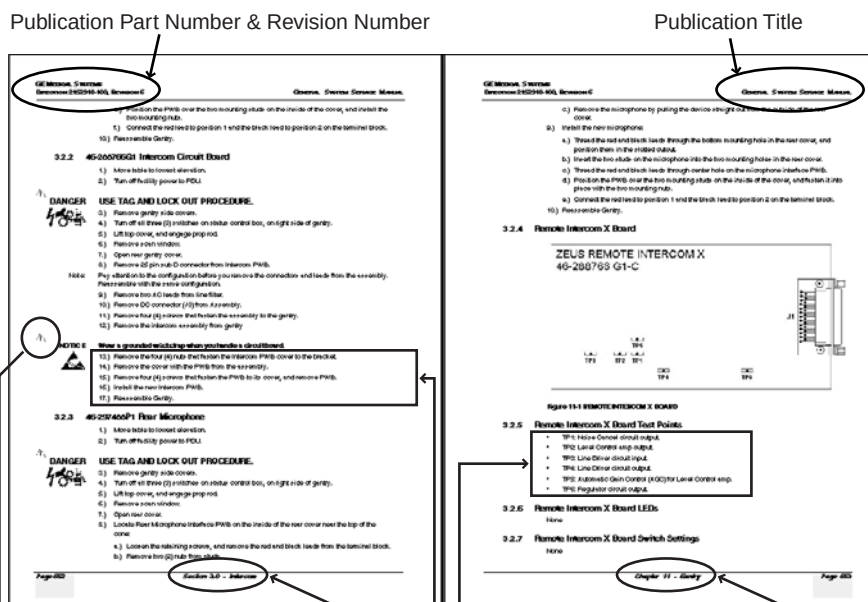
Purpose: Introduces and provides meaning as to the information contained within the chapter, section or subsection, such as used at the beginning this chapter for example.

Note: Conveys information that should be considered important to the reader.

Example: Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.

Comment: Represents "additional" information that may or may not be relevant.

2.2 Page Layout



The current section and its title are always shown in the footer of the left (even) page.

An exclamation point in a triangle is used to indicate important information to the user.

Paragraphs preceded by Alphanumeric (e.g. numbers) characters is information that must be followed in a specific order.

The current chapter and its title are always shown in the footer of the right (odd) page.

Paragraphs preceded by symbols is (e.g. bullets) is information that has no specific order.

Headers and footers in this publication are designed to allow you to quickly identify your location. The document's part number and revision number appears in every header on every page. Odd

numbered page footers indicate the current chapter, its title, and current page number. Even page footers show the current section and its title, as well current page number.

2.3 Computer Screen Output/Input Character Styles

Within this publication different character styles are used to indicate computer input and output text. Character (input, output, and variable) styles are used and applied to the text within a paragraph so as to indicate directions. Computer screen output and input is also formatted using mono (fixed width) spaced fonts.

Example: Fixed Output This paragraph denotes computer screen fixed output. It's output is fixed from the sense that it does not vary from application to application. It's the most commonly used style used to indicate filenames, paths, and text.

Example: Variable Output *This paragraph denotes computer screen output that is variable. Its output varies from application to application. Variable output is sometimes found placed between greater than and lesser than operators. For example: <variable_ouput>*

Example: Fixed Input **This paragraph denotes fixed input. It's typed input that will not vary from application to application. Fixed text the user is required to supply as input.**

Example: Variable Input ***This paragraph denotes computer input that can vary from application to application. Variable text the user is required to supply as input. Variable input sometimes is placed between greater than and lesser than operators. For example: <variable_input>. In these cases, the (<>) operators are dropped prior to input. Exceptions are noted in the text.***

2.4 Buttons, Switches and Keyboard Inputs (Hard & Soft Keys)

Different character styles are used to indicate actions requiring the reader to press either a hard or soft button, switch, or key. Physical hardware, such as buttons and switches, are called hard keys because they are hard wired or mechanical in nature. A keyboard or on/off switch would be a hard key. Software or computer generated buttons are called soft keys because they are software generated. Software driven menu buttons are an example of such keys. Soft and hard keys are represented differently in this publication.

Example: Hard Keys A power switch **ON/OFF** or a keyboard key like **ENTER** is indicated by applying a character style that uses both over and under-lined bold text that is bold. This is a hard key.

Example: Soft Keys Whereas the computer **MENU** button that you would click with your mouse or touch with your hand uses over and under-lined regular text. This is a soft key.

Chapter 1

Position Subsystems



NOTICE

Potential for Data Loss and/or Equipment Damage.

To prevent potential data loss and equipment damage, please do the following:

- When instructed, record data collected from the procedures in this chapter on GE Form e4879. For more information about this form, see [Section 8.0](#) of Chapter 4.
- Only use the Installation manual that arrives with your system. Any other revisions of this manual may not exactly match your system.

Section 1.0

Installer/FE Notices

1.1 Shipping, Warehouse, and Transportation Warning

- This gantry is designed to be moved using the shipping dollies and should not be lifted or moved using a lift truck.
- Do not hoist gantry or table using dollies.

1.2 International Shipments

- Dollies must be used to remove the gantry from the shipping skid and to transport the gantry to the customer's site.
- If lifting is required, refer to the Pre-Installation Manual for instructions.

1.3 On Site Warning (Mechanical Installation Team)

This system requires a gantry bearing gap inspection before electrical calibration is started. See [Gantry Bearing Gap Inspection, on page 40](#).

Section 2.0

Install Table/Gantry Introduction

This chapter describes how to mount, position, and level the CT Scanner subsystems.

Note: Before you start the installation, make sure the site preparation complies with conditions and instructions found in the pre-installation manual. Failure to comply will result in excessive installation delay and potential increased, unrecoverable installation costs. This product is designed to meet specific mechanical installation standards, which should be reviewed prior to installing this system.

2.1 Skill Set Required

NOTICE All Mechanical Installers must meet these minimum requirements.



MECHANICAL

Installation	Safety	OJT with Trainer
Must be trained as FE or Contractor.	Complete all training courses.	Complete training installation with trainer.
Must complete all training Courses.	My Learning verification.	
My Learning verification.		
Tools	Access	Communication
Must have all required tools as listed in this manual.	Must have access to some GE Healthcare web sites and tools.	Must be able to complete all required paperwork and complete iTracs as required.

ELECTRICAL

In-Residence Training.	Safety	OJT with Trainer
Must be trained as FE or Contractor.	Complete all training courses.	Complete training installation with trainer.
My Learning verification.	My Learning verification.	

Only people experienced and trained in the installation process of a GE CT Scanner should perform the work described in this document. A person doing the installation should have:

- Installed a GOC console.
- Completed three BrightSpeed/LightSpeed 4 or 8 or 16 CT installations as a helper, prior to leading an installation. Installation leader must have completed project management, leadership training, and completed all required training prior to becoming a project leader.
- Reviewed the LightSpeed Installation Video, Safety Video, Installation Training CD, and successfully passed all related tests. (Refer to [Table 1-1.](#))

Table 1-1 Installation and Training Media

MEDIA	PART NUMBER
LightSpeed Installation CD	2289471-100
LightSpeed Installation Video	2289468-100
LightSpeed Installation Training CD	2289779-100
GE Safety CD	2290198-100
GE Safety Video	2290199-100

- A working understanding of the mechanical installation process, prior to beginning.
- Completed environmental health and safety (EHS) training, understand and apply lockout/tagout processes, completed Hazcom training, completed Eye & Personal Protection Safety, and have taken Blood Borne Pathogens training.
- All mechanical vendor training and records must be verifiable and available to GEMS.
- Mechanical vendors shall be responsible for training all employees involved in the installation of GEMS CT products.
- It is the mechanical vendors responsibility to require and monitor that all of the above safety requirements are maintained and met on installation sites.

- Mechanical vendor installation teams must have basic measurement skills, system layout skills, and the ability to read and understand blue prints. Must have professional communication skills needed to communicate with customer any changes or additional requirements needed to complete the installation project.

2.2 Floor and Room Preparation

2.2.1 Preparation

Consult your local GE Sales and service representative about your specific needs. It is the purchaser's (buyer) responsibility to provide an approved support structure and an approved method of mounting. General Electric is not responsible for any failure of the support structure or method of anchoring.

The BrightSpeed system has a total floor load of approximately 6500 lbs (2950 kg). About 5175 lbs (2350 kg), including patient (450 lbs (204 kg)), is concentrated in the table-gantry assembly. Refer to the *Pre-installation Manual* for more information.

2.2.2 Flooring

Do not place the scanner on any resilient flooring. Resilient tile or carpeting may slowly yield over a period of time and disturb the alignment of the table to the gantry. Refer to the floor template to determine locations where resilient flooring material should be removed.

Limitations include:

- No part of the floor surface within the table, gantry, or the two interface areas between table and gantry, should be higher than the support areas for the table and gantry.
- The floor structure must withstand the occupied weight of table and gantry, as well as the individual contact area loading of these components.
- The method and placement of anchors, or through bolts, must not reduce the structural strength of the floor.

If you have to remove the gantry covers in order to move the gantry into the room, refer to [Appendix A](#), for the cover removal procedure. Please read the caution statement on [page 117](#) before removing the gantry covers.

2.3 Overview

Procedures in this chapter provide detailed instructions to position, level, and anchor the gantry and table securely for operation. The BrightSpeed system uses adjustable leveling pads to support the gantry and table. The gantry has four (4) primary leveling pads located on the gantry base. The table has five (5) pads used for leveling it.

The process you will be following is:

- 1.) Use the room-layout template to determine the general position of the gantry and table.
- 2.) Move the gantry into position.
- 3.) Level gantry.
- 4.) Use the Laser Alignment Tool to position the table relative to the gantry.
- 5.) Level the table to the gantry, and anchor the system.

Use the template to position the system; however, use the gantry and table to locate and drill the anchor holes. Drill the anchor holes with the system in place.

2.4 Pre-Installation Template

Always use the room-layout template (in two pieces), during installation (see [Table 1-2](#), for part number). The gantry and table will not be properly aligned if existing holes are used. The template shows the location of the gantry and table anchor holes.

This template is shipped with the system. It may also be ordered as FRU.

Table 1-2 Room Layout Templates

System	Part Number
BrightSpeed Elite, Edge, Excel	5160024

2.5 Required Common Tools and Supplies

The following tools and supplies are required for installation of the scanner. Please refer to [Appendix C, on page 155](#), for pictorial descriptions of the tools and supplies.

WRENCHES

- Standard and Metric combination wrench sets
- Standard and Metric Hex Key (Allen wrench) sets
- ½" and 3/8" drive torque wrench: 0-100 N-m (0-100 ft.-lb.) Must be calibrated yearly.

SOCKETS AND EXTENSIONS

- 3/8" and ½" drive ratchet wrenches
- ½" drive 3" & two 6" long extensions
- 3/8" drive 12" long extension
- Standard & Metric 3/8" drive socket sets
- ¾" deep well socket 3/8"
- 1", 1-1/8", 1-¼" & 1-½" sockets for ½" drive
- 3/8" drive universal joint
- Metric hex bit set ¼" or 3/8" drive, including:
 - 14mm hex bit 3/8" or ½" drive (14mm ball hex helpful)
 - 10mm hex bit 3/8" drive

SCREW DRIVERS

- Phillips screwdriver set (small, medium, and large)
- Straight blade screwdriver set (small, medium, and large)

DRILL BITS

- Complete set of standard (U.S.) drill bits
- Metric tap set
- 12 mm concrete drill bit (2203081: HILTI, TE-CX 12/17 or equivalent)
- ½" masonry bit, min. 8" long USA – 18" optional (for rear table hole)
- 3" (76mm) hole saw with 1/4" (6mm) masonry bit (to remove flooring)

POWER TOOLS

- 3/8" or ½" drill, cordless or electric
- Reciprocating Saw (Sawzall or equivalent) and assorted blades.
- Hammer Drill & Bit (8" min, 12" max)
- Sears 17740 Shop vacuum or equivalent, with "HEPA" or dry wall dust filter (Sears part

- number 17918) or equivalent
- 25' Extension power cords

HAND TOOLS

- Ball-Peen Hammer (1lb or 2lb)
- Tongue & Groove Pliers (large)
- Diagonal Cutting Pliers, Large (to cut 1/0 ground)
- Framing Square (e.g., Empire 16" x 24" aluminum square)
- Diagonal Cutting Pliers, Small
- Large pry bar
- 4', 2' & 9" torpedo levels (see Table 1-3: Recommended Levels)
- Laser level (see Table 1-3: Recommended Levels)

Table 1-3 Recommended Levels

9"	Johnson Magnetic Level, model 7500M*
	Johnson Magnetic Level, model 4500
	Stanley Magnetic Level
2'	Johnson Professional Box Beam Level, model 9624*
	Empire Titan Professional Box Beam Level, model 900 series
4' (nominal)	48" Johnson Professional Box Beam Level*
	42" Stanley Contractor Grade Level
	48" Empire Titan Professional Box Beam Level, model 900 series
	48" Stabila Aluminum Box Beam Level, Kit 24816

* Preferred levels

ELECTRICAL TOOLS

- DVM
- Continuity tester

PERSONAL SAFETY EQUIPMENT

- Safety shoes*
- Safety glasses*
- Gloves
- LOTO Kit (supplied)*
- Hearing Protection*
- 6' & 8 ' Step ladder

* These PPE items are absolutely required for every installation job, with NO exceptions.

SYSTEM CLEANERS

Purchase Locally:

- Glass Cleaner or equivalent
- Scrubbing Bubbles Bathroom Cleaner or equivalent
- Grease Cutting Cleaner or equivalent

GE TOOLS

- 5148193 (VCT alignment Kit)
- or
- 5272090 (Complete Laser alignment tool for VCT, Brightspeed Select and RT16)

Note: A box labeled Installation Support Kit is shipped with each system. It contains paint, masking tape, cleaners, towels, and other materials needed to install this CT scanner.

Section 3.0

Delivery and Inventory Procedure

3.1 Delivery Procedure

3.1.1 System Transportation - Temperature Extremes

When transporting the CT system, ensure that the system is not exposed to temperatures or humidity outside the following specifications.

Temperature: -29° to +140° F (-34° to +60° C)

Humidity: 5% to 95%

NOTICE **Component Freezing occurs if CT system is exposed to temperatures below -29° F (-34° C) for a period longer than two days.**

Allow a minimum of 12 hours for the CT system to adjust to ambient room temperature, prior to installation.

3.1.2 Working with the Mover

Follow the instructions provided by your installation specialist regarding working with equipment movers. Help direct movers on where to place equipment and which items you need first.

Generally movers should move all equipment into the customer room. Door removal and other site changes to move equipment should be done only as directed by the install specialist.

For component sizes and weights, refer to the *BrightSpeed Elite, Edge, Excel Pre-Installation* manual.

3.1.3 Floor Protection

It is suggested that the movers use floor protection. Most equipment movers can provide floor protection during the equipment delivery. Installers should provide floor protection for the room.

3.1.4 Equipment Delivery Route

Prior to equipment delivery, review the delivery route with the movers. Refer to the installation specialist for any additional delivery instructions.

3.1.5 Removing Gantry Dollies and Covers

Refer to [Appendix A](#), for the dolly and cover removal procedures. Please read the caution statement on [page 117](#) before removing the gantry covers.

3.1.6 Check for Shipping Damage

Check for the following damage, which may have occurred during shipping:

- Equipment damage: paint scrapes and cover damage.
- Damage to hospital property: floors, door frames and walls.

If damage is found—or items are missing in shipment—notify the appropriate service personnel:

- For item(s) missing in shipment or short shipped, contact the install specialist to have the item(s) shipped.
- Report damaged items:

<http://egems.med.ge.com/edq/home.jsp>.

3.1.7 A1 Breaker

Lock-out and tag-out the A1 breaker now.

Figure 1-1 Sample A1 Breaker

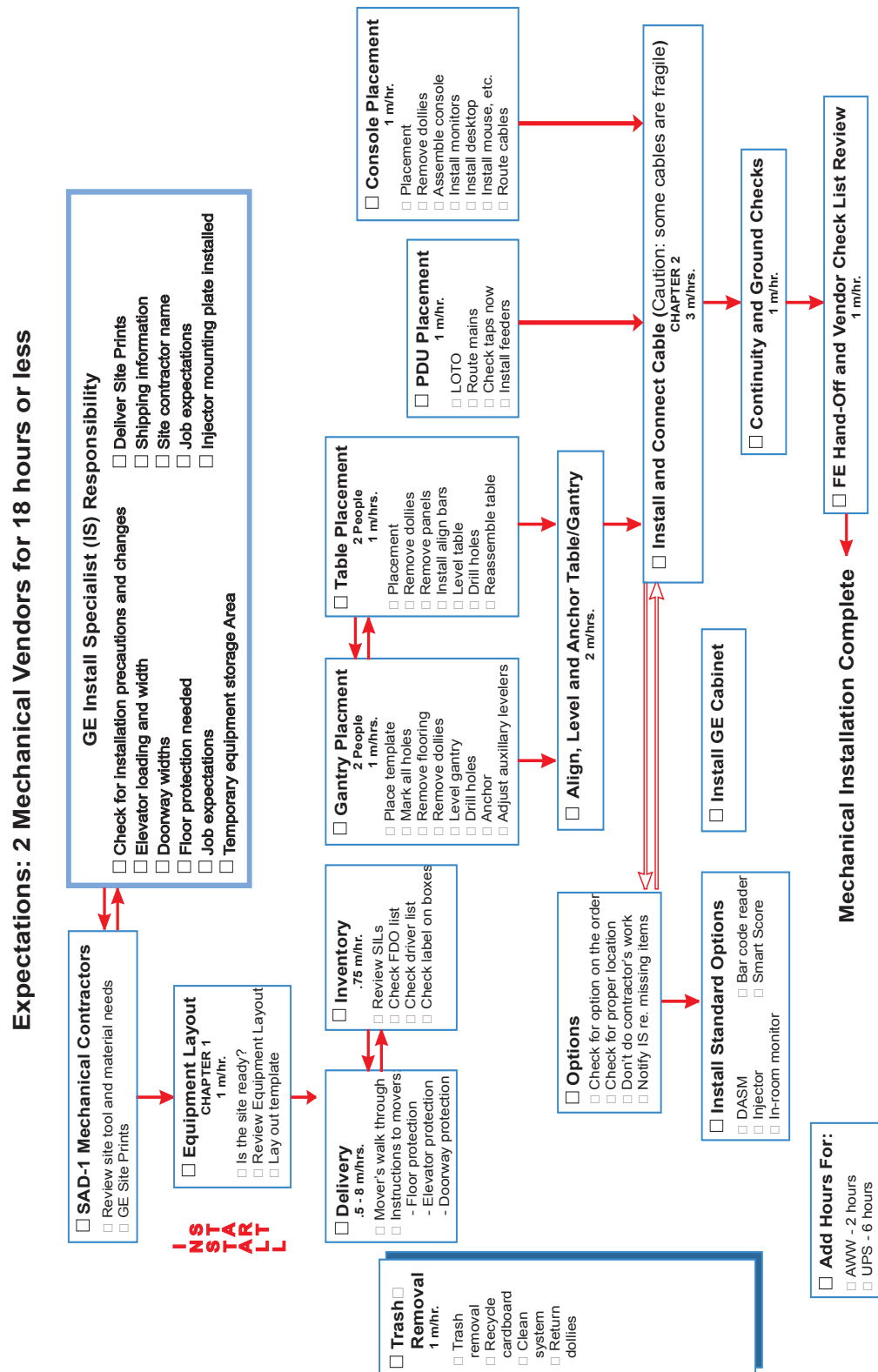


3.1.8 Installation Support Kits

An Installation Support Kit is shipped with every system. Locate this box now and open it. All included materials are to be used during the installation process. These items are to be left ON SITE, for future service needs.

3.2 Mechanical Block Diagram

Figure 1-2 Mechanical Installation Block Diagram



Section 4.0

Install and Level Table/Gantry

4.1 Establish the Room Layout



NOTICE The floor must meet the minimum floor specification in order to begin this procedure. Refer to the *Pre-Installation Manual* for details.

Use the GE print (developed for your site) to establish the room layout. Make sure all the operating and service clearances shown on the print are observed. Using the supplied template, locate the anchor holes. Make sure they clear structural interferences in the floor.

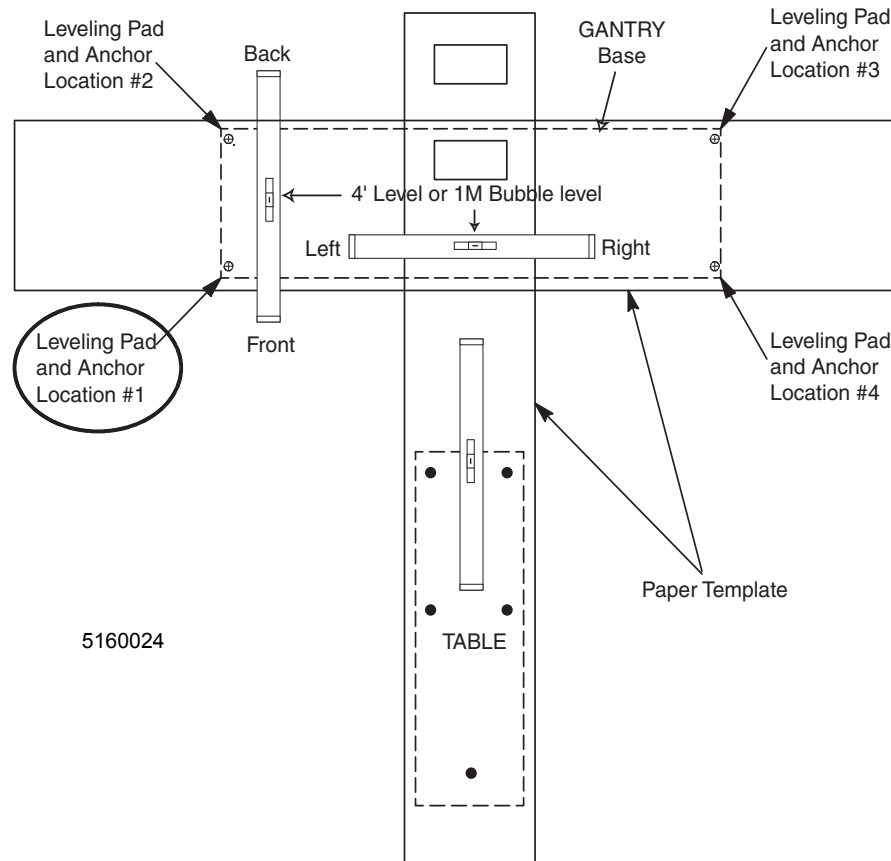
Clean the area. Free the mounting surface of any material that may interfere with the positioning and leveling of the system.

- 1.) Lay out the two (2) pieces of the floor template (P/N 5160024).
- 2.) Start with the Gantry template—align per the GE print.

Note: If the floor has existing anchor penetration, the new anchors must be set at least 4" from the nearest penetration.

- 3.) Place the table template over the top of the Gantry template. Align the scan and table center-lines and secure the templates to the floor. Make sure there are no potential clearance issues.
- 4.) Refer to the ruler of the table template to confirm table travel distance and mark the limitation on the floor if table short footprint setting is needed.
- 5.) Now, check the level of the floor (See [Figure 1-3](#)) across the templates.

Figure 1-3 Hole Locations





NOTICE

- 6.) Scribe a mark (e.g., use a center punch) at each of the gantry's mounting hole locations (there are four (4) of these).
- 7.) Using a center punch, mark the three (3) table mounting hole and two (2) leveler locations.

Positioning requires cutting 7 holes in the floor.

Before you drill or cut any flooring, make sure the appropriate hospital personnel have approved the location of the table/gantry.

- 8.) Cut tiles (or other resilient flooring) around all holes punched in the template for the gantry and table. Use a utility knife with a heat gun, a 3" hole saw with a 1/4" masonry bit or other adequate tools to cut the flooring.
- 9.) Some sites may require sealing of the floor penetrations after the flooring is removed.

- Note:
- Use RTV or other sealant to seal the floor covering, as necessary.
 - All documentation in this manual is based on mounting the table / gantry on a concrete floor only.

4.2 Position the Gantry

4.2.1 Gantry Prep - For Access Greater Than 28"

- 1.) Remove all the transportation packaging, except for dollies, from the gantry.

Note: Some sites require floor protection. Locate and install any required floor protection now.

4.2.2 Gantry Prep - For Access Less Than 28"

- 1.) Remove all the transportation packaging, except for dollies, from the gantry.

Note: Some sites require floor protection. Locate and install any required floor protection now.

- 2.) Remove the blue dolly from the left side of the gantry so that the gantry can be positioned closer to the left side wall:

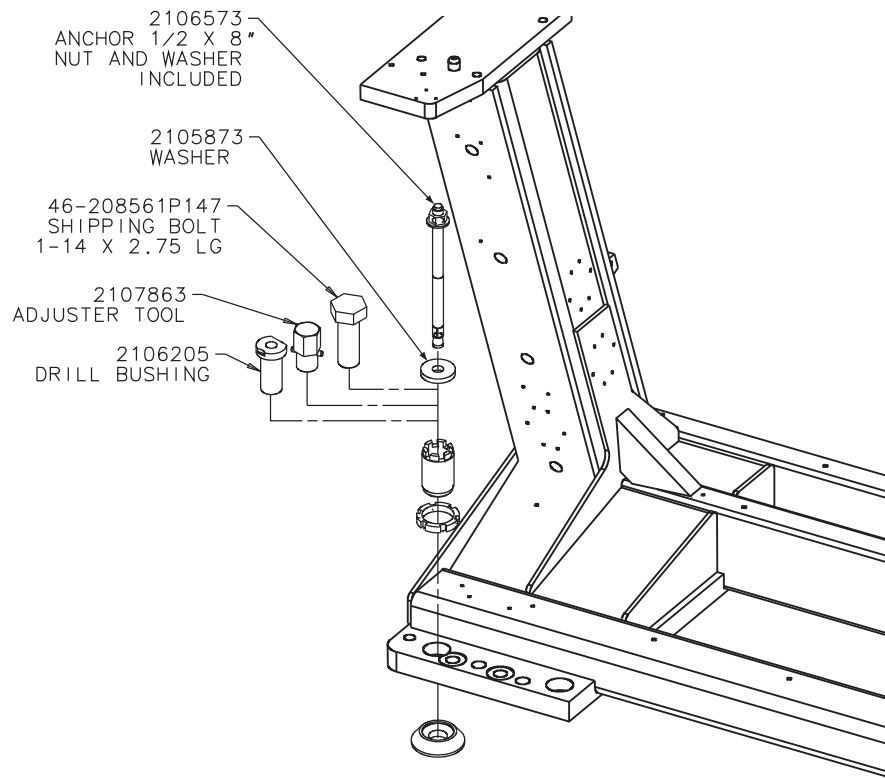
Note: Use Floor Protection for this process.

- a.) Lower the gantry to the floor so that the gantry is resting on the floor.
- b.) Remove the three (3) M14 hex bolts that secure the gantry to the dolly.
- c.) Replace the removed dolly with the shipped black gantry-positioning dolly, and reinstall the three (3) M14 hex bolts.
- d.) Raise the gantry so that it is once again off of the floor.

4.2.3 Gantry Positioning - All Sites

- 1.) Position the gantry over the template appropriately.
 - a.) Locate the four (4) leveling pads, and position each of them beneath its associated adjuster.
 - b.) Use the dollies to evenly lower the gantry, until it is just off of the floor (approximately 1/2" or 12.7 mm). Use a 1/2" ratchet to raise and lower the dollies.
 - c.) Carefully rotate the gantry into the correct position over the template.

Figure 1-4 Gantry Base Installation Hardware



Note: Adjusters are used at each anchor location. Anchor hole ID is 1" (2.5 cm). Void between adjuster and anchor must be filled according to local building codes for seismic application.

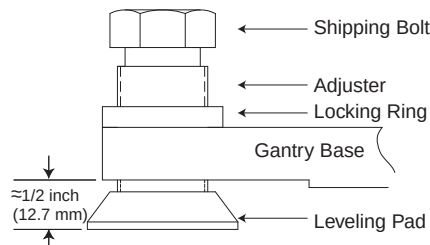
- 2.) Remove the paper templates from the floor and discard properly.
- 3.) Loosen the locking rings and shipping bolts so you can fine-tune the leveling pads to compensate for slight variations in the floor surface.
- 4.) Position the gantry so that the adjusters are centered over their respective holes scribed earlier into the floor.

IMPORTANT:

Make certain to route the gantry power cord under the two rear gantry rails, before removing the gantry shipping dollies.

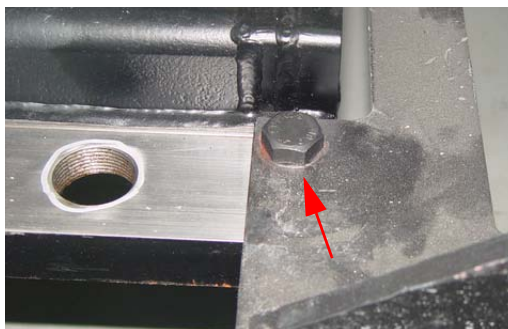
- 5.) Using a 1/2" ratchet, gently lower the gantry until it rests on the floor, over the marked areas.

Figure 1-5 Gantry and Table Base Leveling Pads (Starting Positions)



- 6.) Using a 14mm hex socket, remove the dollies from the gantry by removing the three dolly bolts found at both ends of the gantry ([Figure 1-6](#)).

Figure 1-6 Gantry Dolly Bolts



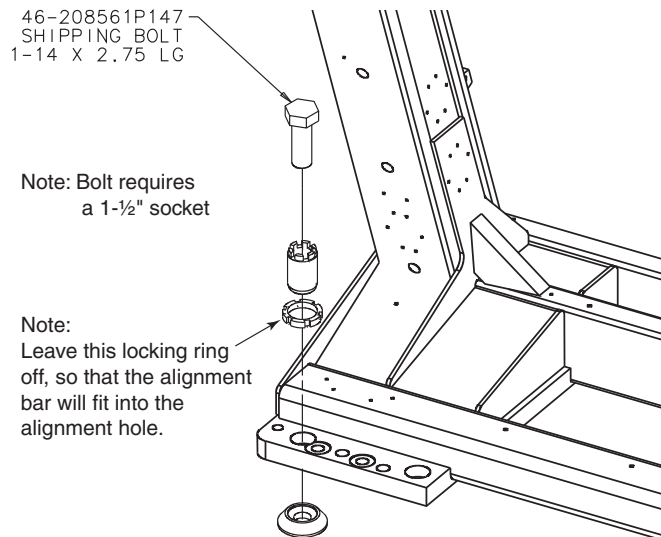
- 7.) Remove the dolly plates on both sides of the gantry. Retain the dolly plates in the service cabinet at the hospital.

Figure 1-7 Dolly Plate



- 8.) Remove the (4) gantry shipping bolts, using a 1½" socket.

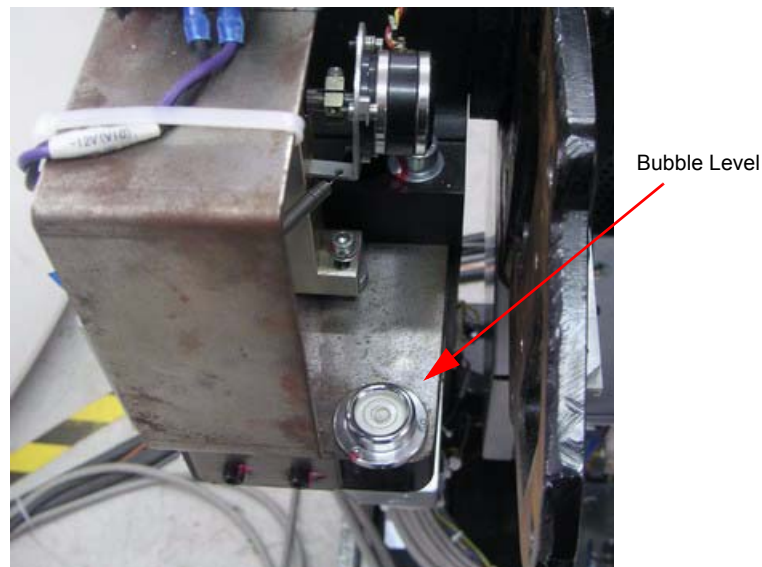
Figure 1-8 Gantry Shipping Bolts



4.3 Level the Gantry

The gantry uses 2 bubble levels that are permanently mounted to machined surfaces on the stationary base to tell when it is level.

Figure 1-9 Gantry Bubble Level



Bubble levels are located on both ends of the gantry stationary base. They're located on the stationary base near a point where the rotating structure pivots mount to the base structure. (See [Figure 1-9.](#)) The gantry is properly leveled when the bubble is centered. (See [Figure 1-11, on page 39.](#))

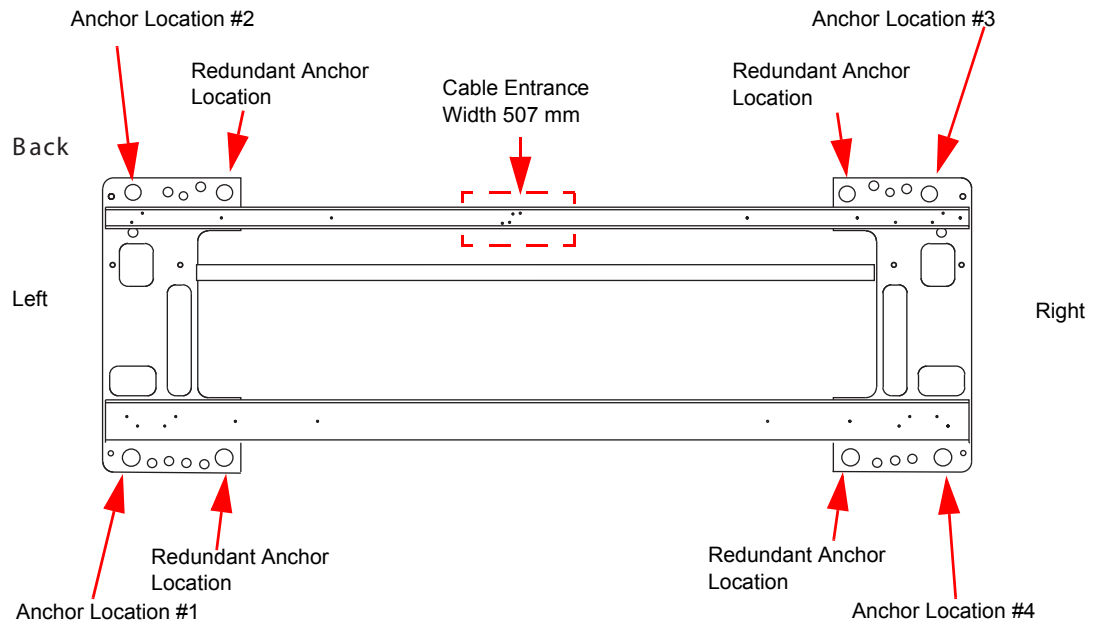
- 1.) Loosen all adjuster lock rings (use a spanner wrench or large channel lock pliers).
- 2.) Systematically turn each of the gantry's adjusters (locations 1, 2, 3 and 4 in [Figure 1-10](#)) until both bubble levels are centered left to right, and front to back.

- Begin by turning each adjuster no more than 1 turn at a time.
- Use the adjuster tool, 1¹/₈" socket, and the ½" drive ratchet to turn each adjuster. (Refer to [Figure 1-4, on page 36.](#))

Systematic Procedure for Leveling gantry follows:

- 1.) Level the left side from front to back by turning adjusters #1 and #2.
- 2.) Level the right side from front to back by turning adjusters #3 and #4.
- 3.) Level the side (right or left) that is higher with respect to the other side. Turn both adjusters on a side equally until that side is level. The side should now also be level.

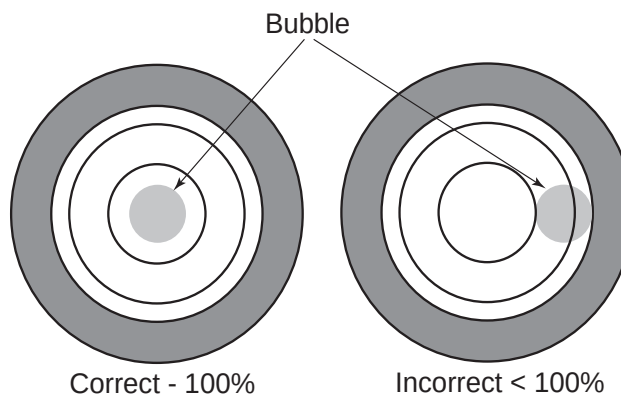
Figure 1-10 Gantry Base “Adjuster” Locations— Top Vie



- 3.) When the bubble levels are centered ([Figure 1-11](#)), each of the four (4) leveling pads should be carrying a portion of the gantry weight. Distribution of the gantry weight prevents the base frame from rocking during normal operation. **DO NOT leave any adjuster un-loaded or floating.**

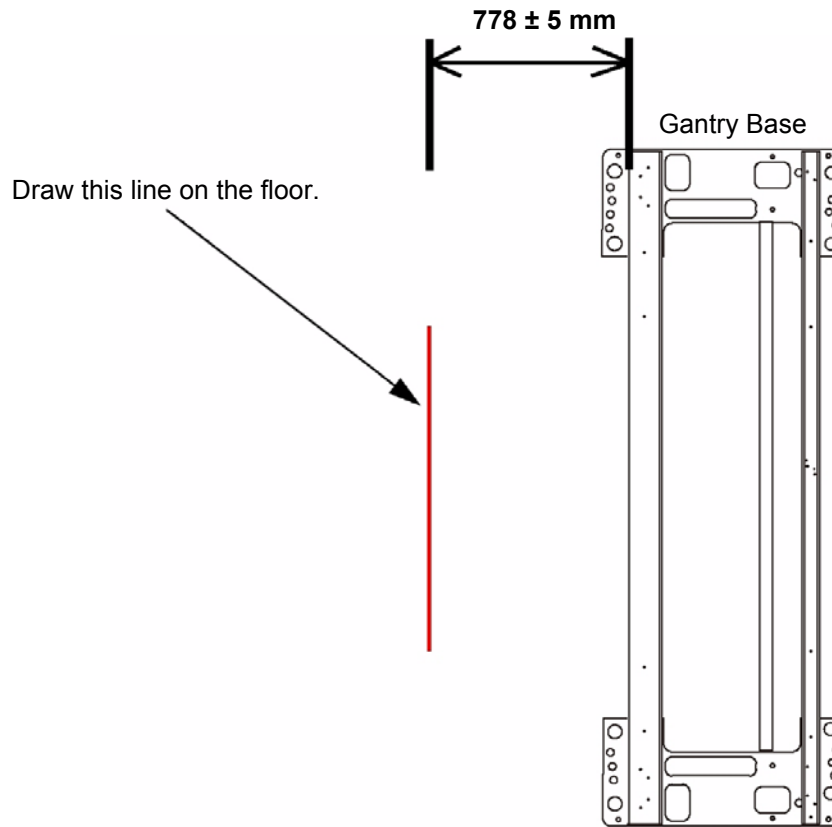
Figure 1-11 Bubble Level Centering

Correct level is 100% of bubble within small circle
Incorrect Level is less than 100% of bubble within small circle



- 4.) Draw a reference line of 778 ± 5 mm position from Gantry Base on the floor as shown in the [Figure 1-12](#). This line should be parallel to the gantry. In a later section, you will move the table against the 778 mm mark.

Figure 1-12 Draw Reference Line



4.4 Gantry Bearing Gap Inspection

All CT systems require a Gantry Bearing Gap inspection before starting electrical calibration. All international gantries are shipped in a wooden shipping crate that should not be removed until it arrives at the installation site. This shipping container is designed to reduce the risk of shipping damage.

4.4.1 Personnel Requirements

REQUIRED PERSONS	PRELIMINARY REQS	PROCEDURE	FINALIZATION
2 (mechanical suppliers or installation team)	15 min	15 min	15 min

4.4.2 Tools and Test Equipment

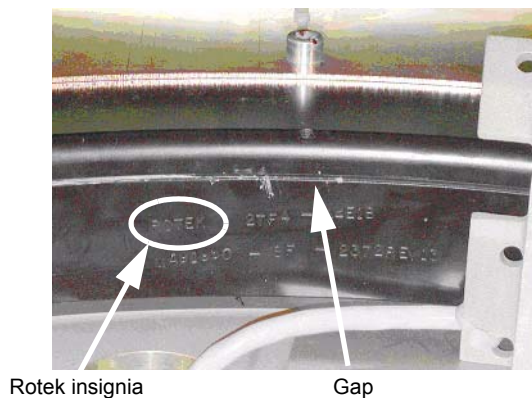
- Standard tool kit
- Inspection document

- 2.5 mm Allen wrench
- Rear cover dollies (2)
- Flashlight

4.4.3 Damage Indicators

On the inside edge of the black-colored bearing assembly, a mark similar to that shown in [Figure 1-13](#) will be seen, if this is a Rotek bearing.

Figure 1-13 Gantry Bearing - Rotek Label

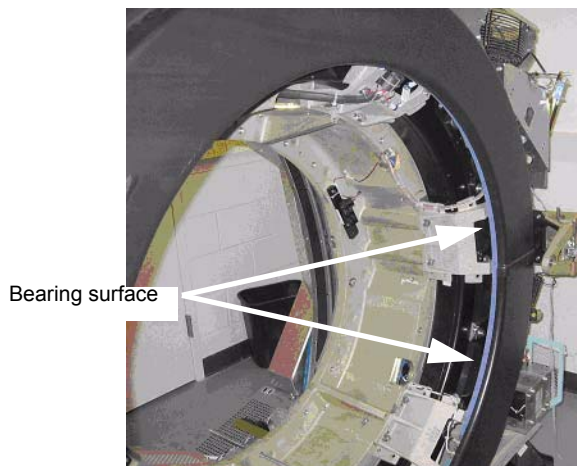


The mark has a serial number in the same format as:

ROTEK 2TF4-44E1B-MA91960-8F-2372-REV13.

The gap to be inspected is shown in [Figure 1-14](#) next to the serial number.

Figure 1-14 Gantry Bearing



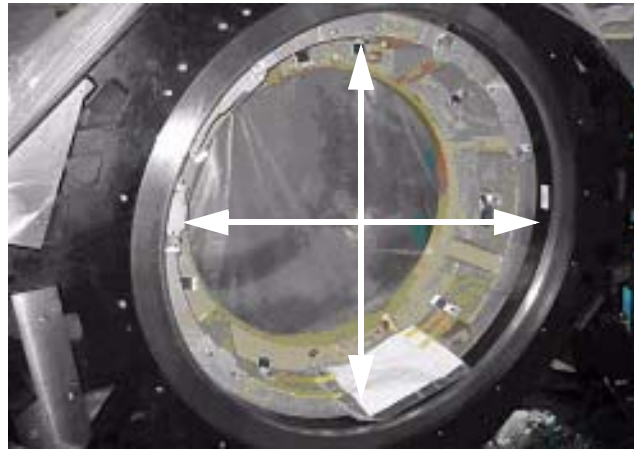
On most systems, a change in the bearing gap does not cause the gantry to make unusual sounds, unless the gap is severe. If the gantry is badly damaged and the gap is severe, it can cause operation issues. Some systems are shipped with shock indicators that must be returned to Milwaukee.

A severe failure may be seen during installation as a problem rotating the gantry.

4.4.4 Procedure

- 1.) Remove the scan window following the procedure in appendix A, [Section 6.0, on page 132](#)
- 2.) Remove the top and rear gantry covers, following the procedures in Appendix A, [Section 2.0, on page 118](#) and [Section 4.0, on page 128](#)
- 3.) Use a 2.5mm hex wrench as a tool to measure the gap at the positions shown in [Figure 1-15](#). The location of gantry components does not matter. Simply measure four (4) locations 90 degrees apart from each other.

Figure 1-15 Inspection Locations



- 4.) If the 2.5mm hex wrench easily fits without effort in the gap, the gap is out of spec. [Figure 1-16](#) shows a gap that is too large. [Figure 1-17](#) shows a gap that is good. Notice that the hex wrench does not fit in the gap in [Figure 1-17](#) but does in [Figure 1-16](#).

Note: Do not use force when putting the wrench in the gap. Either it slips in or it doesn't.

Figure 1-16 Gap Too Large



Figure 1-17 Gap Is Good



- 5.) Replace the top and rear gantry covers, following the procedures in Appendix A, [Section 2.0](#),

on page 118 and Section 4.0, on page 128.

- 6.) Replace the scan window, following the procedures in Appendix A, Section 6.0, on page 132.

4.4.5 Finalization - Mechanical Installers

If the Bearing Gap Inspection passes, complete the signoff on the GE Form e4879, Installation Data verification form, that this inspection was completed.

If the Bearing Gap Inspection fails, contact your site FE.

4.4.6 FE Service Action Required

If the Bearing Gap Inspection fails, the mechanical installer notifies the site FE that the inspection failed.

The site FE should:

- 1.) Open a bearing inspection dispatch.
- 2.) Follow the inspection procedure described in this section.
- 3.) Record the bearing inspection results.

If no damage is found, close this dispatch and continue with the electrical calibration procedures.

If the system is damaged, go to the Equipment Delivery Quality web site and follow their instructions.

To enter a damaged in shipping claim, go to this web site:

<http://egems.med.ge.com/edq/home.jsp>

4.4.7 FE Inspection Completion

- 1.) After the Gantry Bearing Inspection passes, complete the opened service dispatch with the following information:

- Gantry Serial Number
- Gantry Type
- System ID
- Site Name
- Installation date
- Was the Gantry transported to the site in the shipping crate? (Yes/No)
- Was the Gantry lifted or hoisted, were riggers used, or was the Gantry delivered via flatbed wrecker? (Yes/No)
- Number of locations that fail the gap inspection if any: _____

- 2.) Close the service dispatch.

Should any follow-up be required after this inspection, the site engineer will be contacted directly by CT Engineering.

4.5 Install Gantry Alignment Laser and Bracket

4.5.1 Tools and Test Equipment

- Standard tool kit
- Laser Alignment Kit (P/N 5148193 or 5272090)
- Tape measure
- Masking tape

4.5.2 Procedure

NOTICE Use caution while removing the gantry scan window.

- 1.) Rotate the gantry by hand until the collimator face plate is at the 5 o'clock position.

Note: With power OFF, the gantry movement is tight.
DO NOT pin the gantry during this alignment process.

- 2.) With the top and back gantry covers removed, locate the two M10 bolt holes as shown in [Figure 1-18](#). These bolt holes are used to attach the laser tool to the gantry.
 - The bolts can be installed using an 8 mm Allen wrench. Be careful not to bump the alignment light; the mounting space is tight near the alignment light. Tighten bolts until both are snug.
 - Do not drop bolts or the bar on the collimator faceplate. Attach the bar as shown in [Figure 1-18](#).
 - Using a minimum 223 mm (9 in.) level placed on the attached bar, level the bar by rotating the gantry.

Figure 1-18 Alignment Bar Installation Location



CAUTION Potential for injury.

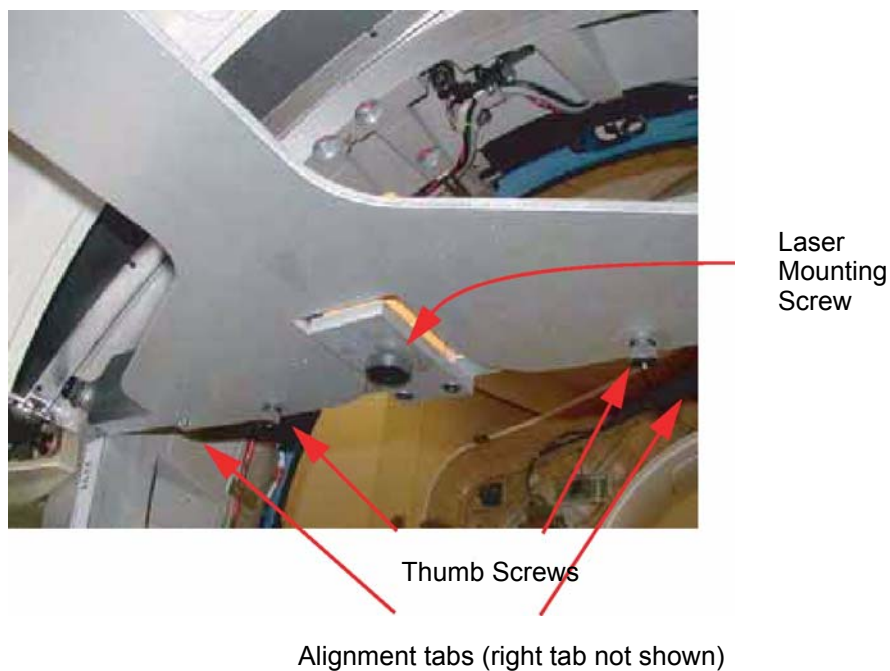


DO NOT look into the laser.

Use appropriate safety procedures when working with lasers.

- 3.) Attach the laser centering plate onto the laser mounting bar as shown in [Figure 1-19](#). The plate is attached from under the alignment bar using two fixed locators and two thumb screws.

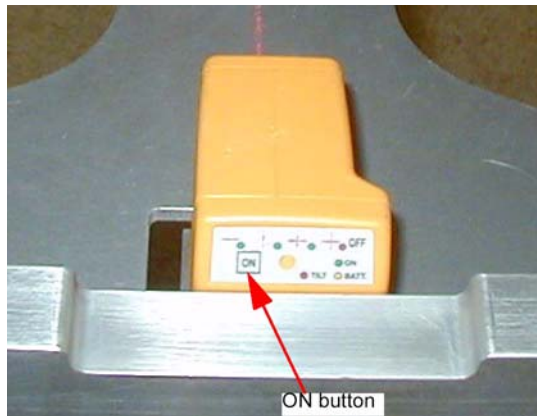
Figure 1-19 Attach Laser Center Plate



- 4.) When done, insert the laser and turn on the laser using the controls on the back. If the laser is loose when mounted, use a 2 in. piece of Velcro loop (fuzzy) section and attach it to the alignment plate over the attachment screw. Remount the laser and it should fit snugly without moving.
- 5.) When pressed, the **ON** button steps through four different beam profiles and “Self-Leveling Off”. Press the **ON** button until the “|” beam shows. It will be used for this operation.

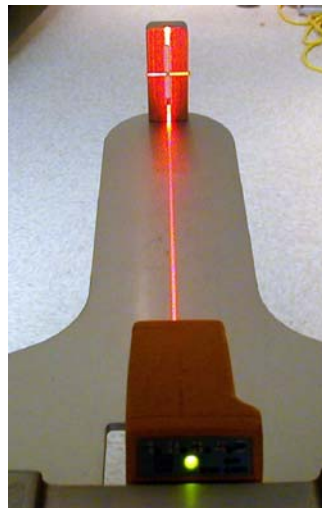
Times pressed	Function	Notes
1	—	Self-leveling on
2		
3	+	
4	Self-leveling off	Do not use

Figure 1-20 Laser ON Button



- 6.) Align the laser by carefully rotating the laser base assembly so that the "I" beam shines through the center of the alignment sight mounted on the end of the alignment plate. Note: The laser beam may be wider depending on the battery life.
- 7.) Use the locking screw on the bottom of the alignment bar to secure the laser to the bar, as shown in [Figure 1-21](#). When done, the laser should fit snugly without moving on the mounting bracket.

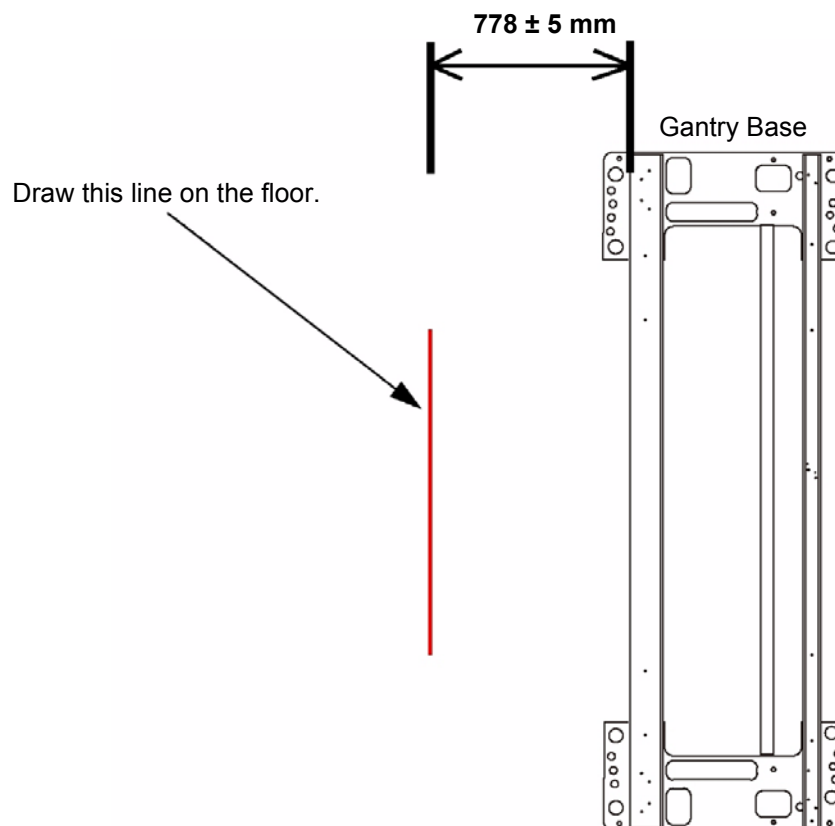
Figure 1-21 Laser Centering



Note: When tightening, the laser may move. Use caution to prevent any movement, as this can result in drilling the table anchor holes in the wrong location.

- 8.) After the laser is centered, notice that the laser beam also appears on the back wall. Place a piece of masking on the wall and carefully mark a line on the tape where the laser appears. This line is later used in the table alignment. This line is also useful in determining if the laser unit moves during the alignment process.
- 9.) Remove the alignment centering plate and store it in the alignment case.
- 10.) Using a chalk line, mark a table center line on the floor along the laser light shining on the floor.

Figure 1-22 Draw Reference Line for Table Center

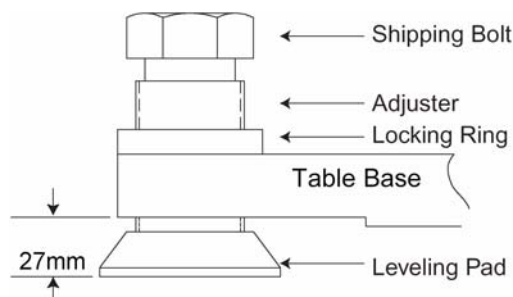


- 11.) Recheck the table-to-gantry reference line for 778 ± 5 mm Z-distance. Refer to [Figure 1-22](#).
- 12.) Turn off the laser but do not remove.

4.6 Position the Table

- 1.) Remove all the transportation packaging and boxes, except dollies, from the table.
- 2.) Wheel the table to its approximate position relative to the gantry, using the marks made earlier.
- 3.) Locate the table leveling pads and position them against the base of the table. Preset leveling pad heights to 27 mm. (see [Figure 1-23](#)).

Figure 1-23 Table Base Leveling Pads (Starting Positions)



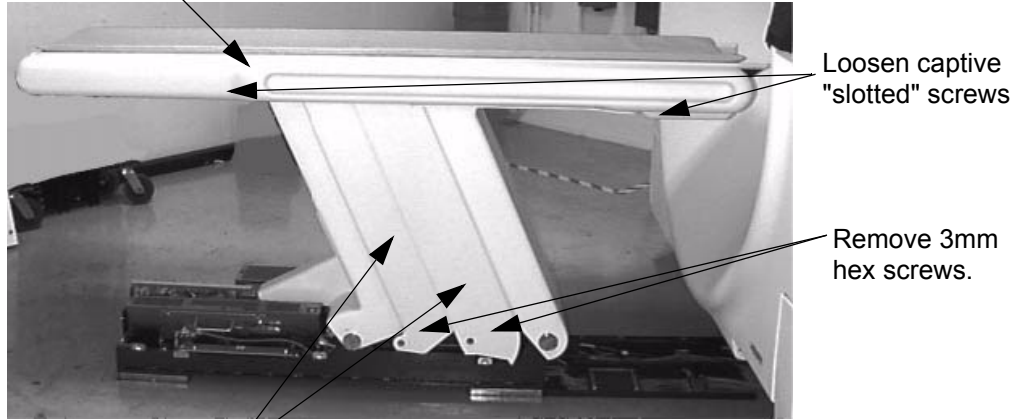
- 4.) Use the dollies to evenly lower the table until it rests on the leveling pads using an $\frac{1}{2}$ " ratchet.

- 5.) Remove the table left and right upper side covers (refer to [Figure 1-24](#)).

Note: The captive screws have a tendency to fall out occasionally.

Figure 1-24 Table Cover Removal

Remove left and right side covers.



Remove the two panels, on both sides.

- 6.) Remove the four side panels, using a 5mm hex key. Without disconnecting the ground straps at the z-channel, carefully lay the side panels on top of the padded cradle.

NOTICE

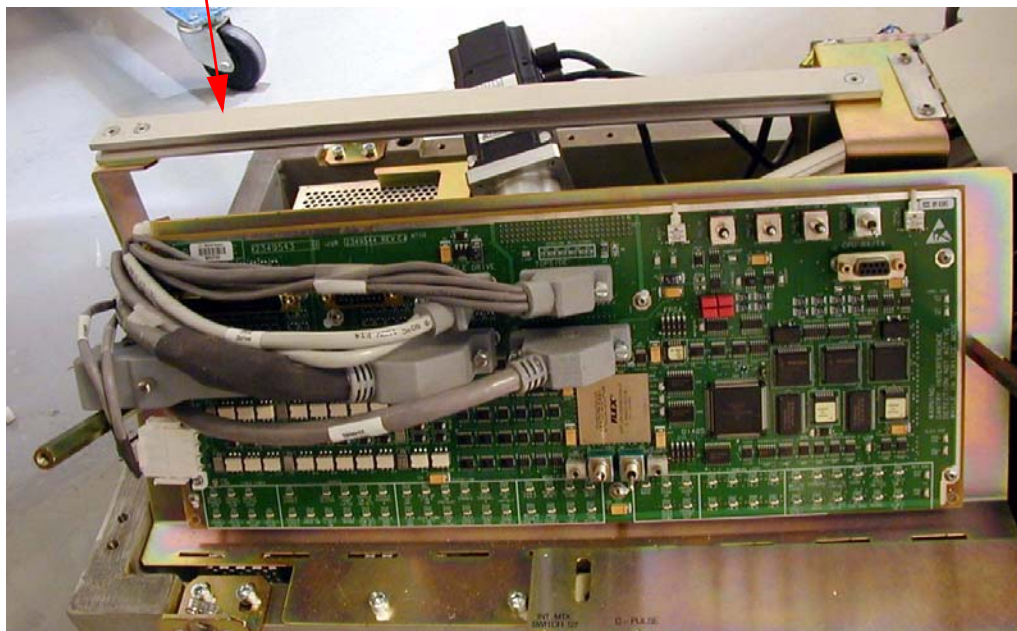
Potential for Equipment Damage.

Take extreme care to prevent damaging the MTCB printed wire assembly. Use some packing material to protect these components.

- 7.) Remove the table base cover center support bar. (See [Figure 1-25](#).)

Figure 1-25 Center Support Bar

Remove this bar.



4.7 Center line on Cradle

- 1.) Using a metric tape measure or other suitable measuring device:
 - a.) Using masking tape, place a 150mm strip on the back of the cradle. See [Figure 1-26](#).

Figure 1-26 Masking Tape



- b.) Determine the cradle center-line by measuring the cradle width. Make two points on the masking tape and make a line. See [Figure 1-27](#).

Figure 1-27 Making the Marker



4.8 Level and Center the Table to the Gantry

4.8.1 Required Tools

- Standard Install Tool Kit
- $\frac{3}{4}$ ", 1- $\frac{1}{4}$ ", 1- $\frac{1}{2}$ " and 1-5/8" sockets
- 8mm, 10mm, and 14mm hex socket bits
- 4 ft. level
- 2 ft. level

4.8.2 Alignment Conditions

- Before you start, turn on the laser and check that the beam is still on the mark placed on the wall. If not, reset the laser.
- If the mark is not present, use a measuring tape and place a 100 mm piece of masking tape on the cradle at the 1000 mm and on the laser line.
- The gantry must be at zero degrees.

4.8.3 Procedure

NOTICE Avoid leaning on the cradle during this procedure.
DO NOT pin the gantry during this alignment process.
This procedure is for systems mounted on 102 mm (4 in.) concrete floors only!

Note: If the floor covering was not properly removed with the glue removed, or the levelers were not centered over the floor cutouts, the leveler may become trapped against the edge of the floor covering, causing the table to become unlevel. If this happens, move the table and enlarge the 102 mm (4 in.) floor cutout for the table. Glue removal is important and aids in moving the table to its final location in accordance with the floor levelness specification.

- 1.) Have the table side panels removed and a ratchet, 1-1/8 in. socket, and a 2 ft. level ready to use.
- 2.) Turn on the laser's "I" beam (vertical beams) by pressing the **ON** button 2 times.

Note: [Step 3](#) through [Step 7](#) are for perpendicular positioning of the cradle to the gantry.

- 3.) The table on the dollies should be resting on the floor, and the laser beam visible on the cradle. The laser light should now shine down the cradle onto the rear vertical target. Moving the table on the dollies by raising and lowering makes it easier to center the table right-to-left.

Note: When using the table dolly to move the table, be sure that the shipping bolts are still attached to the adjuster leveler feet. This prevents the adjuster levelers from gripping on the floor adhesive, making it difficult to move.

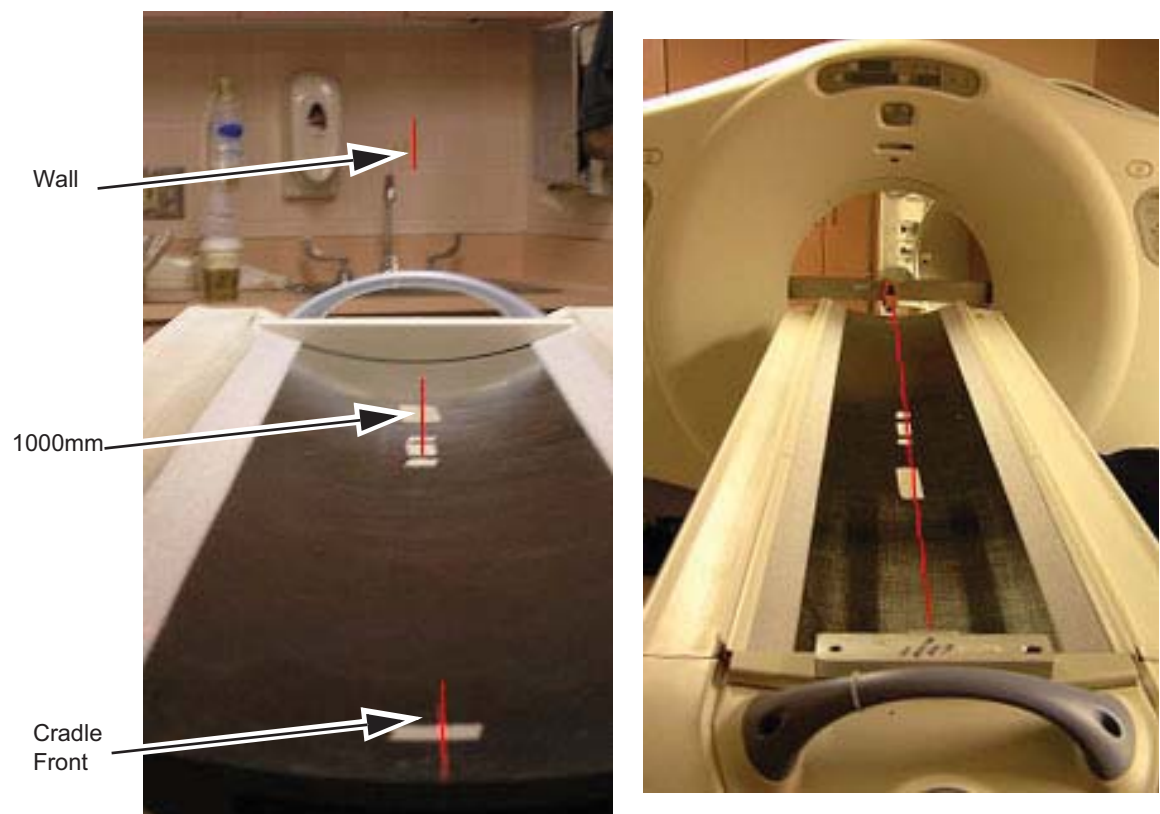
- 4.) Move the table so that the base is roughly centered over the scan center line, the front edge of the table base is on the 778 ± 5 mm line, and the table is resting on the floor. Check that the leveling feet are centered in the cutout circles.
- 5.) Center the cradle on the drive rollers assembly by pushing the cradle into the gantry to its maximum position (1400 mm) and back to just before the latch position six times. The cradle should be centered as shown in [Figure 1-28](#).

Figure 1-28 Centered Cradle



- 6.) Carefully move the table so that the cradle front center line and the back center line are aligned. You may need to raise the table to move it. When aligned, lower the table to the floor.
- 7.) If not already done, measure 1000 mm from the front of the cradle, and place a piece of tape under the laser center line. Carefully mark a line along the laser line when the table is centered.
- 8.) The laser beam should now connect the cradle front centerlines, the 1000 mm cradle center line, onto the rear center line, and finally onto the alignment centering mark placed on the wall, when properly centered along the center line. Use the centering alignment line on the wall to be sure the laser is still centered. If the alignment line on the wall is NOT on the original mark, readjust the laser and repeat the above steps (see [Figure 1-29](#)).

Figure 1-29 Alignment Laser Marks - Table & Wall



Note: [Step 8](#) through [Step 10](#) are for front-to-back and side-to-side leveling of the cradle.

- 9.) The table should be completely on the floor and resting on all 5 levelers.

Figure 1-30 Table Level Alignment

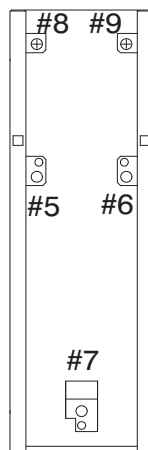
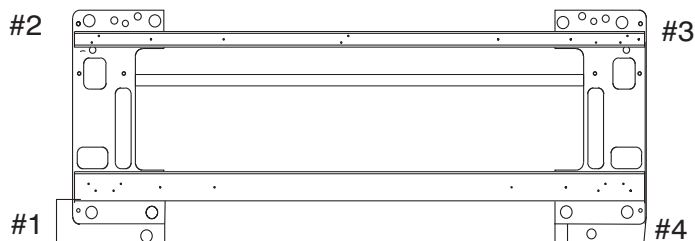


- 10.) Raise or lower the table as needed using the front and rear levelers and level the cradle horizontally in the front and back locations (refer to [Figure 1-30](#)).
- First, horizontally across the cradle at the front of the cradle.
 - Second, horizontally across the cradle at the 1000 mm mark on the cradle.
 - When the cradle is roughly level in both locations, leave the level on the cradle horizontally.
- 11.) Level the cradle front to back using a 4 ft. level placed in the center of the cradle. These two leveling actions often conflict and a few iterations may be required. This process is complete when:
- The cradle is still centered on the front, mid, and rear marks.
 - The cradle is leveled horizontally (across table) front and back (with the bubble centered between the lines.)
 - The bubble is centered between the lines in the Y direction along the length of the cradle.
 - The laser is still centered on the wall center line.
 - The table is still on the 778mm line and the levelers are resting on the flooring.
 - The laser is the same as in [Step 7](#).
- Note: The leveling process may take several iterations of [Step 1](#) through [Step 10](#). Patience and accuracy is required to properly complete this process.
- 12.) When completed, turn off the laser tool.
- Note: Do not remove the table dollies.

4.9 Tighten the Lock Rings

- 1.) Re-check gantry bubble levels.
- 2.) Re-check that bubbles levels on the table.
- 3.) Re-check that adjusters at Locations 1, 2, 3, 4, 7, 8 and 9 carry a portion of the load.

Figure 1-31 Table Base Gantry/Table Level Locations



- 4.) Re-check that gantry "inside" levelers carry a portion of the load.

CAUTION



Eye protection is required when using a hammer and chisel.

- 5.) Tighten the lock rings at Locations #1 through #9 with the spanner where possible. Use a hammer and chisel to tighten the lock rings only where you can not use the spanner.

4.10 Drill the Anchor Holes

WARNING POTENTIAL FOR PATIENT INJURY.



**IMPROPERLY SECURED TABLE MAY TIP, DISLODGING PATIENT.
PROPER ANCHORING IS KEY TO MAINTAINING PATIENT SAFETY DURING
SYSTEM OPERATION.**

4.10.1 Notes to Mechanical Installers

GE-provided floor anchors are designed for use **ONLY** on concrete floors that meet the 4-inch concrete floor requirement. Supplied floor anchors must be installed by a trained contractor, and shall be set to a minimum depth of 3 inches at each anchor point. **ANY** anchors having more than 1-inch of thread showing above the nut, when torque is set to 55 lb.-ft, shall have a second anchor installed in the closest adjacent hole. This is because the minimum anchor engagement length in the concrete was not met. The second anchor shall be installed to the standard depth and torque specification. **Do not cut anchor bolts that extend longer than the 1-inch limit.**

4.10.1.1 Note 1: Alternate Anchoring

If at least four anchors cannot be set for the gantry, and at least three anchors for the table, then the installer must inform GE that the minimum anchoring cannot be met. Additionally, a structural engineering contractor must be engaged to determine the anchoring method and to certify that their anchoring meets the stated GE minimum load requirement and torque specification.

4.10.1.2 Note 2: Non-Concrete Floors

All other anchoring methods — on floor types other than the concrete minimum — must be determined at the customer's expense, by a structural engineering contractor. The anchoring method must be certified to meet the stated GE minimum load requirement and torque specification.

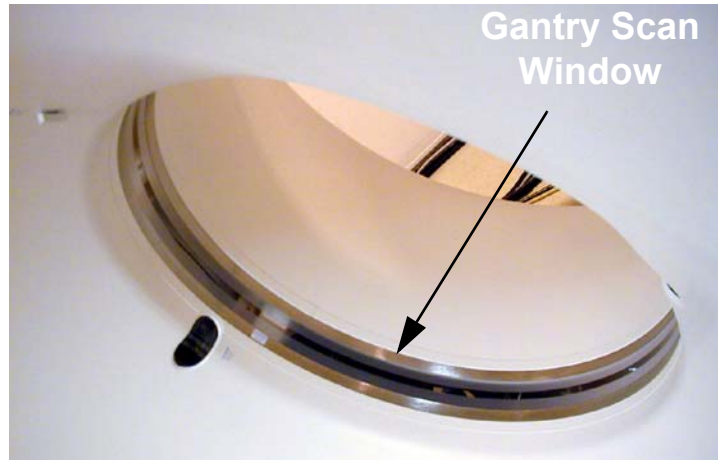
4.10.1.3 Note 3: GE Notification

It is not the role of the mechanical contractor or installer to determine acceptable methods to install or anchor equipment on non-4-inch concrete floors. The appropriate GE contact person shall be notified that the facility's floor type **DOES NOT MEET** the installation mounting requirement for the installation procedure (described in this Installation Manual), and therefore the table-gantry mounting process **CANNOT** continue.

4.10.2 Drilling Procedure

- 1.) Make certain that the gantry scan window is in place, to prevent dust from entering the gantry.

Figure 1-32 Gantry Scan Window



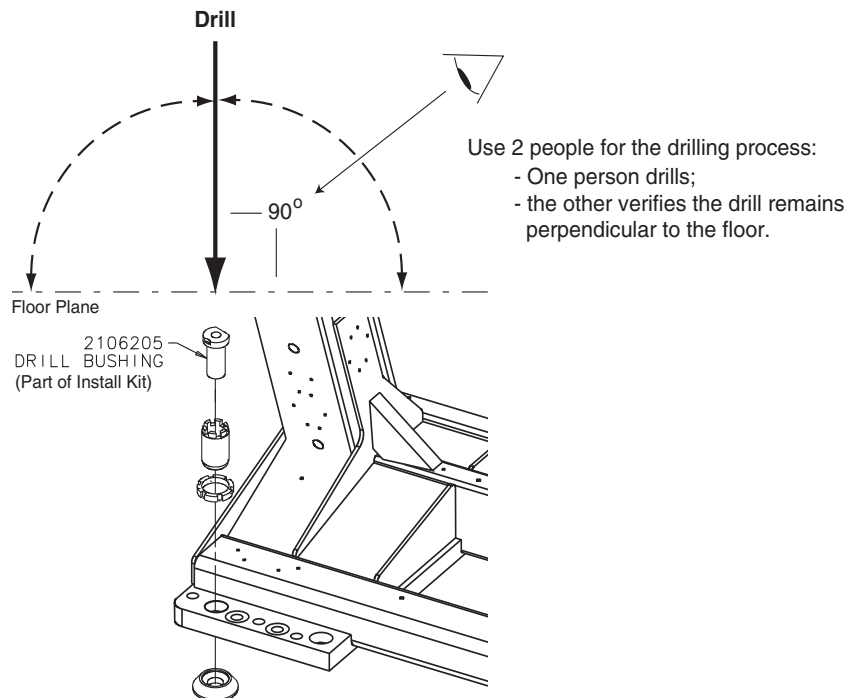
- 2.) Use a piece of tape to mark the drill bit depth of 190 mm ($7\frac{11}{16}$ ") from the tip of the $\frac{1}{2}$ " masonry drill bit.

NOTICE
Potential for
Equipment
Damage from
Dust

You must cover all electronic assemblies in the table base prior to drilling to prevent damage due to the dust created during drilling.

- 3.) Use the $\frac{1}{2}$ " bit to drill all seven (7) anchor holes (refer to [Figure 1-35](#) for locations.)
 - Review [Figure 1-33](#), prior to drilling.

Figure 1-33 Drilling Position



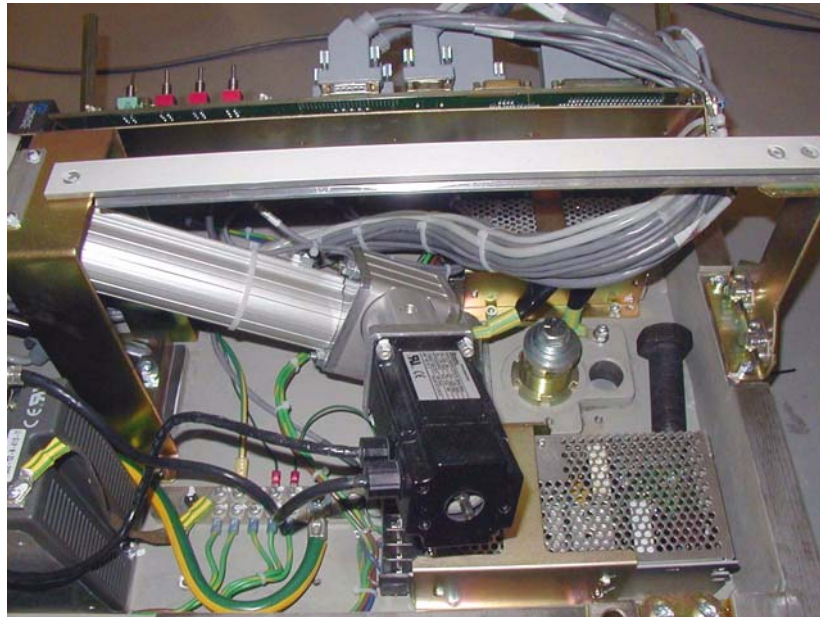
- **Important** - Drill the holes perpendicular to the floor. While one person drills the holes, position a second person to watch the relationship between the drill bit and floor. Make

sure the bit remains absolutely perpendicular to the floor throughout the drilling operation. Always use the mechanical guide supplied with the alignment kit.

- Stop drilling every 15 or 20 seconds and clear the hole of debris. This helps to prevent binding of the drill bit.
- a.) Place appropriate protection to prevent damage and dust contamination to electronic assemblies. Refer to [Figure 1-19, on page 45](#), for placement of dust protection materials.
- b.) Drill each hole until the mark on the drill bit is even with the top of the drill bushing. (195mm)
- c.) Place the drill bushing inside each adjuster, to keep the hole vertical, and centered within the adjuster.
 - * Use the drill bushing to center the anchor holes in adjuster Location #5, #6, and #7, to provide maximum lateral alignment capacity when you center the cradle on isocenter during subsequent system testing.
 - * Take care not to injure yourself on the gantry cover brackets.
 - * Take care to avoid the actuator motor lead when you drill the Location #7 anchor hole. Remove the lead if necessary.

Note: An 18" bit will make drilling anchor location #7 easy, and will prevent component damage (see [Figure 1-34](#)).

Figure 1-34 Table Assembly Showing Actuator Motor and Anchor Location #7



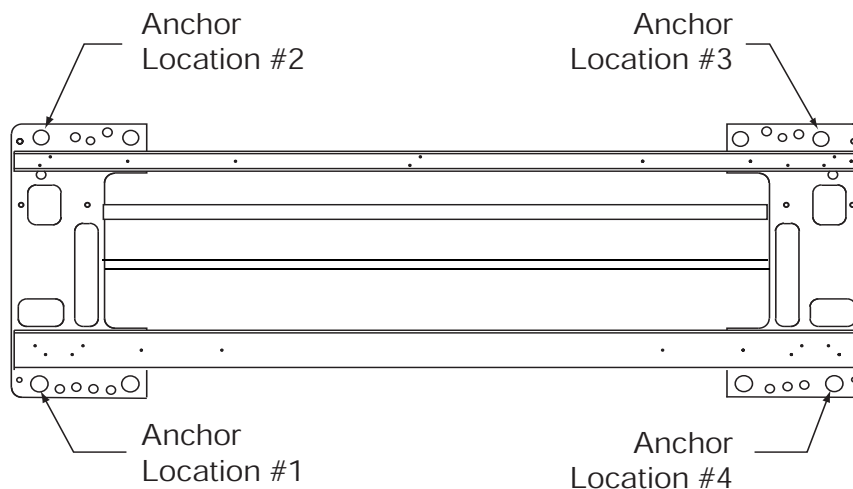
- d.) Stop the drill every 15 or 20 seconds and clear the hole of debris to prevent binding of the drill bit.
- 4.) Vacuum all debris from the inside of the hole:

Note: **A drywall dust filter should be used on the vacuum.**

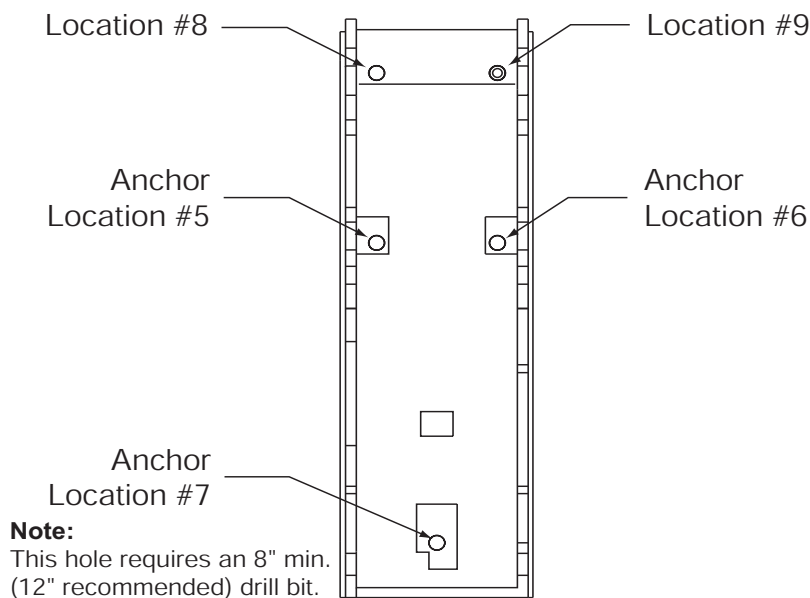
- a.) Place the funnel tip inside the hole; place the vacuum hose in the funnel.
- b.) Continue to vacuum while you drill, if you can, to keep gantry and table as free of dust contamination as possible.
- c.) When you finish clearing the anchor hole, vacuum the debris from the surrounding area.
- d.) Stop drilling frequently to let the drill bit cool.
- e.) All holes must be a minimum of 108 mm deep (see [Figure 1-37, on page 59](#)). Recheck all holes.

- 5.) Do not drill anchor holes at Location #8 and #9. These locations help support the weight, but do not anchor the table into place.

Figure 1-35 Anchor Locations, Standard



Note: Anchor holes in base are 7/8" (2.3 cm).



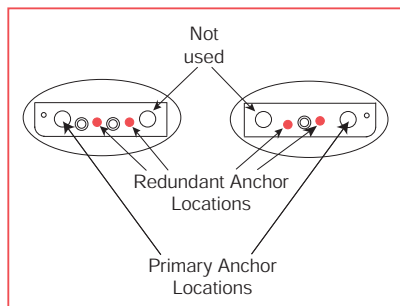
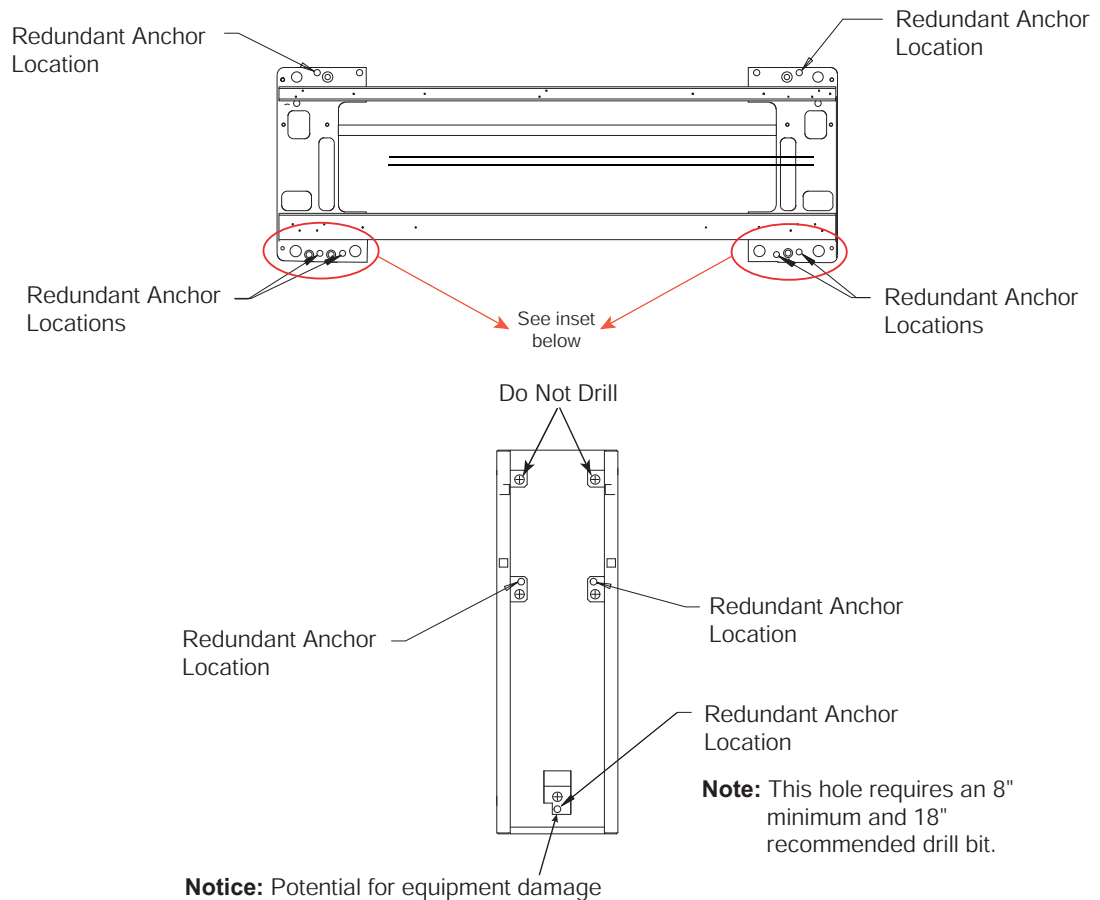
4.11 Gantry & Table Redundant Anchor Holes

If you cannot use one of the adjuster anchor holes due to structural interference, such as reinforcement bars in the concrete, you must use one of the redundant anchor locations, as shown in [Figure 1-36](#).

- The gantry requires a minimum of four (4) anchors, one (1) in each corner.
- The table requires a minimum of three (3) anchors, one (1) at location #5, #6 and #7.

If you must use a redundant anchor hole in the gantry, you must remove the gantry covers. See [Appendix A](#), for gantry cover removal.

Figure 1-36 Table Base Redundant Anchor Locations

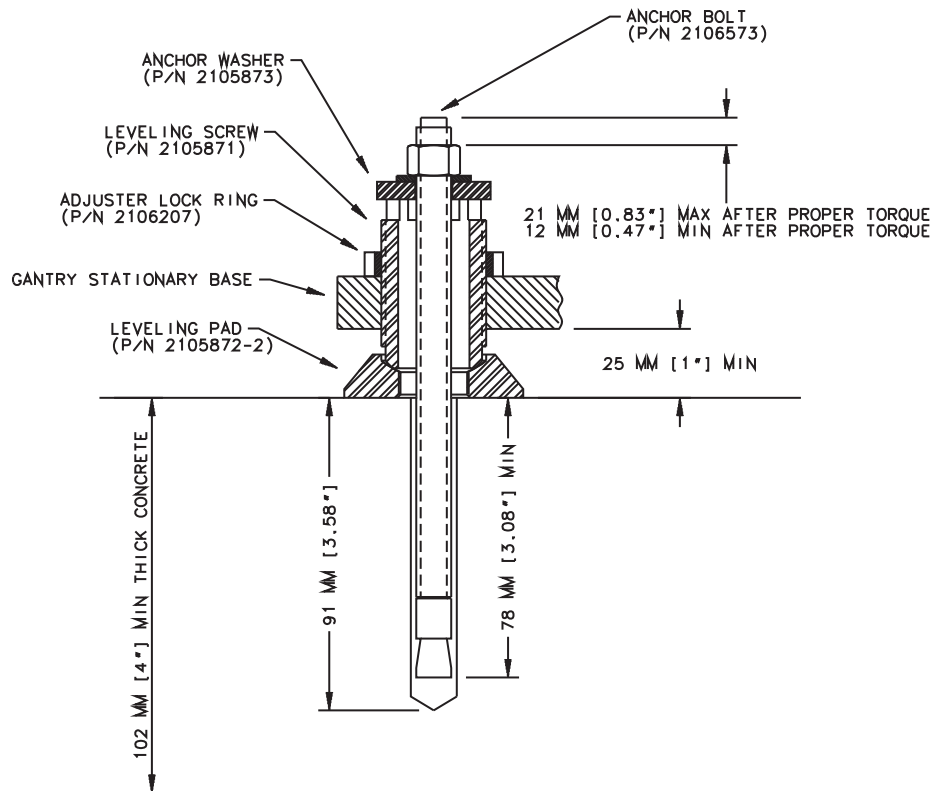


Mark the ½" drill bit with a piece of tape 170 mm (6¹¹/₁₆") from the tip of the bit, and follow the instructions in [Section 4.10, on page 54](#).

4.12 Install the Anchors

Recommended - Use "Hilti Kwik-Bolt II" anchors P/N 2106573 (½" dia. by 8" long) as shipped with the BrightSpeed for this procedure.

Figure 1-37 Table Base Anchor Assembly



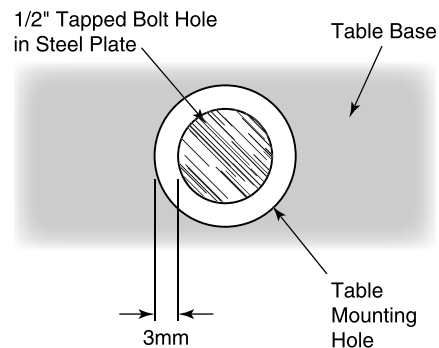
- 1.) Remove the three alignment bars and two attachment bars and repack the install kit.
- 2.) Assemble the anchors before you install them. Refer to [Figure 1-37](#).
 - a.) Remove the nut and washer from the anchor.
 - b.) Add a ¼" thick washer (PN 2105873) under the regular anchor washer.
 - c.) Reassemble the anchor washer and nut and position nut so top is flush with threads of anchor before hammering into hole.

Figure 1-38 Measure Table Base Anchor Thread



- 3.) Fixed site Installations must be centered under mounting holes in table base, with 3mm of space visible surrounding the tapped bolt hole. [Figure 1-39](#).

Figure 1-39 Center tapped holes under mounting holes in table base



Bolt centering is important to provide $\pm 3\text{mm}$ of adjustment for electrical alignment. Always use the drilling centering tool when drilling all bolt holes.

CAUTION



Eye protection is required when using a hammer.

- 4.) Refer to [Figure 1-37](#) for instruction on inserting the anchors in each hole. Using the anchor seating tool, hammer the anchor downward to *completely insert the anchor in the hole*.
- 5.) Once alignment has been verified, torque all mounting bolts. Tighten the Location #1 through #7 anchors and torque to $75 \pm 6 \text{ N-m}$ ($55 \pm 5 \text{ ft.-lb.}$).
- 6.) Replace all the table parts you removed during installation, including covers, ground strap and the center support bar. Check the motor lead.

Note:

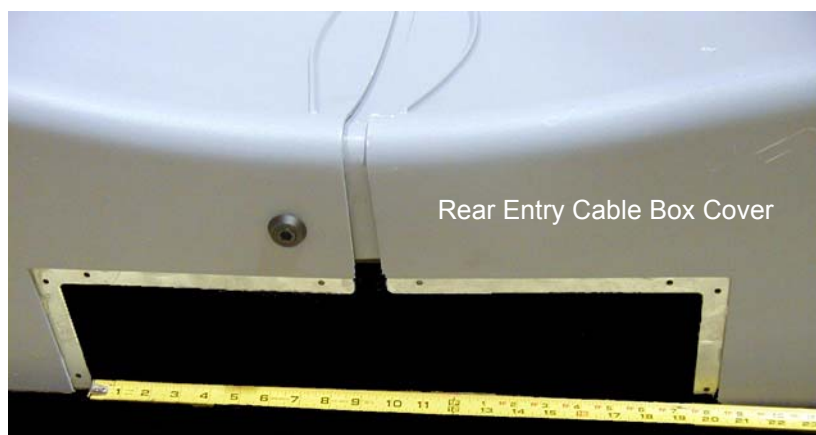
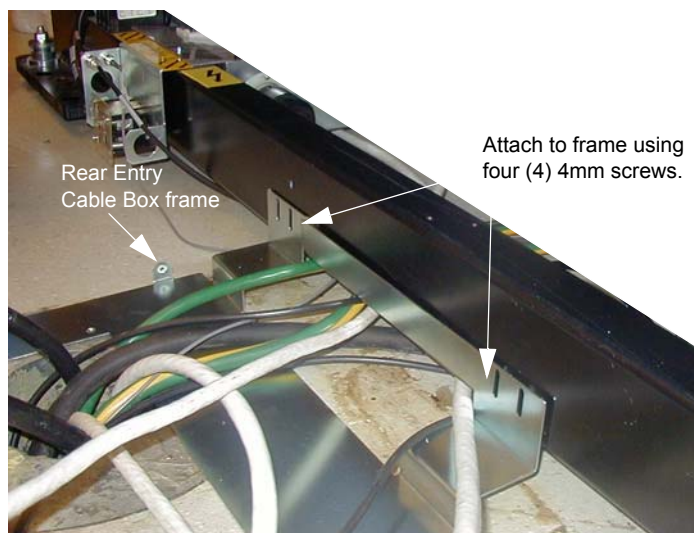
If you cannot replace the lower table cover because the floor interferes, adjust all of the table and gantry levelers by $\frac{1}{2}$ turn increments to raise the table/gantry until the lower table covers clear the floor. (Refer to [Figure 1-40, on page 61](#), for table base cover clearances.) Return to Sections [4.3, on page 38](#), [4.11, on page 58](#) and [4.9, on page 53](#) to level the gantry, level the table, and tighten the locking rings, respectively.

Section 5.0

Rear Entry Cable Box

A rear entry cable box (B7850RC) is used when the cables to the gantry cannot be brought-up inside the gantry base. The box is not supplied with the system and must be ordered separately. The rear entry cable box frame attaches to the gantry base. See [Figure 1-40](#).

Figure 1-40 Rear Entry Cable Box



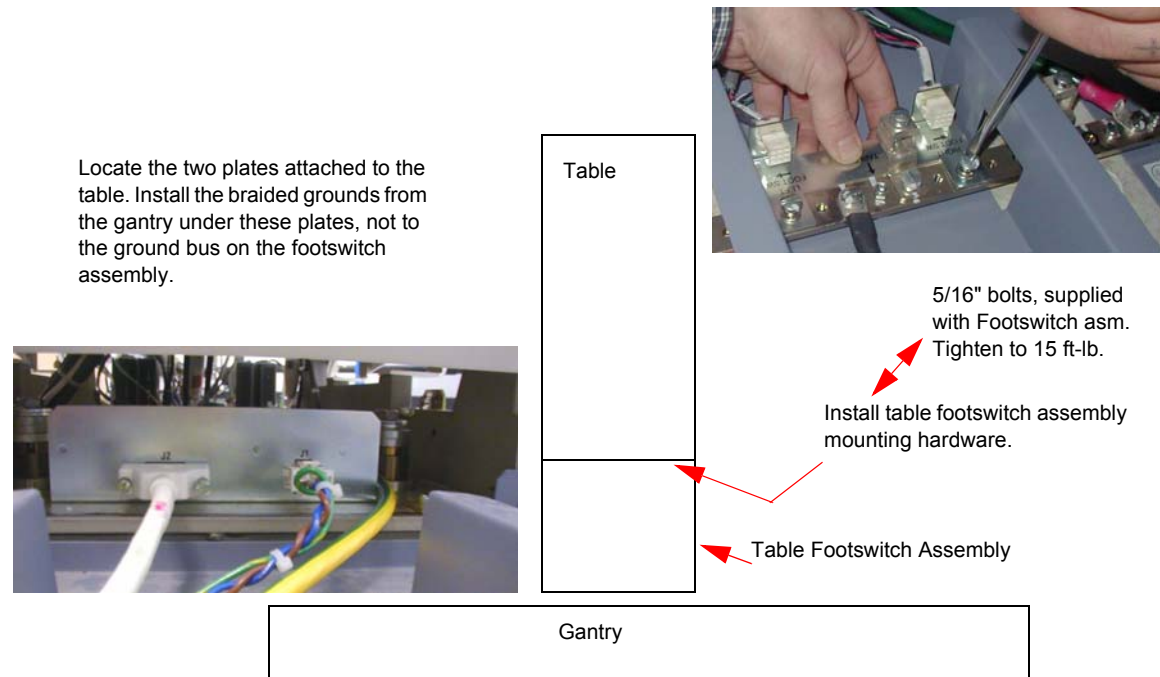
The filler plate may need to be cut in order to fit the rear cable box into the cover opening. Use the supplied white molding around the filler plate.

Section 6.0

Install Table Footswitch Assembly

Install the table foot-switch assembly as shown in [Figure 1-41](#), along with filler strip.

Figure 1-41 Install Table Foot-switch Assembly

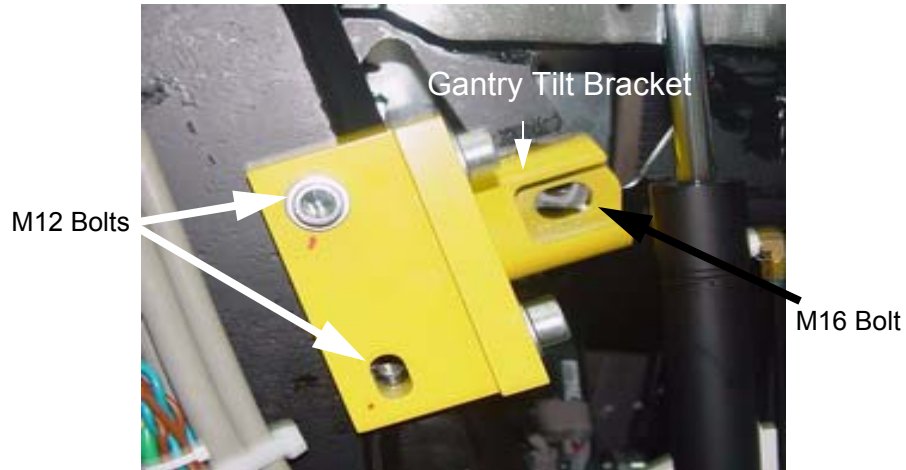


- 1.) Remove foot switch assembly, and locate hardware.
- 2.) Loosen six (6) hex head screws and remove footswitch assembly.
- 3.) Place footswitch cover in a secure place.
- 4.) Locate the supplied 2' velcro in the install support kit, and apply two strips to the bottom of the footswitch assembly.
- 5.) Using the two 5/16" bolts and blue loctite (242), attach the footswitch assembly.
- 6.) Remove the 2" braided strip and gantry cable that are secured to gantry base, and place in the cable tray.
- 7.) Remove three (3) bolts that secure the footswitch bar.
- 8.) Take off the footswitch bar and discard.
- 9.) Locate the new footswitch bar from the install support kit.
- 10.) Install footswitch bar on cover, using the same three screws.

Section 7.0

Remove Gantry Tilt Bracket

Figure 1-42 Gantry Tilt Bracket Removal



- 1.) Refer to [Figure 1-42](#). Remove the M12 bolts using a 12 mm Hex wrench.
- 2.) Loosen the M16 bolt 1-2 turns and check the Gantry tilt bracket, it should be loose to the touch. If loose continue with step 4.

CAUTION



If tilt bracket is not loose, stop and put the M12 bolts back in and tighten tilt bracket back in place. If there is a load on the tilt bracket, removal may cause the gantry to suddenly tilt all the way back due to a possible lack of hydraulic pressure.

- 3.) Check the hydraulic connections for leaks or lack of fluid. You will have to wait until the system can be energized to use the tilt controls to relieve the load on the tilt bracket prior to removal. Do not use force to remove the bracket.
- 4.) If the bracket feels loose, remove the M16 bolt using a 16 mm Hex wrench.
- 5.) Remove the bracket.
- 6.) Close the gantry covers and reinstall the scan window.
- 7.) Store brackets in the gantry base.

Section 8.0

Position the Power Distribution Unit

WARNING



LOCKOUT/TAGOUT IS REQUIRED BEFORE PERFORMING THIS TASK. USE THE SUPPLIED LOTO KIT.

- 1.) Roll the PDU into position on its permanently mounted casters. Leave at least 15.5 cm (6") between the PDU and back wall to allow cooling air to circulate.

Note: Connecting the primary incoming power (steps 2 through 5, below) is performed by the customer's electrical contractor.

Table 1-4 Contractor Connections

Connection or Wall Box	AWG #	Connection From	Connection To PDU	Installed & Checked
TS1	#1	PDB-A	TS1-1	
	#1	PDB-B	TS1-2	
	#1	PDB-C	TS1-3	
	#1/0	GND	N/G (Do NOT connect anything to neutral point.)	

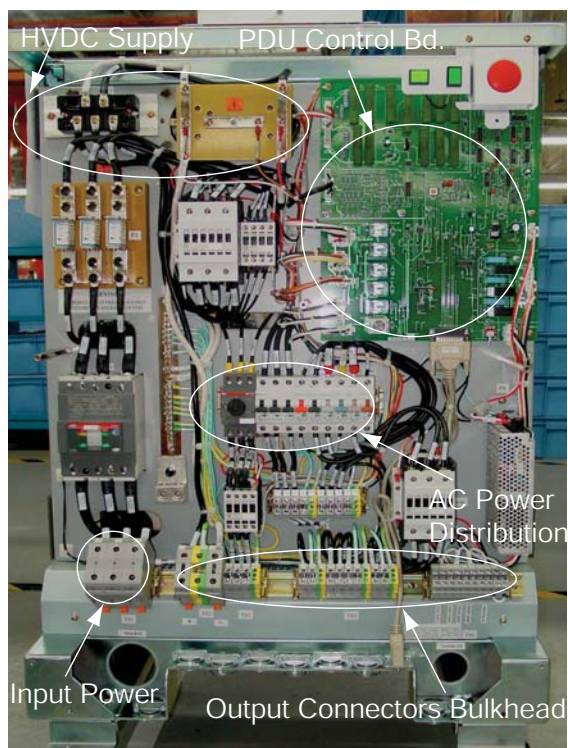
- 2.) Run the main input power conductors and ground through flexible metal conduit (attached between the PDU chassis and room duct-work) so you can move the PDU away from the wall during service.

Figure 1-43 Flexible Conduit for PDU Power



- 3.) Locate the hole cover plate in Box 1 and attach the flexible metal conduit to the PDU.
- 4.) Cut the three phase and 1/0 ground wire to size.
- 5.) Attach cables as shown in [Figure 1-43](#) and [Figure 1-44](#).

Figure 1-44 PDU Area Locations



Section 9.0

Install Operator Console Without FWS Table

9.1 Unpack Console

Figure 1-45 Console boxed on skid



- 1.) Remove all items from the console.
- 2.) Remove all packing materials and discard.
- 3.) Place the step-board under the front edge of the skid and step on it to raise the front edge of the skid as in [Figure 1-46](#).

Figure 1-46 Step-board used to raise front edge of skid



- 4.) Remove the two front cushions from the bottom of the skid. Refer to [Figure 1-47](#)

Figure 1-47 Cushion on bottom of skid



- 5.) Remove the Seismic Brackets from each side of the console. Refer to [Figure 1-48](#). Save brackets for use later if you need to mount the console to the floor.

Figure 1-48 Removing the Seismic Brackets



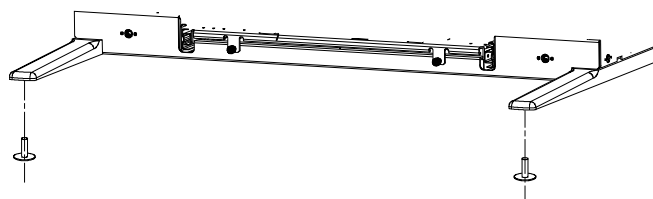
- 6.) Lift up on the strap on the front of the step-board ([Figure 1-46](#)) to lower the skid. Remove the step-board.
- 7.) Ensure the console stabilizers are in line with the notched portion at the front of the skid. This will allow enough clearance to smoothly roll the console down the ramps.
- 8.) Move console to installation location.
- 9.) Install the anti-tip legs of the console: Mount the two anti-tip legs on the console: 2318681-2 (left), 2318681-3 (right), secure three hex screws from console side.

Figure 1-49 Mount anti-tip leg on the console



- 10.) Attach the two pads to the console stabilizers. Refer to [Figure 1-50](#).

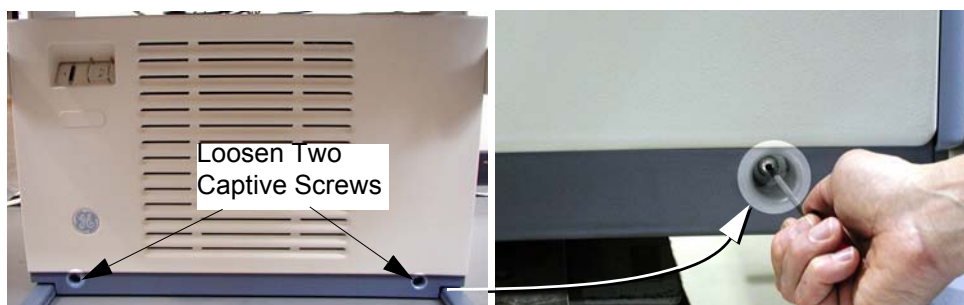
Figure 1-50 Console Stabilizer Pads



- 11.) Screw the pads all the way into the legs

9.2 Remove Console Covers

Figure 1-51 Front Cover Removal - Global Consoles



- 1.) Loosen two captive screws at bottom of console.
- 2.) Rotate bottom of cover outward and upward until it may be lifted free of the console at the top.

9.3 Adjust Table Top Height and Position Console

- 1.) Refer to [Figure 1-52](#). The console normally arrives with the bottom hole on the monitor top aligned with the fourth bolt from the bottom.

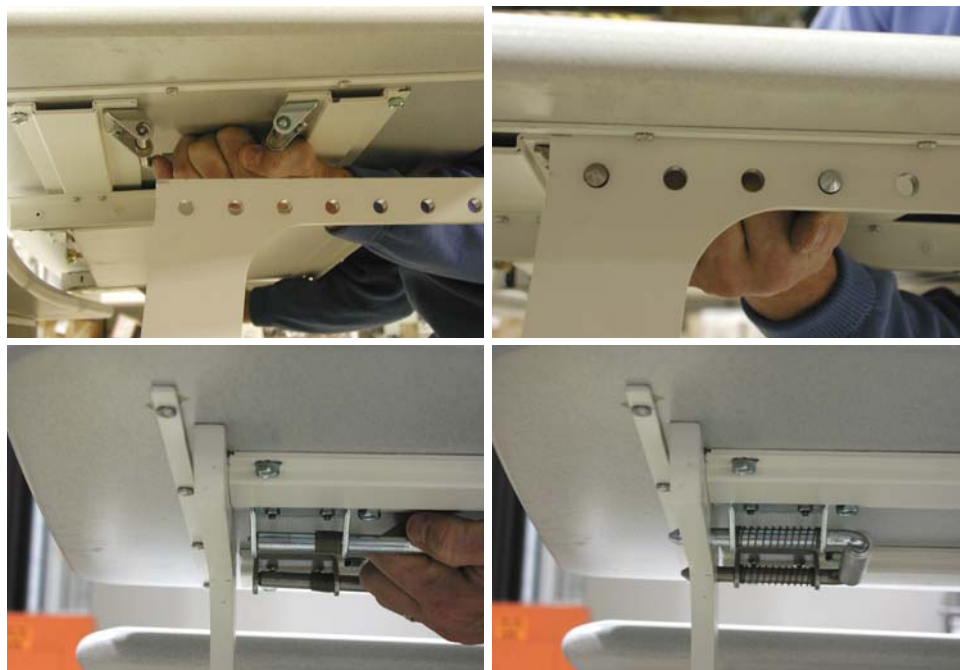
Figure 1-52 Global Console tabletop adjustments



- 2.) For optimum operator comfort, align the bottom hole of the keyboard table top with the fifth hole from the bottom on the console.
Your site may have different requirements and you may have to adjust the monitor top and/or keyboard table top up or down from this position.
 - Always select a table top height that permits the operator to see the patient on the table.
 - Keep the one bolt-hole relationship between the monitor top and the keyboard table top. Fasten the keyboard table top one (1) hole lower than the console table top (Figure 1-52).
- 3.) Install the console side covers.
Side covers are left and right specific.
- 4.) Move the console into its final position, maintaining a 6" (15 cm) minimum distance between the console and the wall.

9.4 Attach Keyboard Table Top

Figure 1-53 Keyboard Table Top Placement

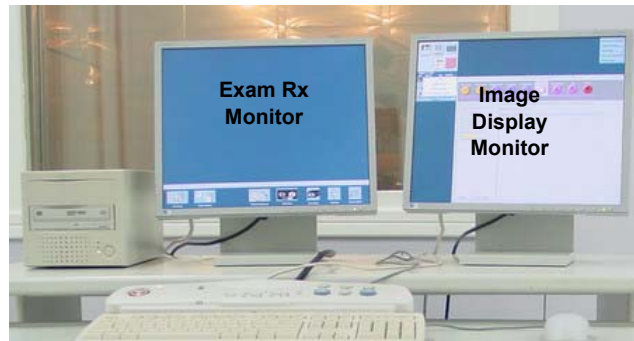


- 1.) Locate the spring loaded latches at either end of the underside of the keyboard table top.
- 2.) Pull latches inward (toward center of table top).
- 3.) Position table on brackets so that latches align with holes.
- 4.) Release latches, making certain that the pins fully engage the bracket holes (pins should protrude beyond outside edge of bracket).

9.5 Monitor and SCSI Tower Placement

- 1.) Locate and unpack the two monitors.
- 2.) Carefully place the monitors onto the monitor arms the console desktop.
- 3.) Locate and unpack the SCSI drive tower.
- 4.) Place the SCSI drive tower on the console desktop. A suggested placement is illustrated in [Figure 1-54](#).

Figure 1-54 Global Console tabletop component placement



Section 10.0

Install Operator Console With FWS Table

10.1 Unpack Console

- 1.) Remove all items from the console.
- 2.) Remove all packing materials and discard.
- 3.) Place the step-board under the front edge of the skid and step on it to raise the front edge of the skid as in [Figure 1-46](#).

Figure 1-55 Step-board used to raise front edge of skid



- 4.) Remove the two front cushions from the bottom of the skid. Refer to [Figure 1-47](#)

Figure 1-56 Cushion on bottom of skid



- 5.) Remove the Seismic Brackets from each side of the console. Refer to [Figure 1-48](#). Save brackets for use later if you need to mount the console to the floor.

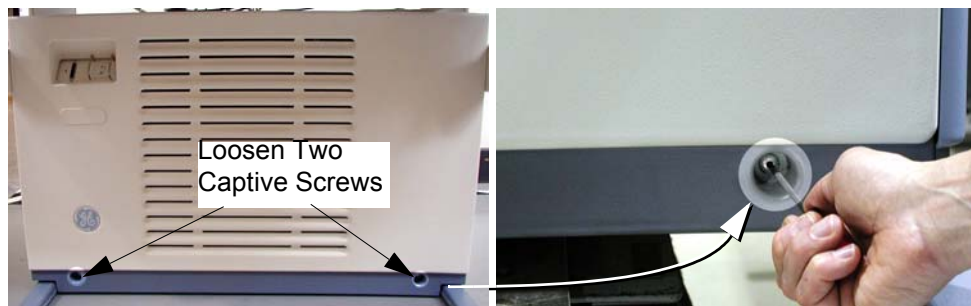
Figure 1-57 Removing the Seismic Brackets



- 6.) Lift up on the strap on the front of the step-board ([Figure 1-46](#)) to lower the skid. Remove the step-board.
- 7.) Ensure the console stabilizers are in line with the notched portion at the front of the skid. This will allow enough clearance to smoothly roll the console down the ramps.
- 8.) Move console to installation location.

10.2 Remove Console Covers

Figure 1-58 Front Cover Removal - Global Consoles



- 1.) Loosen two captive screws at bottom of console.
- 2.) Rotate bottom of cover outward and upward until it may be lifted free of the console at the top.

10.3 Replace Console Table Top With Black Table Top

- 1.) Refer to [Figure 1-59](#). The console normally arrives with the bottom hole on the console table top aligned with the fourth bolt from the bottom.

Figure 1-59 Console Table Top



- 2.) Replace the Monitor table top with the black table top:
 - a.) Remove four (4) screws with 6mm hex tool at both right and left side of the console to remove monitor table top together with the support bracket.
 - b.) Put the table top on the ground with the bracket upward. Remove six (6) screws with 6mm hex tool to remove the bracket from the table top.
 - c.) Put the black table top on the ground with table surface downward. Install the bracket on black table top with original six (6) screws.
 - d.) Install the black table top on the console with the revised order in step a).
- 3.) Install the console side covers. Side covers are left and right specific:
2315656-6: Freedom workspace left side cover for console chassis
2315656-5: Freedom workspace right side cover for console chassis
- 4.) Move the console into its final position, maintaining a 6" (15 cm) minimum distance between the console and the wall.

10.4 Install FWS table and Monitor Arms

- 1.) Assemble FWS table.
Refer to [Appendix B, on page 137](#) for details of FWS and monitor arms assembly.
- 2.) Install LCD Monitor and install monitor arms.
- 3.) Place the FWS table at one side of the console.

NOTICE FWS table should be place side by side with the console considering that the extended cable length used between FWS and Console is limited within 3 metre.

10.5 Peripherals Placement

- 1.) Place keyboard, SCIM
- 2.) Locate and unpack the SCSI drive tower.
- 3.) Place the SCSI drive tower **on the console desktop**.

Section 11.0

Seismic Mounting

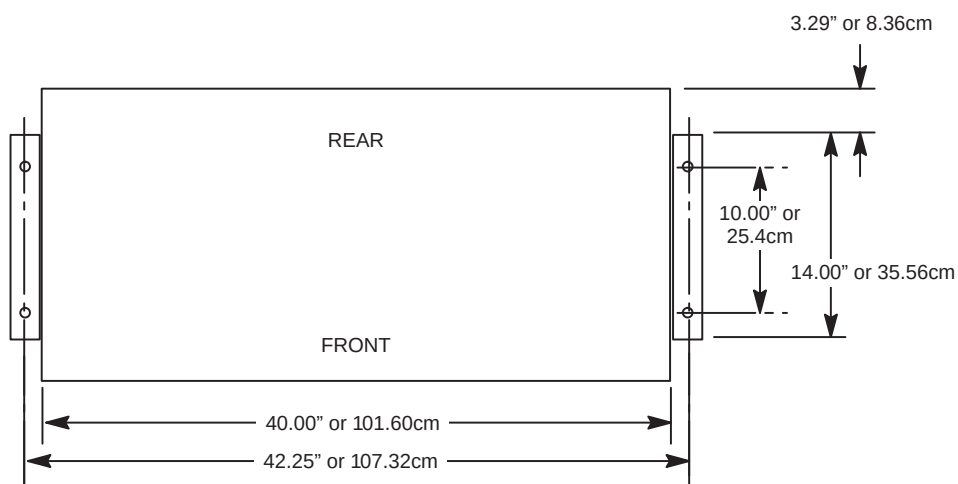
Seismic kits/brackets for console, Power Distribution Unit and Freedom WorkSpace (FWS) are available shipped with the subsystem or in Shipping Collector.

Note: See [Appendix B](#) for FWS seismic kit installation.

11.1 Console

If site specifications require seismic mounting, use ½" bolts to mount the brackets to the floor. Refer to [Figure 1-60](#) for hole placement. The console seismic brackets are shipped with the console. They are used on the console shipping skid.

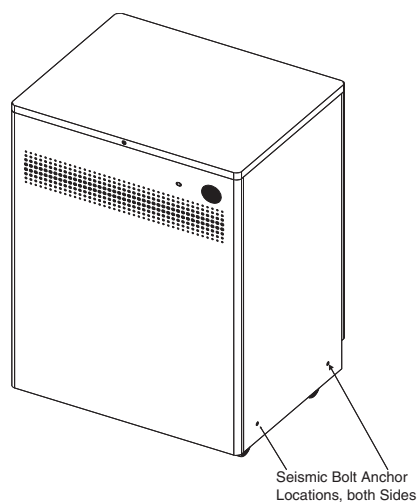
Figure 1-60 Seismic Console Mounting Hole Locations



11.2 Power Distribution Unit

If site specifications require seismic mounting, use the PDU seismic brackets that were shipped with the seismic kit. Refer to [Figure 1-61](#) for hole placement.

Figure 1-61 Seismic PDU Mounting Hole Locations



Chapter 2

Power, Ground & Interconnect Cables



NOTICE Potential for Data Loss and/or Equipment Damage

To prevent potential data loss and equipment damage, please do the following:

- When instructed, record data collected from the procedures in this chapter on GE Form e4879. For more information about this form, see [Section 8.0](#) of Chapter 4.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Note: Use dry cleaning for electro components.

Section 1.0 Introduction

Site use of conduit, floor duct, wall duct, or a raised computer floor, as well as the individual component layout determines the system cable sequence. If your site has floor or wall ducts that will interfere with placement of the table/gantry, it may be important to have the movers unload the cable boxes (8 & 9) first and run those cables while others unload the subsystems.

- Try to run the system cables after the contractor completes the contractor supplied wiring.
- All ground wires and other contractor wiring should be complete to the point of equipment placement.

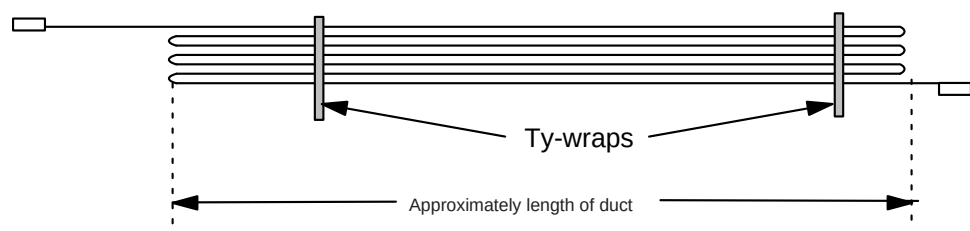


NOTICE Potential for Equipment Damage

Do not store excess cable in the bottom of the PDU or Gantry.

When possible, store excess cable length in a serpentine configuration, approximately one meter long. (**Do NOT coil excess cable.**)

Figure 2-1 Excess Cable Storage Configuration



- Keep signal and control cables away from power cables and power wiring. When you lay cables in a raceway, locate the signal cables in a separate section of the raceway, or a separate conduit.
- Check all connections for tightness.
 - Use suitable tools and judgment.
 - Check all visible connections, especially ground connections.
- Check for reasonable cable routing.
 - Take into consideration necessary take-up distances for equipment maintenance, etc.
 - Try to complete as neat a job as possible.

1.1 System Component Identification

Identify all system cables by the system component designators listed in Table 2-1 on page 78. Each end of a system cable has a label, and may have a color near the connector, (refer to Table 2-2 on page 78) to indicate the component and the jack identifier of the component.

Table 2-1 System Component Identifiers

DESIGNATOR	SYSTEM COMPONENT
CT2	Gantry
CT1	Patient Table
PM	Power Distribution Unit
OC	Operator Console (Console Computer)
WL	X-Ray ON Warning Light

1.2 Cable Color Identifiers

The ends of the cables may be marked with a piece of blue, yellow, red, or orange colored tape to help with the cable installation. [Table 2-2](#) lists the subcomponent, and corresponding color.

Table 2-2 Cable Color Identifiers

SUBCOMPONENT	COLOR
Gantry	Blue
Table	Yellow
PDU	Red
Console Computer	Orange

Table 2-3 BrightSpeed System Interconnect Cables

RUN NO.	DESCRIPTION	PART NUMBER	
		LONG CABLES (KIT 2281840-4)	SHORT CABLES (KIT 2281840-5)
1	Facility MDP to Room Disconnect (A1)	cust. supplied	cust. supplied
2	Room Disconnect (A1) to PDU	cust. supplied	cust. supplied
3	Room Disconnect (A1) to System E-Off	cust. supplied	cust. supplied
4	PDU to Room Warning Light(s)	cust. supplied	cust. supplied
5	PDU to Scan Room Door Switch	cust. supplied	cust. supplied
50	HVDC Power Cable - PDU to Gantry	2343529	2343529-2
51	HVAC Power Cable - PDU to Gantry	2343530	2343530-2
52	LVAC Power Cable - PDU to Gantry	2343528	2343528-2
53	LVAC Power Cable - PDU To Operator's Console	2343531	2343531-2
54	LVAC Power Cable - Gantry to Table	n/a	n/a
55	Ground, PDU to Raceway	2371450	2371450-2
56	Ground, Raceway to Console	2371450-3	2371450-4

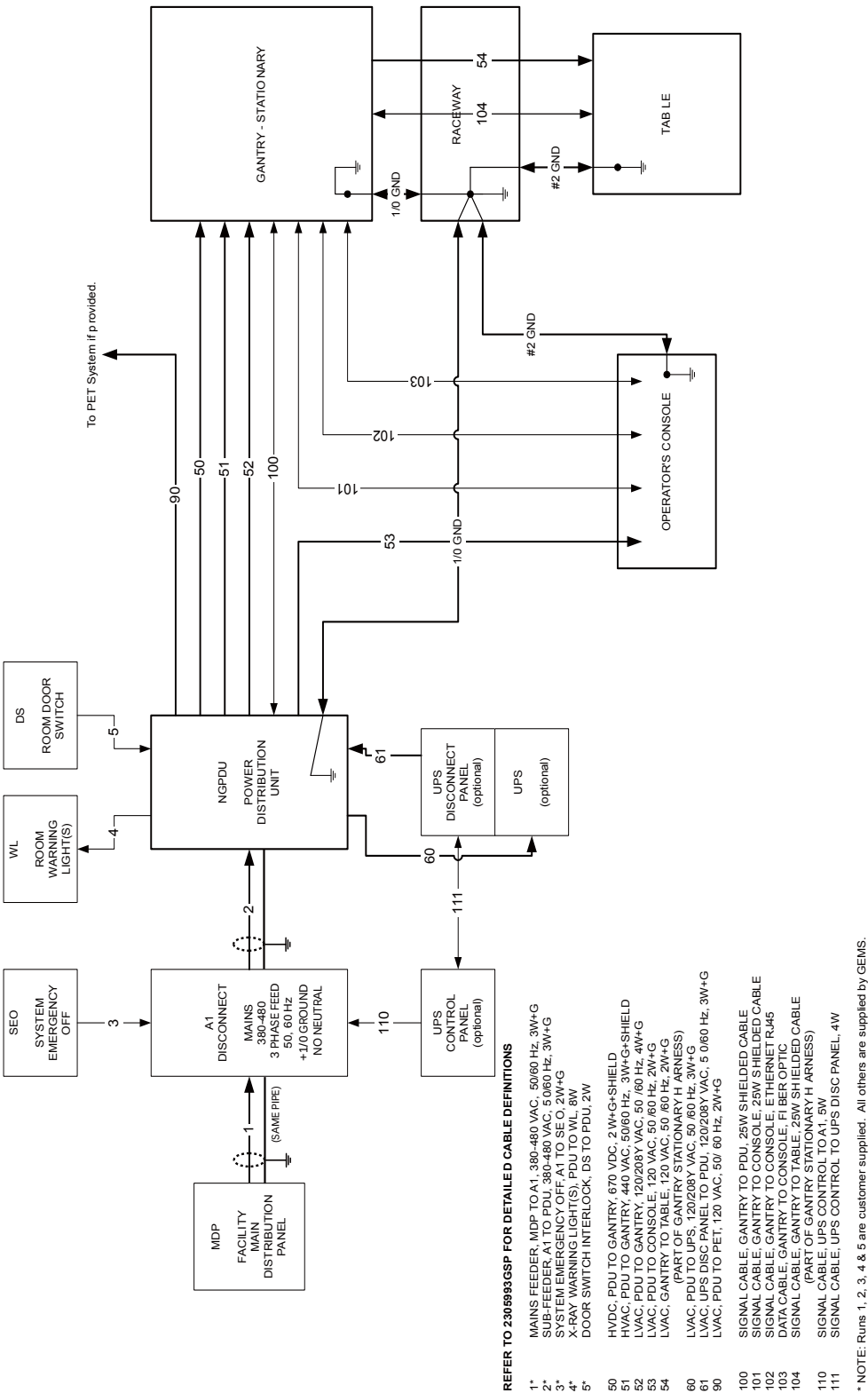
Table 2-3 BrightSpeed System Interconnect Cables (Continued)

RUN NO.	DESCRIPTION	PART NUMBER	
		LONG CABLES (KIT 2281840-4)	SHORT CABLES (KIT 2281840-5)
60	LVAC Power Cable - PDU to Optional UPS	-	-
61	LVAC Power Cable - UPS Disconnect Panel to PDU	-	-
90	LVAC Power Cable - PDU to PET	-	-
100	Signal Cable - Gantry to PDU	5120646	5120646-2
101	Signal Cable - Gantry to Console	5120645	5120645-2
102	Signal Cable (Ethernet) - Gantry to Console	2373436-2	2373436-3
103	Data Cable (Fiber Optic) - Gantry to Console	2117848-2	2117848-7
104	Signal Cable - Gantry to Table	n/a	n/a
110	Signal Cable - UPS Control to Room Disconnect (A1)	-	-
111	Signal Cable - UPS Control to UPS Disconnect Panel	-	-

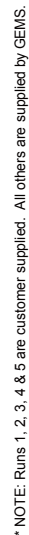
Section 2.0

System Interconnect Diagram

Figure 2-2 System Interconnect Diagram



2 - Install Power



Section 3.0

Contractor Connections

Table 2-4 Contractor PDU Connections

CONNECTION OR WALL BOX	AWG #	CONNECTION FROM	CONNECTION TO PDU	INSTALLED AND CHECKED
A1	#1	Load - T1	TS-1 L1	
	#1	Load - T2	TS-1 L2	
	#1	Load - T3	TS-1 L3	
	#1/0	GND	TS-1 GND (Do NOT connect anything to neutral point.)	
WL (Warning light) See Figure 2-33 , on page 105.	#14	LV Source -1	TS6 1	
	#14	LV Source -2	TS6 2	
	#14	X-Ray ON Light -1	TS6 3	
	#14	X-Ray ON Light -2	TS6 4	
	#14	Sys-ON Light -1	TS6 5	
	#14	Sys-ON Light -2	TS6 6	
	#14	Ready Light -1	TS6 7	
	#14	Ready Light -2	TS6 8	
DS (Scan Room Door Switch)	#14	Door SW-1	TS6 9	
	#14	Door SW-2	TS6 10	

Note:
Add #2 ground
wire.

IMPORTANT: Add AWG #2 ground wire from Table frame to Table/Gantry raceway ground bar (as shown in [Figure 2-2](#)).



WARNING



WORK WITH THE ELECTRICAL CONTRACTOR TO BE SURE EXTERNAL POWER SOURCE IS TURNED OFF.

Section 4.0 Console Connections

4.1 SCIM, Keyboard, Trackball & Mouse Installation

NOTICE Use OC Long Cable Kit (P/N: 5160577) for Freedom Workspace (FWS) table.

- 1.) Route the keyboard cable under the SCIM, as shown in [Figure 2-4](#).

Figure 2-4 SCIM control with keyboard cable routed through SCIM



NOTICE

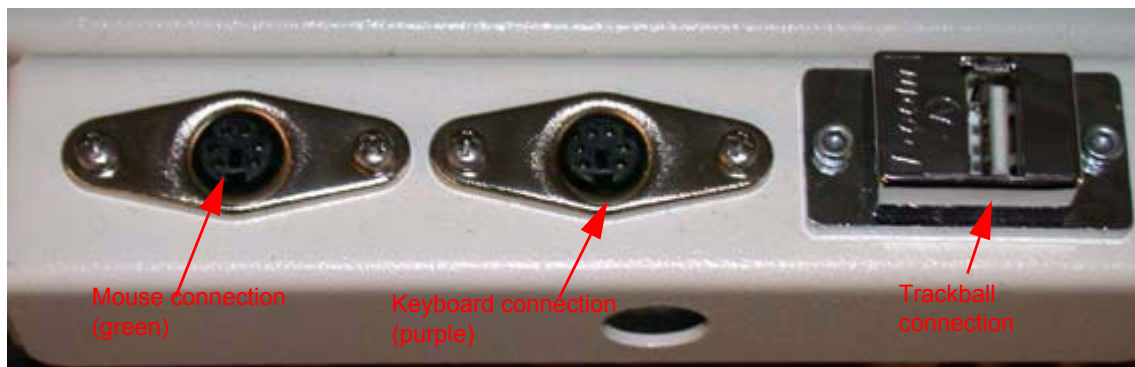


Potential for equipment damage

Never connect a mouse or keyboard with the host computer powered “ON”. Doing so can destroy components within the host computer.

- 2.) The keyboard and mouse connect to the ports on the top of the console front bulkhead. The keyboard connects to the port on the right, the mouse connects to the left port ([Figure 2-5](#)).

Figure 2-5 Global Console front bulkhead, showing keyboard and mouse connection



- 3.) Connect the SCIM cable to the SCIM as shown in [Figure 2-6](#) (note the cable routing).

Note: Make sure the SCIM connector fits snug. Some molding may need to be removed to allow the cable to fit snug.

Figure 2-6 SCIM bottom, showing cables and keyboard mounting bracket



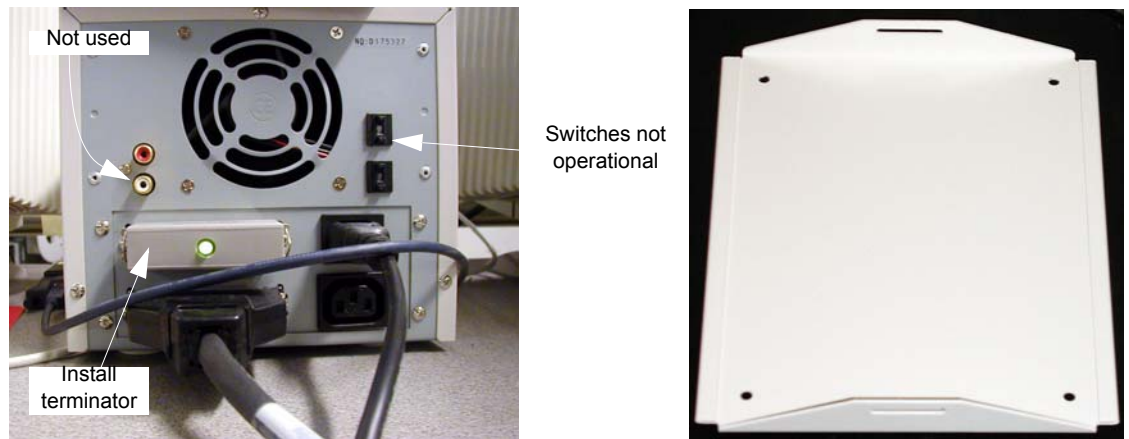
- 4.) Select and install the proper overlay with the appropriate language for your system: (1) with Tilt or (2) without tilt.
Verify that none of the buttons get caught and stuck under the overlay. Pay close attention to the prescribed tilt button on systems with the tilt feature.
- 5.) The keyboard should attach to the SCIM using the supplied Velcro strip and fit snugly against the SCIM when finished, as shown in [Figure 2-7](#).

Figure 2-7 SCIM connected to the keyboard with the US English tilt overlay installed



4.2 Connecting the SCSI Tower

Figure 2-8 SCSI tower: Connections (left) and Optional Seismic Mounting Bracket (right)



- 1.) Connect the SCSI cable to the rear of the SCSI tower.

Note: The cable is included in the SCSI Cable Kit and is the same as the DASM cable.

- 2.) Connect the SCSI terminator to the rear of the SCSI tower.
- 3.) Connect the power cable to the rear of the SCSI tower.

4.3 Connecting the Monitor

NOTICE Use OC Long Cable Kit (P/N: 5160577) for Freedom Workspace (FWS) table.

- 1.) With the monitor and computer switched off, connect the video signal cable to the monitor's video input (see [Figure 2-9](#)).

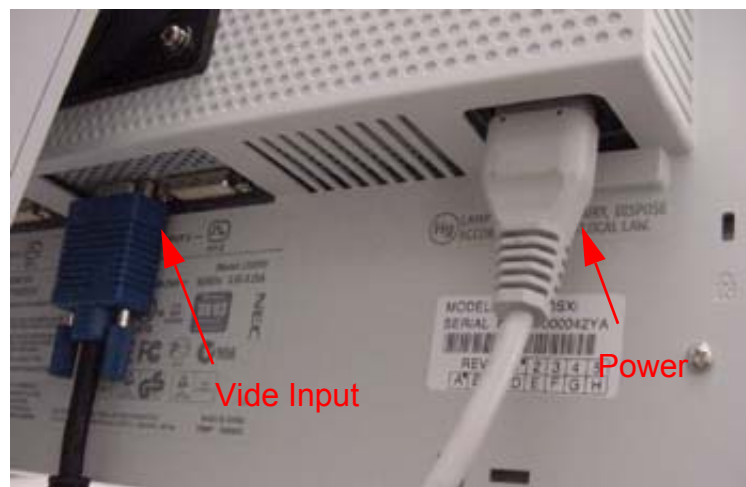
NOTICE **Equipment Damage Possible**



Do not touch the video signal cable connector pins as this might bend them. When connecting the video signal cable, check the alignment of the HD15 connector. Do not force the connector in the wrong way, or the pins might bend.

- 2.) Connect the power cord to the monitor, then connect it to console power panel outlet J10 or J12.

Figure 2-9 Monitor Connections (NEC Model 1980SXi)



4.4 Power Panel Connections

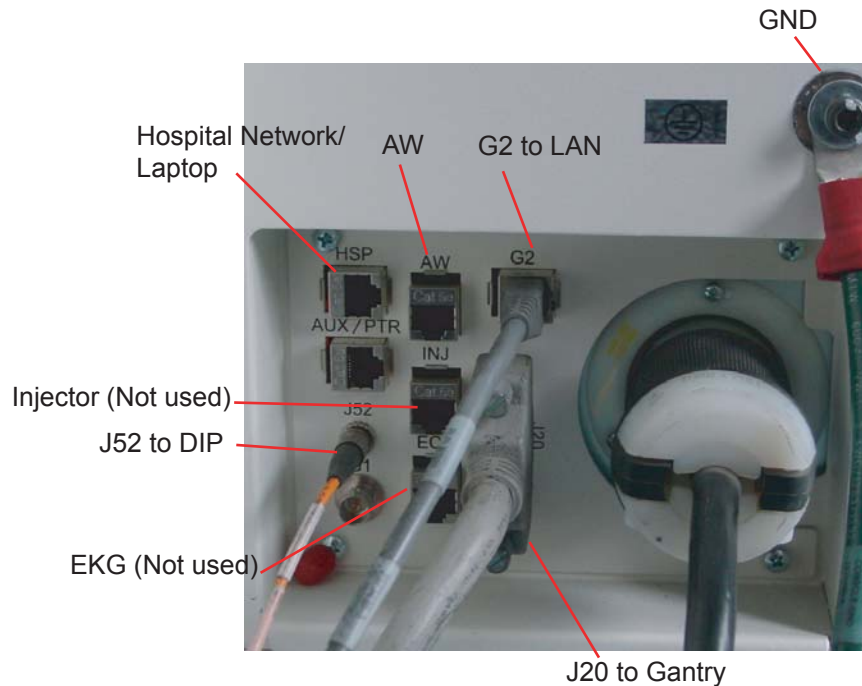
Note: Console power is single phase power. Outlet assigned is not critical.

- 1.) Connect the console power cable to the console power panel.
- 2.) Connect console component power cords as listed in [Table 2-5](#). (“J numbers” increment from top to bottom, left to right)

Table 2-5 Power Panel Outlet Assignments - GC-GRE

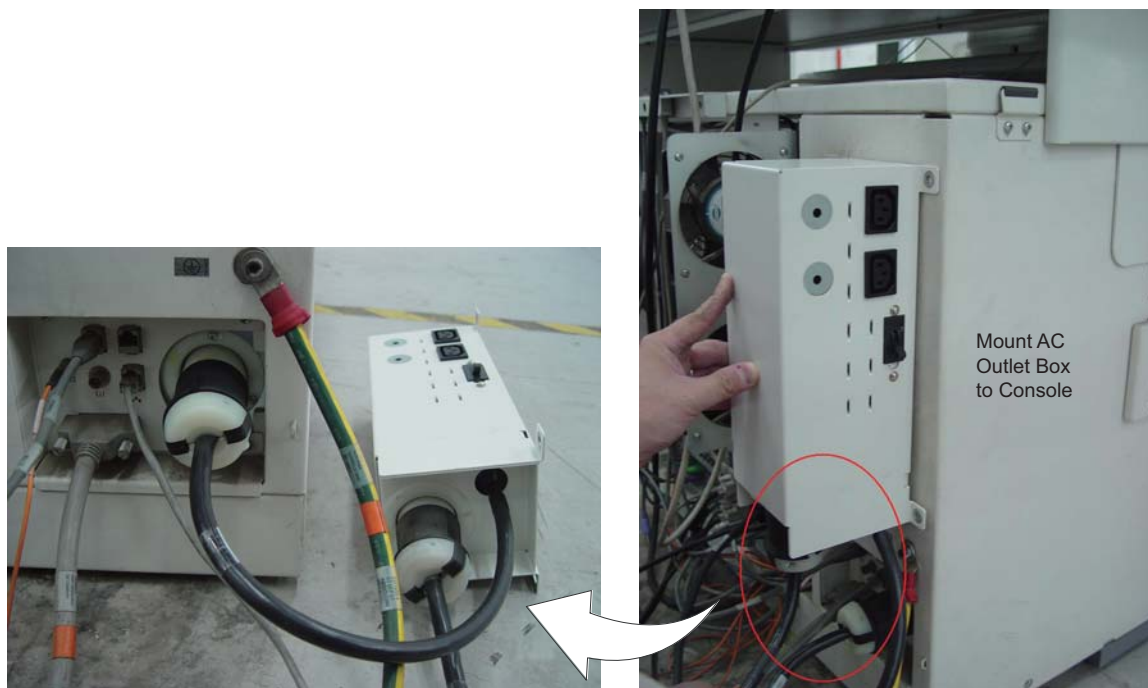
J#	DEVICE	J#	DEVICE
J1		J9	Host Computer
J2		J10	Power Monitor
J3	Modem	J11	DARC
J4	IG 3	J12	Power Monitor
J5	IG 1	J13	ICOM
J6	IG 2	J14	Power Media Conv
J7		J15	Fan
J8		J16	

Figure 2-10 Power Connections



Note: If Integrated Injector is installed, AC Outlet Box must be used for power connection. Refer to [Figure 2-11](#) for cable connection.

Figure 2-11 Power Connection (with AC Outlet Box)

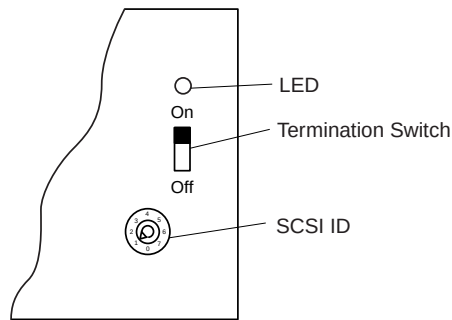


4.5 Installing a DASM (Hardware Option)

If you have a digital DASM to install, complete the following steps to connect it to the computer.

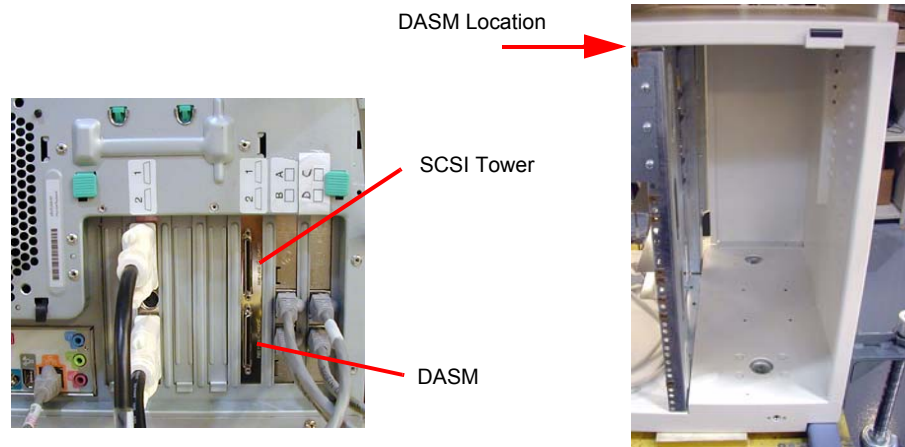
- 1.) Turn the DASM over so you are looking at the bottom.
- 2.) Set the DASM SCSI ID to 1.
- 3.) Set the DASM termination to ON.

Figure 2-12 DASM Bottom



- 4.) Set the RS232/RS422 switch to the RS232 position, for a digital DASM.
- 5.) Refer to [Figure 2-13](#). Attach the SCSI cable supplied with DASM kit to the back of the DASM.

Figure 2-13 Rear view of HP xw8200, showing DASM SCSI cable connection



- 6.) Attach the free end of the SCSI cable to the rear of the Host Computer (see [Figure 2-13](#)).
- 7.) Plug DASM power cord into J10 (or any open slot).
- 8.) Attach remaining cable(s) from camera to DASM and install the DASM to the right of the slide-out tray.

Note: See your *Software Load Procedures* manual for instructions on configuring cameras.

4.6 Modem Option

If you have a modem to install, do it now. Place the modem on top of the console desktop. Hook up the power, phone, serial line (for GOC4/LCGOC) or USB line (for All-In-One Console) as shown in drawing which is located in the back cover of the console.

NOTICE If global Modem with Series port is used on All-In-One Console, connect the Series-USB converter cable to modem at first, then connect to USB port on the Host Computer inside of the All-In-One Console.

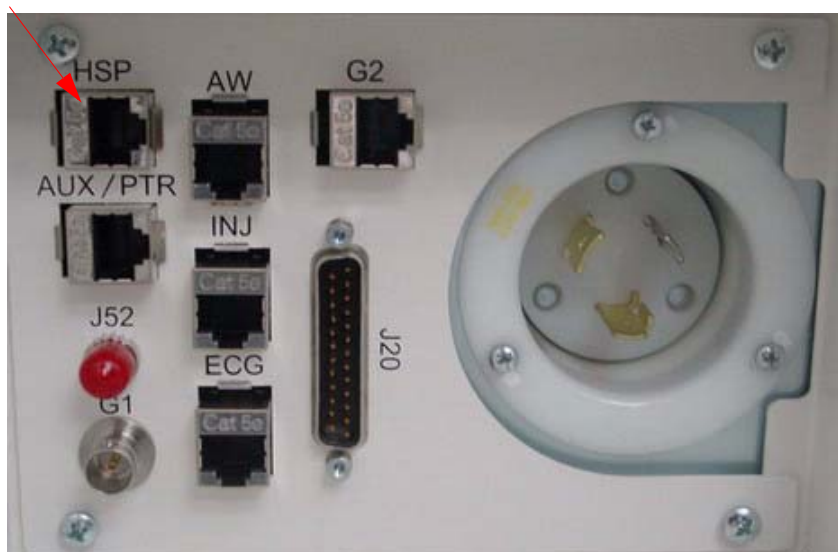
NOTICE If the USB Modem is used on All-In-One Console, there is no power cable for USB Modem. Only connect the USB cable to USB port on Host Computer inside of the All-In-One Console.

4.7 LAN Connections

Plug LAN cable into HSP port of rear bulkhead on console.

Figure 2-14 LAN Connection

LAN Connection



4.8 GOC4/LCGOC Interconnect

Figure 2-15 GOC4/LCGOC Interconnect

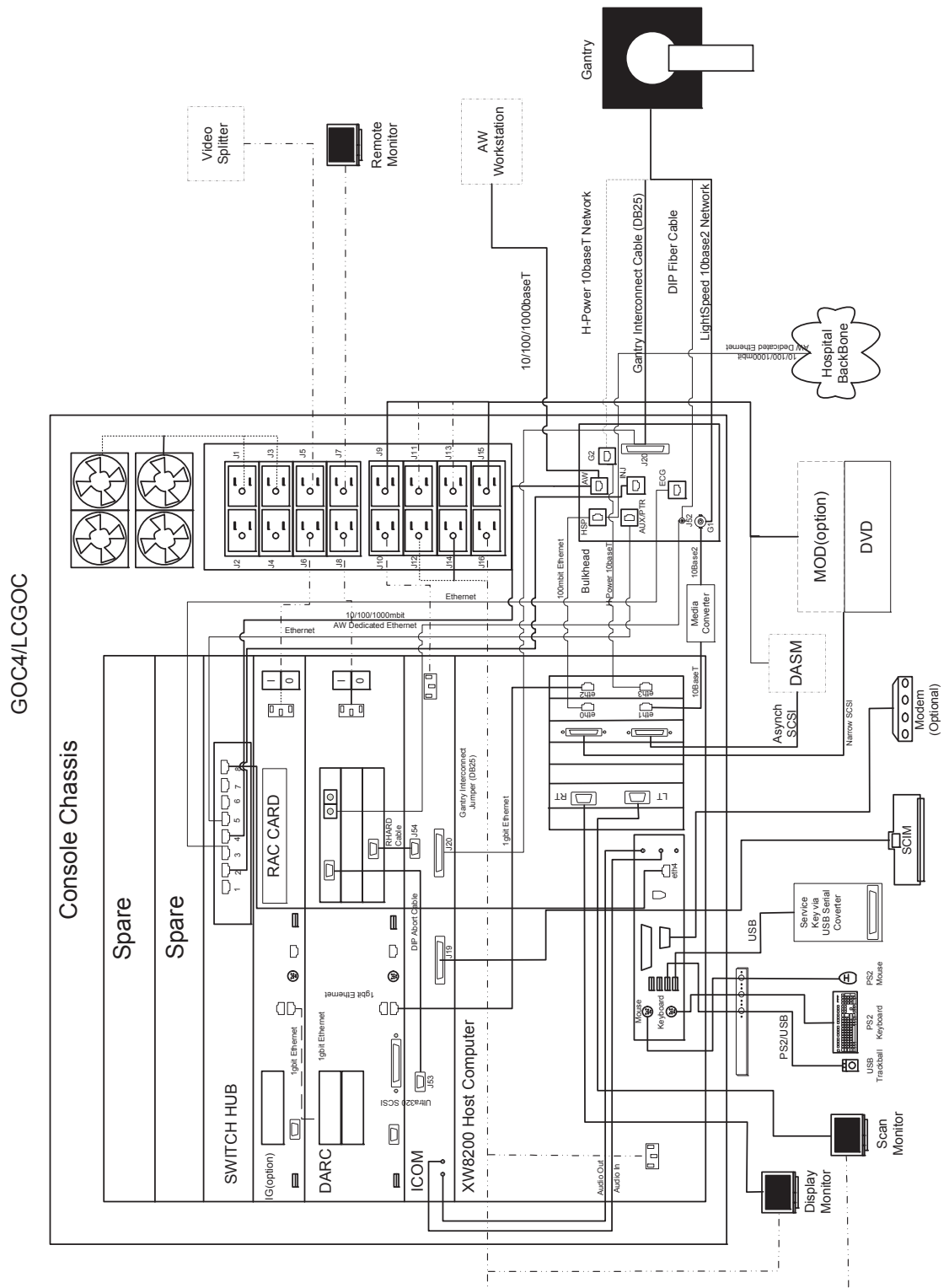
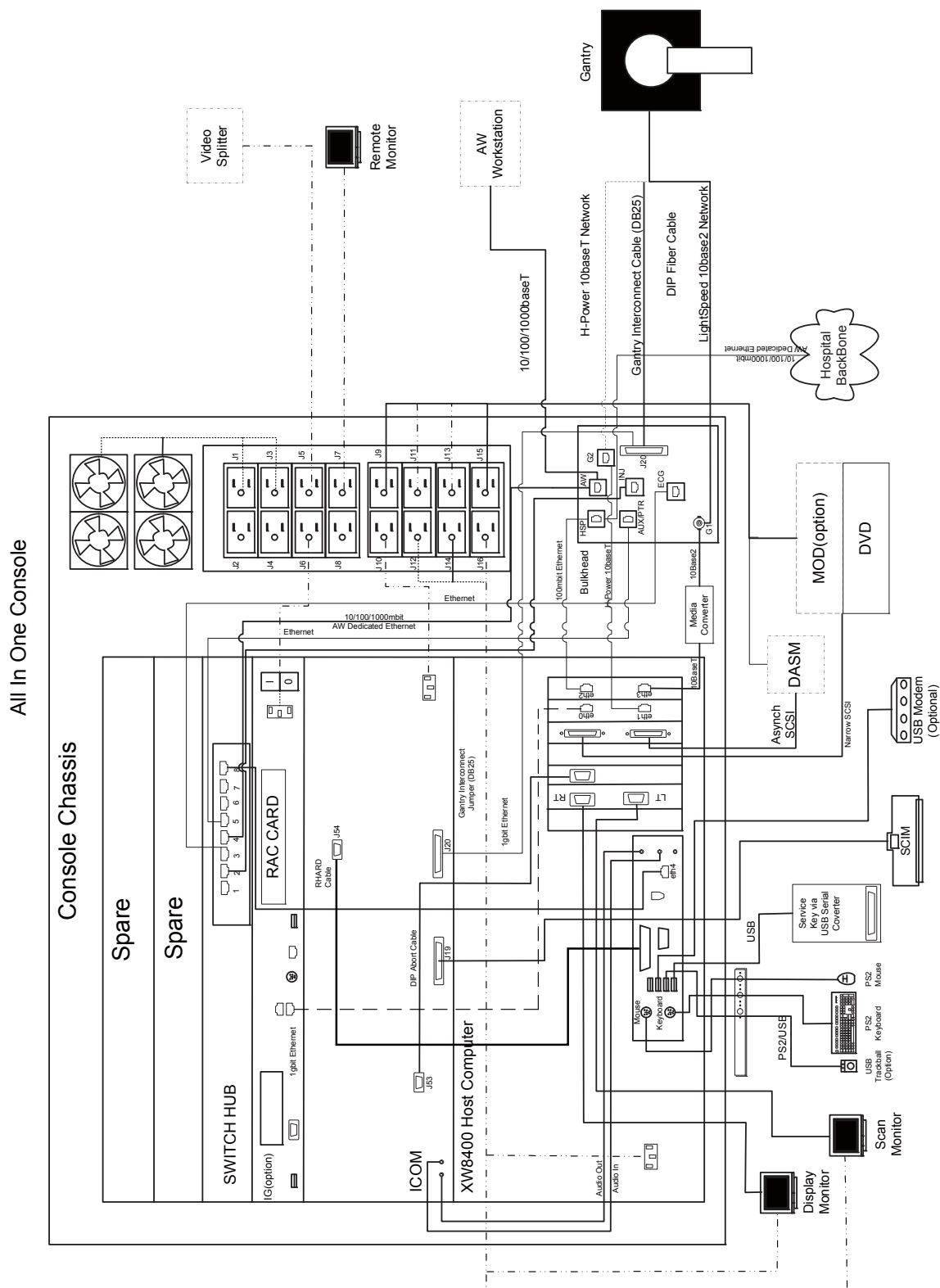


Figure 2-16 All-In-One Console Interconnect



Section 5.0

Install Options

5.1 Install Optional Bar Code Reader

Follow the installation procedures in the Bar Code Reader box.
When finished neatly dress all cables.

5.2 Install Optional Remote Monitor

Follow the installation procedures in the Remote Monitor box.

5.3 Install Cardiac Gating IVY Monitor and Stand Option

Refer to the instruction shipped with the option. This option is attached to the Gantry Option Interface, and should only be mounted on the non-motor side of the gantry. Neatly dress all cables along the gantry base so that the base covers fit properly.

Note: For Gantry with GOB, Monitor power will plug into the wall. For Gantry with IPC board, Monitor power will plug into the Gantry or Console.

5.4 Install Respiratory Gating Option

Refer to the instruction shipped with the option. This option is attached to the Gantry Option Interface, and should only be mounted on the non-motor side of the gantry. Neatly dress all cables along the gantry base so that the base covers fit properly.

Note: For Gantry with GOB, Monitor power will plug into the wall. For Gantry with IPC board, Monitor power will plug into the Gantry or Console.

5.5 Install Injector Option

Follow the instructions shipped with the option. If this is a ceiling-mounted option, check that the plate is installed correctly with the holes in the correct location.

Note: For Gantry with GOB, Injector power will plug into the wall. For Gantry with IPC board, Injector power will plug into the Gantry

5.6 Customer Accessories (Head Holders and Extender)

Open the boxes and installed the appropriate language warning labels.

The head holders ship with shims installed to assure proper fit. Check that shims are included and a pair is installed. The holder should fit snugly. Follow the procedure in Section 4.0 of Chapter 3 to install Head Holders.

5.7 Install Service Cabinet

The service cabinet you receive may ship disassembled. Assembly takes about 1.5 hours.

- 1.) Assemble the cabinet following the instructions located in the cabinet's shipping box.
- 2.) When you complete assembly, place the cabinet in the location shown on the site print.
- 3.) Place all service materials shipped with the system in the service cabinet.

5.8 Install UPS

Follow the instructions shipped with the UPS option. The option ships with two sets of instructions, a GE set and a Powerware set.

The GE set instructs you to install the connections between the UPS and the PDU and between the UPS and the A1.

Note: A GE A1 Disconnect with UPS controls is required for this option.

The Powerware set instructs you to internally connect the batteries and do a power-up check. Refer to both manuals for additional guidance.

WARNING LOCKOUT/TAGOUT IS REQUIRED WHEN WORKING IN THE A1 DISCONNECT.

Section 6.0

Gantry Cable Connections

Please refer to [Figure 2-2](#) for complete system interconnect details.

Table 2-6 Gantry Cable Connections

TO	FROM	CABLE DESCRIPTION
Gantry Power Pan	PDU	HVDC
Gantry Power Pan	PDU	440VAC
Gantry Power Pan	PDU	120VAC
Gantry Power Pan	Console	Fiber - <i>Take extreme care when you install the fiber optic DAS data cable. Do not step on, kink, or sharply bend this fragile DAS cable.</i>
Gantry Power Pan	Console	LAN
TGPU (J9)	Console	Control
TGPU (J11)	PDU	Control

NOTICE



- 1.) If using a rear cable entry box (B7850RC), install it now, before routing cables to gantry.

Potential for equipment damage.

Observe correct polarity when connecting the high voltage DC power. Reversing these leads will result in serious equipment damage. The HVDC positive conductors have red insulation and are labeled “ONE.” The HVDC negative conductors have black insulation and are labeled “TWO.” Lead “ONE” must be connected to lead “ONE,” and lead “TWO” must be connected to lead “TWO.”

Observe correct phase rotation when connecting the axial motor power. Phases one, two and three should be connected top to bottom.

- 2.) Install the cables to the gantry power pan. The power pan is located on the rear of the gantry at its base. See [Figure 2-17](#), [Figure 2-18](#), and [Figure 2-19](#) for connections.

Note:

The gantry 120VAC cable may not fit under the gantry frame. Install this cable before gantry placement—or remove the power plug—to route it under the gantry.

Figure 2-17 Gantry Power Pan Connections

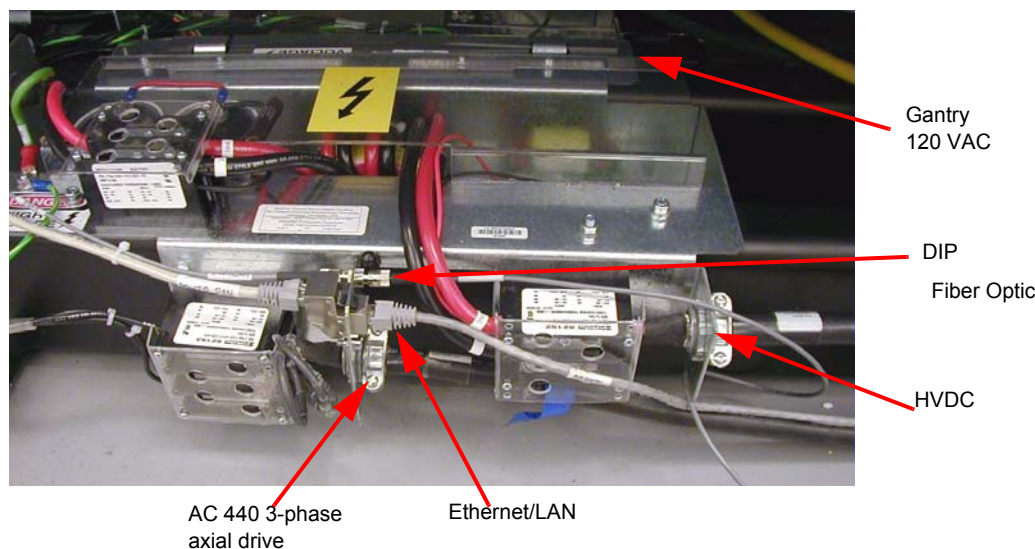


Figure 2-18 Gantry Power Pan

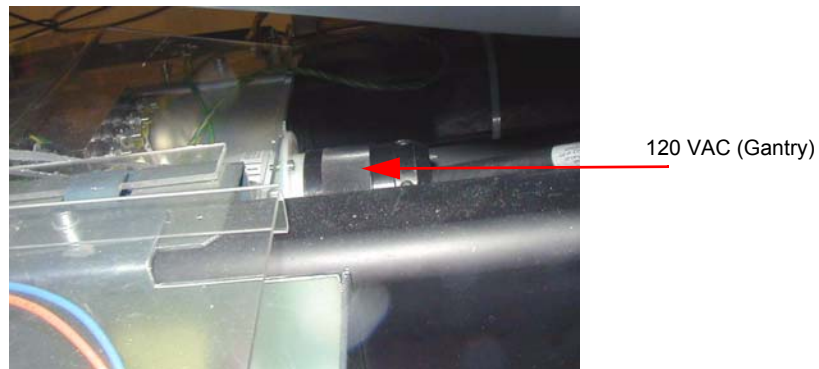
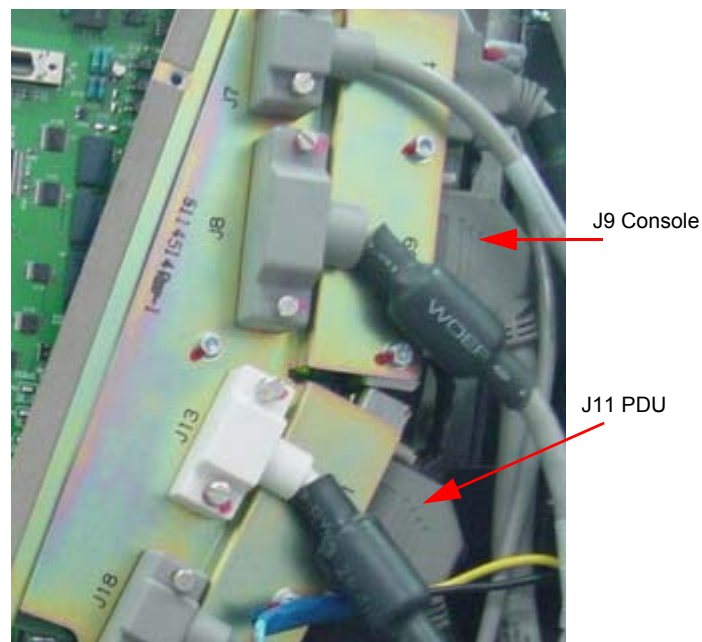


Figure 2-19 TGPU Connections

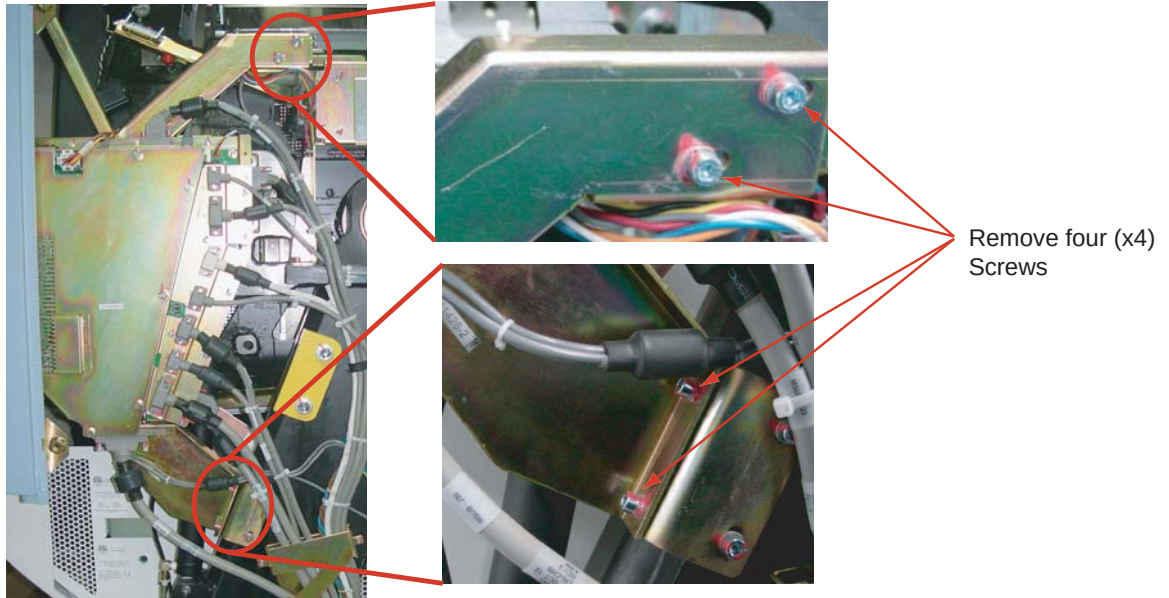


3.) Install cables to the gantry TGPU.

Note: If it's difficult to connect the cables to TGPU, follow below steps to take the TGPU outside the gantry frame (not remove the TGPU), so that it is easier to attach the cables to TGPU.

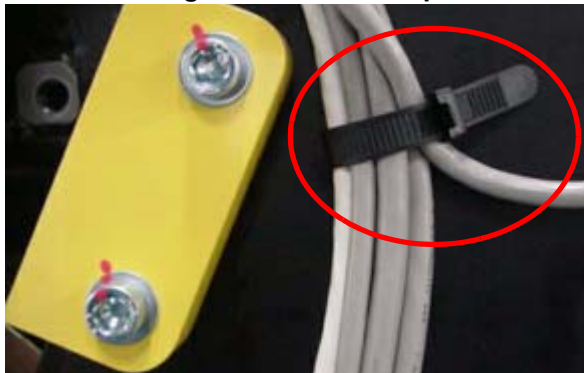
- a.) Remove the four (4) screws which secure the TGPU assembly on the gantry frame. See [Figure 2-20](#).

Figure 2-20 TGPU Assembly On The Gantry Frame



- b.) Loosen the cable clip. See figure below.

Figure 2-21 Cable Clip



- c.) Slightly rotate the TGPU outside, so that the cable receptacles on the TGPU is easier to access.
- d.) Connect the cables on TGPU, then install the TGPU assembly with reversed order through step c) to a).

- 4.) Route J9 and J11 cables behind all cables at this area. See [Figure 2-22](#).

Figure 2-22 Route Cables



Route J9, J11 cables

Add a cable tie-wrap here



tie cables with hydraulic pipe

Section 7.0

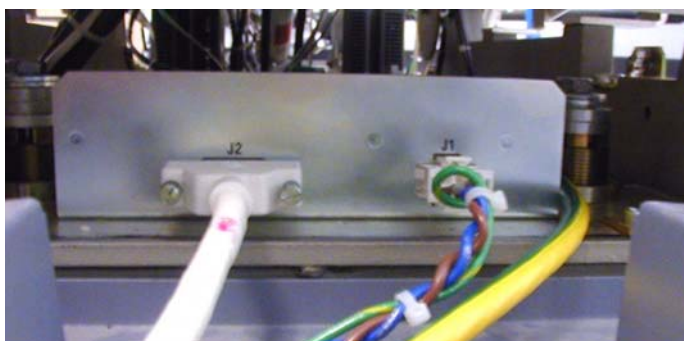
Table Connections

Pull and connect the following cables:

Table 2-7 Cables Connected to Table

J#	CABLE DESCRIPTION
J1	120 VAC
J2	Signal Cable

Figure 2-23 Table Bulkhead Connections



Section 8.0

PDU Cable Connections & Configuration



CAUTION

Do not work in an energized PDU. When working on the PDU, follow this simple rule: Always tag and lock out power to the PDU at the “main” disconnect. Failure to do so can result in electrocution or death.

Do apply power to the PDU until all work has been completed and all PDU covers are in their proper place.

8.1

NGPDU

As seen in [Figure 2-24](#), a number of cables must be installed throughout the PDU. Specific details on each connection can be found in the sub-sections that follow. Use [Figure 2-24](#) for reference. The PDU has been designed to have cables routed into the PDU from its behind and/or from beneath it.

Figure 2-24 PDU Cable Connections - Front

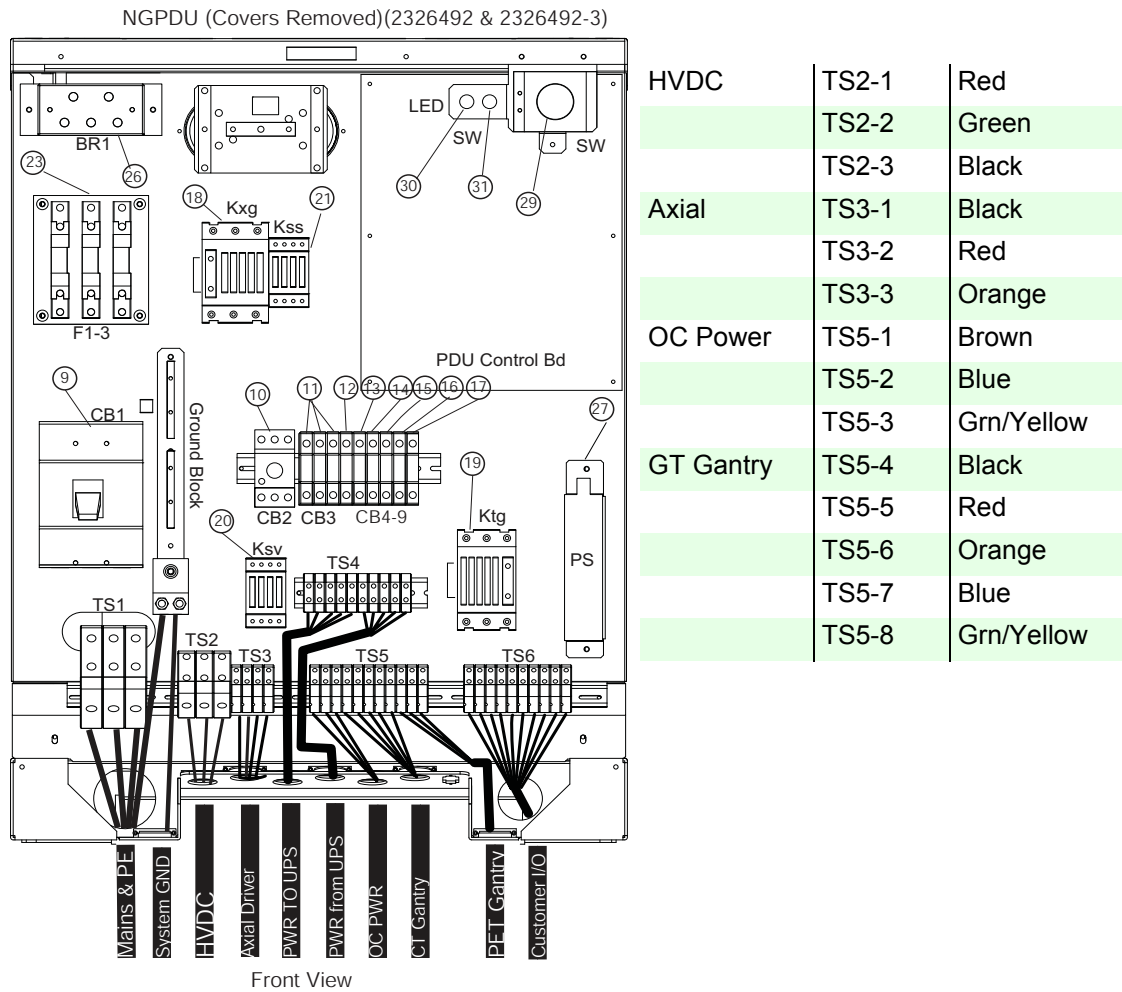
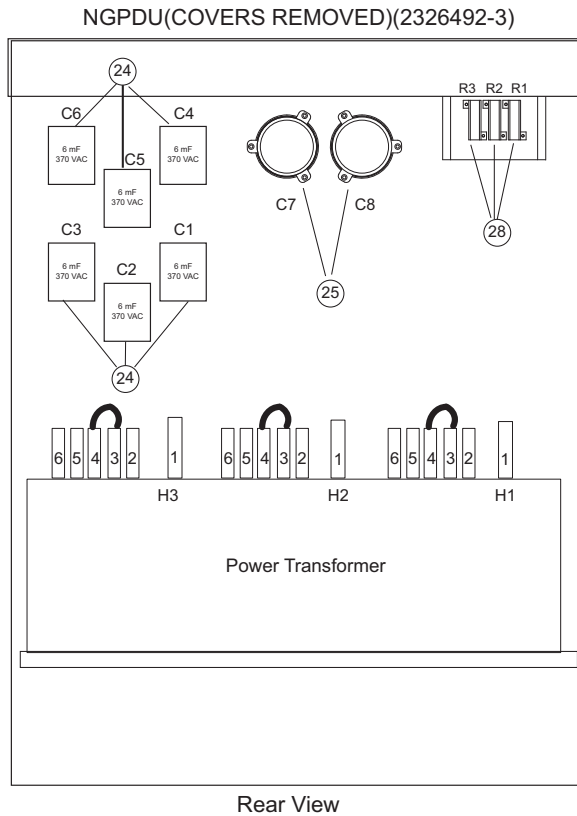


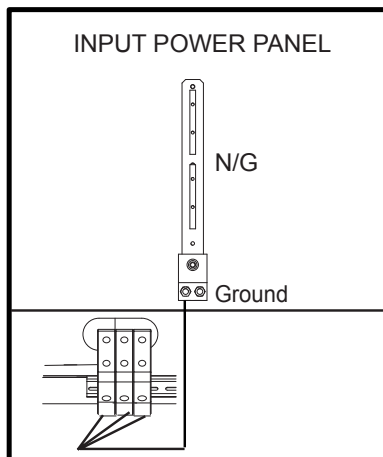
Figure 2-25 PDU Cable Connections - rear



8.1.1 Panel - 380 - 480VAC Mains “TS1” Input Power Connection

- 1.) Remove the TS1 panel front cover.
- 2.) Strip the wires to fit securely on the power block.
- 3.) Observe incoming phases (L1, L2 and L3) and insert bare leads into power block.
- 4.) Insert “vault” ground into PDU “vault” ground lug.
- 5.) Tighten all fasteners securely and replace the TS1 front panel.

Figure 2-26 Input Power Panel Connections

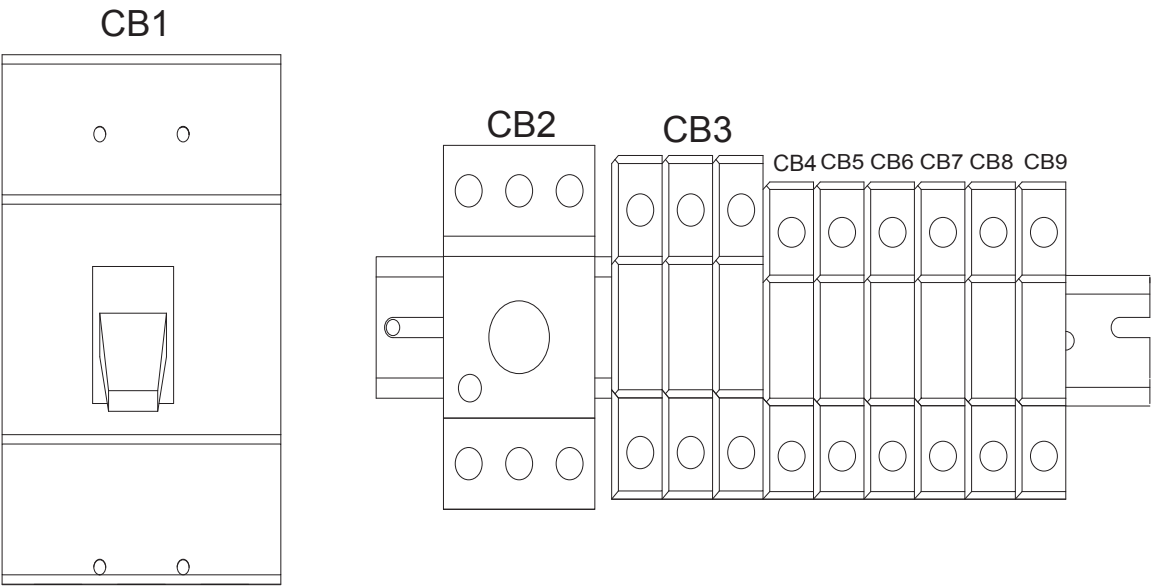


When Mains power is available to the PDU, the “TS1” power light will be illuminated.
(See [Figure 2-24.](#))

8.1.2 Panel - Circuit Breakers

Place the circuit breakers in the “off/down” position during installation, even with Mains incoming power tagged and locked out. After you have completed work on the PDU, you may return the circuit breakers to the “ON” positions.

Figure 2-27 Circuit Breaker Panel



By design, when CB3 is in the “OFF” position, circuit breakers 4, 5, 6, and 7 are switched “OFF”. CB3 is essentially in series with these breakers.

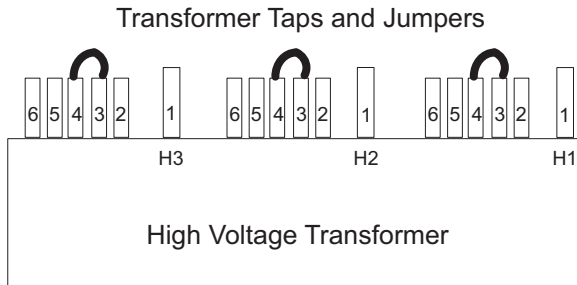
Table 2-8 Panel Circuit Breaker Descriptions

CIRCUIT BREAKER	DESCRIPTION
CB3	Fully Winding Protection (Master power of CB 4, 5, 6, and 7)
CB4	CT Gantry Service Outlets
CB5	CT Gantry rotating loads
CB6	Table & CT Gantry Station Loads
CB7	Operator Console
CB8	PET Gantry
CB9	NGPDU Control Power Supply

8.1.3 Transformer (480VAC) Taps

Verify that the transformer taps are set properly. The transformer taps are set to 480 VAC operation at the factory. The taps should be set as shown in [Figure 2-28](#).

Figure 2-28 PDU Tap Positions for 480 volt operation

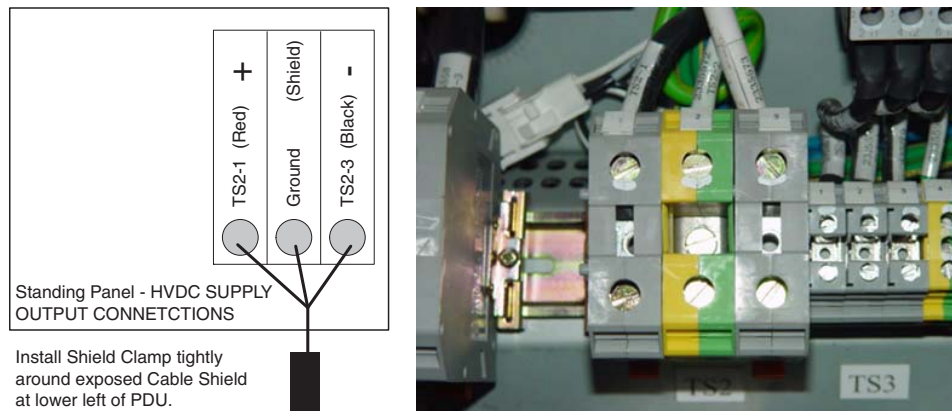


Note: Taps should be shipped as shown for 480 VAC only. For all others, you must move the taps to correct position according to [Figure 2-28](#)

8.1.4 HVDC Connection

Connect the internally shielded HVDC cable to TS2 on the standing panel. See [Figure 2-24](#) for the location of the connector and [Figure 2-29](#) for details. Observe polarities and grounds.

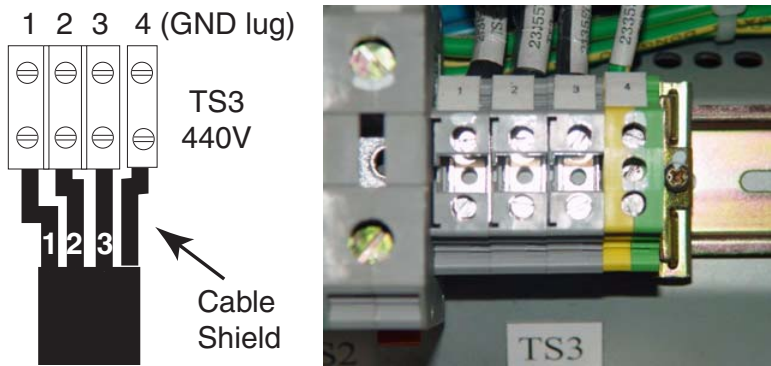
Figure 2-29 HVDC Connection



8.1.5 440V Connection

Connect the internally shielded 440V cable from the gantry to TS3 on the panel. See [Figure 2-24](#) for the location of the connector and [Figure 2-30](#) for details. Observe the labels on the cable leads for proper identification and orientation.

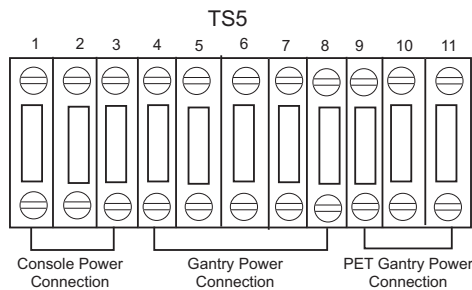
Figure 2-30 440VAC Connection



8.1.6 Gantry & Console Power Connections (120V)

Both Gantry and Console power cables come pre-terminated. Simply plug the Gantry power cable into “J4” and the Console power cable into “J5” as shown in [Figure 2-31](#).

Figure 2-31 Gantry & Console Power Connections



8.1.7 Console Power Cable Re-termination

Power cable re-termination should be used as a last result. Short and long cables are available.

- 1.) Carefully remove the power plug and **record the color of the wires** in [Table 2-9](#). The terminals are labelled X, W and G on the plug.

Table 2-9 Console Power Cable Termination

TERMINAL	X	W	G
Description	Hot	Neutral	Ground
Color			

- 2.) Cut the cable to desired length and dress ends.
- 3.) Re-install the power plug, according to the orientations recorded in [Table 2-10](#).
- 4.) Verify that less than 1 ohm of resistance exists between the following connections:

Table 2-10 Resistance Verification Points

FROM PDU PLUG	TO CONSOLE PLUG END	
CB1 -11 (Black) (J5 Phase X (Brown))	Phase X (Brown)	☐ Check box when complete
A3 Neutral Buss Bar (Blue) (J5 -13 Phase W)	Phase W (Blue) Neutral	☐ Check box when complete
A3 Ground Buss Bar (J5 -22 Ground Green)	Ground (Green) Ground Green screw	☐ Check box when complete

8.1.8 PDU Control Cable

The PDU control cable comes pre-terminated and should not be re-terminated in the field. Excess cable length must be stored. Simply plug the cable into “J2” on the “A4” panel. Secure it by using the fasteners intergrated into cable’s connector shell.

8.1.9 System Ground Connection

Connect the ground wire (green with a yellow strip) from the Table/Gantry raceway ground bus to the system ground lug in the PDU. See [Figure 2-24, on page 99](#), and [Figure 2-32, below](#).

Figure 2-32 PDU System Ground Connection

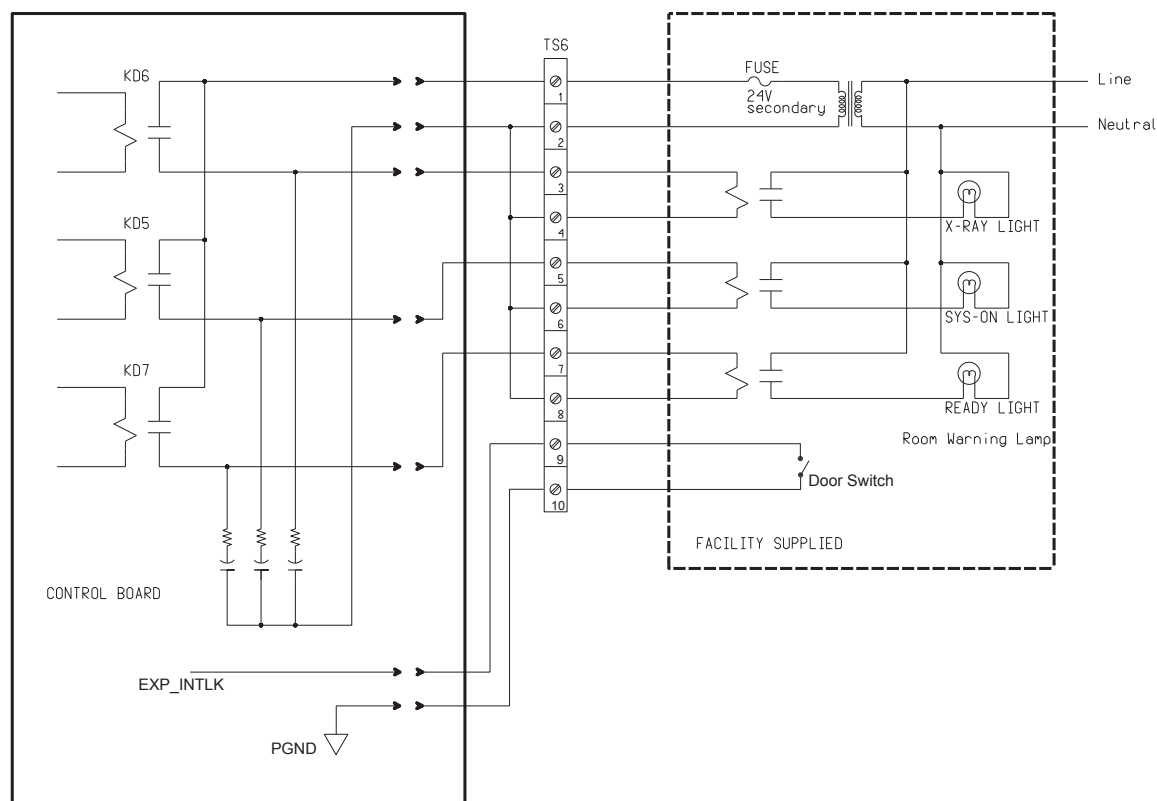


8.1.10 Warning Light & Door Interlock Connections

8.1.10.1 Warning Light Configuration & Connection

- 1.) Warning Light is controlled by signals from the system.
- 2.) This step is site specific. The PDU by default is configured for “no” external warning light connection. If you have external warning lights, see [Figure 2-33](#) for proper connection.

Figure 2-33 Typical TS6 Warning Light & Door Interlock Connections



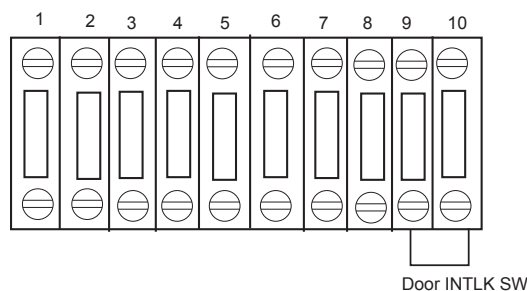
It is recommended that you use the four (4) wire method of adding a X-ray warning light to a room, as shown in [Figure 2-33](#). When using this method, you:

- Minimize EMC interference.
- Increase contact life of the relay used in the PDU.

8.1.10.2 Door Interlock Connections

Door interlocks are used to prevent X-Rays from being generated when the scan room door is open. The Door Interlock circuitry in the PDU is shipped from the factory engaged. This means the system cannot generate X-ray until disengaged. A short must exist between pins 9 & 10 for X-ray to be generated. Using a small piece of wire, short pins 1 and 2 together. See [Figure 2-34](#).

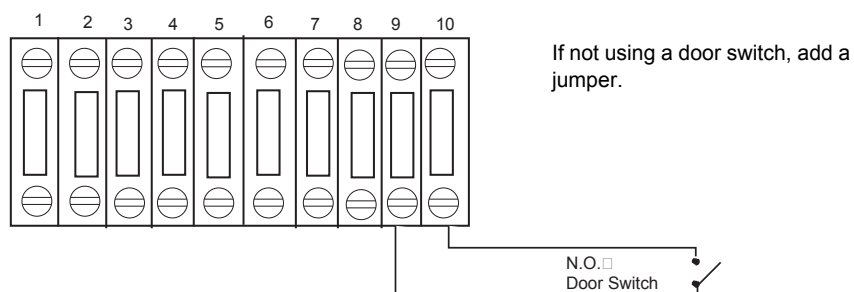
Figure 2-34 Without a Door Interlock



If jumper is not in place, exposures will not be made. Check this jumper if you get scan interlock errors.

To use the system with a door interlock, wire a normally open switch between pins 1 & 2 that is attached to the interlock.

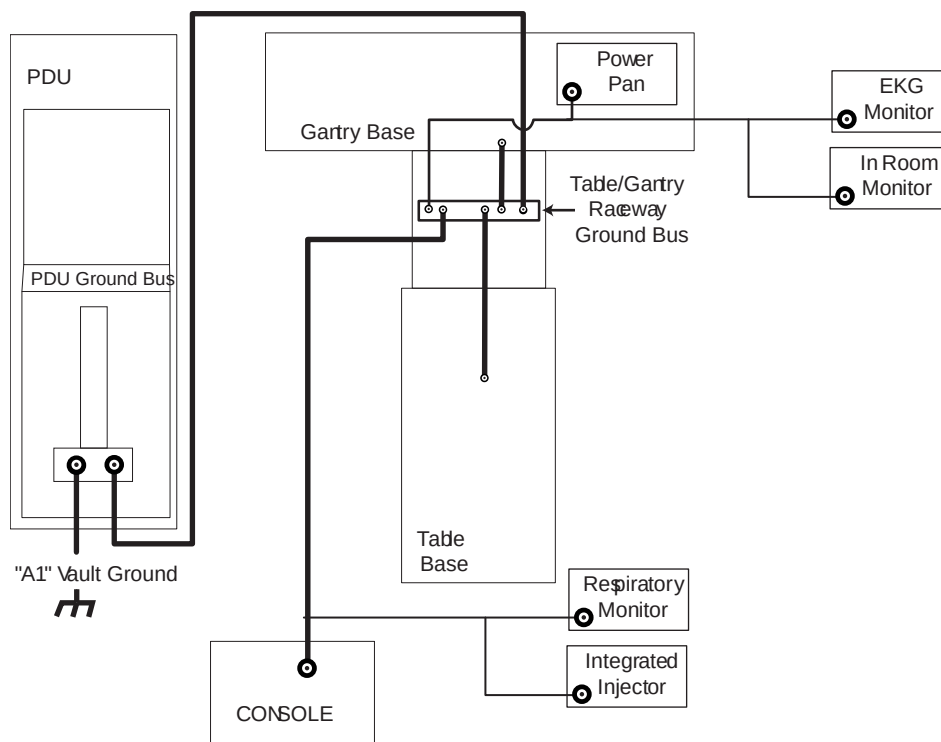
Figure 2-35 With a Door Interlock



Section 9.0 System Ground Connections

As seen in [Figure 2-36](#), the Table/Gantry raceway ground bus is used to centralize all system grounding. The system ground is tied to vault ground at the PDU, through its chassis.

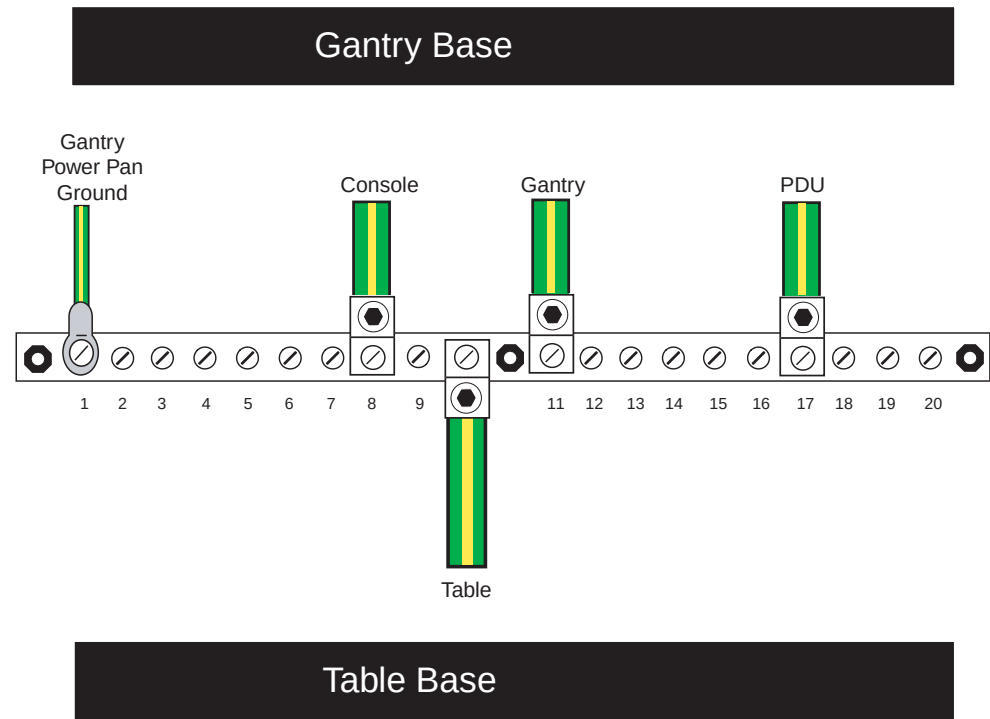
Figure 2-36 CT System Ground Connections



The gantry is tied to system ground at a number of points. It is important that all of these ground connections are securely made. See [Figure 2-37](#).

Check that the terminals will not be loosened, by moving or swinging the cables rather strongly by hand.

Figure 2-37 Table/Gantry Raceway Bus - Grounds



Various types and sizes of wire are used to ground the system. Please use the type and sizes specified in [Table 2-11](#), below.

Table 2-11 System Ground Connections

AWG #	CONNECTION TO	CONNECTION TO
#1/0	PDU	Power Main
#1/0	Gantry (Power Pan)	Raceway
#2	Console	Raceway
#1/0	Gantry	Raceway
#2	Table (frame)	Raceway
#1/0	PDU	Raceway

Chapter 3

System Continuity & Ground Checks



NOTICE **Potential for Data Loss and/or Equipment Damage**

To prevent potential data loss and equipment damage, please do the following:

- When instructed, record data collected from the procedures in this chapter on GE Form e4879. For more information about this form, see [Section 8.0](#) of Chapter 4.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Note: Use dry cleaning for electro components.

Section 1.0

System Continuity & Ground Checks (Mechanical Contractor)

Use this section to check cable and ground connections.

1.1 Tools Required

- Digital VOM
- 10m (30 ft) of #18 wire
- 600 VAC meter leads

1.2 Procedure

Reference [Figure 3-1: Front View of NGPDU, with Covers Removed](#) on page 111 and [Figure 3-2: Gantry Power Pan](#) on page 111.



WARNING



USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES; LOCK OUT WALL POWER.

- 1.) Remove all System Power at the A1 Mains Disconnect Panel. Follow Lockout/Tagout procedures.
- 2.) Put the UPS in the Service Position.
- 3.) Remove the PDU A3 input power panel cover.
- 4.) Verify, with a voltmeter, that mains power is disconnected.
- 5.) Verify that less than 1 ohm of resistance exists between the following ground connections:

Table 3-1 Mains Connections to PDU

FROM	TO	
Wall ground connection	PDU Cabinet	☐ Check box when complete

6.) Verify that less than 1 ohm of resistance exists between the following connections:

Table 3-2 Resistance Verification Points

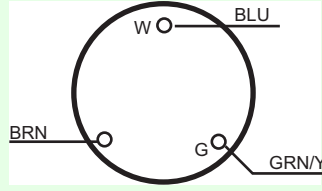
FROM	SIGNAL NAME (COLOR)	TO	
PDU TS2-1	+HVDC (Red)	Gantry HV Power Pan TS1-1	☐ Check box when complete
PDU TS2-2	HVDC Ground (Green/Yellow)	Gantry Power Pan Chassis	☐ Check box when complete
PDU TS2-3	-HVDC (Black)	Gantry HV Power Pan TS1-2	☐ Check box when complete
PDU Ground Bus	HVDC shield	Gantry HVDC cable shield	☐ Check box when complete
PDU TS3-1	Axial drive 440vac (Black)	Gantry HV Power Pan TS2-1	☐ Check box when complete
PDU TS3-2	Axial drive 440vac (Red)	Gantry HV Power Pan TS2-2	☐ Check box when complete
PDU TS3-3	Axial drive 440vac (Orange)	Gantry HV Power Pan TS2-3	☐ Check box when complete
PDU TS3-4	Axial drive ground (Green/Yellow)	Gantry Power Pan Chassis	☐ Check box when complete
PDU Ground Bus	Axial drive shield	Gantry 440 VAC cable shield	☐ Check box when complete
PDU TS5-1	120vac Phase A (Brown)	Console Power Plug: 	☐ Check box when complete
PDU TS5-2	120vac Neutral (Light Blue)		☐ Check box when complete
PDU TS5-3	Ground (Green/Yellow)		☐ Check box when complete
PDU TS5-4	120vac Phase A (Black)	Gantry LV Power Pan A1J1 Filter - L1	☐ Check box when complete
PDU TS5-5	120vac Phase B (Red)	Gantry LV Power Pan A1J1 Filter - L2	☐ Check box when complete
PDU TS5-6	120vac Phase C (Orange)	Gantry LV Power Pan A1J1 Filter - L3	☐ Check box when complete
PDU TS5-7	120vac Neutral (Light Blue)	Gantry LV Power Pan A1J1 Filter - N	☐ Check box when complete
PDU TS5-8	Ground (Green/Yellow)	Gantry Power Pan Chassis A1J1 Filter Ground Stud	☐ Check box when complete

Figure 3-1 Front View of NGPDU, with Covers Removed

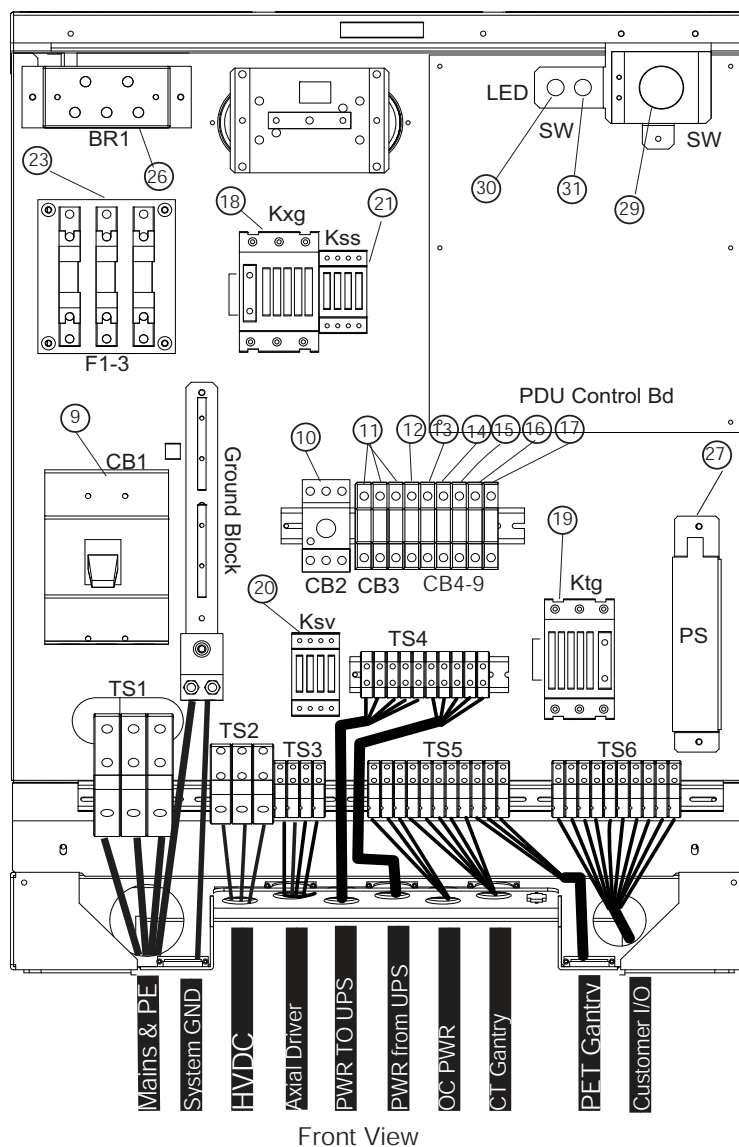
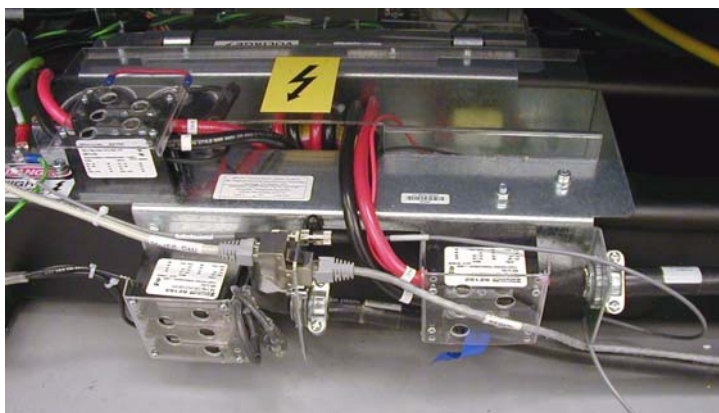


Figure 3-2 Gantry Power Pan





WARNING



TURN OFF ALL PDU CIRCUIT BREAKERS.

- 7.) Set an ohmmeter to the lowest scale. Check between the following points for shorts to ground. Verify no continuity exists between the following points:

Table 3-3 No Continuity Verification Points

FROM PDU	TO A1 BREAKER BOX	
TS2-1 (+HVDC) (Red)	vault ground	☐ Check box when complete
TS2-3 (-HVDC) (Black)	vault ground	☐ Check box when complete
TS3-1 (440vac output) (Black)	vault ground	☐ Check box when complete
TS3-2 (440vac output) (Red)	vault ground	☐ Check box when complete
TS3-3 (440vac output) (Orange)	vault ground	☐ Check box when complete

- 8.) Leave the metal cover off the PDU A3 input power panel until you complete the checks in the next section.

Section 2.0

Site Ground Continuity Check

Use an ohmmeter to verify the presence of **less than 1.0 ohm of resistance** between each of the following points:

Table 3-4 Resistance Verification - Site Ground

FROM	TO	
PDU Ground Bus	Vault Ground	☐ Check box when complete
PDU Ground Bus	Table/Gantry raceway ground point	☐ Check box when complete
Table/Gantry raceway ground point	Gantry Chassis	☐ Check box when complete
Table/Gantry raceway ground point	Table Chassis	☐ Check box when complete
Table/Gantry raceway ground point	Operator Console Chassis	☐ Check box when complete
All Display or Computing Options (if any)	Operator Console Chassis	☐ Check box when complete

Section 3.0

Axial Head Holder Shim Installation

3.1 Overview

This procedure applies to following Table Types:

GT2000	All head holders
GT1700	All head holders
HP 1600	All head holders
PET Table	All head holders

3.2 Requirements

Table 3-5 Personnel Requirements

REQUIRED PERSONS	PRELIMINARY REQS	PROCEDURE	FINALIZATION
1	10 mins	15 mins	5 mins

Table 3-6 Tools and Test Equipment

ITEM	QTY	EFFECTIVITY	PART#	MANUFACTURER
Stand FE Tool Kit	1	-	-	-

Table 3-7 Replacment Parts

ITEM	QTY	EFFECTIVITY	PART#	MANUFACTURER
Shim Kit	1	-	-	-

NOTICE



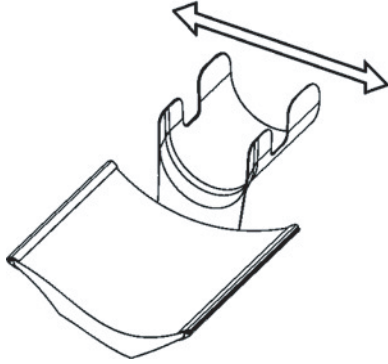
Understand and Follow All General Table Safety Procedures.

3.3 Required Conditions

Check head holder for a tight fit. If the head holder fit is loose, follow this procedure.

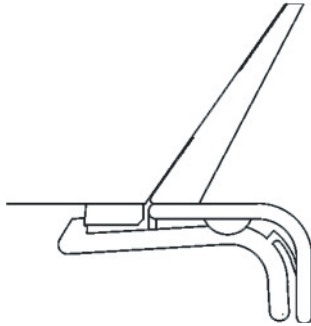
Introduction:

- Some Axial Head Holders have a large free-play in the horizontal direction which could potentially lead to motion and therefore image artifacts.
- Installation of the 2327335 rubber shim kit can minimize this motion.

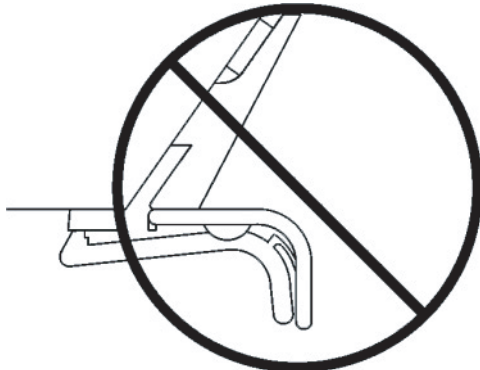


Notes before Selecting Shim Thickness:

- While selecting the best shim size, do not attach the rubber shim yet using the adhesive on the back. It is best to use a piece of tape to hold on the shim in order to see if the size is correct.
- Selecting a shim size that is too thick may result in:
 - Difficulty latching the head holder properly. The head holder must latch so that a patient is not injured.
 - Damage to the plastic latch or the plastic screws that secure it.



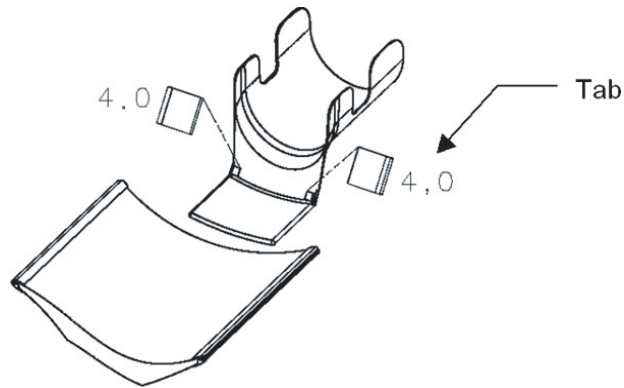
Correct - Head Holder is latched onto first step of plastic latch mechanism (The head holder does not need to be latched onto the second step)



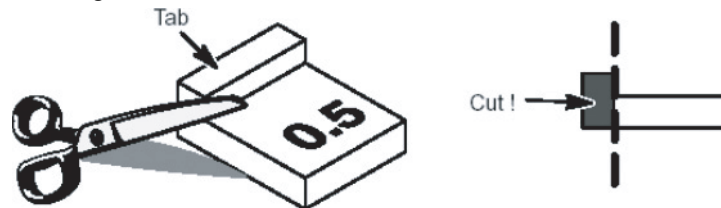
Wrong - Head Holder is NOT latched after installing shims

3.4 Procedure

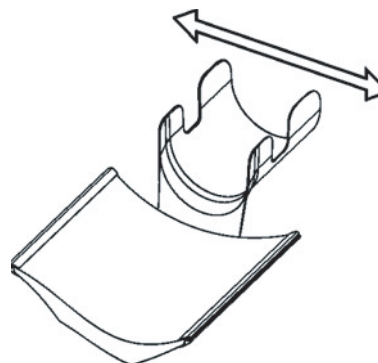
- 1.) First place the two 4.0mm shims (thickest size) onto both edges of the head holder as shown (use a piece of tape to temporarily secure them)
 - The shim must be placed with the tab facing out
 - The thickness is printed on the shim



- 2.) Insert the head holder into the cradle
- 3.) Check if the head holder is latched onto the cradle at the first step of the plastic latch mechanism. (The head holder does not need to be latched onto the second step)
- 4.) Check if the head holder has a small free-play in the horizontal direction. If the rubber is too thick, repeat steps 1-4 using a thinner shim (3.5, 3.0...0.5mm) until the head holder is latched (without excessive force) and fits securely in the cradle. If the thinnest shim (0.5mm) is too tight, the tab can be cut off to reduce the thickness



- 5.) Clean off the surfaces where the shims will mount using alcohol.
- 6.) Peel off the paper from the back of the selected shims and attach with the tabs facing out. Hold each shim with your fingers for a few seconds to attach it to the head holder.



3.5 Finalization

Review latching of head holder with customer after installation.

Section 4.0

Mechanical Installation Completion Section

Complete the *Mechanical Installation* section of the GE e4879 and turn it over to your site FE. The electronic file for the checklist is found on the BrightSpeed Series Service Information CD-ROM 5193754-200. The electronic checklist is also located in the downloadable forms directory of the CD-ROM.

Appendix A

Removal & Installation of Covers

Section 1.0 Gantry Side Covers

1.1 Side Cover Removal

CAUTION



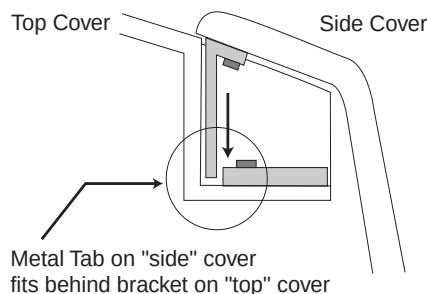
- 1.) Lower table to home (lowest) position.
Potential for injury if covers removed and power is left "ON".
Always remove the right side cover first, and turn OFF power at the Service Switch Panel.
- 2.) Use an 8mm Hex wrench, turn the latch $\frac{1}{4}$ turn to latch and unlatch the side cover from the front and rear covers.

Figure A-1 Side Cover Latches



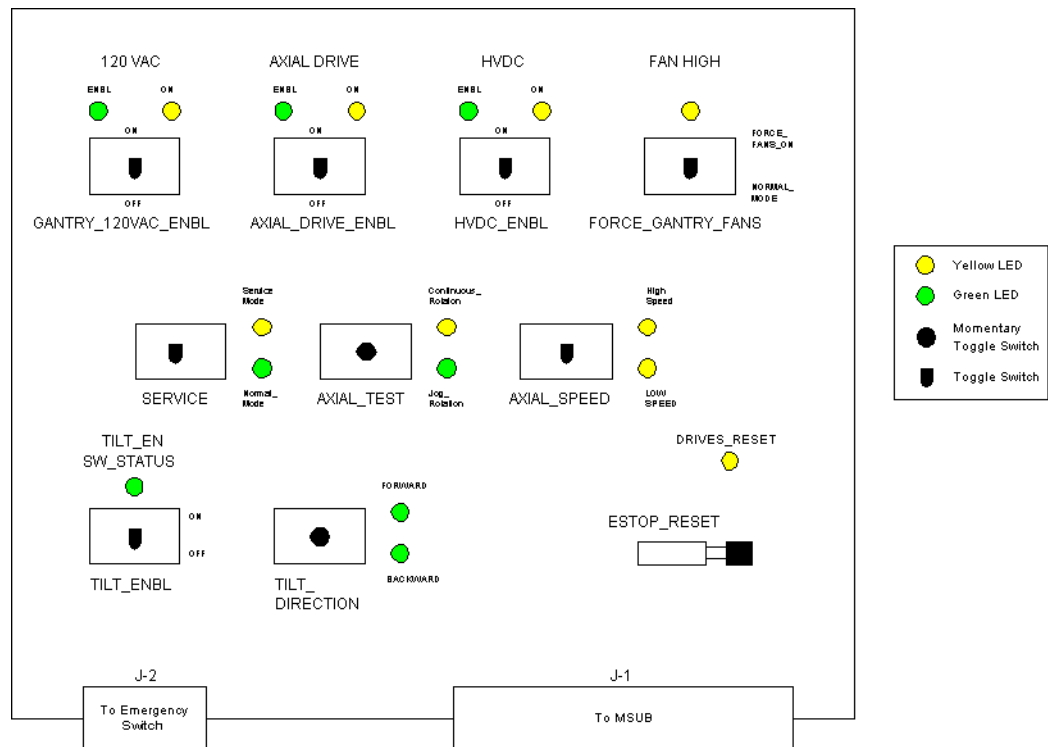
- 3.) Remove the right side cover by lifting it upward to release the two (2) latches, located on the top edge of the cover.

Figure A-2 Side and Top Cover Clasp



- 4.) Turn OFF the **HVDC**, **120 VAC**, and **AXIAL DRIVE** power switches on the gantry service switch panel (see [Figure A-3](#)).

Figure A-3 Gantry Service Switches



- 5.) Repeat steps 1-3 for the left side cover.

1.2 Side Cover Installation

- 1.) To install a side cover, place it over the top cover and let the two (2) side cover latches slide behind the metal tabs, located on the top cover. See [Figure A-2](#).
- 2.) Use Hex wrench to secure the side cover to front cover by turning the bolts a quarter turn. See [Figure A-1](#).

Section 2.0 Gantry Top Covers

2.1 Top Cover Removal



CAUTION

Potential for Shock.

Voltage may be present. Potential for injury if covers removed and power is left ON. Always remove the right side cover first, and turn OFF power at the service switches.

- 1.) Remove the gantry right side cover (see Gantry Side Covers Removal and Re-install).

NOTICE



Always turn OFF the HVDC before the 120 VAC. Turning OFF 120 VAC power before HVDC power can result in equipment damage.

- 2.) Turn OFF the **HVDC**, **120 VAC**, and **AXIAL DRIVE** power switches on the gantry service switch panel (see [Figure A-3](#)).
- 3.) Disconnect the power cable that connects the fan on the top cover to the gantry.
- Note: The cooling fans are now mounted on the top covers, one on each.
- 4.) Loosen the 2 Phillips screws which secure the top cover with the front and rear covers.

Figure A-4 Top Cover Mounting Screws



- 5.) Take the end of the top cover nearest to the side cover and tilt upwards.
- 6.) Slide the cover down to disengage the tab from the mounting bracket. See [Figure A-5](#).

Figure A-5 Top Cover Tabs and Bracket



- 7.) Lift the cover clear and repeat the above steps for the other cover.

2.2 Top Cover Installation

The top cover consists of two (2) pieces. Install the front and rear gantry covers, if not already installed. See [Section 3.0 on page 120](#), and [Section 4.0 on page 128](#).

- 1.) Take one of the top covers and align the tabs on the cover with its associated bracket. Lift and slide the cover into place. Position the cover to fully engage the fan interlock switch. Secure the cover using 2 Phillips screws.
- 2.) Take the other top cover and align the tabs on the cover with its associated bracket. Lift and slide the cover into place, while being sure to engage the fan interlock switch. Secure the cover using 2 Phillips screws.
- 3.) Connect the cable from the fans to the gantry.

NOTICE



Always turn OFF the HVDC before the 120 VAC. Turning OFF 120 VAC power before HVDC power can result in equipment damage.

- 4.) Turn on the three (3) power switches.
- 5.) Ensure fans work properly.
- 6.) Re-install the gantry side covers.

Section 3.0

Gantry Front Cover

NOTICE Potential for cover damage.



Front and rear cover removal and installation can be safely accomplished by one (1) person using the dollies provided with the system. Failure to use these dollies will significantly increase the likelihood of damage to the covers. Do not lean covers against walls.

3.1 Front Cover Dolly Setup

DANGER



DO NOT USE DOLLIES ON UNEVEN SURFACES SUCH AS STEPS OR ELEVATOR THRESHOLDS. THE DOLLIES ARE DESIGNED TO BE USED ON FLAT LEVEL FLOORS WITHIN THE SCANNING SUITE ONLY. MISUSE CAN RESULT IN PERSONAL INJURY OR DAMAGE TO COVERS OR OTHER FACILITY ITEMS.

CAUTION



Rotating arms on the stand are supposed to be stiff. If they fall freely, tighten the tensioning nuts. Loose rotating arms will reduce the stability of the dollies when supporting the front cover. Do not lubricate.

- 1.) Arrange Dolly sections for assembly. The base and post can be assembled only one way.
Refer to [Figure A-6](#) and [Figure A-7](#).
 - The base uses two (2) palm screws to clamp the four (4) legs in the open or usage mode.
 - The base also uses the same palm screws to prevent the legs from falling in storage mode.
 - The top post can be inserted in either base and is keyed for proper engagement.
 - The top post locking pin prevents the sections from separating during usage.

Figure A-6 Front Cover Dolly in Storage Mode

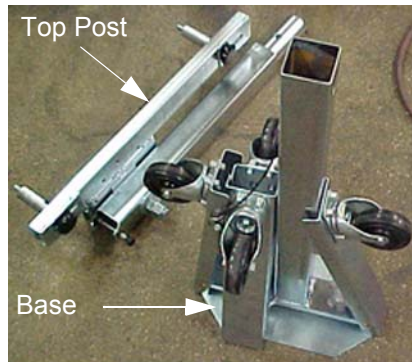
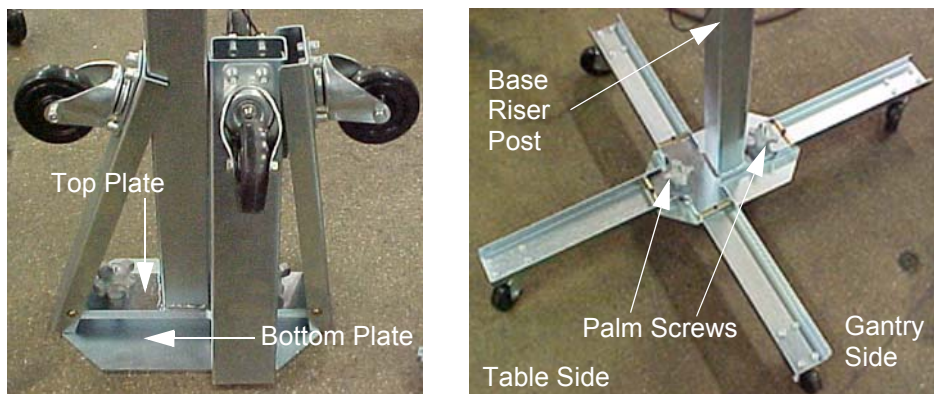


Figure A-7 Front Cover Dolly Base Assembly



- 2.) Unfold the base legs by loosening both palm screws to the top of their travel.
- 3.) Carefully unfold the legs so that the castors touch the floor.
- 4.) Tighten the palm screws to clamp the legs between the base top and bottom plates.

Note: Lifting the base by the riser post while leaving the castors on the floor will ease palm screw tightening. Reference [Figure A-7](#).

WARNING



ENSURE BOTH PALM SCREWS ARE TIGHTENED SECURELY AND THE LEGS ARE CLAMPED TIGHTLY BETWEEN THE BASE TOP AND BOTTOM PLATES. FAILURE TO DO SO WILL RESULT IN INSTABILITY DURING FRONT COVER HANDLING.

- 5.) Insert top post into the base riser post. Align the key for complete engagement.
- 6.) Insert top post locking pin to secure both top and bottom sections.
- 7.) Reverse above steps to disassemble.

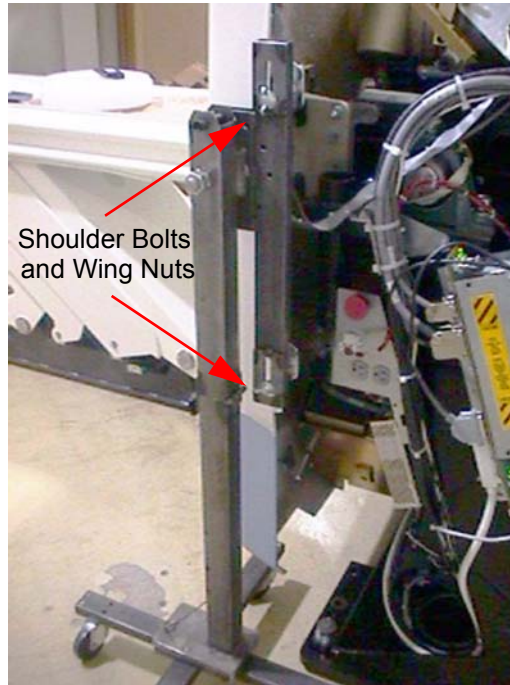
Note: For base storage only one (1) palm screw needs to be tightened. This will engage the bottom base plate and the leg ends preventing the legs from unfolding during transport and storage.

3.2 Removal

- 1.) Position the table at its lowest position.
- 2.) Remove gantry side and top covers, if you have not already done so. See [Section 1.0 on page 117](#). Make sure that the three (3) power switches have been turned off. See [Figure A-3](#).

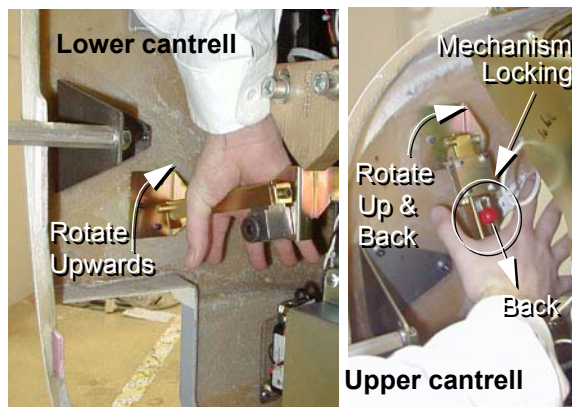
- 3.) Assemble the front cover dolly.
 - a.) Tighten the two (2) shoulder bolts to the gantry securely. This will make cover installation easier. See [Figure A-8](#).

Figure A-8 Front Side Dolly



- b.) Attach side dolly to the shoulder bolts and secure assembly with two (2) wing nuts.
 - c.) Repeat steps a and b to assemble the other side dolly.
- 4.) Detach front cover J3 and J2 and front cover BKHD J1 cables.
- 5.) Remove front cover
 - a.) Disengage upper and lower cantrell brackets on both sides of the cover.
 - 1.) Using steady but firm pressure, lift each of the lower cantrell brackets from their associated retainers. See [Figure A-9](#).

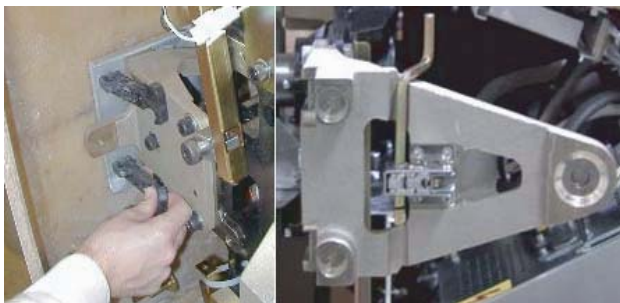
Figure A-9 Releasing cover brackets



- 2.) Disengage the locking mechanism on the upper cantrell brackets by using your thumb to slide the trigger (red lever) back. This will release the locking mechanism and allow the cantrell to be rotated upwards with steady and firm pressure.

- b.) Disengage the rubber retaining straps on both sides. You may find it helpful to lift “up” on the cover to align the stud while attaching the rubber retaining straps.
- c.) Also lift and rotate cover locking arm to unlocked position.

Figure A-10 Rubber retaining straps and Cover Locking mechanism



NOTE: The rubber retaining straps at the right side of the gantry are different from the current HP60 system.

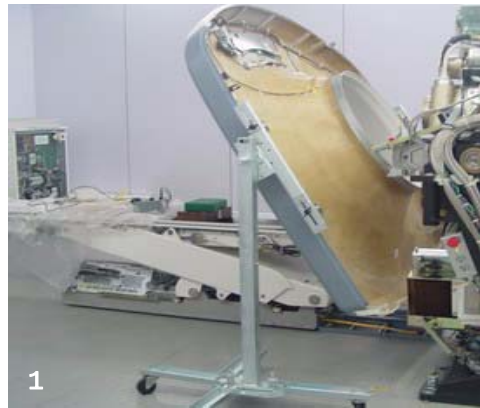


- 6.) Rotate front cover away from gantry.
 - a.) Move front cover away from gantry giving enough space (about 5 feet) between front cover and gantry.
 - b.) Pull the locking pin and rotate front cover away from gantry. Place locking pin in one of the side dolly perforations. See [Figure A-11](#).

Figure A-11 releasing Front Cover Dolly Hinge



Figure A-12 Front Cover Removal Sequence



1. Upper Left>Rotate cover to make room for step 2.
2. Upper Right>Rotate cover to clear the table. Roll the cover to foot end of table
3. Lower Left> Rotate the cover upside down to provide clear work area.
Note: Remove Gantry Display and Control Panel in position 2.

- 7.) Rotate the cover horizontally and move it back and over the table to a safe location. Once in a safe location, you may over-rotate the cover full vertically but upside down.
- 8.) Remove the gantry display and one (1) of the cover's control assemblies, and place them into the service positions.
 - a.) Remove the gantry display and place it into its service position.
 - * The gantry display is held in place with (6) thumb screws. Use a flat-blade screwdriver to remove the Display. Reference [Figure A-13](#).

Figure A-13 Gantry Display Removal



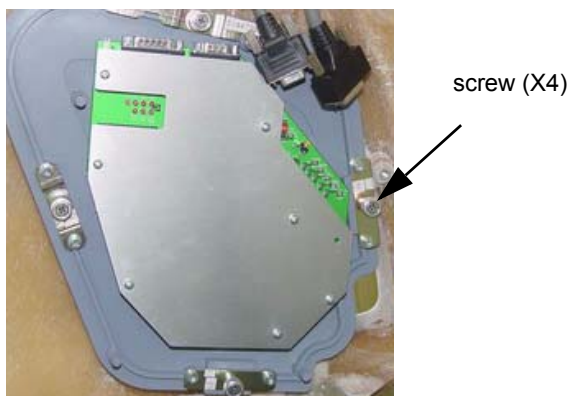
- * Place the Display in the bracket on the right side of the gantry. See [Figure A-14](#).

Figure A-14 Gantry Display Service Mounting Location



- * Disconnect the cabling at the right rear gantry cover. Only (1) cable will connect to the Gantry Display. Connect the cable taken from the rear cover to the display.
- b.) Remove a gantry control and place it into its service position.
- 1.) Loosen the screws that mounts the gantry control. Keep one hand on the control panel at all times to prevent it from dropping to the floor.

Figure A-15 Gantry control panel removal



- 2.) Align the ball studs with their associated receivers and snap into place.

Figure A-16 Control panel service position



- 3.) Connect cable to terminator located on the cantrell arm. Reference [Figure A-17](#).

Note:

There are 3 cables, each of which is unique. The ribbon cable is not used in the Service configuration. The other 2 cables will only fit in the terminator or the control panel, not both.

Figure A-17 Gantry Service Mode Cable Terminator



3.3 Installation

- 1.) Remove the gantry display and control assembly from their service positions and re-attach them to the gantry cover.
 - a.) Disconnect cables from Display and Gantry Control Panels.
 - b.) Install Gantry Display in front cover. Secure the 6 thumb screws. With a flat-blade screw driver gently tighten past finger tight.
 - c.) Install the gantry control panel making sure the screws are secure within the receivers.
 - d.) Re-attach cables.

- 2.) Rotate gantry back to its vertical position.

NOTICE

Potential for front cover damage.

When you put rotate the gantry back to its vertical position, make sure not to scratch the front cover with the edge of the table cradle.

- 3.) Attach the front cover.

- a.) Align the studs on both sides of the front cover with each associated receiver. Receiver is located on the gantry frame.

Figure A-18 Cover stud and Mounting bracket receiver



- b.) Insert the stud on one side into its associated receiver and attach the rubber retaining straps. Then insert the stud on the other side into its associated receiver and attach its rubber retaining straps.

You may find it helpful to lift “up” on the cover to align the stud while attaching the rubber retaining straps.

- 4.) Re-attach upper and lower cantrell brackets on both sides.
 - a.) Remove upper Cantrell brackets from service position and rotate them into position over their associated retaining pins. See [Figure A-19](#).

Figure A-19 Service position of upper and lower cantrell brackets

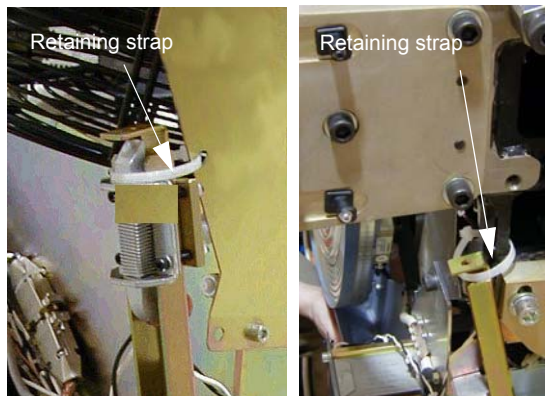
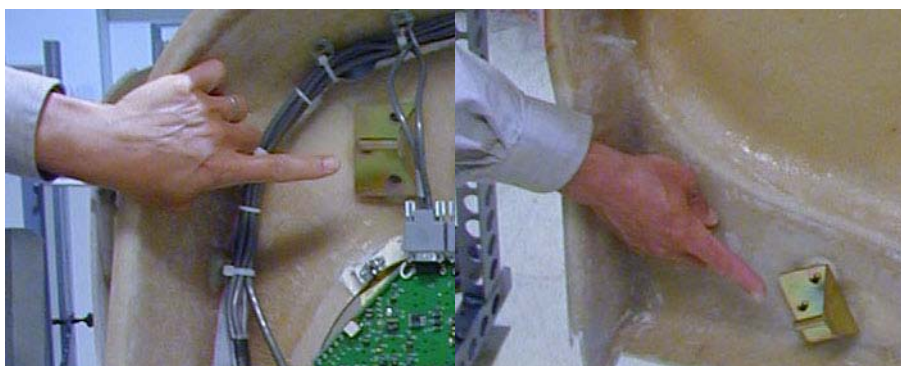


Figure A-20 Cover retaining pins (top and bottom)



Press down firmly on the bracket and snap it into place. The locking mechanism on each upper bracket should lock the bracket securely into place. Do this on both sides. See [Figure A-21](#).

Figure A-21 Locking the cover brackets into place



- b.) Remove lower cantrell brackets from service position (see [Figure A-19](#)), and rotate them into position over their associated retaining pins. Press down firmly on the bracket and

snap it into place. See [Figure A-21](#).

Note: Mis-adjustment of the cantrell brackets can cause misalignment of the top and side covers. The upper and lower cantrell brackets do not require adjustment during normal use.

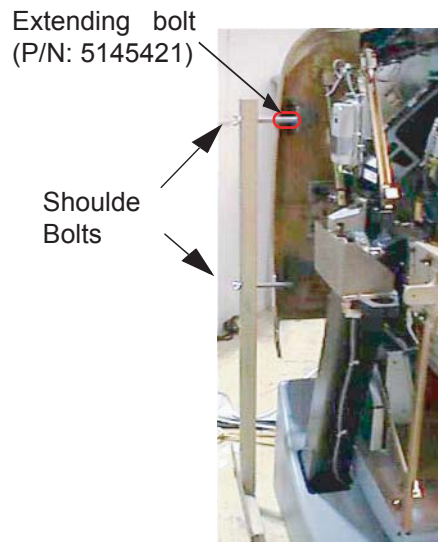
- 5.) Remove dolly, disassemble and store safely away for later use.
- 6.) Re-attach cables to cover.

Section 4.0 Gantry Rear Cover

4.1 Removal

- 1.) Assemble the rear cover dolly.
 - a.) Tighten the two (2) shoulder bolts to the rear cover.

Figure A-22 One side of the Rear cover dolly



- b.) Fit side dolly through the shoulder bolts and secure assembly with two (2) wing nuts. See [Figure A-22](#).
 - c.) Repeat steps a and b for the other side dolly.

Potential for injury if covers removed and power is left "ON".

- 2.) Disconnect cables on the right side of the rear cover.
- 3.) Remove rear cover.
 - a.) Disengage upper and lower cantrell brackets on both sides of the rear cover.
 - 1.) Using steady but firm pressure, lift each of the lower cantrell brackets from their associated retainers. See [Figure A-9](#).
 - 2.) Disengage the locking mechanism on the upper cantrell brackets by using your thumb to slide the trigger (red lever) back. This will release the locking mechanism and allow the cantrell to be rotated upwards with steady and firm pressure.
 - b.) Disengage the rubber retaining straps on both sides.



4.2 Installation

- 1.) Position cover in back of gantry
- 2.) Attach the rear cover
 - a.) Align the studs on both sides of the rear cover with the receivers located on the gantry frame.
 - b.) Insert the stud on one side into its associated receiver and attach the rubber retaining straps. Then insert the stud on the other side into its associated receiver and attach its rubber retaining straps.

Note: You may find it helpful to lift “up” on the cover to align the stud while attaching the rubber retaining straps.

- 3.) Reattach upper and lower cantrell brackets on both sides.
 - a.) Remove upper cantrell brackets from service position and rotate them into position over their associated retaining pins. Press down firmly on the bracket and snap it into place. The locking mechanism on each upper bracket should lock the bracket securely into place. Do this on both sides.
 - b.) Remove lower cantrell brackets from service position and rotate them into position over their associated retaining pins. Press down firmly on the bracket and snap it into place.

Note: Adjustment of the cantrell brackets can cause misalignment of the top and side covers. The upper and lower cantrell brackets do not re-quire adjust during normal use.

- 4.) Remove dolly, disassemble and store safely away.
- 5.) Re-attach cables to cover.
- 6.) Re-install the mylar (scan) window. Carefully bend the scan window and place it into the channel (groove) provided in the covers.

Figure A-23 Installing the mylar window



Section 5.0 Gantry Cover Alignment Adjustment Guide

5.1 Overview

This section explains the adjustment capability available for the Gantry covers and guidelines for performing adjustments.

5.2 Tools Required

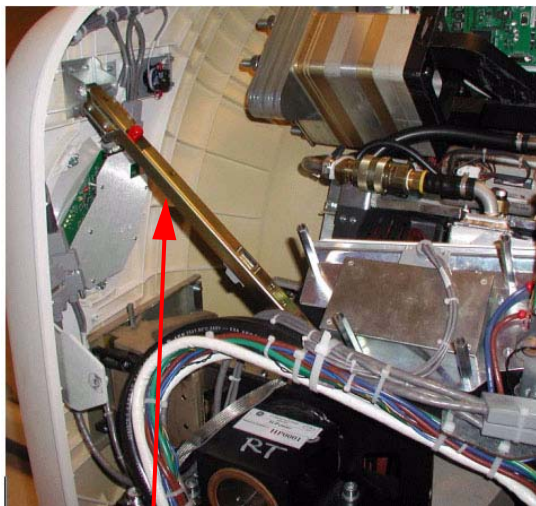
3mm, 5mm, 8mm, Hex wrenches

5.3 Adjustment Guide

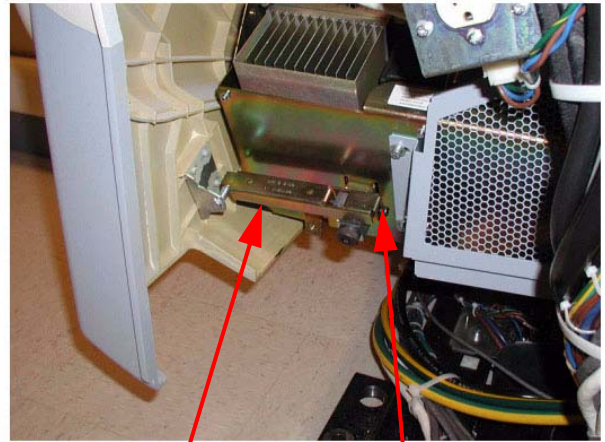
0.0.1 Front/Rear Cover adjustment

The front and rear covers are held in a fixed vertical position by the cover locking mechanisms at the vertical middle of each cover on each side. The only adjustment capability is found on each of the cover cantrells (see [Figure A-24](#), one near each corner of the front and rear covers. Adjustments are best done at zero tilt.

Figure A-24 Cantrells



Top Cantrell same
for all top latches



Bottom Cantrell same
for all bottom latches

Adjustment screw on
end of all brackets

- 1.) Remove the Gantry side covers using a 8mm hex wrench to release the bottom latches. Refer to [Section 1.0 on page 117](#), for further details.
- 2.) Disable gantry rotation, 120VAC and HVDC via the service switch panel.
- 3.) Disconnect the fan power via the white molex connector (one for each top cover). Remove the top covers by lifting up and pulling to the side of the gantry.
- 4.) Remove the scan window to avoid creases or kinks during cover movements.
- 5.) Using a 5mm hex wrench, adjust the bolt on the gantry end of the cover cantrells (see [Figure A-24](#)) to lengthen or shorten the cantrell arm that will push or pull the cover corner. The distance between the inside edges of the front and rear covers will be typically $27" + 1/8"$ (689mm). Distance should remain the same as measured along the inside edges from top to bottom with both covers hanging level as seen with a level on the inside vertical edge.

0.0.2 Scan Window

Position the scan window in the opening to verify that it seats all the way around with no gaps. The scan window should seat easily if the front and rear covers have been adjusted properly. If not, go back and readjust such that the scan window seats properly. There is no easy solution other than adjusting and looking for fit.

0.0.3 Top Cover Adjustment

The top covers rest in a top bracket that is mounted to the stationary gantry frame and subsequently just rest on top of the front/rear covers. There are also two alignment fingers on the outer edge of the top covers that need to sit in brackets on the front/rear covers.

Figure A-25 Top Cover Bracket



Screws on both sides
allow adjustment up/
down only

- 1.) Set the top covers back on the gantry and look for proper fit. The top covers should rest on the front/rear covers all along the edges. If they do then skip the rest of this section.
- 2.) If the top covers do not rest on the gantry at the top then lift them back off and use a 5mm hex wrench to loosen the bolts that hold the top cover bracket for adjusting vertical alignment to allow the top covers to rest on the front and rear covers.
- 3.) The brackets on the front/rear covers for the fingers on the outer edge of the top covers are adjustable using a 3mm hex wrench and sliding the bracket till it lines up with the top cover finger. (See [Figure A-26](#))

Figure A-26 Top Cover Fingers



- 4.) Verify the top cover tapered edges fit over the lip along the top edge of the front/rear covers. If not, go back to Section 0.0.1 and readjust the cantrells as necessary.
- 5.) When done adjusting the top covers, reconnect the fan power molex connections.

0.0.4 Side Covers

The side covers just hang from the top covers by two fingers that are not adjustable. When set into the top covers, the side covers should hang straight down with the tapered edges fitting over top of the lip of the front/rear covers.

0.0.5 Finish

- 1.) Re-Enable gantry rotation via the Axial Enable switch on the service switch panel.
- 2.) Replace the side covers and latch using a 8mm hex wrench on the side cover bottom latches.
- 3.) Run an axial scan to ensure proper system operation prior to turning system back over to the customer.

Section 6.0 Gantry Scan Window

6.1 Remove Scan Window

- 1.) Grab the window at the top and pull firmly downward.
- 2.) Continue to pull until the top of the scan window makes contact with the bottom portion of the scan window.
- 3.) Hold the top and bottom portions of the scan window together, grasp both sides of the scan window, move them together and lightly pull upward, until you can free the window from between the front and rear covers.

Figure A-27 Scan Window Removal



6.2 Install Scan Window

- 1.) Install the front and rear covers.
- 2.) Deform the scan window, as shown in [Figure A-28](#), and nest the scan window at the bottom of the opening between the front and rear covers, ([Figure A-29](#)) with the rivets in the 6 o'clock position installation position. Remember the rivets must be in the 12 o'clock position when the mylar window is fully installed.
- 3.) After you complete the initial seating of scan window, let the window slowly unfold, and work both sides of the window into position, starting at the bottom and finishing at the top.
- 4.) Make sure you position the window with the rivets at the 12 o'clock position, and the mylar window slit at either the 3 or 9 o'clock position.

Figure A-28 Install Scan Window

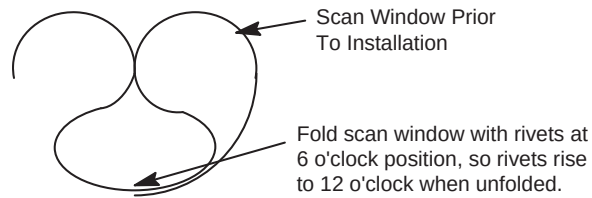
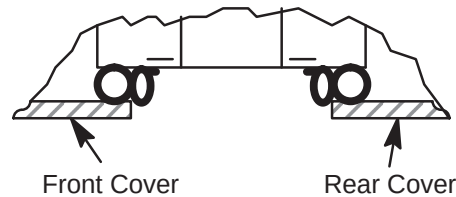


Figure A-29 Scan Window Nested Between Front and Rear Cover



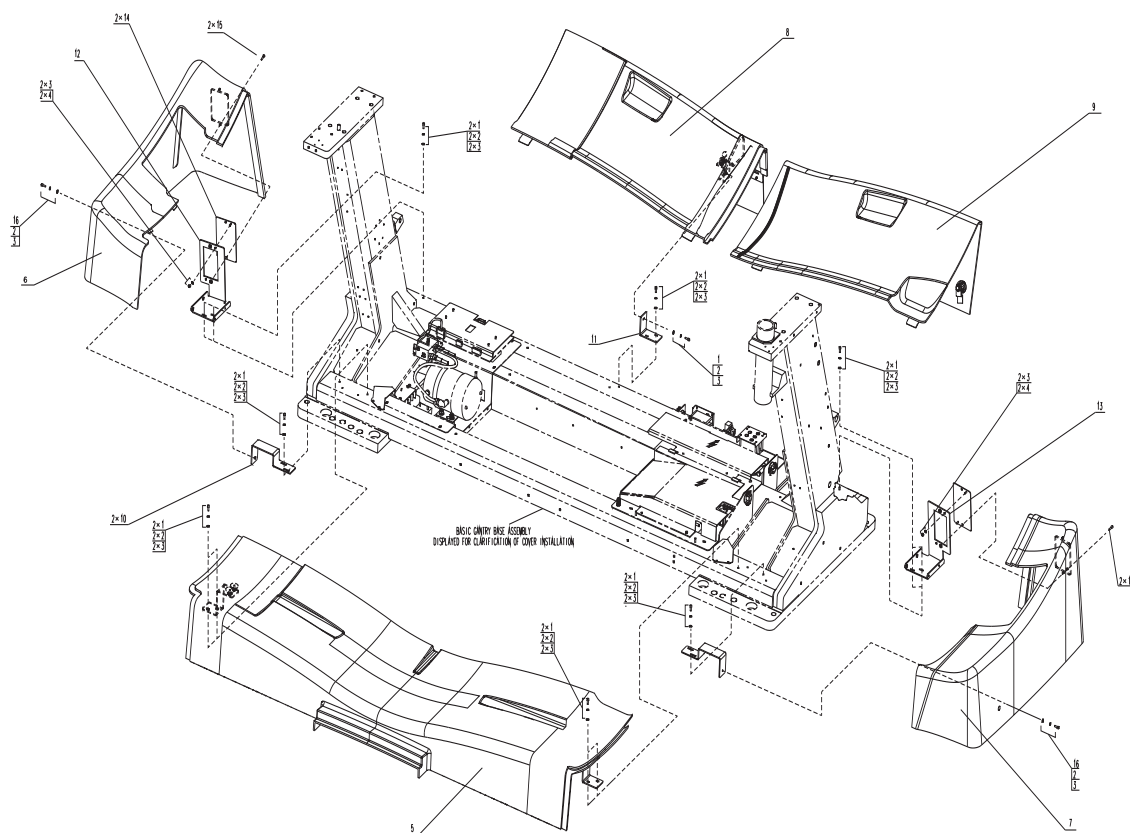
Section 7.0 Gantry Base Covers

Refer to [Figure A-30](#) for the following assembly sequence.

Note: Tighten means torque to 2.3 Nm

- 1.) Position cover (item 5) on gantry base with bracket slots aligned to gantry holes. Center cover left to right and attach with (4) hardware (items 1, 2, 3) as shown and tighten.
- 2.) Assemble two (2) bulkheads (item 14) to two (2) brackets (item 13) using (4) hardware items 3 and 4. Assemble two (2) brackets (item 10) and two (2) brackets (item 13) to gantry base using eight (8) hardware items 1, 2 and 3. Finger tighten hardware with bracket moved outward to end slots. Install side covers (item 6 and 7) on base pushing brackets (item 11 and 13) inward until properly aligned with front cover. Remove side covers, tighten fasteners and replace side covers using one (1) hardware (item 16, 2 and 3) and two (2) item 15 on each cover and tighten.
- 3.) Assemble last bracket (item 11) loosely to gantry base with two (2) hardware (item 1, 2 and 3). Install rear cover (item 8) to base properly aligned to side cover (item 6). Attach rear cover to bracket with hardware (items 1, 2 and 3) tightening all fasteners. Lock latch.
- 4.) Place cover (item 9) on gantry base, aligned to covers (item 7 and 8). Lock both latches.

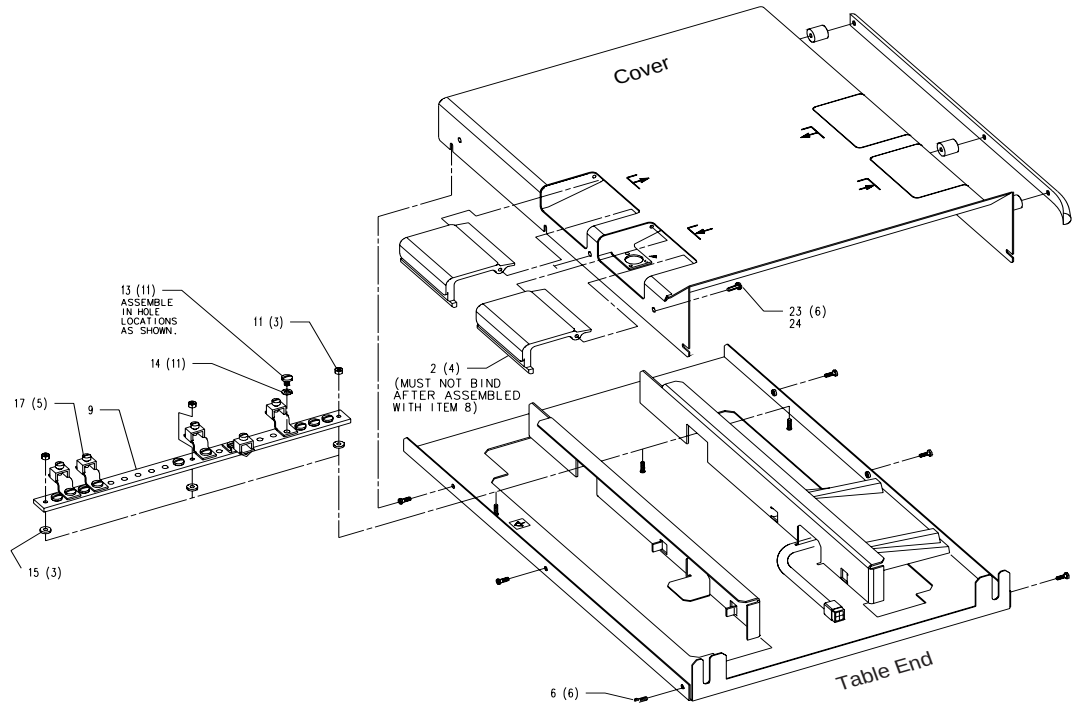
Figure A-30 Gantry Base Covers



Section 8.0

Footswitch Covers

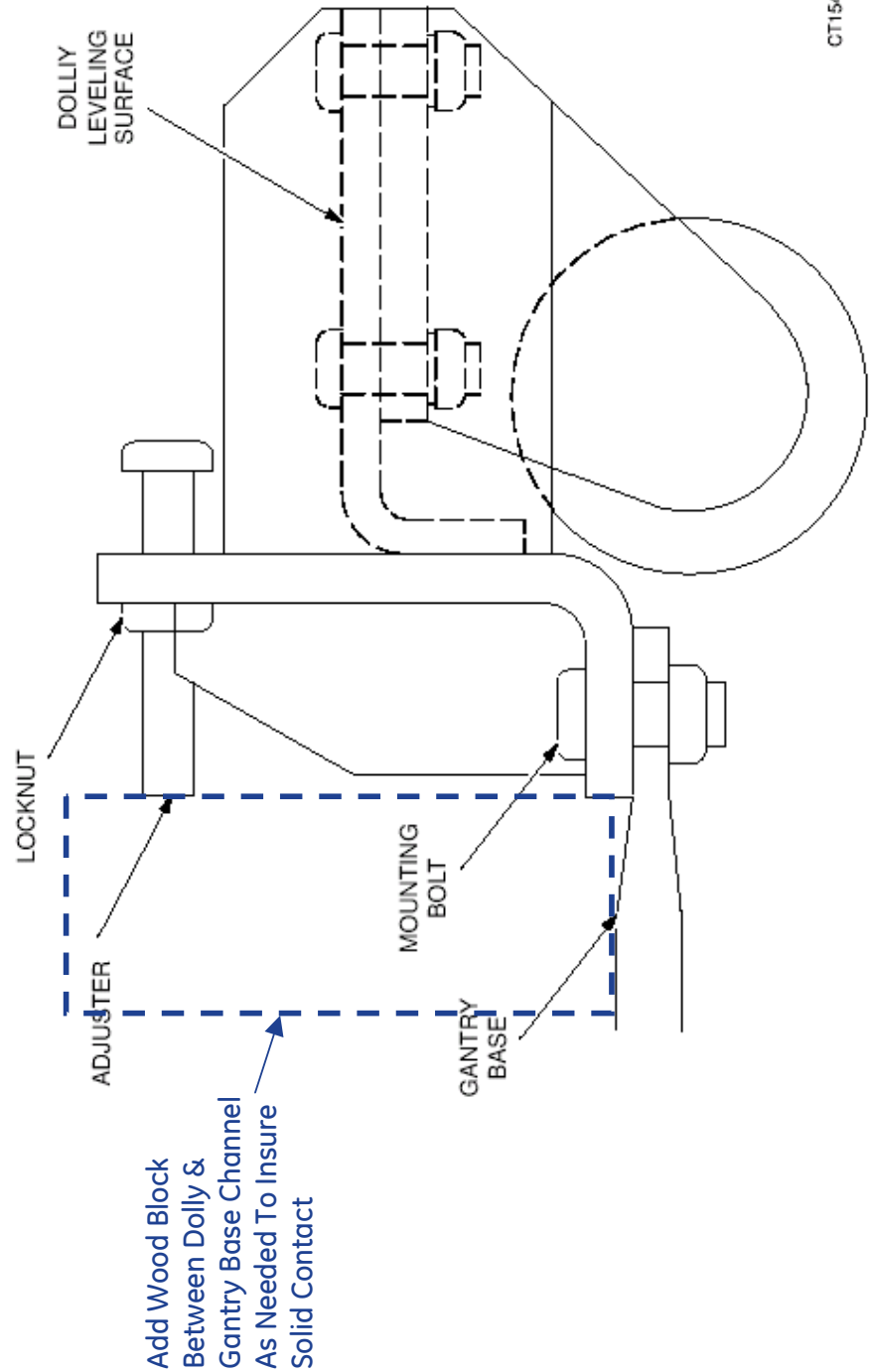
Figure A-31 Table Footswitch Assembly



Section 9.0

Gantry Auxiliary (Mini) Dolly Installation w/Wood Block

CT Gantry Auxiliary (Mini) Dolly Installation Instruction With Wood Block



Appendix B

FWS Assembly and Adjustment

Section 1.0 Introduction

Freedom WorkSpace (FWS) is an optional console table with better ergonomic performance. The monitor arm is designed to adjust quickly and easily.

There are two FWS: part 5160735 and part 5168666. They are different in height adjustment. Refer to respective install procedure during installation.

Before assembly go through this section and checklist with the FWS to have an overview.

- Assemble worksurface
 - Assemble chain channel and chain with table base (only applicable for part 5160735)
 - Assemble table base and worksurface
 - Assemble drawer
- Install monitors
 - Mount pole onto table surface
 - Install monitor with monitor arms
 - Install monitor arms on the pole
 - Route cables
- Adjust the monitors for customer use
- Install Seismic Kit

Figure A-32 Overview of Assembled FWS w/o Peripherals

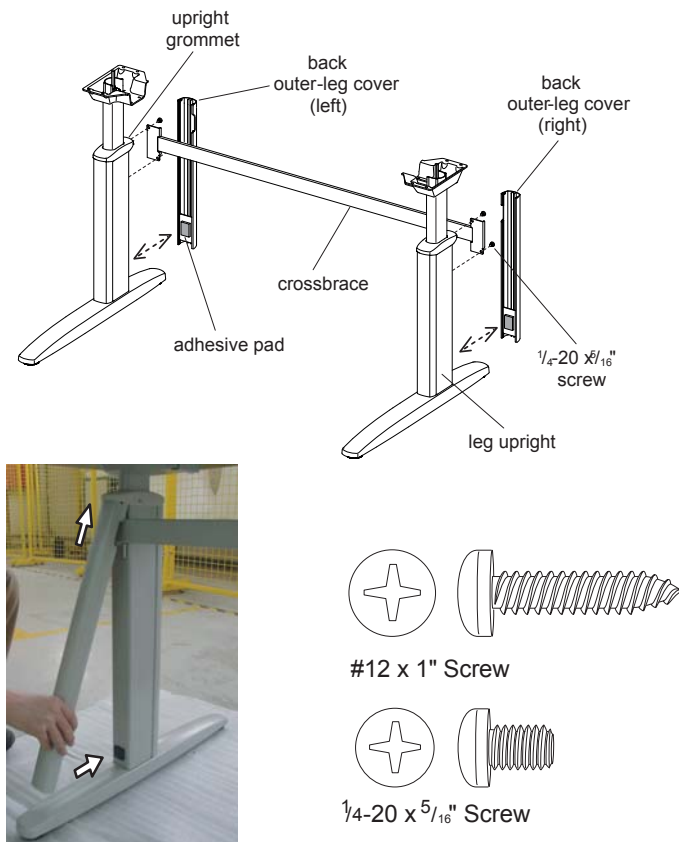


Section 2.0 Procedure

2.1 Assemble Worksurface (For Part 5160735)

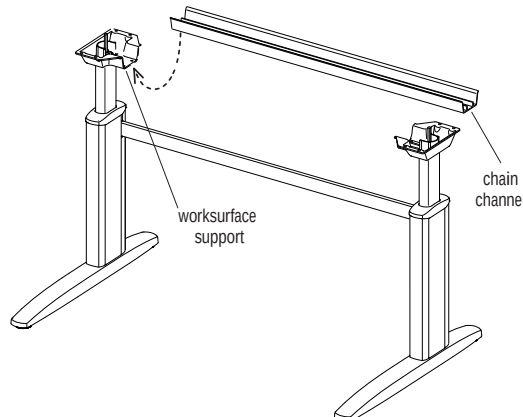
- 1.) Carefully remove the base assembly from the packaging material. With the leg uprights supported, move the legs apart to attach the crossbrace.
- 2.) Secure the crossbrace to the leg uprights with four 1/4 - 20 x 5/16" screws (Figure A-33).
- 3.) Replace the back outer-leg covers to the leg uprights (Figure A-33).
 - a.) Remove the adhesive pad paper on the legs.
 - b.) Position the top of each cover up into the upright grommet.
 - c.) Press the bottom down to secure with the adhesive pad.

Figure A-33



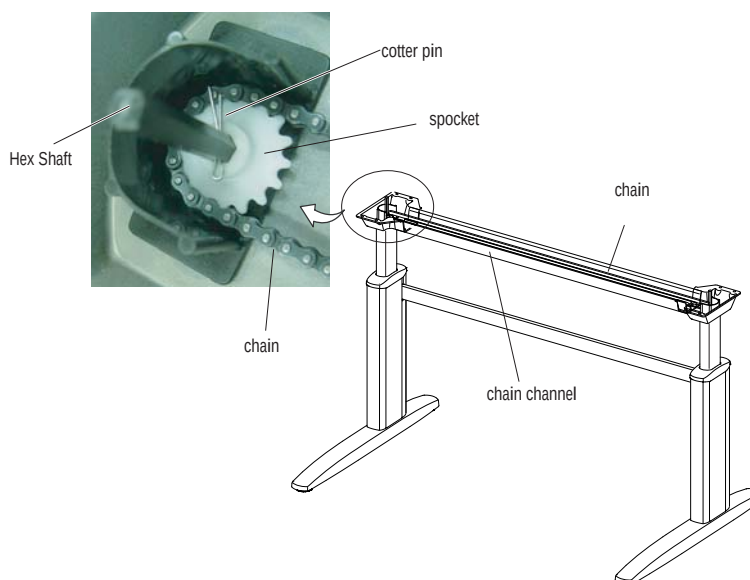
- 4.) Slide the chain channel into the two openings in the worksurface supports (Figure A-34).

Figure A-34



- 5.) Insert hand crank on one hex shaft, adjust the legs height and ensure the chain channel is horizontal.
- 6.) Install chain (Figure A-35).
 - a.) Pull out the cotter pin.
 - b.) Slide the sprocket off over the hex shaft.
 - c.) Engage the sprocket into the chain and reinstall the sprocket.
 - d.) Insert the cotter pin and spread the ends to secure.
 - e.) Repeat the steps on the other side.
 - f.) Insert hand crank on one hex shaft and rotate to ensure the worksurface is adjustable.

Figure A-35



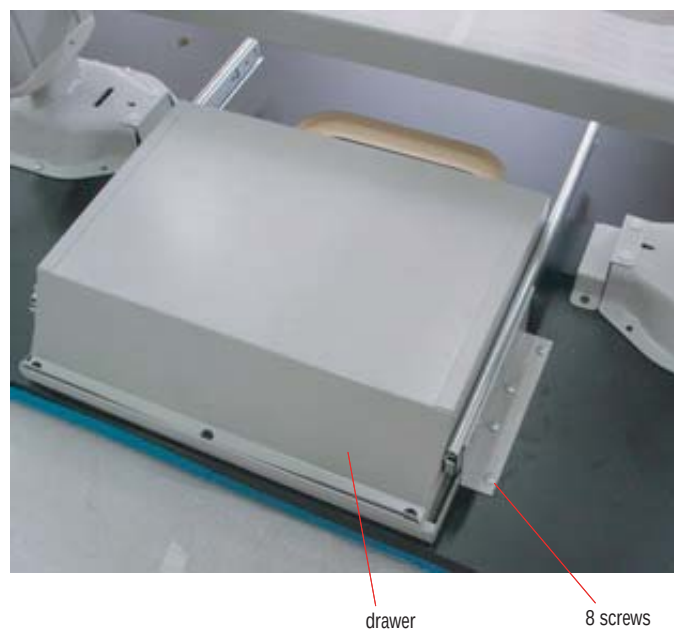
- 7.) Place the worksurface upside down on a clean, soft surface to prevent scratching. Invert the table base onto the worksurface, taking care to align the holes with the pre-drilled holes in the worksurface. Secure the table base to the worksurface with eight #12 x 1" screws (Figure A-36).

Figure A-36



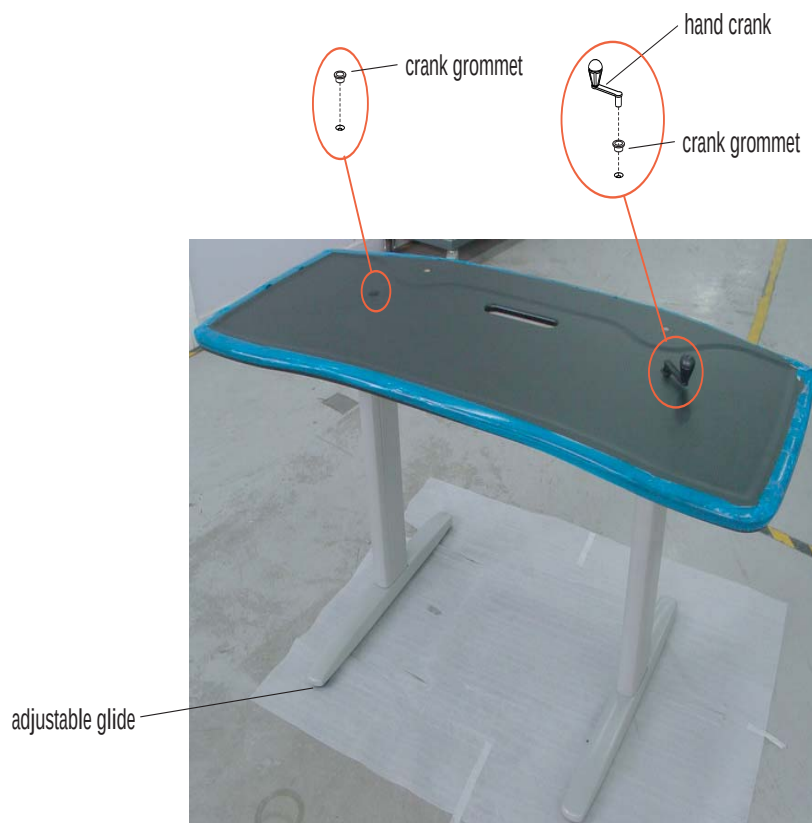
- 8.) Install two Chain Channel Brackets using four metal screws and two plastic push screws. ([Figure A-36](#))
- 9.) Find the 8 drawer screws inside the drawer. Install the drawer. ([Figure A-37](#))

Figure A-37



- 10.) Carefully turn the table assembly to the upright position. Insert crank grommets into holes in the worksurface as illustrated. Insert crank into one grommet and place a filler cap into the other side. Adjust worksurface to desired height with hand crank (Figure A-38).

Figure A-38



- 11.) The four adjustable glides may be raised or lowered to accommodate varying floor surfaces. Lift the leg off the floor and turn the glide clockwise or counterclockwise to adjust (Figure A-38).

2.2 Assemble Worksurface (For Part 5168666)

- 1.) Carefully remove the base assembly from the packaging material.
- 2.) Attach the crossbrace with the two inner-leg cover and secure the crossbrace to the down leg with three M6 x 20 screws (Figure A-40).

Figure A-39

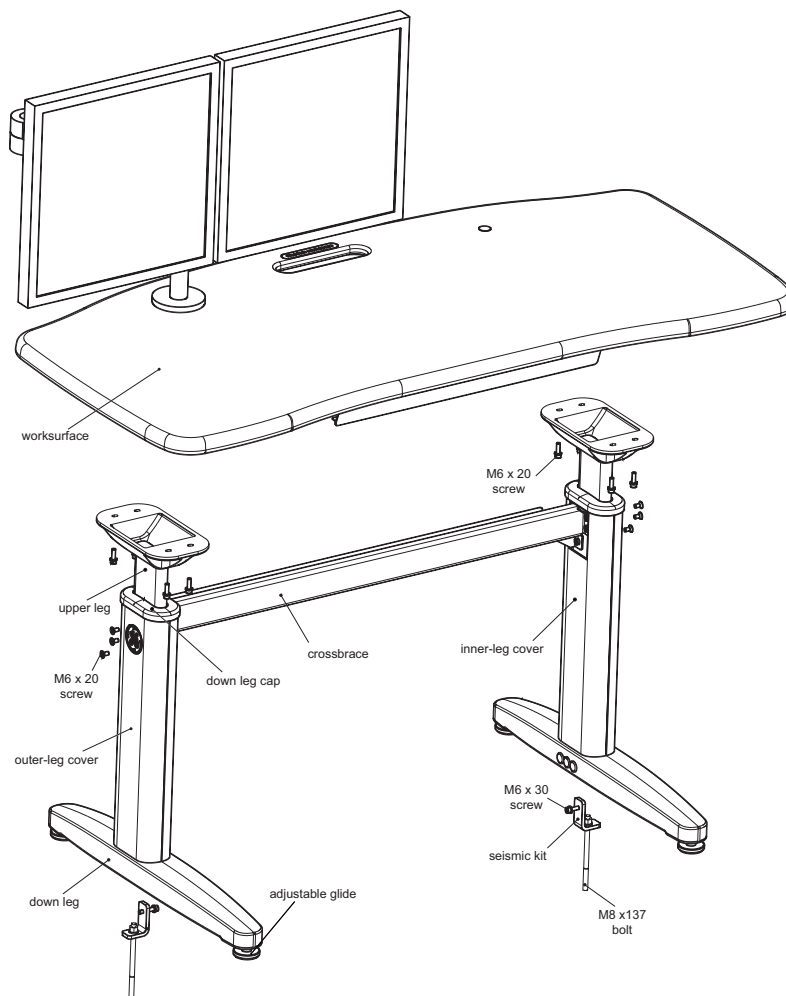


Figure A-40



- 3.) Press the inner-leg cover to secure with the adhesive pad ([Figure A-41](#)).

Figure A-41



- 4.) Install upper legs to down legs.
 - a.) Install down leg cap to the down legs ([Figure A-42](#)).
 - b.) Insert upper legs into down legs, adjust to desired height by attaching the upper legs' screws to appropriate down legs' holes. [Figure A-43](#) shows an example of attaching upper legs at the fifth and seventh holes from the bottom upward. In this case, the height of worksurface would be 785mm.
 - c.) Secure the upper legs with four M6 x 20 screws each leg.

Figure A-42



Figure A-43



- 5.) Insert outer-leg cover and inner-leg cover into down leg slot. Then push down the upper cover to fix them. ([Figure A-44](#)).

Figure A-44



- 6.) Place the worksurface upside down on a clean, soft surface to prevent scratching. Invert the table base onto the worksurface, taking care to align the holes with the pre-drilled holes in the worksurface. Secure the table base to the worksurface with eight M6 x 20 screws (Figure A-45).

Figure A-45



- 7.) Carefully turn the table assembly to the upright position. Insert hole cover into one of the two holes in the worksurface, the other hole would be used for monitor installation.
- 8.) The four adjustable glides may be raised or lowered to accommodate varying floor surfaces. Lift the leg off the floor and turn the glide clockwise or counterclockwise to adjust.

2.3 Install Monitors

Note: For more information, refer to the materials shipped with the FWS.

Figure A-46 (Part 5160735)

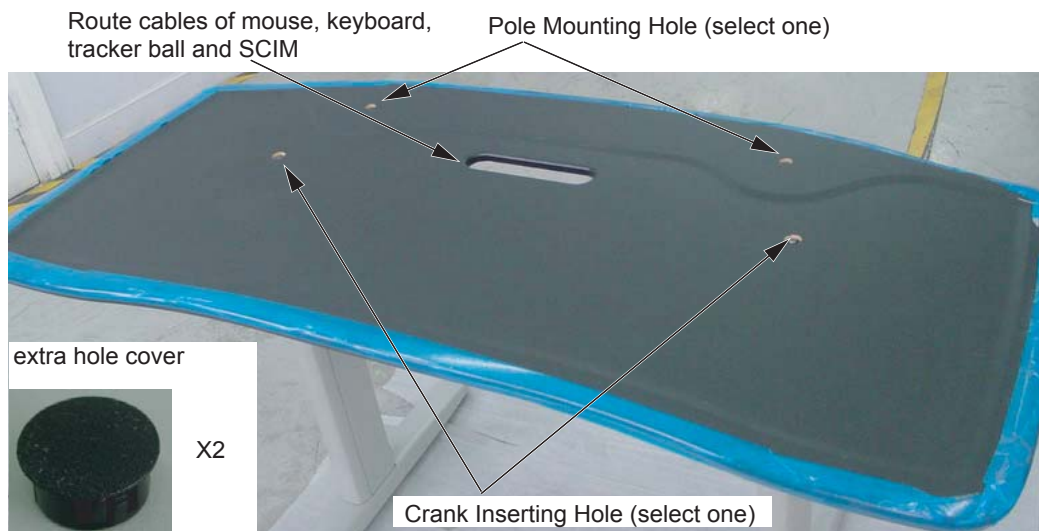


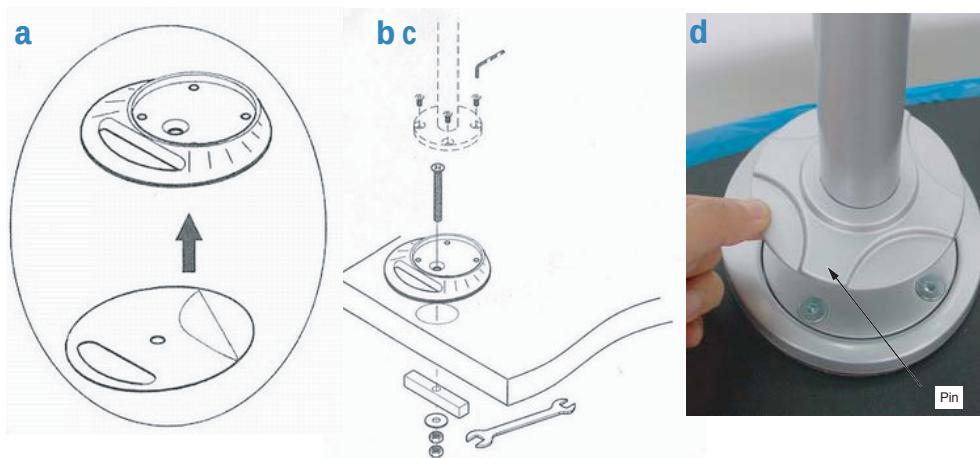
Figure A-47 (Part 5168666)



- 1.) Mount the pole on the worksurface. (See [Figure A-46](#), [Figure A-47](#) and [Figure A-48](#))
 - a.) Adhesive the pad under the grommet mount.
 - b.) Use large bolt to go through parts.
 - c.) Fix the clamp on the desk with 4 screws.

d.) Put the screw plate cover through the pole. Check the pin position with the screw.

Figure A-48



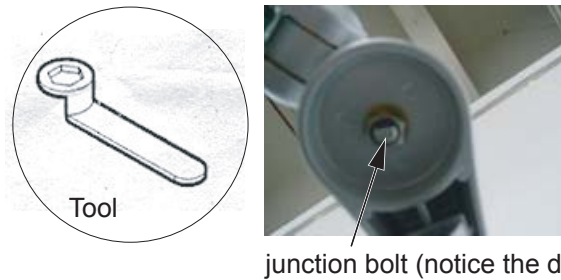
- 2.) Change configuration of the monitor arms for site.
 - a.) Three configurations for customer to select, see pictures below:

Figure A-49



- b.) If B or C is selected, disengage the not-to-used arm by unscrewing the junction bolts and reconfigure the arms.

Figure A-50



junction bolt (notice the direction during reconfiguratio)

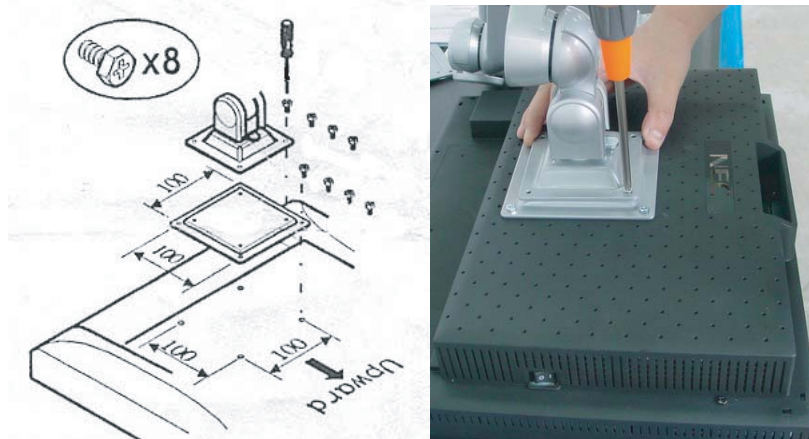
- 3.) Secure cable covers onto monitor arms. Cable covers should open backward the customer.

Figure A-51



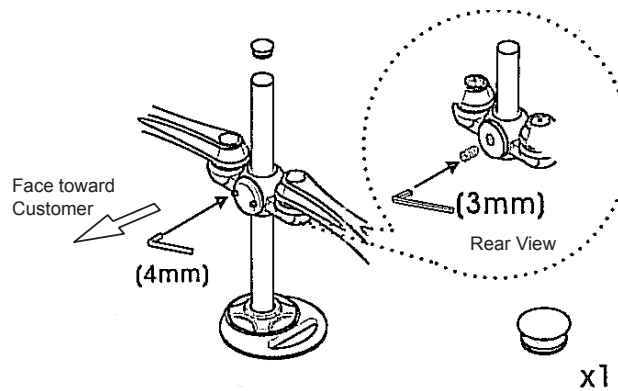
- 4.) Mount the monitors to the arm.

Figure A-52



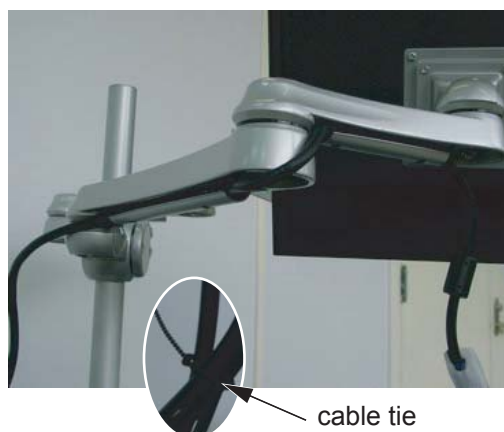
- 5.) Mount arms with pole. ([Figure A-53](#))
- Loosen two screws slightly. Put the arm into the pole and adjust it to the suitable position. Middle joint mount (two large allen screws) of the monitor arms should face toward the customer.
 - Tighten up the pole with 4mm and 3mm allen keys.
 - Put in the plug on top of the pole.

Figure A-53



- 6.) Put extra hole covers onto the holes not used. ([Figure A-46](#))
- 7.) Route cables.
- Extended Cables Kit (5160577) for FWS is used to connect monitors, keyboard, mouse, SCIM and trackball. Connection is same with short cables, see [Section 4.0 on page 83](#), Console Connections.
 - Thread monitor cables through cable covers.
 - Use cable tie to wrap the cables together or wrap the cables with pole.

Figure A-54



- 8.) Route cables (for FWS Part 5168666), thread cables through worksurface and use cable tie to wrap them together to the back slot of crossbrace ([Figure A-55](#)).

Figure A-55

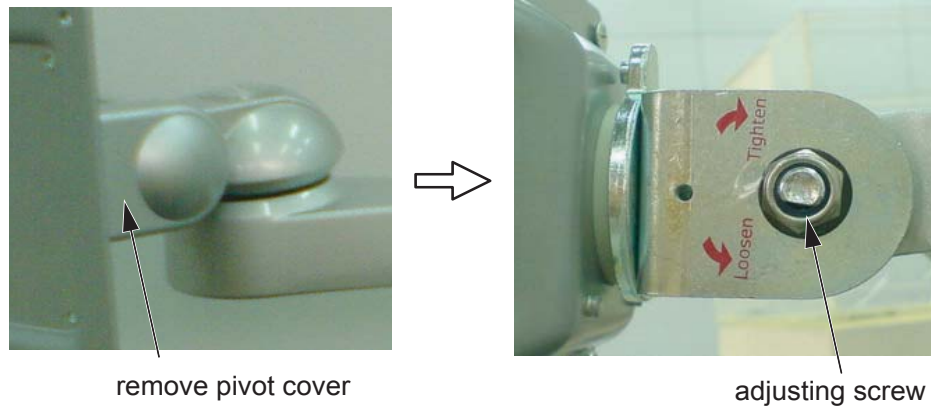


2.4 Monitor Adjustment

- 1.) Position the arms for ergonomic viewing. For optimum ergonomic viewing set top of monitor screen 1" below eye height. With monitor and arm properly supported, loosen screw and adjust up or down to desired height. Retighten to secure arm to pole.
- 2.) Adjust monitor tilt: Tilt monitor back and forth through entire pivot range of motion. If it does not stay in place or movement in one direction is stiff, pivot needs adjustment - see below (pivot cover is removed). Adjust until monitor stays in place and tilting forces, up or down, are equal.
Increase tilt lifting force: Turn screw clockwise.
Decrease tilt lifting force: Turn screw counterclockwise.

Figure A-56

side view



2.5 Install Seismic Kit (For Part 5160735)

If site specifications require seismic mounting, follow below steps:

- 1.) Insert the seismic bracket (x4) into the gap above the adjustable glide, at the bottom of leg.
- 2.) Use 1/4" bolts to mount the brackets to the floor. The seismic brackets are shipped with FWS table.

Figure A-57



2.6 Install Seismic Kit (For Part 5168666)

If site specifications require seismic mounting, follow below steps:

- 1.) Use M8 x 137 bolts to mount the brackets to the floor at a distance 916mm (Figure A-58).
- 2.) Mount table base to brackets with two M6 x 30 screws (Figure A-59).

Figure A-58

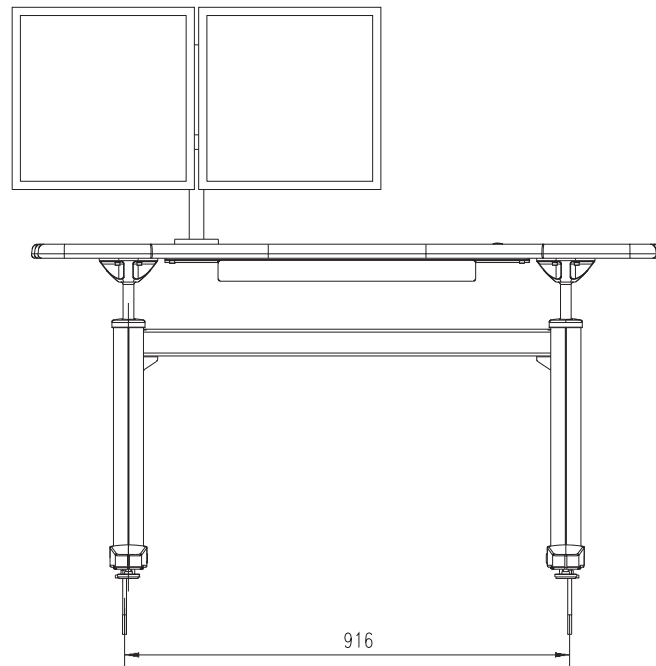
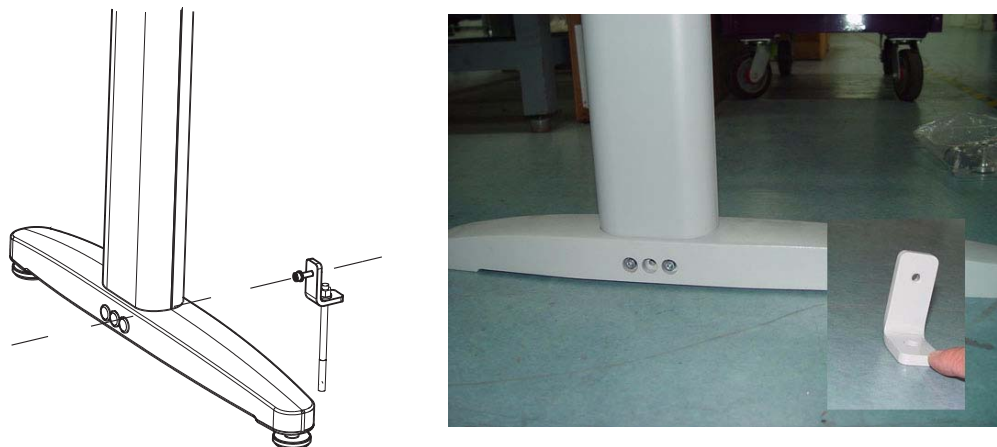


Figure A-59



Appendix C

Pictorial Representation of Required Tools

Use the following guide as a reference, if you are unsure of a tool listed in [Section 2.5, on page 28](#).

Table B-1 Required Tools

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Adapter		Sears Industrial: $\frac{3}{8}$ " to $\frac{1}{2}$ " (9-4258)
Ball-Peen Hammer		Sears Industrial: 1lb/2lb (9-38465)
Canned Air		Miller Stephenson: Aero Duster (MS-222N)
Clamp on Amp Meter		Sears Industrial: 9-WTAD105
Combination Wrench Set		Sears Industrial: U.S. Standard & Metric (9-44048)
Cordless Screwdriver		Sears Industrial: 9-MU65401
Deep Well Socket		Sears Industrial: $\frac{3}{4}$ " X $\frac{3}{8}$ " (included with 9-34496)
Dental Pick		
Diagonal Cutting Pliers		Sears Industrial: Small (9-45077)

* Part Numbers given for reference only. GEMs does not endorse any tool brand name.

Table B-1 Required Tools (Continued)

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Drill		Sears Industrial: $\frac{3}{8}$ " or $\frac{1}{2}$ " (9-27859)
Drill Adapter		Sears Industrial: 3" X $\frac{3}{8}$ " (9-APSZ24)
Drill Bit Set		Sears Industrial: U.S. Standard (9-66084)
DVM		Sears Industrial: 9-82028 Sears Industrial: 9-FL873
Extension for Ratchet Wrench		Sears Industrial: 3" X $\frac{1}{2}$ " (9-44133)
Gloves		Sears Industrial: Large (9-40502)
Hammer Drill		Sears Industrial: $\frac{1}{2}$ " (9-27205)
Hex Bit Set		Sears Industrial: $\frac{1}{4}$ " (9-SK45508)
Hex Key (Allen Wrench) Set		Sears Industrial: U.S. Standard (9-46284)

* Part Numbers given for reference only. GEMs does not endorse any tool brand name.

Table B-1 Required Tools (Continued)

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Level		Sears Industrial: 4' (9-39856)
Masonry Bit		
Open-End Wrench (Thin or Standard Tappet)		Snap-on: 10mm (SRSM10) & 21mm (LTAM2124)
Ratchet Wrench		Sears Industrial: $\frac{3}{8}$ " (9-43175)
Reciprocating Saw with Blades		Sears Industrial: 9-MU650921
Safety Glasses		Sears Industrial: 9-18650
Safety Shoes		
Screwdriver Set		Sears Industrial: Phillips & Straight (9-41505)
Socket Set		Sears Industrial: Standard $\frac{3}{8}$ " (9-34496)
Sockets		Sears Industrial: 1 $\frac{1}{8}$ " X 1 $\frac{1}{2}$ " (9-47516)

* Part Numbers given for reference only. GEMs does not endorse any tool brand name.

Table B-1 Required Tools (Continued)

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Step Ladder		Sears Industrial: 6' (9-WN6006)
Tongue & Groove Pliers		Sears Industrial: Large (9-CL440)
Torpedo Level		Sears Industrial: 9" (9-39829)
Torque Wrench		Sears Industrial: $\frac{3}{8}$ " (9-WR3470)
Universal Joint		Sears Industrial: $\frac{3}{8}$ " (9-4435)
Vacuum Cleaner		Sears Industrial: 8 Gal (9-17780)
* Part Numbers given for reference only. GEMs does not endorse any tool brand name.		

AppendixD

Option Cables

Run all option cables that connect to the Table, Gantry or Console.
For example the remote monitor (through the splitter) and SmartScore.
For more information, refer to “LightSpeed Linux CRT & LCD Monitor Option”

Figure C-1 Remote Monitor Splitter



Figure C-2 SmartScore



AppendixE

Regulatory and Service Clearances

Section 1.0 Regulatory Code Description

Egress: 29 CFR 1910 Subpart E (OSHA) and NFPA 101 (Life Safety Code) define the minimum requirements for means of egress. The requirement most applicable to equipment installation and room layout is minimum width of exit access. Under OSHA 1910.37(f)(6), the minimum width of exit access shall in no case be less than 28 in. from any potentially occupied point in the room.

Under NFPA 101 (2006 edition) 7.3.4.1, the minimum width of any means of egress is 36 in. However, NFPA allows this to be reduced to 28 in. around furniture or equipment, provided that a 36 in. clearance would otherwise be available without moving permanent walls.

Electrical Clearance: 29 CFR 1910 Subpart S (OSHA) and NFPA 70E (Standard for Electrical Safety in the Workplace) define minimum clearance requirements for the workspace around electrical equipment. Under both OSHA 1910.303(g)(1) and NFPA 70E (2004 edition) 400.15, a minimum clear space of 36 in. depth (with minimum 30 in. width and 78 in. height) must be provided in front of electrical equipment with parts operating at 600 volts or below and likely to require examination, adjustment, servicing, or maintenance while energized.

This safety clearance requirement applies to all GEHC equipment. Although 36 in. is the minimum clearance for most installations, the standards require an increased minimum clearance distance where parts operate above 150 volts (but still below 600 volts) under the following circumstances:

- If the wall or surface directly facing the electrical equipment is grounded (e.g. brick, concrete, or tile) or includes grounded protrusions (such as medical gas ports, metal door or window frames, water sources and metallic sink structures, metallic cabinetry, electrical disconnects or emergency off panels, air conditioners or vents), then a 42 in. clearance depth is required.
- If the possibility exists of exposed and unguarded live parts on both sides of the workspace (for example if a power distribution unit were positioned on the wall directly facing the GEHC equipment), then a 48 in. clearance depth is required.

Section 2.0

Terms and Definitions

EGRESS

The path of exit from within any room. U.S. regulatory requires a minimum of 28 inches (711.2 mm) of continuous and unobstructed space including trip hazards along the path of exit.

WORK SPACE

This is the dimensional box required for safe inspection or service of energized equipment. It consists of depth, width, and height. The depth dimension is measured perpendicular to the direction of access. U.S regulation is minimum of 36 inches (914.4 mm). Additional conditions can increase the minimum requirement. CT defines this as the envelope of the component superstructure. For the NGPDU it is with the front panel removed. For the gantry and table, it is with the patient or external covers removed.

SERVICE ACCESS WIDTH

This is the width of the working space in front of the equipment, a minimum of 30 inches (762 mm), or the width of the equipment whichever is greater.

HEAD CLEARANCE

This is the height dimension of "Work Space". The height of the workspace measured from floor at the front edge of equipment to ceiling or overhead obstruction(s), 78 inches (1981.2 mm) or height of equipment, which ever is greater.

GROUNDING WALL

Any wall that can be electrically conductive to earth ground. Masonry, concrete, or tile, are considered conductive. Additional commonly found aspects of a wall should also be considered as grounded. This is not an all-inclusive list:

- Medical Gas ports
- Metal door and window frames
- Water sources and metallic sink structures
- Metallic wall mounted cabinets
- A1 disconnect panel
- Equipment Emergency Off panels
- Industrial equipment such as air conditioners and vents
- Expansion joints

The following are not considered as grounded elements of a common wall:

- Standard wall outlet
- Light switches
- Telephones
- Communication wall jacks

Section 3.0 Regulatory and Service Clearances

3.1 Regulated Minimum Working Clearance by Major Subsystem

- Requirements apply to equipment operating at 600V or less, where examination, adjustment, servicing, or maintenance is likely to be performed while live parts are exposed.
- Direction of Service Access is defined as perpendicular to the surface of the equipment being serviced.
- Required regulatory clearance distances must be maintained and may not be used for storage. This includes normal system operation as well as service inspection or maintenance..

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access: front of console	914 mm (36 in.)	There are no exposed live part hazards with the cover in place. If the console is placed under a counter, the front edge of the console must be even with the vertical edge of the console workspace. Note: This component is typically serviced from the front with access to the rear.
Service access width: Front of console	762 mm (30 in.)	This is the width of the workspace in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.
Head clearance	1981.2 mm (78 in.)	This is the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s). A minimum of 1981.2 mm (78 in.) or the height of the equipment, whichever is greater, is required.

Table 3-2 Console Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Front of NGPDU)	914.4 mm (36 in.)*	There are no exposed live part hazards with the cover in place. This component is typically serviced from the front with access to the rear. *If exposed live parts of 151 - 600 volts are present, 1219 mm (48 in.) is required on both sides of the workspace with the operator between. *If the opposite wall is grounded and exposed live parts of 151 - 600 volts are present, 1067 mm (42 in.) is required.

Table 3-3 NGPDU Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Service Access Width (Left-Right of workspace)	762 mm (30 in.)	This is the width of the working space in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.
Head Clearance	1981 mm (78 in.)	This is the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s). A minimum of 1981 mm (78 in.) or the height of the equipment, whichever is greater, is required.

Table 3-3 NGPDU Subsystem

- For the gantry and table, distances are measured from the enclosure, not the finish covers.

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (All Sides)	914 mm (36 in.)	If exposed live parts of 151 - 600 volts are present, 1219 mm (48 in.) on both sides of workspace with the operator between is required. If the opposite wall is grounded and exposed live parts of 151 - 600 volts are present, 1067 mm (42 in.) is required.
Service Access Width (Left-Right of workspace)	762 mm (30 in.)	This is the width of the working space in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.

Table 3-4 Gantry Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Table Head or Foot)	914 mm (36 in.)	There are no exposed live parts hazards with the cover in place. This component is typically serviced from all four sides. this is the width of the workspace on each side of the equipment. A minimum of 914.4 mm (36 in.), or the width of the equipment, whichever is greater, is required.
Direction of Service Access (Table Sides)	914 mm (36 in.)*	*This distance can be reduced to 711 mm (28 in.) provided a written and signed approval is obtained by the local team from the local AHJ (Authority Having Jurisdiction). The signed document must be on file with GE.

Table 3-5 Table Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Table Foot)	711 mm (28 in.)	For the front gantry cover removal, a minimum of 457 mm (18 in.) is allowed only if an unobstructed egress space of 711 mm (28 in.) is maintained around the equipment for room exit. This also means no trip hazards exist along the path of egress.
Service Access Width (Left-Right of workspace)	762 mm (30 in.)	This is the width of the working space in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.

Table 3-5 Table Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Front of UPS)	914.4 mm (36 in.)*	There are no exposed live part hazards with the cover in place. This component is typically serviced from the front with access to the rear. *If exposed live parts of 151 - 600 volts are present, 1219 mm (48 in.) is required on both sides of the workspace with the operator between. *If the opposite wall is grounded and exposed live parts of 151 - 600 volts are present, 1067 mm (42 in.) is required.
Service Access Width (Right side and length of UPS)	762 mm (30 in.)	This is the width of the working space in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.
Head Clearance	1981 mm (78 in.)	This is the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s). A minimum of 1981 mm (78 in.) or the height of the equipment, whichever is greater, is required.

Table 3-6 UPS Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Front of A1 Disconnect)	914.4 mm (36 in.)*	There are no exposed live part hazards with the cover in place. This component is typically serviced from the front with access to the rear. *If exposed live parts of 151 - 600 volts are present, 1219 mm (48 in.) is required on both sides of the workspace with the operator between. *If the opposite wall is grounded and exposed live parts of 151 - 600 volts are present, 1067 mm (42 in.) is required.
Service Access Width (Right side and length of A1 Disconnect)	762 mm (30 in.)	This is the width of the working space in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.
Head Clearance	1981 mm (78 in.)	This is the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s). A minimum of 1981 mm (78 in.) or the height of the equipment, whichever is greater, is required.

Table 3-7 A1 Disconnect Subsystem

Section 4.0

Minimum Room Size (Limited Access)

The CT Gantry Left Side Limited Access Initiative provides the capability to reduce the minimum room size for CT Systems while still meeting all installation requirements and specifications. This adds left side flexibility, allowing the CT system to be sited in rooms with widths 22 in. smaller than the current minimum room width. In the minimum room configuration, left-side access and egress may be restricted for sites with less than 28 in. of left side clearance. Refer to your site's installation print for your room's detail.

4.1 Regulatory Caution

Site prints are required for all system installations including relocation and moves. CT room layout as shown on your site print shall meet all regulatory requirements as described in the installation manual. Additional room components such as cabinets reduce room size. Equipment not shown on the site print may void the caution statement, making the room non-compliant. Actual site measurements before installation will be taken to determine room size and compliance. For minimum room size sites, where the gantry distance between the wall and side cover is between 14 in. - 27 in. the left side cannot be considered or used for egress. In this configuration, egress around the left side of the gantry will be restricted. Unobstructed egress route shall be provided around the table foot end or behind the back of the gantry. If millwork is required or will be installed later, consider room placement locations that allows for regulatory compliance and clearances.

4.2 Operational Caution

In this minimum room layout (14 in. to 27 in.) the customer should consider workflow, customer access for patient care, and critical-care operations space requirements. Additionally, there may be limited equipment access on the gantry left side when loading patients or when positioning patient equipment in the room between the gantry and the wall. Detailed customer installation tasks are detailed in the product PIM. Chapters 1-4.

4.3 Recommended Room Size

This room configuration offers the most flexibility for future upgrades. It has sufficient workspace and space to add millwork, while meeting all regulatory requirements. This room would be compatible with most two-step future installations.

4.4 Typical Room Size

This room configuration allows for some future upgrades. It has sufficient workspace but limited space to add millwork and meet all regulatory requirements. This room may be compatible with some two-step future installations.

4.5 Minimum Room Size

This room configuration allows for no future upgrades. It has limited workspace and no in-room millwork, but meets all regulatory requirements. This room is not compatible with two-step future installations.

Section 5.0

System Specifications (BrightSpeed)

5.1 Recommended Room Size

3962 x 6299 mm (13' x 20'8")

Same Regulatory requirements apply

5.2 Typical Room Size

3912 x 5868 mm (12'10" x 19'3")

Same Regulatory requirements apply

5.3 Minimum Room Size

3353 x 6096 mm (11' x 20')

Same Regulatory requirements apply, with the addition of no energized left side service.

5.4 Rooms w/ Less Than 28 in. Egress Clearance around Table Foot End

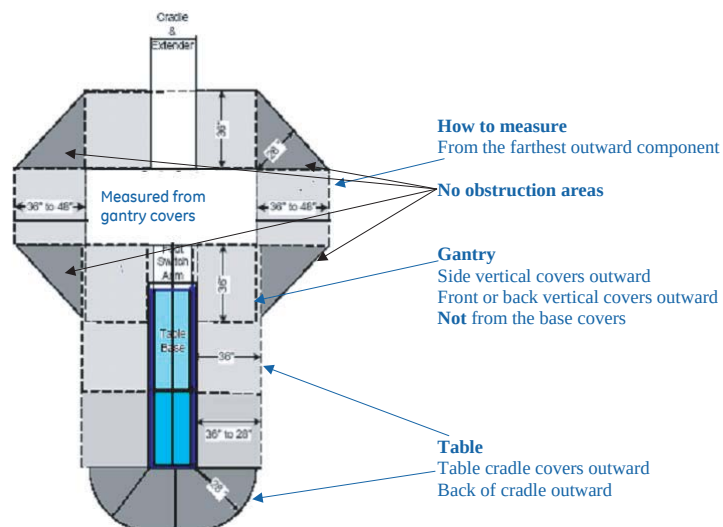
Egress requires a clear unobstructed route out of the room, either around the back of the gantry or around the back of the table. If your egress route is not around the back of the table, maintain 18 in. of clearance between the back of the table, with a continuous width of 126 in. on each side to any obstruction so that the front cover can be removed. Refer to the Pre-Installation manual for more details on service clearances.

Exceptions

Rooms smaller than 11' x 20' or 20'8" will require construction to meet the minimum

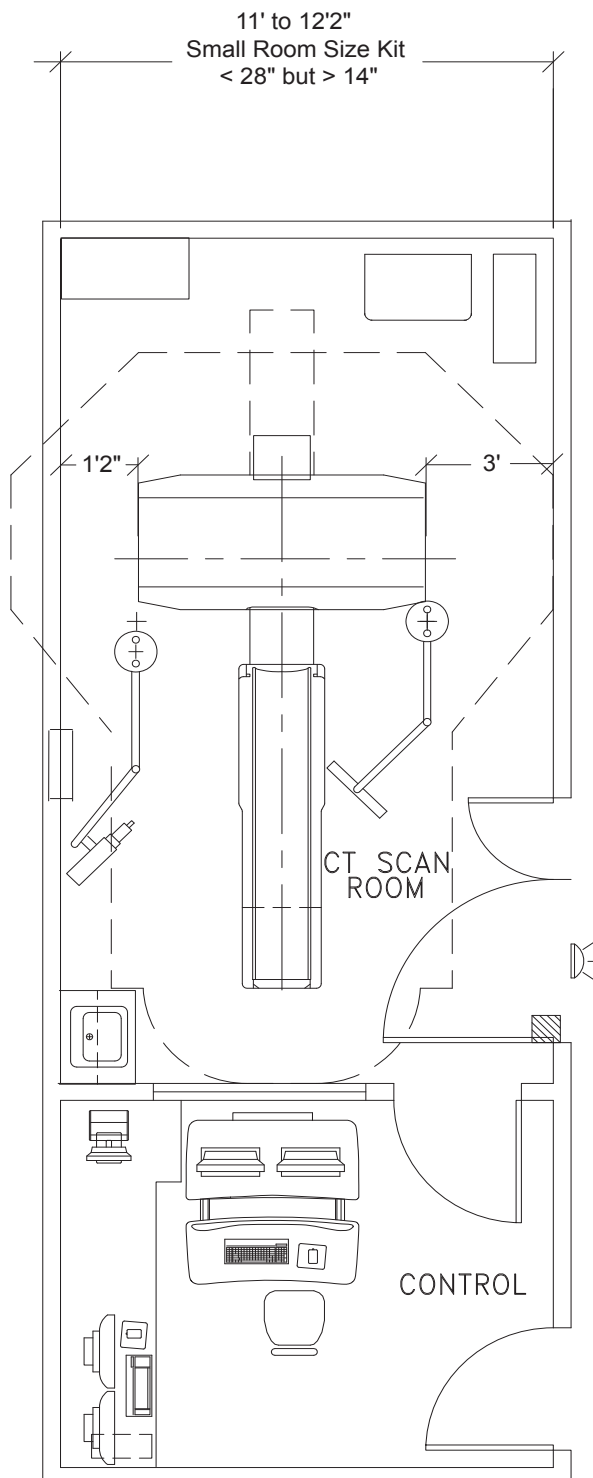
Section 6.0

How to Measure

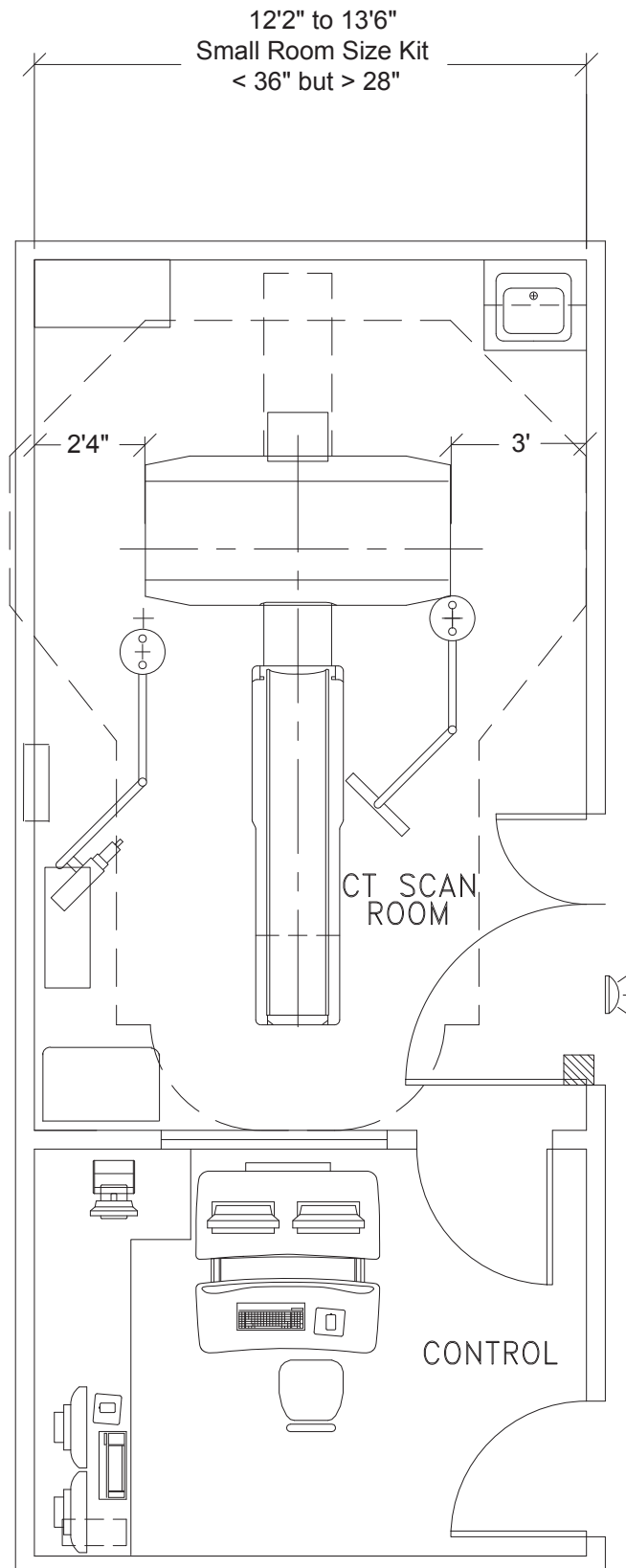


6.1 Minimum Room Size & Requirement Layouts

Room A - Less than 28 in. but greater than 14 in. measured from the covers to the left sidewall. In this configuration service, egress and workspace are compromised around the gantry.

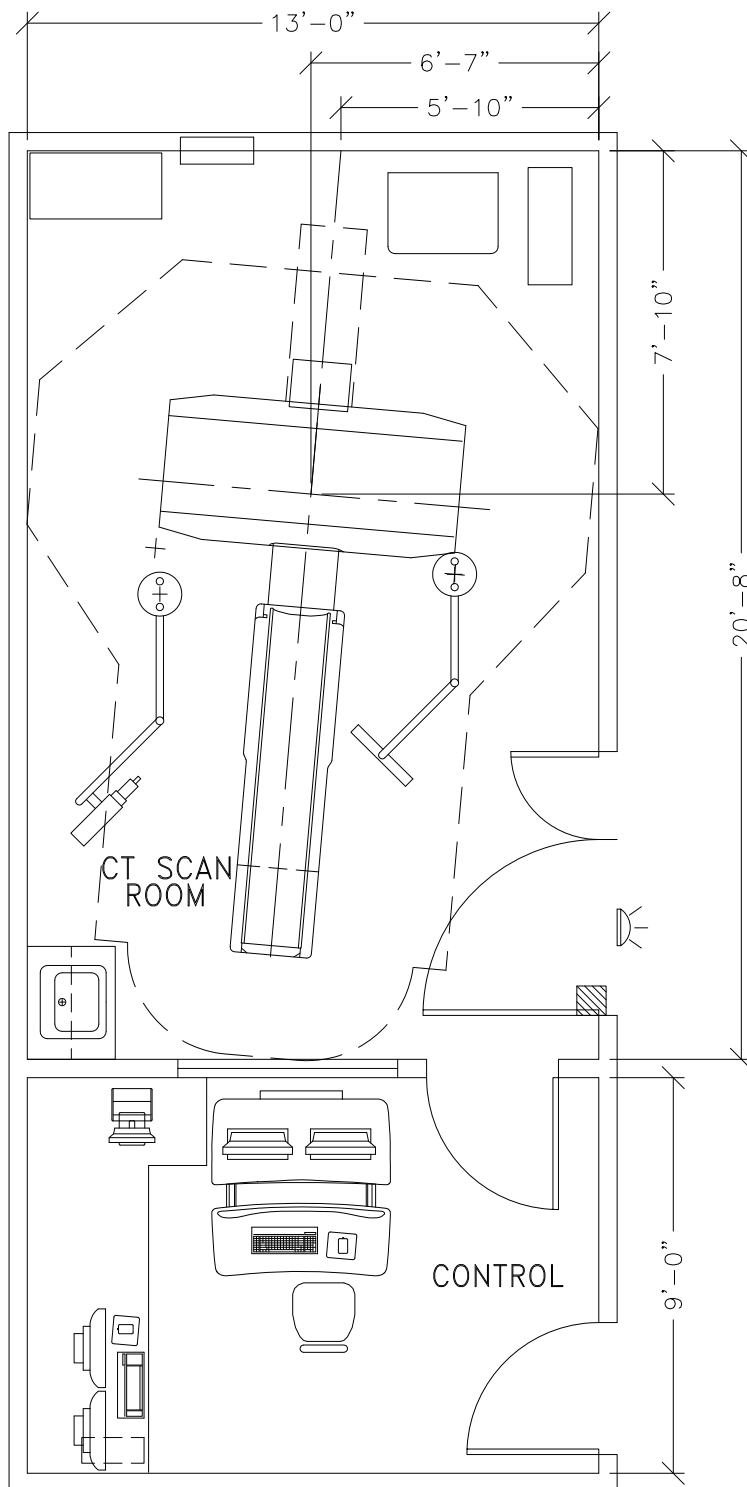


Room B - Less than 36 in. but greater than 28 in. measured from the covers to the left sidewall. In this configuration service, egress and workspace are acceptable around the gantry.



6.2 Typical Room Size & Requirement Layouts

Note: Your room layout may meet the Typical or Recommended room requirements but look different than the room shown above. Contact your sales person to have a detail room layout completed for your site.



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GE Healthcare

BrightSpeed Elite, Edge, Excel Installation Manual

Book 2 of 2: Electrical Calibration, Integration & Testing

OPERATING DOCUMENTATION



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Book 2 of 2: Electrical Calibration, Integration & Testing

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Chapter 4

Electrical Introduction



CAUTION

Shock Hazard.



Voltage Present.

No service on left side while energized.



NOTICE

To prevent potential data loss and equipment damage, please do the following:

- When instructed, record data collected from the procedures in this chapter on GE Form e4879. For more information about this form, see [Section 8.0](#) of Chapter 4.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0 Introduction

Use the continuity and ground checks to verify the system power connections have not shorted to ground and that the ground and neutral connections are intact.

Section 2.0 Review Mechanical Hand Off Material

Complete the Mechanical Hand Off checklist:

- ☐ All options were installed. If not, contact your install specialist.
- ☐ Check for short ships.
- ☐ Review cable connections with mechanical team.
- ☐ Complete mechanical vendor evaluation.

Section 3.0 Training

This product requires a trained FE to proceed with the calibrations in this section.

Section 4.0

Required FE Common Tools and Supplies

4.1 FE Calibration and Service Tool List

Note: Items with a “check” (✓) are included in the Install Support Kit.
Special CT Tools used for mechanical alignments:

- | | |
|---|---|
| ✓ 5mm hex bit for 3/8" drive 6" long
(Snap-On FAML5E or equivalent) | ✓ 10mm open-end thin wrench
(Snap-On SRSM10 or equivalent) |
| ✓ 10mm hex bit for ratchet wrench (ball end)
(Snap-On FABM10E or equivalent) | ✓ 21mm open-end thin wrench
(Snap-On LTAM2124 or equivalent) |
| ✓ 2.5mm Allen hex bit for 1/4" | ✓ 14mm ball hex socket bit for 3/8" drive |
| • Standard FE Tool Kit | ✓ 14 mm hex socket bit for 1/2" drive |
| • Torque Wrench Kit | |

4.2 Electrical Tools

These tools must be calibrated yearly.

- Fluke 87 DVM or equivalent
- Clamp on amp meter

4.3 Image Quality Calibration Tools

- QA Phantom (2144715)
- IQ Cal poly phantom:
 - 35cm (2144721)
 - 48cm (2144721-2)
- Phantom Holder

4.4 Detector Service Tools

A DAS/Detector Service Kit (PN 2344539) is included in the Install Support Kit.

- Aero Duster
- AMAX Contact Cleaner

4.5 Optional Tools

Needed for warranty recalibration:

- GE HV Bleeder
- Scope with 10X probes
- Beckman CT231 clamp-on Amp probe: 46-194427P228 (Fits up to #2 size wire)
- Beckman CT232 clamp-on Amp probe: 46-194427P270 (Fits #1 size wire and larger)
- Scope Probe-to-bleeder Cable 46-219921G1 (Belden #8422 cable, 30 ft)
- Polaroid type 52 film and developer
- Radial dial indicator (mm or inches) and mounting bracket
- Caliper Dial indicators (mm or inches)
- Ground Rod Tester Clamp on Amp Probe

4.6 Safety Materials

Note: Items with “checks” (√) are included in the Install Support Kit

- √ Lockout/Tagout kit, or equivalent
- √ Nitrile Rubber Gloves
- Safety Glasses

4.7 Cleanliness

Any dirt on the surface increases leakage current on the filter or converter cards and causes the DAS to fail the drift spec. Wear Nitrile gloves (part number 2207303-6 [large] and 2207303-7 [extra large]) when you handle the DAS, because fingerprints on the board can cause problems during humid conditions. Use only clean, new Nitrile gloves. Do not use hospital grade gloves.



NOTICE **NEVER USE AN ERASER TO CLEAN ANY PART OF THE DAS.**
DUST COVERS REQUIRED for installations on construction sites.

Section 5.0

Requirements/Assumptions

- This procedure will be performed by an appropriately trained engineer.
- All stations in a suite must have the same SUITE ID.
- You need the name of all hosts in the suite and their corresponding Internet/Ethernet numbers.
- You need the Internet (IP) addresses the first time you execute a reconfig on the system.
 - When you connect the system to a network, contact the system administrator to obtain the IP addresses for all the computers in the suite.
 - You also need an IP address for each gateway (second) ethernet board in any OC or IC.
 - You can use the default internet number on stand-alone systems (stand-alone = not connected to any network).
- For more detailed information and instructions regarding Network Integration, see [Chapter 7 on page 249](#).

Section 6.0

FE Workflow

- 1.) Review mechanical hand-off material.
- 2.) Obtain required FE common tools and supplies.
- 3.) Perform electrical power-on and ground checks.
- 4.) Gather all customer information needed for reconfiguration.
- 5.) Perform computer integration.
- 6.) Complete Table/Gantry integration.
- 7.) Complete the calibration process.
- 8.) Perform the Table/Gantry Alignment procedure.
- 9.) Perform tube warm-up and fast calibration.
- 10.) Complete tomographic plane indication.
- 11.) Run image series tests.
- 12.) Run system functional test.
- 13.) Create system state DVD.
- 14.) Perform the Patient Touch Leakage Test.
- 15.) Perform the CT System Chassis Leakage Test, as required by local code.
- 16.) Complete installation and verification of any customer options.
- 17.) Complete and return GE Form e4879 Installation Data Verification, for all installations.

Section 7.0

Checklists for Completed Installation

Complete the installation tasks listed below and check the appropriate boxes here and on the GE e-4879 form to verify the completion of these tasks. Section 8.1 contains an explanation of the GE e-4879 form, which the FE must complete and submit for ALL installations.

7.1 System-Level

7.1.1 General

- ☐ HVAC system is operational and environmental data reported on the GE e-4879 form.
- ☐ System realignments completed, if required.
- ☐ Broadband installed and operational.
- ☐ Power and ground audit completed.

7.1.2 Optional and Regional

- ☐ Seismic mounting kit installed, if required in your area.
- ☐ Generator recalibration completed, if necessary.
- ☐ Collimator recalibration completed, if necessary.

7.2 Site Clean Up

- ☐ All DVDs for customer options placed in the GE service cabinet.
- ☐ All system software and service tools placed in the GE service cabinet.
- ☐ System cleaned and nicks touched-up with paint.
- ☐ Installation site cleaned and all trash properly disposed.

7.3 Dolly Return

- ☐ Return of dollies arranged and dolly pick-up confirmed.

7.4 Options

Check the appropriate boxes here and on the GE e-4879 form to verify the installation and proper functionality of all customer-ordered options.

- ☐ Injector installed and operational.
- ☐ Advantage Windows Workstation installed and functional tests completed.
- ☐ Advantage 4D installed and functional tests completed.
- ☐ Filming/Camera/DASM installed and operational.
- ☐ Modem installed and functional tests completed.
- ☐ UPS installed and functional tests completed.

- ☐ Network items installed and functional tests completed.
- ☐ Customer software options installed and operational.
- ☐ Teleradiology connections completed. See Section 10.0 of Chapter 7.
- ☐ Remote monitor installed and operational.
- ☐ Bar code reader installed and operational.
- ☐ Cardiac monitor and stand installed and operational.

7.5 Paperwork (Final Activities)

- ☐ GE e-4879 form completed; see Section 8.1. (Required for installations in ALL countries.)
- ☐ FDA 2579 form completed; see Section 8.2. (Required ONLY for U.S. installations.)
- ☐ Any PQRs or PSRs that you encountered have been reported.
- ☐ All FMIs for system completed, if necessary.
- ☐ All dispatching activities (03-04-10 codes) completed.
- ☐ Customer acceptance checks completed.
- ☐ System transfer completed and appropriate GE Healthcare personnel notified.
- ☐ All outstanding customer installation issues have been addressed.

Section 8.0

GE and Regulatory Forms

Field Engineers must complete and submit the documents listed in Section 8.1 for ALL installations, regardless of the country. In addition, for installations performed within the United States, Field Engineers must ALSO complete and submit the documents listed in Section 8.2.

8.1 All Countries

8.1.1 GE e-4879 Form

The Field Engineer should:

- 1.) Locate the GE e-4879 form on the Service CD.
- 2.) Complete the form.
- 3.) E-mail the completed form to the following:
 - a.) HHS Administrator

8.1.2 Product Locator Cards

The Field Engineer should:

- 1.) Enter the Product Locator Card information on the Product Locator Web site. Go to the following address to access the site: http://gib.gehealthcare.com/gib/gib_entry.jsp
- 2.) Leave ONE (1) Product Locator Card (or a copy) at the customer site for EACH piece of equipment installed there.

Note: CT Manufacturing completes GE HHS Data Sheets and provides them to the HHS Administrator.

8.2 U.S. Installations Only

8.2.1 FDA 2579 Form

The Field Engineer should:

- 1.) Download the FDA 2579 form from the HHS Support Central Web site:
http://supportcentral.ge.com/products/sup_products.asp?prod_id=16442
- 2.) Complete the form.
- 3.) E-mail the completed form to the HHS Administrator.

Note: Do NOT print this form after completion. The HHS Administrator will e-mail a printable version to the FE for customer site records.



NOTICE

Some states require a State Registration Number to complete this form. For any questions concerning your state, contact the HHS Administrator or check the HHS Support Central Web site. Some states may also require additional information and test information. For instructions, contact the Project Manager of Installation.

8.2.2 System Chassis Ground Leakage Test Form

The Field Engineer should:

- 1.) Locate the System Chassis Ground Leakage Test form on the Service CD.
- 2.) Complete the form, if required in your area.
- 3.) Forward the results as instructed on the form.

Chapter 5

Electrical Integration and Safety Verifications



CAUTION

Shock Hazard.



Voltage Present.

No service on left side while energized.



NOTICE

To prevent potential data loss, please do the following:

- When instructed, record data collected from the procedures in this chapter on GE Form e4879. For more information about this form, see [Section 8.0](#) of Chapter 4.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Electrical Power ON & Ground Checks



WARNING



THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES.
USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

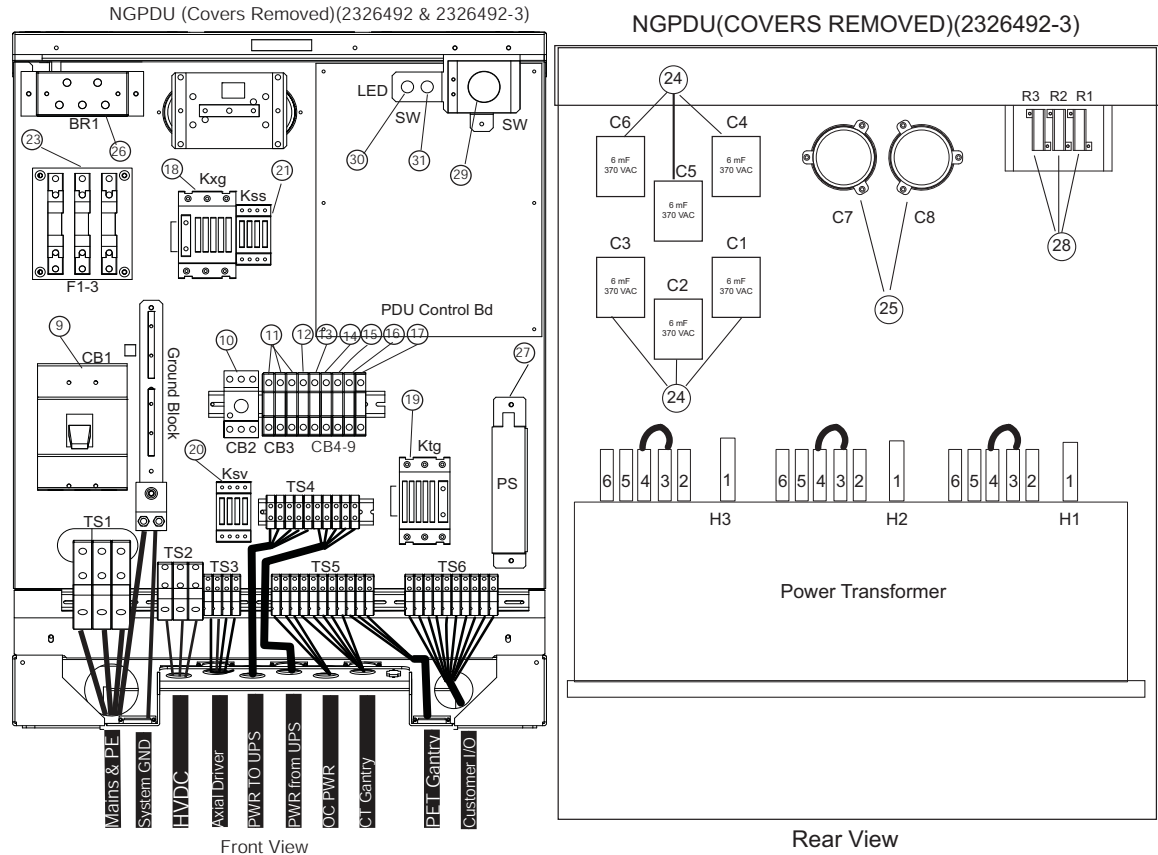


Figure 5-1 NGPDU

1.1 Required Tools

- Multimeter with a rating of at least 1000 volts
- Multimeter leads with a rating of at least 1000 volts

1.2 Initial PDU Configuration



WARNING

THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

1.2.1 Circuit Breakers

Set all circuit breakers to OFF

1.2.2 Relay Board

- 1.) Set SW to the normal position.
- 2.) When system is already, three lamps are both lighting (refer to [Figure 5-2](#)).

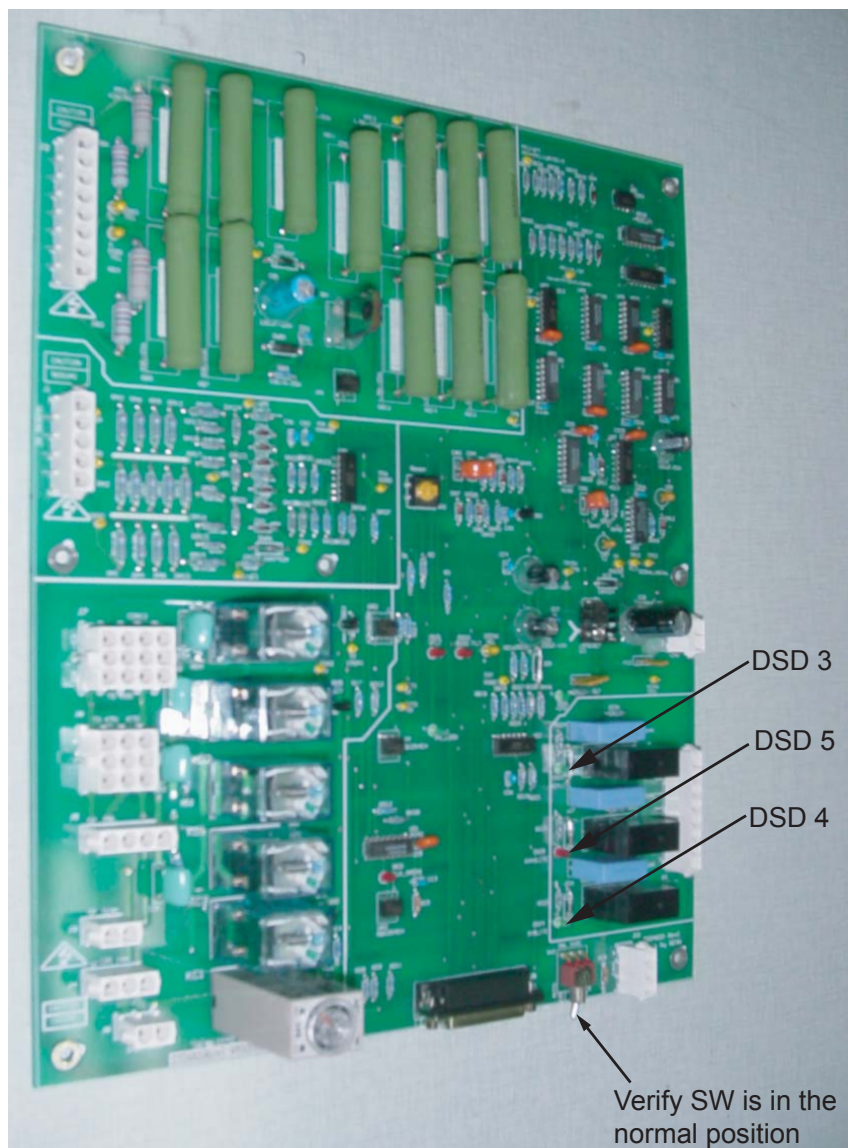


Figure 5-2 NGPDU Control Board

1.2.3 Power Switches

Turn OFF all power switches on all subsystems

- Gantry
- DAS
- Power Pan Breaker
- Table
- Console

1.2.4 Hardware and Connection Check

Use this step to check mechanical connections and tighten anything that may have shaken loose during shipment. Verify **all** hardware and connections in the PDU are securely fastened.

1.2.5 Covers

Install, or verify the presence of, all the lexan safety covers.

1.3 Suite Emergency Off Checks

-  **WARNING**  **VERIFY ALL PERSONNEL HAVE CLEARED THE SYSTEM BEFORE YOU TURN ON WALL POWER.**
- 1.) Turn wall power ON to the PDU.
 - 2.) **Press the suite emergency off button and verify it turns off wall power to the PDU.**
[Typically, this red palm button is located on the wall close to the console, within the scan suite.]
 - 3.) Verify that all “Emergency Off” button are working properly.
 - 4.) Leave power “OFF”

1.4 Line Transformer Settings

-  **WARNING** **MAINS VOLTAGE MAY VARIABLE FOR DIFFERENT COUNTRY (REGION). MAKE SURE TO SET THE APPROPRIATE PDU TAPS CONNECTIONS.**

1.4.1 Requirements

- 1.) The PDU is shipped configured for 480VAC.
- 2.) Complete only if your site uses a voltage other than 480VAC.
- 3.) If PDU is configured for 480VAC, go to [1.5](#). Otherwise, proceed to [Section 1.4.2](#).

-  **WARNING**  **MAKE SURE YOU TURNED OFF, TAGGED AND LOCKED THE MAIN WALL POWER BEFORE YOU CHANGE TAPS. FAILURE TO DISCONNECT POWER AT MAIN INPUT MAY RESULT IN ELECTROCUTION. TURN OFF WALL POWER TO CONNECT OR MOVE METER LEADS, OR TO REMOVE OR INSTALL COVERS.**

-  **NOTICE** **TAPS SHOULD BE SHIPPED AS SHOWN FOR 480VAC ONLY. FOR ALL OTHERS, YOU MUST MOVE THE TAPS. THE TAP CHECK SHOULD BE COMPLETED BY THE MECHANICAL INSTALLER.**

1.4.2 Line Input Conditions

- 1.) Monitor the No Load Line to Line Voltage at L1, L2, L3, during the workday. Do not record this data during “brown out” conditions.
- 2.) After you determine the nearest nominal line, verify the tap connections match (refer to [Table 5-1](#) and [Figure 5-3](#) for tap locations).

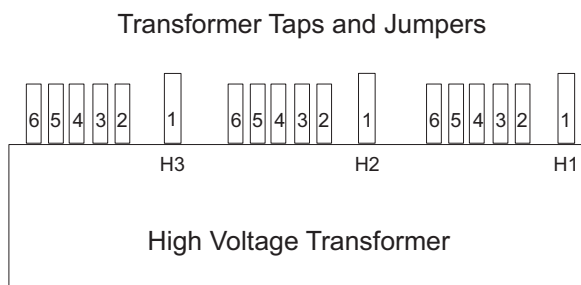


Figure 5-3 PDU Tap Positions (Rear)

- 3.) Verify that the No Load Line to Line Voltage never falls outside the corresponding minimum and maximum values listed in [Table 5-1](#).
- 4.) Use a 0-750 AC voltmeter of 3/4% accuracy to measure the line-to-line voltages at L1, L2, and L3.
 - Verify the highest line-to-line voltage does not exceed 1.02 times the lowest voltage.
 - **Example:** If the lowest voltage equals 474, the highest voltage should not exceed $474 \times 1.02 = 483.5$ volts.



WARNING

THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

NO LOAD		TAP CONNECTIONS		
Line to Line Voltages		(All 3 phases must have same the configuration)		
Nominal	Maximum Range (10%)	Phase A Connection	Phase B Connection	Phase C Connection
480V*	432 to 528*	3-4*	3-4*	3-4*
460V	414 to 506	3-5	3-5	3-5
440V	396 to 484	3-6	3-6	3-6
420V	378 to 462	2-4	2-4	2-4
400V	360 to 440	2-5	2-5	2-5
380V	342 to 418	2-6	2-6	2-6
240V**	216 to 264**	1-4**	1-4**	1-4**
220V**	198 to 242**	1-5**	1-5**	1-5**
200V**	180 to 220**	1-6**	1-6**	1-6**
* Factory Default				
** 2326492-3 PDU only				

Table 5-1 PDU Line Tap Connections

1.5 System Power Up



CAUTION

Verify all personnel have cleared the system before you turn on wall power.

- 1.) Turn ON A1 breaker panel.

- 2.) Turn ON all system power switches and breakers (table, gantry, PDU, console).

SUB-SYSTEM POWER-UP

- 1.) Turn ON switch S3 in the table (120vac 24hr power).
- 2.) Turn ON the Gantry **120VAC**. (Light should turn on.)
- 3.) Turn **AXIAL DRIVE ENABLE** ON. (Light should turn on.)
- 4.) Turn **HV DC ENABLE** ON. (Light should turn on.)
- 5.) Push the Service Switch Panel reset button (see [Figure 5-4](#)).

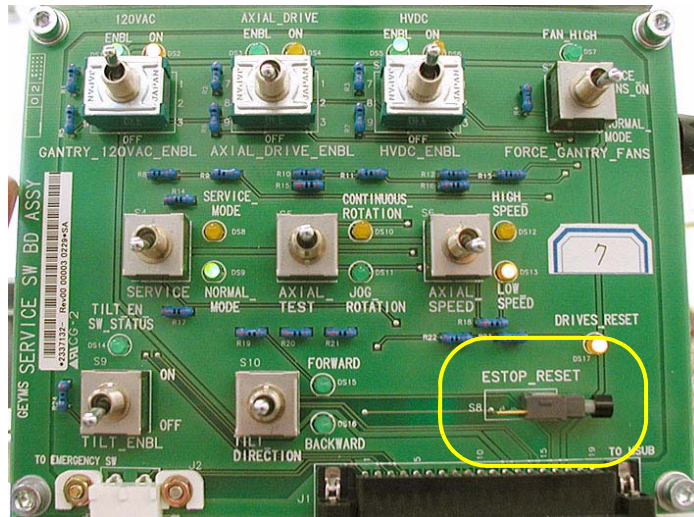


Figure 5-4 Service Switch Panel

AXIAL ENABLE SWITCH TEST

- Note:
- 1.) Turn OFF axial drive enable switch **AXIAL_DRIVE** on the Service Switch Panel.
- For the initial condition, do not leave the tube at the 2:30 position.
- 2.) Clear the gantry area for rotation.
 - 3.) Press the alignment light push button.
 - 4.) Verify that the gantry did not rotate.
 - 5.) Turn ON axial drive enable switch **AXIAL_DRIVE** on the Service Switch Panel.

ROTATION SAFETY CHECKLIST

- 1.) Turn OFF axial drive enable switch **AXIAL_DRIVE**.
- 2.) Turn OFF **HVDC** enable switch.
- 3.) Press red **E-STOP** button.
- 4.) Manually rotate the gantry 360 degrees. (Keep one finger on the Gantry button.)
 - Listen for any interference between the rotating and stationary parts. (Correct any interference problems.)
 - Listen for any loose parts. (Tighten parts as needed.)
- 5.) Turn ON axial drive enable switch **AXIAL_DRIVE**.

 **WARNING** MAKE SURE THERE ARE NO OBSTRUCTIONS AROUND THE GANTRY. PRESSING THE ALIGNMENT LIGHT PUSHBUTTON WILL CAUSE THE GANTRY TO ROTATE.

- 6.) Press the alignment light push button.
- 7.) Verify that the gantry rotates.
- 8.) Perform a 4 second X-ray OFF scan.



NOTICE

During the scan, it may be necessary to enter the scan room, to obtain a better listening position. If so, keep a finger on one of the four E-STOP buttons (on the gantry), to quickly stop the gantry, if necessary.

- a.) From the console, click on the SERVICE DESKTOP icon. 
- b.) Select DIAGNOSTICS.
- c.) Select DIAGNOSTIC DATA COLLECTION
- d.) Set the scan time to 4.00 seconds
- e.) Leave the door open. (This makes it easier to hear any loose or interfering parts.)
 - * Listen for any interference between the rotating and stationary parts. (Correct any interference problems.)
 - * Listen for any loose parts. (Tighten parts as needed.)
- 9.) After completing the 4 second scan, repeat steps 9a through 9e, with the following scan times:
 - 2.0 second scans
 - 1.0 second scans
 - 0.7 second scans
 - 0.5 second scans

1.6 Emergency Stop Check

- 1.) Use the gantry push-buttons to advance the cradle about 0.5m (2ft) from the home position.
- 2.) Press one of the E-STOP buttons on the gantry.
- 3.) Make sure the TABLE POWER shuts off, and the green LED flashes.
- 4.) Depress one of the table elevation buttons, to verify the emergency stop disabled table elevation.
- 5.) Depress one of the cradle drive buttons, to verify the emergency stop disabled the cradle drive.
- 6.) Press one of the RESET buttons to turn on X-RAY DRIVES POWER. (120 VAC LED stops flashing.)
- 7.) Press the other E-STOP button on the gantry.
 - a.) Make sure the Table Power shuts off.
 - b.) Manually move the cradle to the home position to make sure the cradle clutch released.
 - c.) Make sure the cradle latches securely in the home position.
- 8.) Press one of the RESET buttons to turn on X-RAY DRIVES POWER.
- 9.) Press one of the four table tape switches to make sure the table down motion stops. Repeat with the three remaining table tape switches.
- 10.) Press the console emergency stop switch; make sure the Table Power shuts off.

11.) Press one of the **RESET** buttons to turn on X-RAY DRIVES POWER. (See [Figure 5-4](#)).

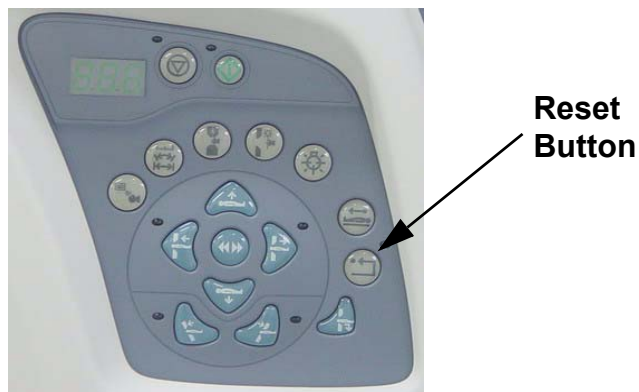


Figure 5-5 Reset buttons on Gantry and Service Switch bank.

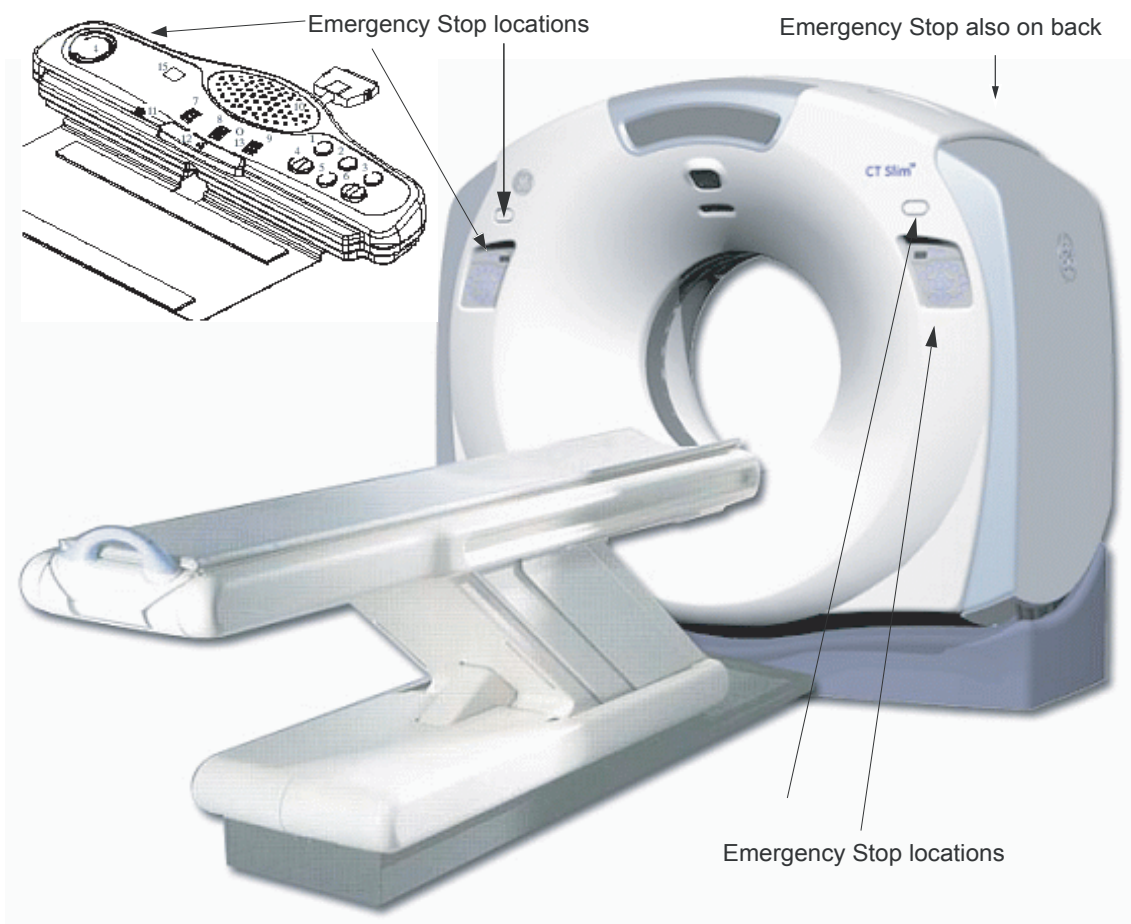


Figure 5-6 Emergency Stop button positions

Note: Emergency Stop buttons are located on the front and rear of the gantry (8 in all). They are also located on both sides of the table base (4 in all) as noted in [Figure 5-6](#).

Section 2.0 Computer Integration

2.1 Introduction

This Section describes the reconfiguration, system state restore, options, and monitor adjustment procedures.

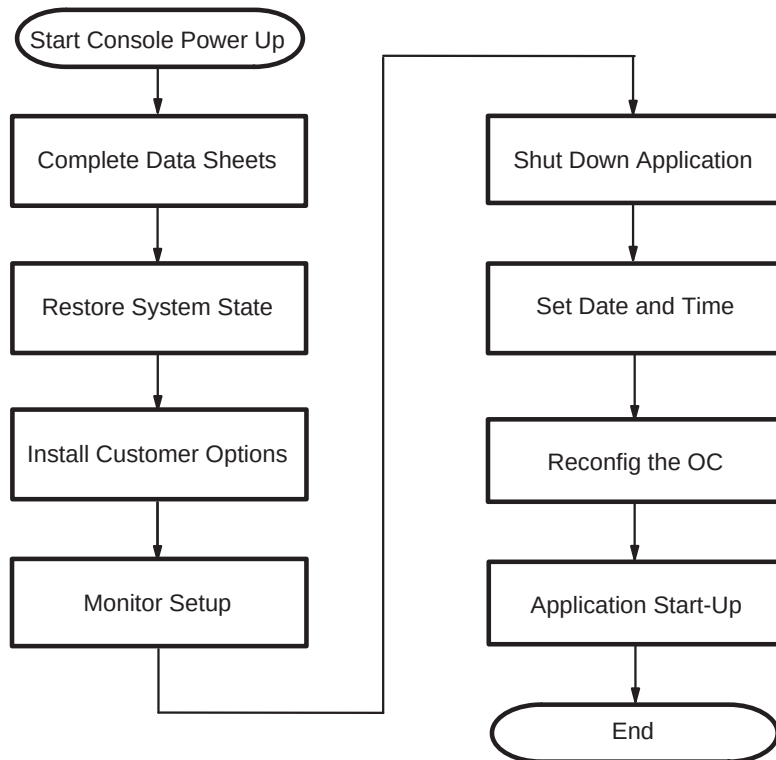


Figure 5-7 Computer Integration Process Overview

2.2 System Configuration Data Sheets

For convenient removal and use during installation, System Configuration Data Sheets appear in [Appendix G](#). Please locate and complete to them at this point during installation.

2.3 Restore System State

Tools Required: None

Your system should have a system state DVD, located in the Gantry accessories' box. The system state DVD contains:

- Collimator Characterization
- Phantom Calibrations
- Gen Cal
- Other Data

The installation process uses all the system state files. At this time, use the system state DVD to restore all files.

If you cannot locate an existing system state DVD, you must recalibrate your system.



- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.
- 3.) Select SYSTEM STATE.
- 4.) Insert the DVD into the DVD drive.
- 5.) Select CHARACT.
- 6.) Select CALS.
- 7.) Select RESTORE to restore the system characterization and phantom calibration files to the system.

Note: **Restore State can take as long as ten minutes, although typical times average about three minutes. When Restore State completes, dismiss the tool, and proceed to the next section.**

If any error should occur during the restore process, see the Software Load Procedure manual (Load From Cold) for information regarding possible error messages and their recovery.

- 8.) Click NO for Reset Scan Hardware popup message.
- 9.) Select DISMISS.

2.4 Install Customer Options

Tools Required: None

2.4.1 Software

Note: **Your system may have a DVD that contains customer purchased options. If your system has an options DVD, install it at this time--otherwise skip this section.**

Ensure that the options DVD is NOT write protected at this time. The initial install requires that the DVD be write enabled; subsequent installs can be done with the DVD write protected.



- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the CONFIGURATION icon.
- 3.) Select INSTALL OPTIONS.

4.) Select **INSTALL OPTIONS**.

An Options Window appears (Figure 5-8):

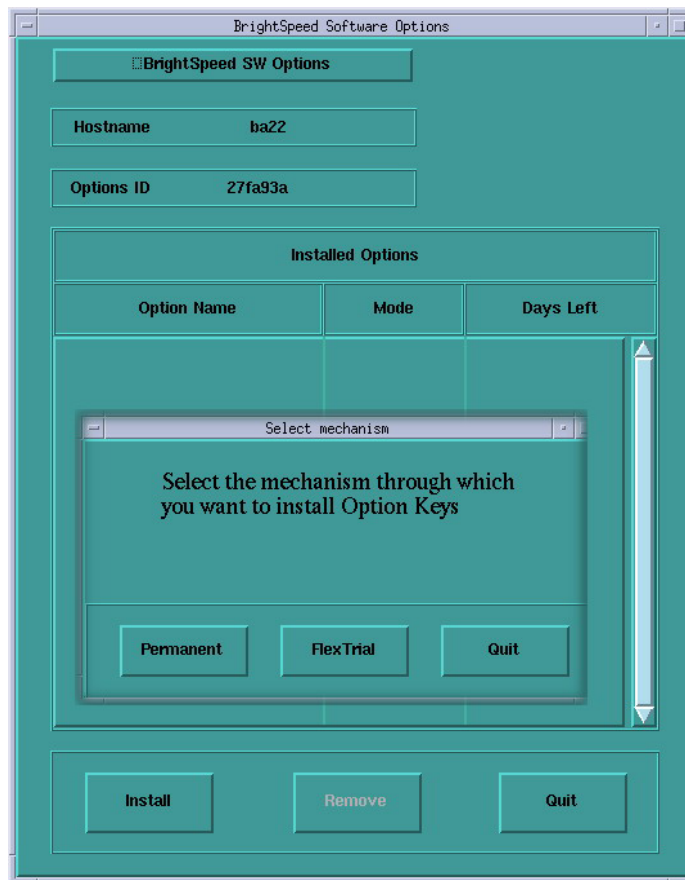


Figure 5-8 Options Window when First Selected

- Check the FDO to see what options were ordered.
- Compare FDO options to those on the Option DVD.
- If different contact your local sales representative.

- 5.) Insert the options DVD into the DVD drive and click on OK. (If you do not have an options DVD, click on OK anyway, wait for the abort pop up, then abort the process.)

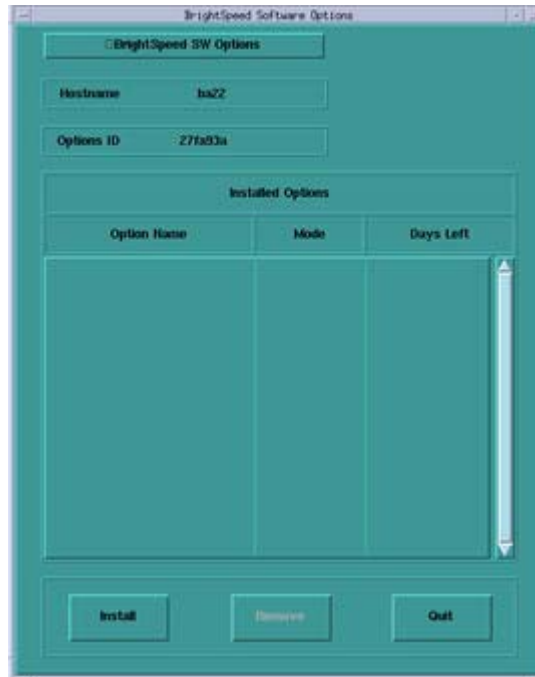


Figure 5-9 Example: Options Window

- 6.) Select all of the options in the left-hand column to install the corresponding software.
- 7.) Select INSTALL. A box may appear while the options are loading. When an option is displayed in the Installed Options list, then installation of that option is complete. Note that some options take a fraction of a second to install, while options like 3D may take a half minute (due to the fact that they are installing software).
- 8.) After the options are installed, select QUIT.
- 9.) Select OK.
- 10.) Remove the DVD and write protect the side with options.
- 11.) When the system prompts to Reboot, click YES, and reboot the system to complete the installation.

2.4.2 Camera

Tools Required:

- Small flat blade screw driver
- Data collected from data sheets ([See Camera Application Configuration on page 298.](#))
- Software Load Procedures manual
- System Service manual.

Note: If a DASM is required, the DASM hardware must be installed before proceeding.
For details on camera configuration, refer to the Software Load Procedures manual.
For details on troubleshooting the camera, refer to the System Service manual.



- 1.) Click on the SERVICE DESKTOP icon.



- 2.) Select CONFIGURATION icon.

- 3.) Select INSTALL CAMERA.
- 4.) Select ADD.
- 5.) Select DASM or DICOM.
- 6.) Follow procedures on the screen.
- 7.) Return to Home Page.

2.5 Shut Down Application

Tools Required: None

Standard Level

If Applications is currently running, you must shutdown system applications.

- 1.) Click on the SERVICE DESKTOP icon. 
- 2.) On the desktop toolbar select UTILITIES icon. 
- 3.) Select APPLICATIONS SHUTDOWN (to bring down applications only).

Super User Level

- 1.) Open a UNIX Shell window.
- 2.) Type: **su** - ENTER at the prompt.
- 3.) Type the root (super user) password: **#bigguy**

2.6 Reconfig the OC

Tools Required: None

Note: The document collector box that arrived with your system contains the *Software Installation Procedures* manual, which documents the reconfiguration procedure in more detail.

2.6.1 Overview

On the following screens, you should make the changes necessary, pressing the corresponding button at the top of the screen to move from screen to screen. When you are done, you can either press the ACCEPT button to start the reconfiguration process, or press the QUIT button to exit without changing the system configuration.

While the reconfiguration is going on, messages are displayed in a shell window that closes when reconfiguration is complete. Should you later want to review the reconfiguration output, it is logged to the following file:

`/var/adm/install.log.YYYYMMDDWWHHMMSS`

Where

`YYYYMMDDWWHHMMSS` is the Date/Time that the reconfiguration was started.

To view the file, type: **`more /var/adm/install.log.YYYYMMDDWWHHMMSS`**

It is possible to abort the reconfiguration while entering information on the reconfiguration screens. Simply press the QUIT button at the top of the screen. There is NO safe way to abort the reconfiguration after pressing the ACCEPT button. If the entries made in the screens were incorrect, DO NOT try to stop the reconfiguration, instead wait for it to complete, and rerun reconfig, entering the correct parameters.

2.6.2 Procedure

- 1.) Change directory to scripts:
Type: **cd /usr/g/scripts** ENTER at the prompt.
- 2.) Launch the Install utility:
Type: **reconfig** ENTER at the prompt.
The OC displays the Install Utility Window as shown in [Figure 5-10](#).

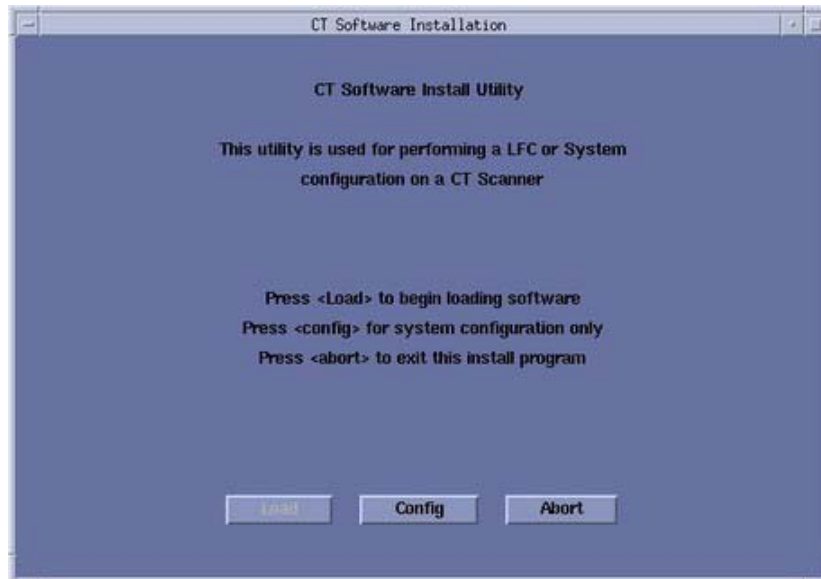


Figure 5-10 Install Utility Window

- 3.) Click on the CONFIG button.
The OC displays the System Configuration - System Settings Screen as shown in [Figure 5-11](#).

Comment: The following pages show the screens that are used to change the configuration of the system. These screens are the same as those used for the Software Configuration during Load From Cold. The actual screens will vary depending on the current configuration of your system.

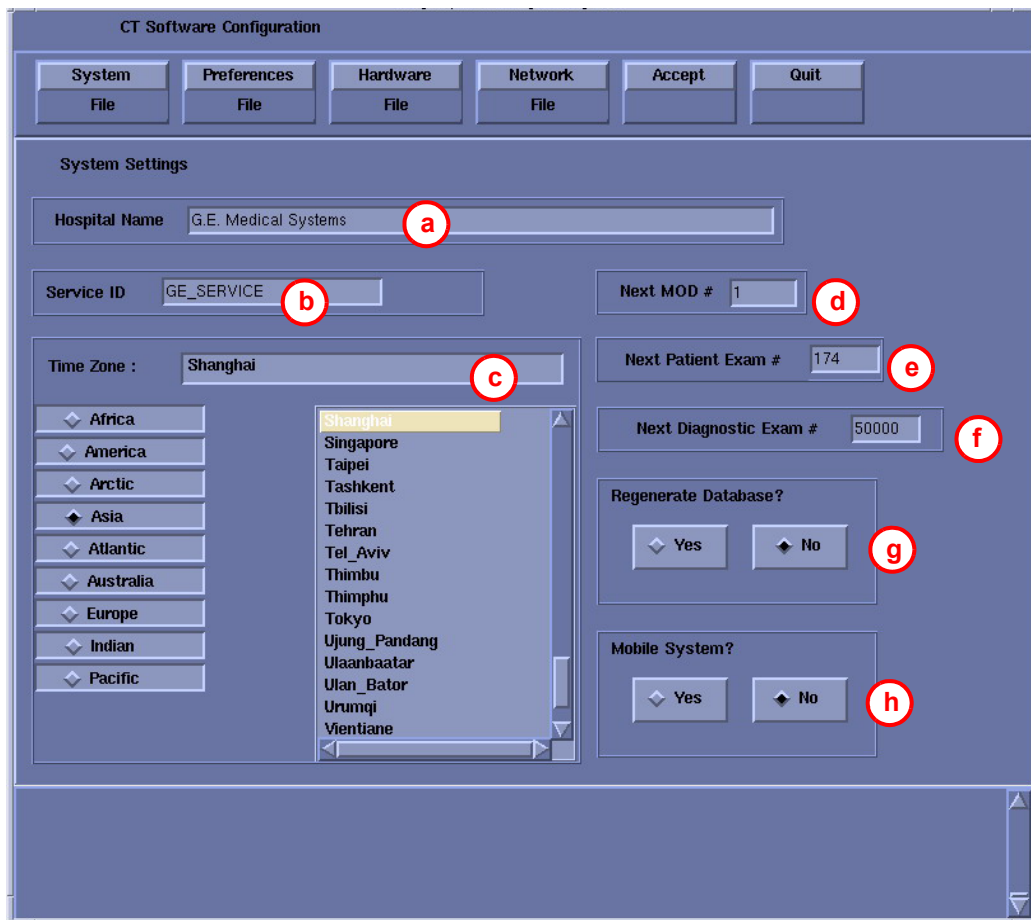


Figure 5-11 System Settings Screen

4.) Configure System Settings

- a.) Hospital Name (Figure 5-11, item 1) configures the name that will show up on images produced by this scanner. *Example: ST MARYS HOSPITAL*
- b.) Service ID (Figure 5-11, item 2) is issued by the Service organization.
Example: 262785CT2 (no spaces)
- c.) Select the Time Zone for the site.

Note: Use the scrollbar at the bottom of the time-zone selection list to view the entire description of the time-zone you are about to select, to ensure that you are selecting the correct time-zone for your location.

If the time-zone of your location is not in the list above, select one of the universal times in the selection menu. In this case, automatic changes for daylight savings time will not take effect. See Load from Cold manual, if you require more information regarding time-zone setting & selection.

- d.) Next MOD # *Customer selected.*
 - e.) Next Patient Exam # configures the next Exam number the scan user interface will use. At initial system installation, type: **1**
 - f.) Next Diagnostic Exam # *Customer Selected.*
 - g.) Regenerate Database
 - h.) Mobile System Select to tell the software if this CT is in a mobile environment or not.
- 5.) Select the PREFERENCES button to display the Preference Settings Screen as shown in Figure 5-12.
- 6.) Regenerate Database (refer to Figure 5-12) determines whether the Scan Database will be regenerated during reconfiguration. You should select NO.

Note: If YES is selected, then all scan data will be deleted on the Scan Data Disk. To avoid accidental regen, NO is selected by default each time reconfig is run. Regenerating the Scan Database destroys all the scan and calibration files on the Scan Data Disk. Make sure that all images have been reconstructed before regenerating the Scan Database.

This procedure is normally only needed when a Scan Data Disk is replaced or reformatted. See Load from Cold manual, if you require more information regarding Scan Database Regeneration.

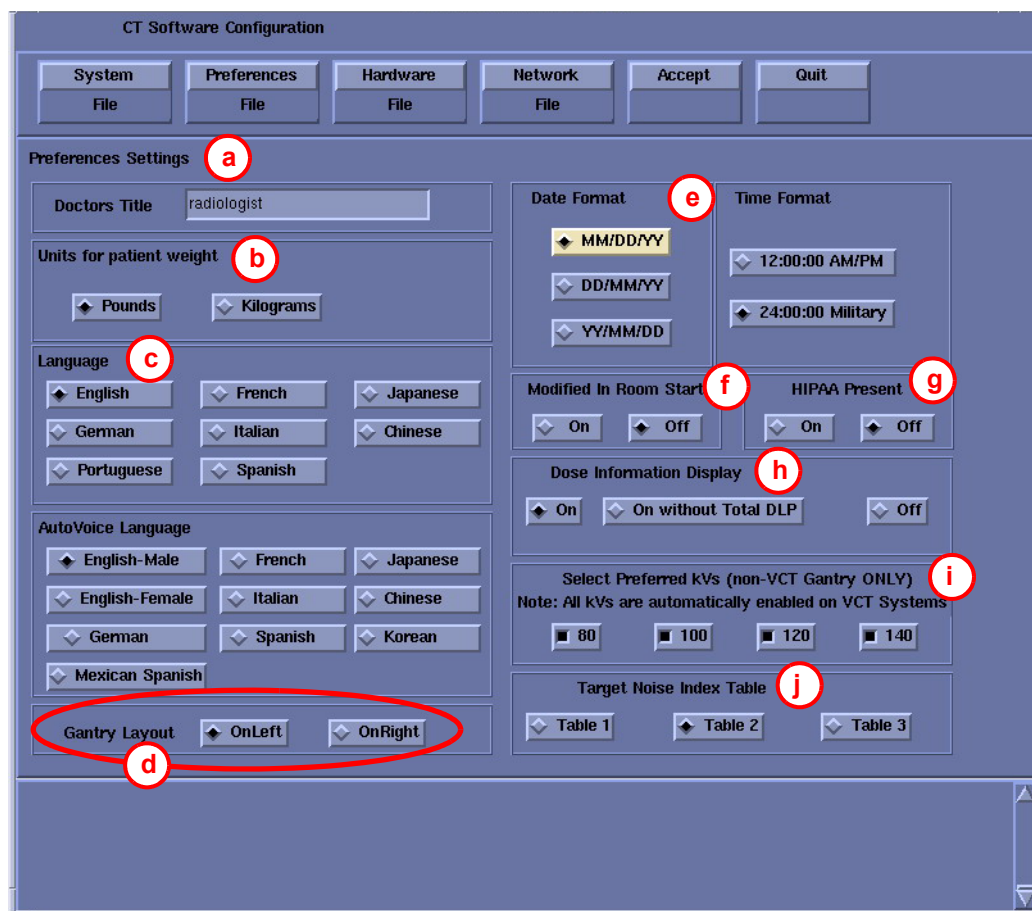


Figure 5-12 Preferences Setup Screen

7.) Configure Preferences Settings

- a.) Doctors Title Enter the title for the Doctor. (eg. radiologist)
- b.) Units for Patient Weight Tells the software whether pounds or kilograms are being used.

Note: Use Direction 5221102-1EN (found in the keyboard collector kit shipped with the system) to complete the language selection.

- c.) Language configures the language to be displayed on the Application screens.
 - 1.) Review the language matrix (in direction 5221102-1EN) and identify the appropriate language for your country.
 - 2.) Ask the radiology manager, or equivalent, to agree to the user interface (UI) language and keyboard requirements listed for that country (per 5221102-1EN), OR choose another language for the UI and keyboard.
 - 3.) Record the information on GE form e4879.

Note: d.) Select Gantry Layout according to gantry/table layout in the hospital.
Select ONLEFT if gantry is laid to the left of table (viewed from the control room), otherwise, select ONRIGHT.

- e.) Date Format configure the format in which the date will be displayed on the images. Time Format configure the format in which the time will be displayed on the images
- f.) Modified In Room Start: **Select ON for Japan sites, OFF for other sites.**

- g.) Set HIPAA Present to [OFF] unless the customer requests HIPAA to be on.
- h.) Select the site preferred Dose Information Display option for the site to use in monitoring calculated Patient Dose:
 - * Select ON (full CTDIw Display)
 - * Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or
 - * Select OFF (no CTDIw Display)
- i.) Preferred Fast Cal KV configures the preferred kV that the Fast Cal Routine will calibrate (80, 100, 120, 140 in the Selected Preferred Fast Cal KV field). The default selections are 120 and 140.
- j.) Select Target Noise Index Table: default is [TABLE 2].

Comment:

These kVs should include all kVs that the site uses for patient scanning. Deselecting All Preferred FastCal KVs is the same as selecting ALL the Preferred FastCal KVs

- 8.) Select the HARDWARE button to display the Hardware Settings Screen (Figure 5-13).

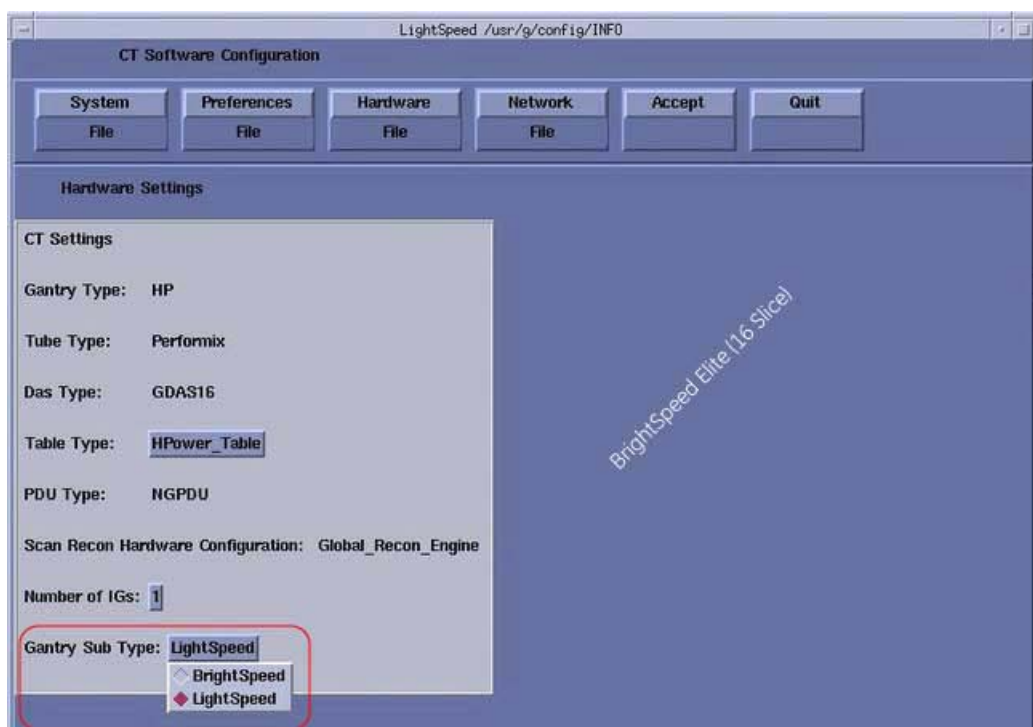
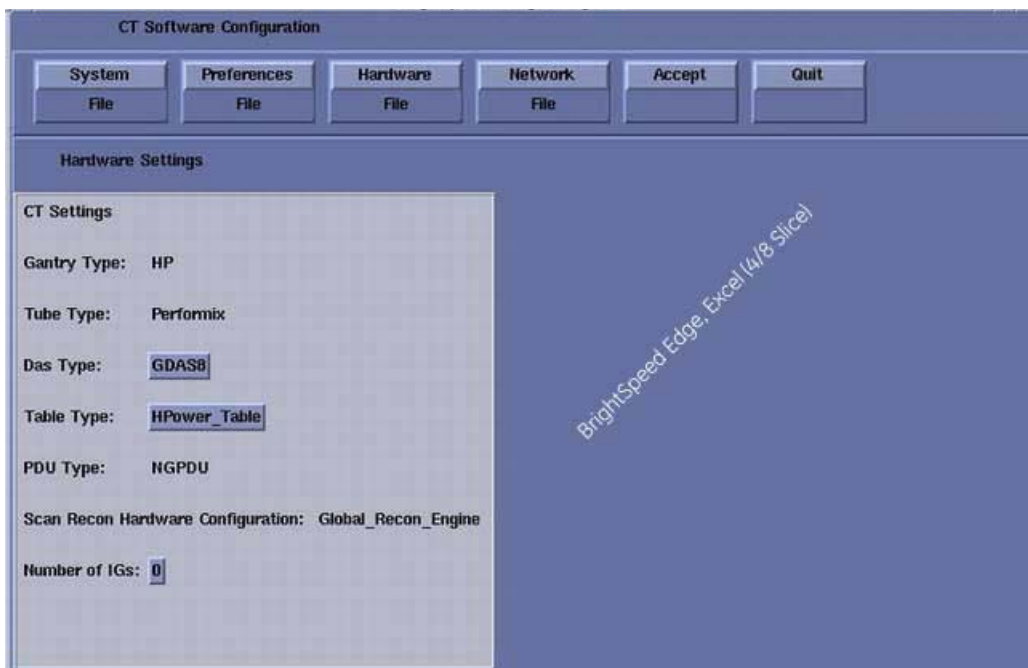


Figure 5-13 Hardware Settings Screen (example only - actual screen may vary)

- 9.) Configure Hardware Settings
Verify correct hardware type is shown. If not correct, select correct DAS Type (**4/8 Slice** only), Table Type and Numbers of IGs.
16 Slice only: Select Gantry Sub Type (SLIM GANTRY or NORMAL GANTRY) to switch between LightSpeed HP60 16 slice and BrightSpeed Elite. LFC is NOT required.

10.) Select the **NETWORK** button to display the Network Settings Screen as shown in [Figure 5-14](#).

Figure 5-14 Network Settings Screen

11.) Configure Network Settings

Comment: This screen provides the ability to declare the CT system on a hospital network. Key information such as Host Name, IP Address, Net Mask (for CT systems on a subnet) must be obtained from the hospital network administrator.

See [Chapter 7](#) for more information and complete details of setting the Hospital/System Network Configuration.

a.) Enter the Suite Name.

The Suite Name must start with a letter, followed by three alphanumeric characters. Total must be four characters long. The name of the OC interface will be <Suite Name>, within the scanner's subnet.

Typically, you should use su01 or ct01 ("su" and "ct" must be lowercase), unless the customer prefers a different suite name.

b.) Enter the hospital provided Host Name.

The Host Name identifies the network hostname and AE Title of the CT system to the hospital's network. The Host Name:

- * **MUST NOT** exceed 16 Characters.
- * **MUST** only contain the following characters: A through Z, a through z, 0 through 9, - and _.

Comment: The Host Name is typically stmary or ct01.

c.) Enter the hospital provided IP Address for the system.

d.) Enter the hospital provided Net Mask (if the CT system is on a subnet).

e.) Enter the Broadcast Address.

Comment: The Broadcast Address should be the same as the IP Address except for the bits of the host id portion (last digit group) set to 1's or 0's depending on the configuration of the network. The standard default is 1's, but older Sun OS machines used 0's.

For example:

If the IP Address is 192.100.9.17, the Broadcast Address should be 192.100.9.255 if the network is configured to use 1's to specify the Broadcast Address.

If the network contains genesis based scanners or other Sun OS 3.5 or 4.1 computers, the Broadcast Address should be 192.100.9.0.

- f.) Enter the hospital provided Default Gateway IP Address in the Default Gateway field (if applicable). If the site network does not use a default gateway, leave the field blank.
- g.) Select NIS (a.k.a. Yellow Pages database) Advanced Option—only if requested by the hospital network administrator—as follows:
 - * Select ADVANCED OPTIONS button on the Network Settings Screen.
 - * Select USE NIS button.
 - * Enter the hospital provided Domain Name (from the hospital network administrator).
 - * Review all pages to be sure the information entered is correct before proceeding to the next step.

- 12.) Select ACCEPT on the System Configuration Screen.

Comment:

The system loads the CT Application Software, OS patches, kernel changes and configures the system on the OC.

The loading process takes approximately 15 minutes. While the load is going on, the results are displayed in a shell window that closes when the loading process is complete. All the window output is logged to a file named: _____

- 13.) When the loading process and configuration changes are complete, the system displays a prompt to reboot. Click on YES. (See [Figure 5-15](#).)



Figure 5-15 Reboot Screen

- 14.) The system will automatically login as ctuser after the reboot. Select OK on the Autostart Disabled popup message.

2.7 Check/Set Date and Time

Tools Required: None

If date and time need to be corrected:

- 1.) Open a Unix Shell window to check the current date.
Type `{ctuser@hostname} date` ENTER at the prompt.
- 2.) If the date is correct, proceed to the next section; if it is incorrect, continue this procedure.

Note: You must set the date and time on the Host computer with the Application software down.

- 3.) Open a Unix Shell window and login as root:
 - a.) Type `{ctuser@hostname} su -` ENTER at the prompt.
 - b.) Type the password: `#bigguy`

- 4.) Set the date and time.
 - a.) Type `{root@hostname}# setdate` ENTER
 - b.) Follow the instructions of the individual time-entry prompts, which will appear in the following sequence:
 - * Note: Type "q" at any time to quit. Press ENTER to continue.
 - * Note: TO BE ACCURATE, this tool will prompt you to enter the "Second." Watch your clock or PC carefully to enter the proper value, and hit [ENTER] at the right second to set the accurate time. Enter to proceed. Press ENTER to continue.
 - * Enter the current Year (1980 - 2030) [2007]:
 - * Enter the current Month (1-12) [04]:
 - * Enter the current Day (1-30) [14]:
 - * Enter the current Hour (Military Time) (0-23) [18]:
 - * Enter the current Minute (0-59) [13]:
 - * Enter the current Second (0-59) [00]:
 - * Updating the time on the OC and DARC, Please Wait...
 - * PING darc (172.16.0.2) 56(84) bytes of data.

2.8 Save System State

- 1.) Insert the DVD into the Peripheral Tower DVD drive.
 - 2.) Select: SERVICE DESKTOP.
 - 3.) If reloading software, select: UTILITIES. If upgrading from earlier version software, select: PM.
 - 4.) Select: SYSTEM STATE.
- Note: System State Save may be under Utilities or PM.
- 5.) Select ALL to save all data.
 - 6.) Click SAVE.
 - 7.) If applicable, click OK when the following message appears: System State Media Status: Please insert a DVD or MOD into the drive and press Save again.
 - 8.) When completed select DISMISS.
 - 9.) Label and date the disk including the suite name.
 - 10.) Close the Service Desktop window at the upper left corner of the screen.

2.9 Applications Start-Up

Open the Console shell window, and type: `st` ENTER. The applications desktop appears on the monitor.

2.10 GRE Console Boot-up Flow Chart

Figure 5-16 GOC4/LCGOC Boot-up Flow Chart

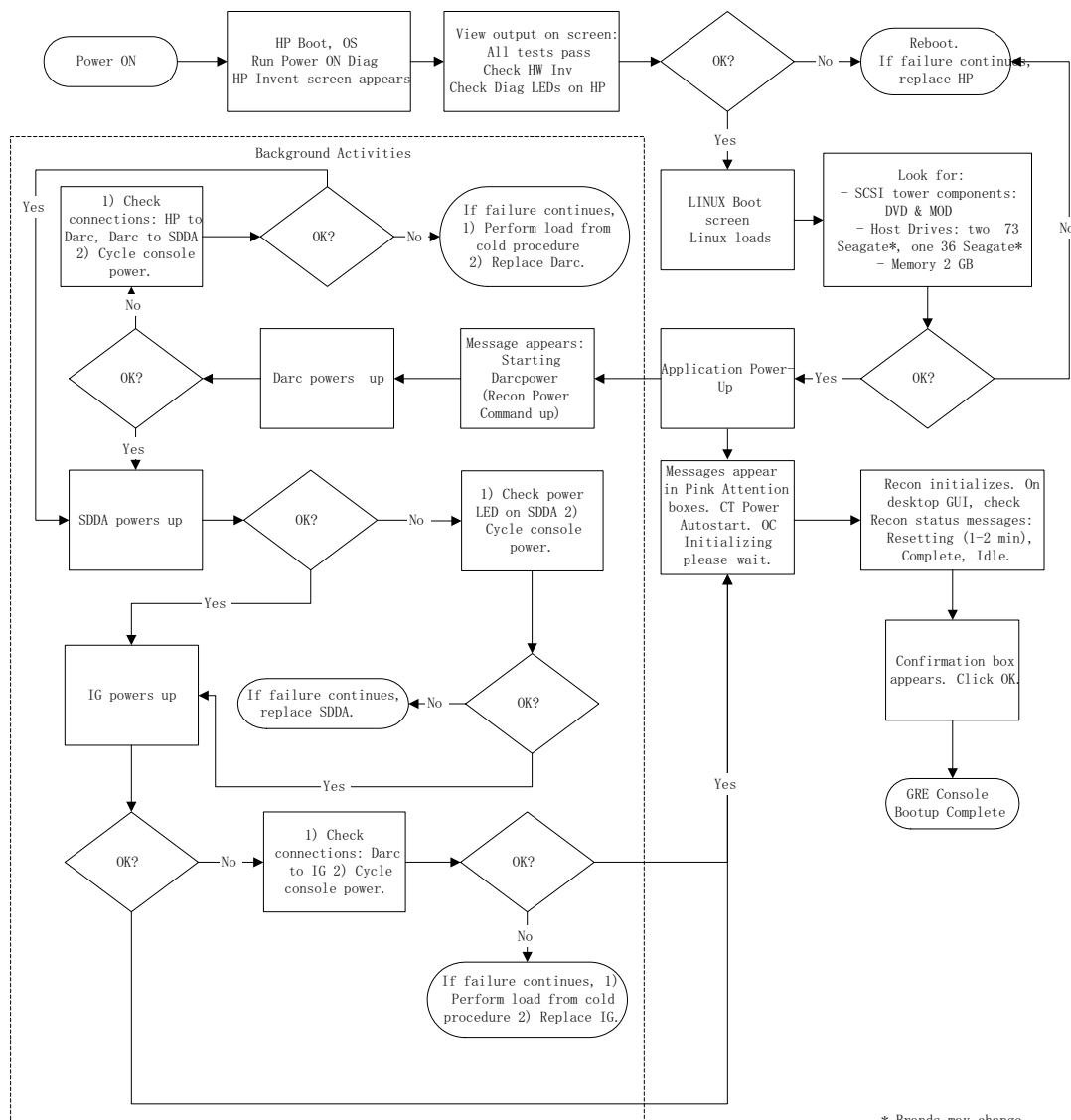
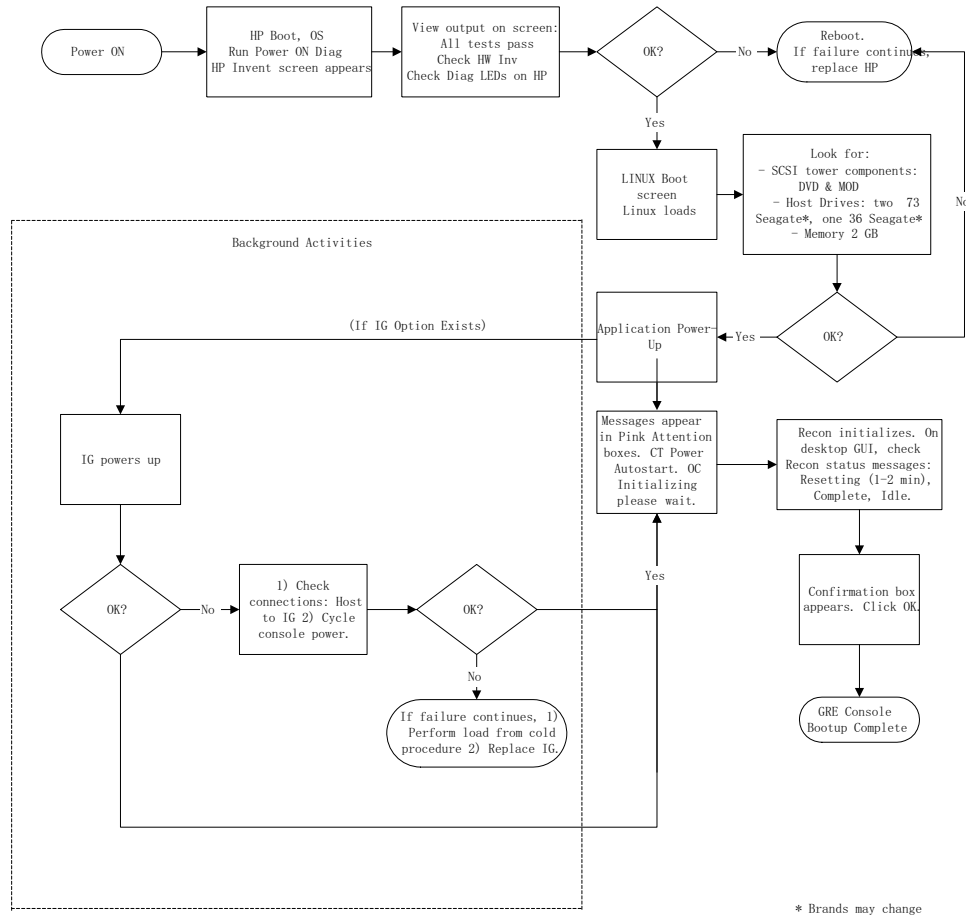


Figure 5-17 All-In-One Console Boot-up Flow Chart



2.11 Adjust Monitor



- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the IMAGE QUALITY icon.
- 3.) Select INSTALL SMPTE IMAGE and wait approximately 3-4 minutes for SMPTE image to install. (When complete the following message will display: SMPTE and QA images have been successfully copied)
- 4.) Press ENTER to exit the Service Desktop.



- 5.) Click the IMAGEWORKS icon
- 6.) Display the SMPTE pattern. Use the browser to select Exam 1000, which contains the SMPTE pattern, and enlarge the image to full screen display.
- 7.) Select Viewer.
- 8.) Select 1:1 format.
- 9.) Increase the monitor's contrast to maximum.

Note: Adjust monitor contrast until the operator sees the anatomical structure (window raster)

- 10.) Increase the Brightness to maximum.
- 11.) Decrease the Brightness, until the raster just fades into, and matches, the monitor screen background. At this point, the 5% and 95% patches should be just visible.
 - If additional tweaking is required to attempt to match the monitor image to the filmed image, use only the brightness control.
 - If the LCD image exhibits any tearing or smearing of the alphanumeric characters, then reduce the contrast setting slightly until the tearing/smearing is just eliminated. The optimum setting for contrast is the highest setting that does not cause tearing/smearing of the alphanumeric characters.

You should always finish up by displaying and filming images of anatomy (typical heads and bodies), and asking the technologist to compare the LCD image to the film image.

Section 3.0 Table Gantry Integration

3.1 Introduction

Use these procedures to functionally check every part of the table/gantry subsystem.

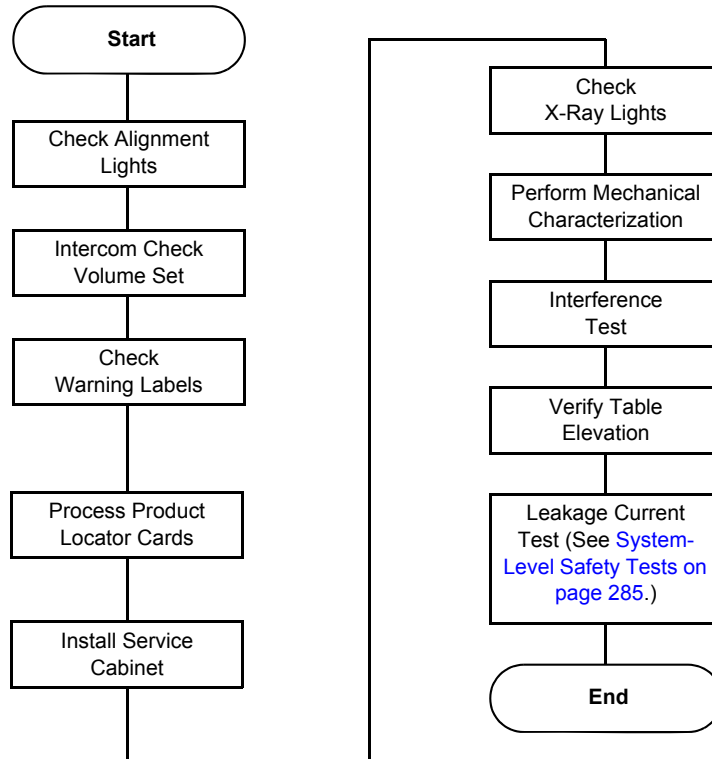


Figure 5-18 Table Gantry Integration Process Overview

Required Tool

- Multimeter

3.2 Check Alignment Lights

3.2.1 Room Light Adjustment

Adjust the scan room lights to normal customer operating levels.

3.2.2 Turning the Alignment Lights ON



CAUTION Verify all personnel have cleared the system. The Gantry rotates during this check.



- 1.) Turn ON the AXIAL DRIVE ENABLE and HVDC ENABLE switches (located on the service switch board).
- 2.) Turn on the alignment light switch on the gantry control panel. The gantry will rotate and the alignment lights will turn ON.



CAUTION LASER EYE INJURY!
NEVER STARE DIRECTLY INTO THE LASER BEAMS WHEN YOU OPERATE THE ALIGNMENT LIGHTS. STARING INTO THE BEAMS CAN CAUSE PERMANENT EYE DAMAGE.

3.2.3 Internal Axial Lights

- 1.) Place a sheet of plain white paper over the output port of each light.
- 2.) Verify that the two laser lines coincide and appear as a single line.

Note: GE designed the internal axial lasers on the current CT system to shine *down* on the collimator. Do NOT adjust the internal alignment lights at this time. The tomographic plane tests use the QA phantom to check the internal axial lasers alignment to the collimator.

3.2.4 External Axial to Internal Axial Distance

Note: Ensure that cradle is level.

- 1.) Raise the table to its highest elevation.
- 2.) Extend the cradle until you see both the internal and external laser lights shining on the cradle.
- 3.) Place a metric rule on the right edge of the cradle, and measure the distance from the internal axial laser line to the external axial line. Verify this distance equals 240.0 mm +1.0 mm.
- 4.) Place the rule on the left edge of the cradle, and measure again.
- 5.) Leave the cradle in its current position, and lower the table to the minimum elevation.
- 6.) Measure the distance between the internal and external lights on both edges of the cradle, as above. Verify the distance remains equal to 240.0 mm +1.0 mm.

3.2.5 Coronal Lights

- 1.) Place a sheet of plain white paper at the left side of the patient opening, in front of the coronal laser light. Verify the two coronal lines coincide.
- 2.) Move the paper to the right side of the patient opening. Verify the two coronal lines coincide.
- 3.) Place the paper in the center of the gantry opening. Use a level to verify that the coronal lines are horizontal.

3.2.6 Turn Lights OFF

Press the alignment light button on the gantry control panel, again, to turn the lights OFF.

3.3 Autovoice/Intercom Check

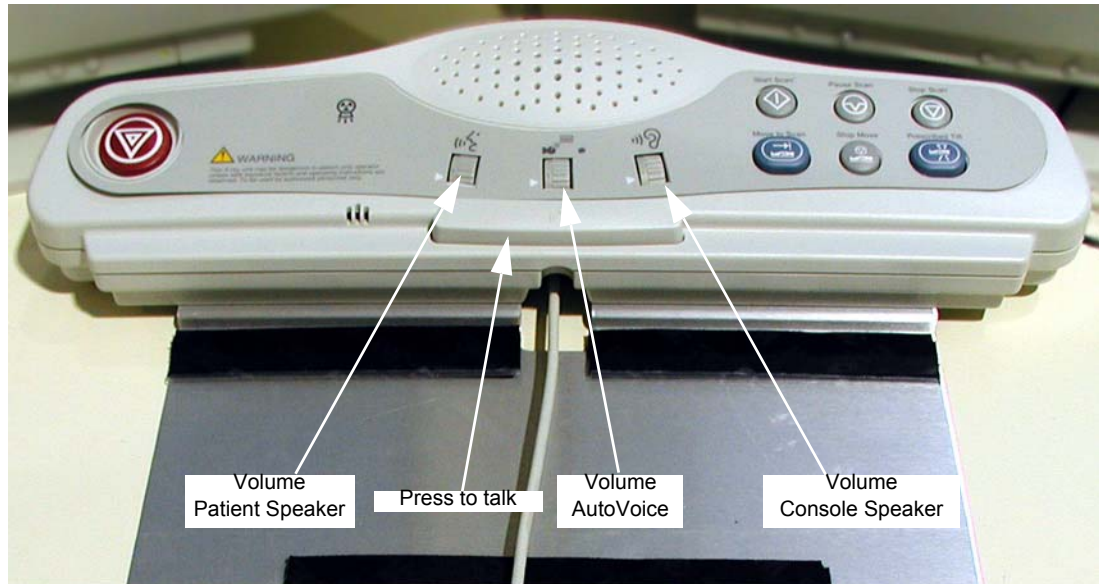


Figure 5-19 SCIM Volume Controls

3.3.1 Requirements

Two people are required to complete this procedure.

3.3.2 Patient Speaker

- 1.) To adjust the volume of the patient speaker in the table, adjust the left-most volume thumb wheel on the SCIM while speaking into the console microphone. (Press the bar on the SCIM to talk; release the bar to listen.)
- 2.) The patient should be able to clearly hear the operator.

3.3.3 Operator Console Speaker

To adjust the console speaker volume:

- 1.) Have an assistant speak into the gantry microphone.
- 2.) Adjust the SCIM console volume knob until you can clearly hear the patient.

3.3.4 Autovoice Volume

- 1.) On the Scan Desktop, select PROTOCOL MANAGEMENT.
- 2.) Select AUTO VOICE RECORD.
- 3.) Click the 3.4 button, to the right of "FF2. Inspiration".
- 4.) Click the PLAY button, to play the Inspiration AutoVoice message.
- 5.) Adjust the center volume thumb wheel while Autovoice is playing, to set the volume for the gantry speaker.
- 6.) Repeat steps 4 and 5 as necessary to achieve satisfactory volume.
- 7.) Select DONE, then select QUIT.

Note: If a satisfactory volume can not be achieved, refer to the system service manual and review the intercom module setup procedure.

3.4 CT System X-Ray ON Indicators, Cautions & Warning Labels

3.4.1 Check And Install System Warning Labels

All labels are installed in English and present on PDU, Console, Table, Gantry and Accessories. Add the labels listed below ([Table 5-2](#) and [Section 3.5](#)) for the appropriate language for the country in which this system is installed. Additionally, apply any other warning labels if present, on equipment where appropriate.



NOTICE Do not cover English labels already on the system.

Subsystem	Component	Label(s)
Console	SCIM	SCIM overlay warning label
	Keyboard	Function key overlay label
Gantry	Scan Window	Laser warning label
	Laser Window	Laser warning label
	Front Cover	Laser warning label Information labels
	System GIB	System Global Installation Base (rating) label
Table	Front Side Cover	Pinch Hazard warning label - each side of cover
	Rear Side Cover	Pinch Hazard warning label - each side of cover
	Back Cradle Pan	Pinch Hazard warning label - each side of cover
NGPDU	Front Cover	Emergency OFF label Gantry Enable label Power ON label
Accessories	Table Foot Extender	Warning label
	Coronal Head Holder	Warning label
	Sagittal Head Holder	Warning label
	Accessory Tray	Warning label
	IV Pole	Caution label

Table 5-2 System Warning Labels

3.4.2 Documentation - Verification

When finished update GE form e4879 (US Only) and the installation completion form that all appropriate language labels were installed and present.

3.5 Check Warning Labels

Note: The labels on the system and the system manuals must comply with the country law, as listed in Direction 5221102-1EN (found in the keyboard collector kit shipped with the system) regardless of the user interface (UI) language that is chosen. Compliance to the law must be completed prior to releasing the system to the customer.

Note: Do not cover English labels already on the system.

3.5.1 On SCIM

- 1.) Make sure the SCIM overlay warning label appears in the correct location on the SCIM.
- 2.) Record this information on Form 4879. For more information about this form, see [Section 8.0](#) of Chapter 4.

3.5.2 On Gantry

- 1.) Check that all laser warning labels are present on the gantry near the laser opening.
- 2.) There should also be warning labels on the lower right side of the gantry front cover.
- 3.) Record this information on Form 4879. For more information about this form, see [Section 8.0](#) of Chapter 4.

3.5.3 On Laser

- 1.) Make sure all laser warning labels appear in the correct location on the outside of the gantry.
- 2.) Obtain and install replacements for any missing labels.



Figure 5-20 Laser Warnings and Precautions

3.6 Process Product Locator Cards

- 1.) Collect the product locator cards shipped with the system. There should be approximately 28 product locator cards with the average system.
- 2.) Update the online product locator web site with the required hospital information.
- 3.) Confirm that the serial numbers on the cards shipped with the system match those found on the web site for that GON number.
- 4.) Update as required. Place the cards in a plastic bag, then place them in the service cabinet.

3.7 Verify Service Cabinet Installation

Verify that the service cabinet is installed and that the shipped service materials are in the cabinet. If the cabinet was not installed, install it at this time by following the Service Cabinet Installation procedure in [Section 5.7](#) of Chapter 2.

3.8 Check X-Ray Lights

Perform several scans following the steps below. Verify that the X-ray ON lights are ON during the scans. When done, check the boxes in [Table 5-3](#).

- 1.) Make sure the axial drive enable and HVDC enable switches are ON.
- 2.) If you are not on the Service Desktop, click on the Service Desktop icon.
- 3.) Select DIAGNOSTICS.
- 4.) Select DIAGNOSTIC DATA COLLECTION.
- 5.) Set the scan time to 2.00.
- 6.) Set the kV to 80.
- 7.) Set the mA to 40.
- 8.) Press ACCEPT RX.
- 9.) Press **START SCAN** button when flashing.
- 10.) Record this information on Form 4879. For more information about this form, see [Section 8.0](#) of Chapter 4.

Light On	Warning Light Locations
☐	SCIM
☐	Gantry Front
☐	Gantry Back (Use a mirror to view)
☐	Room Light (outside of the room)

Table 5-3 X-ray Light Chart

3.9 Mechanical Characterization

The relationship of table height to ISO center and internal-to-external landmarks must be characterized for proper interference matrix functionality.

Note: Do NOT perform tilt characterization.

3.9.1 Alignment Light Characterization

- 1.) Start the Mechanical Characterization tool from the Calibration tab on the Common Service Desktop.
- 2.) Select the CHARACTERIZE ALIGNMENT LIGHTS button from the interface.
- 3.) Follow the on-screen instructions.

3.9.2 Table Height Characterization

- 1.) Select the CHARACTERIZE TABLE HEIGHT button from the interface.
- 2.) Follow the on-screen instructions.

Note: If the table height is less than 21mm or greater than 25mm, relative to ISO, you must adjust the table height using the table leveling pad and adjusters. Raise or lower all four adjusters equally to achieve desired results. Note down the value of distance "V" which will be used in Section 3.11.

3.10 Short Footprint Setting

Normally, the table cradle can travel up to 1861 mm from scan central line. The Short Footprint function can limit this distance to a value shorter than 1861 mm (in 1 mm increments).

Note: If you moved the cradle into the set maximum distance while the table is not at the highest position, then the system will inhibit the table upward operation.

Note: During the table characterization procedure, or while operating the cradle with the service switches on the MTCB board, the cradle is enabled to move up to 1861 mm from scan central line, regardless of the short footprint set distance.



CAUTION **Potential for injury to a person.**
Small space present.

The IN-limit position of Cradle short foot print mode should be set in order not to pinch patient between the cradle edge and scanning room wall.



NOTICE **It is recommended that safety clearance from cradle IN-limit to wall should be no less than 100 mm.**

- 1.) Attach the cradle extender on the cradle.
- 2.) Launch the MECHANICAL CHARACTERIZATION tool from the Service Desktop, select CALIBRATION tab.
- 3.) Select SHORT FOOT PRINT.
- 4.) Follow the on screen instructions.
- 5.) After the setting, verify that you can not move the cradle inward further from the set position, with the following conditions:
 - a.) The table is at the highest position.
 - b.) The table is at a lowest position where scanning is possible.

3.11 Interference Test

PREREQUISITES

- Be sure that the System State was restored from **DVD** per Section 2.0.
- Reset the hardware to download the new characterization values before performing the table/gantry interference tests in this section.

CONFIGURATION

- 1.) Table flashed with latest SW
- 2.) Table must have elevation and cradle characterized
- 3.) Table must be mechanically aligned to gantry
- 4.) Table must have had the table-gantry characterization completed
- 5.) Verify the Table extender is installed.

TEST OUTLINE

The following tests verify the proper tilt and table interference matrix on the H-Power gantry.

- [Verify Table Elevation – Section 3.11.1](#)
- [Position Tilt, Move Table to Interference Limit – Section 3.11.2.1](#)
- [Position Table, Move Tilt to Interference Limit – Section 3.11.2.2](#)

- [Tilt Limits When Table Below Scan Plane Lower Limit – Section 3.11.2.3](#)

REQUIREMENTS

The following requirements are tested in this series of tests.

- 1.) No motion shall cause the table to hit the gantry (or gantry to hit the table)
- 2.) Requirement #1 shall include the use of the table extender.
- 3.) No tilt motion shall cause the gantry tilting frame to touch the stationary base covers for any tilt angle.

LIMITATIONS

These requirements will only be met when the table is NOT in service mode.

INTERPRETING TEST RESULTS

If test results indicate that elevation and/or tilt display readings fail to meet specifications, DO NOT adjust the limit switches. Instead, re-characterize and/or adjust tilt speed. Refer to Appendix C. Perform elevation and cradle first, then repeat the test. If it still fails, perform tilt.

3.11.1 Verify Table Elevation

Note: "V" means distance from table height to ISO.

#	TEST	EXPECTED RESULTS
3.11.1-1	Move the cradle to home position. Push the table down gantry push-button to lower the table to the minimum height.	Elevation Display should read $475.0 + V \pm 3$ mm
3.11.1-2	Push the table up gantry push button to raise the table to the maximum height. If the mechanical alignment of the table/gantry is not correct, as is often the case during manufacturing staging, this value may be out of range. Most of the following tests will still be valid: Those that might have some variation are indicated in bold type .	Elevation Display should read $V \pm 3$ mm. If mechanical alignment of the table/gantry is not correct, this value can be as low as 0 mm and as high as 40 mm.

Table 5-4 Table Elevation Tests

3.11.2 Tests

3.11.2.1 Position Tilt, Move Table to Interference Limit

The following tests verify the table interference limits at different tilt locations.

- Note:
- "I" means top of gantry tilts toward the table base
 - "S" means top of gantry tilts away from the table base.
 - "V" means distance from table height to ISO

Important: For all tests, make sure there is 2.5 cm of clearance between the gantry and table. Also make sure that the Gantry Tilting frame covers do not touch the stationary base covers, for all tilt angles used in this test.

#	TEST	EXPECTED RESULTS
3.11.2.1-1	Move cradle to home position and set internal landmark.	Cradle position on display should read 0.0

Table 5-5 Position Tilt, Move Table to Interference Limit Tests

#	TEST	EXPECTED RESULTS
3.11.2.1-2	Raise table height to maximum height, set the internal landmark, move cradle into gantry 1m.	Table elevation on display should read $V \pm 3\text{mm}$. (This value will be the same as in 3.11.1-2 above.) Cradle position on display should read 1000.0 mm.
3.11.2.1-3	Tilt the gantry to I30.0, then lower table until motion stops.	Table elevation on display should read $100 \pm 3\text{mm}$. Tilt display should read I30.
3.11.2.1-4	Tilt the gantry to I23.0, then lower table until motion stops.	Table elevation on display should read $155 \pm 3\text{mm}$. Tilt display should read I23.
3.11.2.1-5	Tilt the gantry to I20.0, then lower table until motion stops.	Table elevation on display should read $170 \pm 3\text{mm}$. Tilt display should read I20.
3.11.2.1-6	Raise the table elevation to maximum height.	Table elevation on display should read $V \pm 3\text{mm}$. (This value will be the same as in 3.11.1-2 above.)
3.11.2.1-7	Tilt the gantry to S30.0, then lower table until motion stops.	Table elevation on display should read $156 \pm 3\text{mm}$. Tilt display should read S30.
3.11.2.1-8	Tilt the gantry to S23.0, then lower table until motion stops.	Table elevation on display should read $182 \pm 3\text{mm}$. Tilt display should read S23.
3.11.2.1-9	Tilt the gantry to S20.0, then lower table until motion stops.	Table elevation on display should read $192 \pm 3\text{mm}$. Tilt display should read S20.
3.11.2.1-10	Raise the table to 154 mm.	Table elevation on display should read 154mm.
3.11.2.1-11	Tilt gantry to S30 and verify the table height can be adjusted from 154 to 25 mm. (This value will be the same as in 3.11.1-2 above.)	Tilt display should read S30. Table lower limit should be $154 \pm 3\text{mm}$. Upper table limit should be $V \pm 3\text{mm}$. (This value will be the same as in 3.11.1-2 above.)
3.11.2.1-12	Set the table height to 98 mm.	Table elevation on display should read 98mm.
3.11.2.1-13	Tilt gantry to I30 and verify the table height can be adjusted from 98 to 25 mm. (This value will be the same as in 3.11.1-2 above.)	Tilt display should read I30. Table lower limit should be $98 \pm 3\text{mm}$. Upper table limit should be $V \pm 3\text{mm}$. (This value will be the same as in 3.11.1-2 above.)

Table 5-5 Position Tilt, Move Table to Interference Limit Tests

3.11.2.2 Position Table, Move Tilt to Interference Limit

The following tests verify the tilt interference limits at different table heights.

- Note:
- "I" means top of gantry tilts toward the table base
 - "S" means top of gantry tilts away from the table base.
 - "V" means distance from table height to ISO

Important: For all tests, make sure there is 2.5 cm of clearance between the gantry and table.

#	TEST	EXPECTED RESULTS
3.11.2.2-1	Move cradle to home position and set internal landmark. Set gantry tilt to zero.	Cradle position on display should read 0.0. Gantry tilt on display should read 0.0.

Table 5-6 Position Table, Move Tilt to Interference Limit Tests

#	TEST	EXPECTED RESULTS
3.11.2.2-2	Raise table height to maximum height, set the internal landmark, move cradle into gantry 1m.	Table elevation on display should read $V \pm 3$ mm. (This value will be the same as in 3.11.1-2 above.) Cradle position on display should read 1000.0 mm.
3.11.2.2-3	Lower table until height is 115mm. Tilt the gantry top away from the table ("S") until it stops.	Table elevation on display should read 115 mm. Tilt display should read $S30 \pm 0.5^\circ$.
3.11.2.2-4	Tilt the gantry top toward the table ("I") until it stops.	Table elevation on display should read 115 mm. Tilt display should read $I28 \pm 0.5^\circ$.
3.11.2.2-5	Tilt the gantry to 0. Lower table until height is 200mm. Tilt the gantry top away from the table ("S") until it stops.	Table elevation on display should read 200 mm. Tilt display should read $S17 \pm 0.5^\circ$.
3.11.2.2-6	Tilt the gantry top toward the table ("I") until it stops.	Table elevation on display should read 200 mm. Tilt display should read $I14.5 \pm 0.5^\circ$.
3.11.2.2-7	Tilt the gantry to 0. Lower table until height is 210 mm. Tilt the gantry top away from the table ("S") until it stops.	Table elevation on display should read 210 mm. Tilt display should read $S13.5 \pm 0.5^\circ$.
3.11.2.2-8	Tilt the gantry top toward the table ("I") until it stops.	Table elevation on display should read 210 mm. Tilt display should read $I12 \pm 0.5^\circ$.

Table 5-6 Position Table, Move Tilt to Interference Limit Tests

3.11.2.3 Tilt Limits When Table Below Scan Plane Lower Limit

The following tests verify the table and tilt interference limits when the table height is below the scan plane.

- Note:
- "I" means top of gantry tilts toward the table base
 - "S" means top of gantry tilts away from the table base.
 - "V" means distance from table height to ISO

Important: For all tests, make sure there is 2.5 cm of clearance between the gantry and table.

#	TEST	EXPECTED RESULTS
3.11.2.3-1	Set gantry tilt to zero. Move cradle to home position, lower the table all the way, and set the internal landmark.	Cradle position on display should read 0.0 Gantry tilt on display should read 0.0. Table height should read $475.0 + V \pm 3$ mm. (This value will be the offset from 500 mm by the same amount as the upper table limit varies from 25 mm in 3.11.1-2 above.)
3.11.2.3-2	Tilt the gantry forward and backwards and verify the following tilt limits: S3.5 and I30.	Gantry tilt on display should read $S3.5 \pm 0.5^\circ$. Gantry tilt on display should read $I30.0 \pm 0.5^\circ$.
3.11.2.3-3	Set Gantry tilt to 0. Then, using the gantry push-buttons, move the cradle in towards the gantry until it is stopped.	Gantry tilt on display should read 0.0. Cradle should stop at $45.0 \text{ mm} \pm 6 \text{ mm}$ from the home position. (This value will not be valid if the table/gantry mechanical characterization is not correct as described in 3.11.1-2 above. Insure that the front of the table is at least 25 mm from the gantry.)

Table 5-7 Tilt Limits When Table Below Scan Plane Lower Limit Tests

#	TEST	EXPECTED RESULTS
3.11.2.3-4	Set gantry tilt to zero. Move cradle to home position.	Cradle position on display should read 0.0 Gantry tilt on display should read 0.0.
3.11.2.3-5	Raise the table to a height of 386 mm and verify the following tilt limits: S5.0 and I30.0.	Table height should read 386 mm. Gantry tilt on display should read $S5.0 \pm 0.5^\circ$. Gantry tilt on display should read $I30.0 \pm 0.5^\circ$.
3.11.2.3-6	Set Gantry tilt to 0. Set the internal landmark. Then, using the gantry push-buttons, move the cradle in towards the gantry until it is stopped.	Gantry tilt on display should read 0.0. Cradle should stop at $59.0 \text{ mm} \pm 6 \text{ mm}$ from the home position.
3.11.2.3-7	Set gantry tilt to zero. Move cradle to home position.	Cradle position on display should read 0.0. Gantry tilt on display should read 0.0.
3.11.2.3-8	Raise the table to a height of 242 mm and verify the following tilt limits: S12.0 and I30.0.	Table height should read 242 mm. Gantry tilt on display should read $S12.0 \pm 0.5^\circ$. Gantry tilt on display should read $I30.0 \pm 0.5^\circ$.
3.11.2.3-9	Set Gantry tilt to 0. Raise the table to 210 mm. Then, using the gantry push-buttons, move the cradle in towards the gantry until it is stopped.	Gantry tilt on display should read 0.0. Cradle should go all the way through the gantry bore to the full-extended position.
3.11.2.3-10	Set gantry tilt to zero. Move cradle to home position, lower the table all the way and set the internal landmark.	Cradle position on display should read 0.0 Gantry tilt on display should read 0.0. Table height should read $475.0 +V \pm 3 \text{ mm}$. (This value will be the offset from 500 mm by the same amount as the upper table limit varies from 25 mm in 3.11.1-2 above.)
3.11.2.3-11	Tilt the gantry top toward the table to a tilt of 30 degrees	Display should read I30.
3.11.2.3-12	With the table down all the way, move the cradle in until it stops.	Cradle position should be $359.0 \pm 3 \text{ mm}$.
3.11.2.3-13	Move the table to the home position, raise the table to a height of 386 mm, set the internal landmark, and then move the cradle in until it stops.	Cradle position should be $330 \pm 50 \text{ mm}$.
3.11.2.3-14	Raise the table to a height of 98 mm, then move the cradle in.	Cradle should go all the way through the gantry bore to the full-extended position.

Table 5-7 Tilt Limits When Table Below Scan Plane Lower Limit Tests

Chapter 6

Image Quality



CAUTION Shock Hazard.
Voltage Present.
No service on left side while energized.



NOTICE To prevent potential data loss, please do the following:

- When instructed, record data collected from the procedures in this chapter on GE Form e4879. For more information about this form, see [Section 8.0](#) of Chapter 4.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Introduction

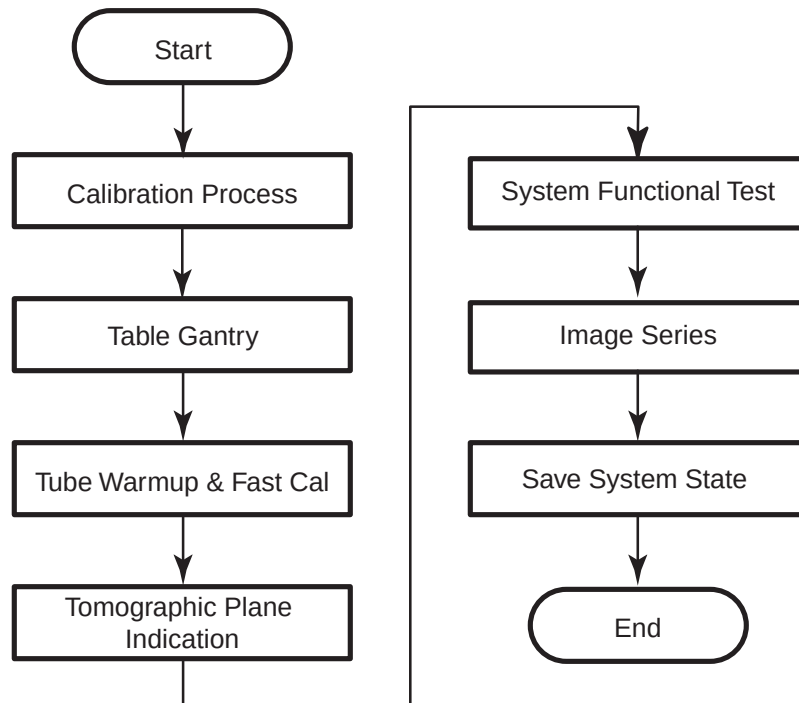


Figure 6-1 System Test Overview

Section 2.0

Calibration Process

2.1 Reference Procedures

Do not perform these procedures ([Scanning with Service Protocols](#) or [Center Phantom](#)) until instructed to do so in other sections of this chapter.

2.1.1 Scanning with Service Protocols

Locate the Manufacturing and Installation protocols under Infant area 20.

Note: **Manufacturing and Service share this Protocol list. Different product option offerings also use this list. Carefully follow the scan section instructions, and verify you acquired the images with the correct technique before filling out the data sheet. Otherwise you may troubleshoot an image problem that only exists because you used the wrong technique.**

- 1.) Select the NEW PATIENT icon on the left monitor
- 2.) Enter a Patient ID (e.g., **getest**).
- 3.) Click on the box labeled PEDIATRIC.
- 4.) Select a service protocol from the list, to display the corresponding view edit screen.
Optional Method: Enter the protocol number into the Protocol Number Field on the Exam Rx Screen.

2.1.2 Center Phantom

- 1.) Select SCANNER UTILITIES icon on the left monitor
- 2.) Select CENTER PHANTOM.
- 3.) Follow the on-screen procedures.
- 4.) The phantom center spec is $\pm 5\text{mm}$.
- 5.) Select QUIT, when the phantom is within specification.
- 6.) Level the phantom both front to back and side to side. (use a 6" level)

2.2 Prepare the QA Phantom

Note: **The QA phantom is shipped water filled.**

- 1.) Locate the multi-language sticker packet in the QA phantom shipping box.
- 2.) Attach the sticker with the customer's language to the face of the phantom hanger bracket.

2.3 Calibration Process Introduction

If your system has a factory supplied state DVD-RAM, you used it to load the system calibration files during the Restore System State ([2.3 on page 199](#) of this manual).

Section 3.0

Table/Gantry Alignment Procedure

3.1 Time & Personnel

Required Persons	Preliminary Reqs	Procedure	Finalization
2 (FE or mechanical supplier)		45 minutes labor on-site	

3.2 Tools and Test Equipment

- 1 mm wire

3.3 Preparation

- All table mechanical alignment procedures completed.
- The table perpendicular alignment test passed.
- Table anchors are in place and within specification.
- The table is level in all directions.

3.4 Procedure

TABLE GANTRY PREP

- 1.) Check that the table cradle is level in all directions. Correct, if necessary.
- 2.) Drive the table to its highest elevation ISO with the phantom holder removed.
- 3.) Check the scan window for proper installation.

VERIFY TABLE DRIVE CONSISTENCY

- 4.) Drive the table cradle in and out five times to seat the rollers.

CRADLE SETUP

- 5.) Turn on the alignment lights.
- 6.) Advance the end of the cradle to the black dot on cradle.
- 7.) Tape a 100 mm section of 1 mm wire on the cradle that aligns with the white cradle center line.
- 8.) Using the gantry keypad, set an internal landmark, and then advance the cradle 1000 mm.
- 9.) Tape second 100 mm section of 1 mm wire on the cradle that aligns with the alignment light.

SCANNING SETUP

- 10.) From the application screen Select *NEW PATIENT*.
 - a.) Fill out patient ID: GE Test
 - b.) Name: Alignment
- 11.) From the Protocol screen:
 - a.) Select *SERVICE*,

- b.) Select *IMAGE QUALITY*,
- c.) Select *PERPENDICULAR ALIGNMENT*.
- 12.) The red boxes should disappear from the screen. If not, reset an internal landmark.
- 13.) Select *CONFIRM*, then press the START SCAN button when lighted.

IMAGE REVIEW

- 14.) On the Service screen, select *IMAGE WORKS*
 - a.) Locate the scanned examination in the Examinations column.
 - b.) Highlight the Alignment scans
 - c.) Select *VIEWER*.
 - d.) Select *FORMAT*, and select the two-in-one format horizontal display view.
- 15.) Click on image 1 and select the grid. With the grid and image displayed, visually compare image 1 to image 2.
 - For close inspection, you may need to use the zoom function to see a difference.
 - Visually compare image 1 and image 2 to verify the centering wire appears in the center of the grid. As shown on the screen, the wire is 1 mm. Use the measure tool to determine the alignment difference. Move the table until both are within ± 2 mm of center.

MOVING THE TABLE

- 16.) The adjustment is likely to require a very small movement. Use a suitable tool to move the table the required distance.
- 17.) Rescan to confirm each movement trial. This can take a few trials to move the table to a position that is within the specification.
- 18.) The plastic accessory edges of the cradle are installed to allow cradle accessories to be used. These edges, although visible, should not be used to determine cradle center. Edge-to-edge difference can be greater than the alignment specification.
- 19.) Repeat above steps until both images are visually aligned on the screen.

FINALIZATION

- 20.) Use a calibrated torque wrench to tighten the anchors to 75 ± 6 N-m (55 ± 5 ft.-lb.). Confirm that the torqued anchor still meets the anchor installation specifications:
 - a.) Maintain 1 full thread of adjustor showing above the lock ring or table base plate.
 - b.) Have not more than 1 in. of anchor showing above the nut. Do not cut off any access.
 - c.) Using a permanent marker, draw a line on the nut and base. Use this line to determine whether the anchor loosened over time.
- 21.) Reinstall all table components removed to access the anchors.

Section 4.0 Tube Warm Up and Fast Cal



- 1.) Select  to warm up the tube.
- 2.) Select FAST CALIBRATION from the Daily Prep menu.

Note: **Use the default Fast Cal selections determined by the system configuration. (The system defaults to all four kV stations, but you can choose kV stations to calibrate during reconfig.)**

- 3.) Run the selected air calibrations.
- 4.) When the calibration process completes, click on QUIT.

Section 5.0

Tomographic Plane Indication

- 1.) Place the QA phantom on the phantom holder.
- 2.) Center the Phantom (refer to procedure [2.1.2 on page 228](#))
- 3.) Turn ON the internal alignment lights, and drive the phantom into the gantry opening, until the line on the phantom lines up with the internal laser lights.
- 4.) Verify that BOTH internal axial lasers line up along the line on the QA phantom. If not, check table/gantry, cradle, and/or laser alignment.
- 5.) Center the phantom in the scan plane with the calibration program. (See [2.1.2 on page 228](#) for details on the phantom centering procedure.)
- 6.) Select the service protocol TOMO PLANE INDICATION. (See [2.1.1 on page 228](#) for details on scanning with service protocols.)

or

Manually select the scan parameters in [Table 6-1](#).

Scan Type	kV	mA	SFOV	Thickness	Scan Time	Start Loc.	End Loc.	Algorithm	Interval
Helical	120	200	Small	1.25HQ	1.0sec	I3.0	S3.0	Bone detail	0.2

Table 6-1 Tomographic Plane Indication Scan Parameters

- 7.) Display the image series, and locate the scan plane indicator, the longest bar in the bar pattern on the right side of the phantom. The right side of the phantom corresponds to the side of the image labeled **L** on the display screen.
- 8.) On the HHS Data Sheet, record the scan location (shown on the image annotation) of the image with the darkest scan plane indicator (darkest long bar).
- 9.) If your system meets all the installation and alignment specifications, the image at scan location zero (S0.0) should contain the scan plane indicator. If scan location S1.0 or scan location I1.0 has the darkest bar, the system still meets the specification. The scan plane deviation should equal $S0.0 \pm 1.0\text{mm}$. If necessary, adjust the internal alignment light position to meet the $S0.0 \pm 1.0\text{mm}$ requirement.
- 10.) Repeat the Tomographic Plane Indication test with the external alignment lights.
 - a.) Use the external alignment light, and press the external landmark.
 - b.) Verify the external light lines up along the black line on BOTH the left and right sides of the QA phantom.
 - c.) The scan plane indication must fall within the $S0.0 \pm 1.0\text{mm}$ specification.
- 11.) Initial below.

Section 6.0

Image Series

IMPORTANT: Run ALL Image Series Tests in **Auto Mode**. Manual procedures are provided in this section as **REFERENCE ONLY**.

6.1 Scan Protocol

The person who acquires the image series has the responsibility to review the images and verify they meet the specifications listed on data sheets. Responsibilities also include means and standard deviation measurements and keeping a record of failures that occur during the image series. Unless otherwise stated, use the following scan parameters during the image series acquisition:

- Scan FOV equal to display FOV (Field of View)
- 512x512 matrix size

Note: **Consider any image series scan that does not meet specifications as failing.**

For means and standard deviations, 90% of the slices must pass. Any failure on a particular technique requires at least ten additional slices to evaluate effectively.

Systems with metal-free cradles have a phantom holder with a perpendicular adjustment (Z-axis) knob on it. **Each time you change phantoms**, make sure you use a bubble level, and the Z-axis knob on the phantom holder, to level the phantom.

6.2 Data Recording: Means and Standard Deviation

Any failure on a particular technique requires at least a ten additional slices to evaluate effectively. For means and standard deviations, 90% of the slices must pass.

- Record means to two decimal places, and round to the nearest one-tenth, (one decimal place) when you compare the resulting values to the spec.
- Record standard deviations to two decimal places, then round off to one decimal place, to compare it to the spec.
- Average standard deviations: Use two decimal places to average the values, then round off to one place.

Before you record the means and standard deviations, check the image data sheets to determine whether to average the means and standard deviations, or record them slice by slice. Make sure you record all the required image data on the HHS data sheets.

6.3 Term Definitions and Screens

Xc - Mean CT number for the specified center coordinates of the phantom image.

AvXc - Average Mean CT number for the center of the phantom image: Average the mean CT value for all specified center coordinates of all slices in an exam.

Xo - Mean CT number for the outside of the phantom image: Average the mean CT value for all specified outside coordinates of one slice.

AvXo - Average outside mean CT number for the number of slices in an exam.

AvSDc - Average image noise about the center image coordinate (measured as the standard deviation) of all slices in an exam.

AvSDo - Average image noise (standard deviation) for the outside of a phantom: Average of all outside coordinates of all the slices in an exam.

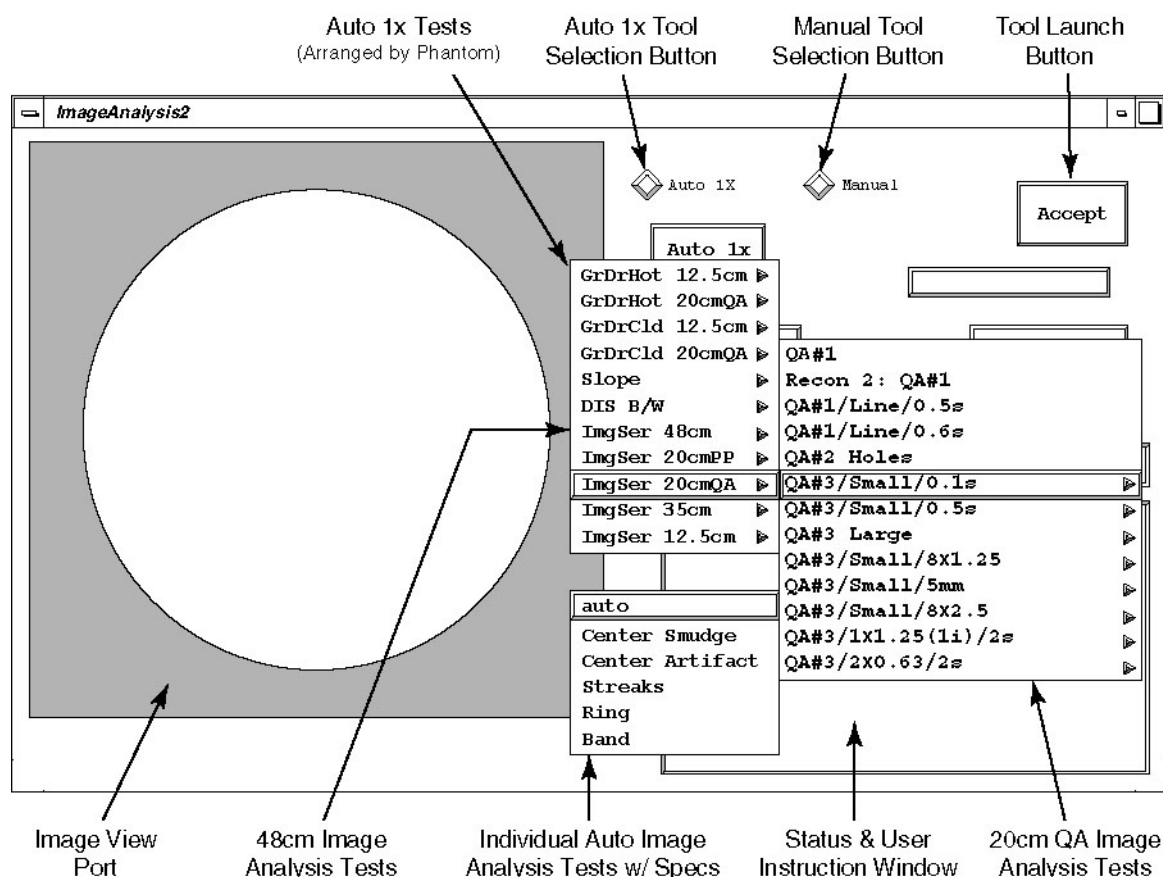


Figure 6-2 Image Analysis Tool User Interface - Auto 1x Test Pull Down Menu

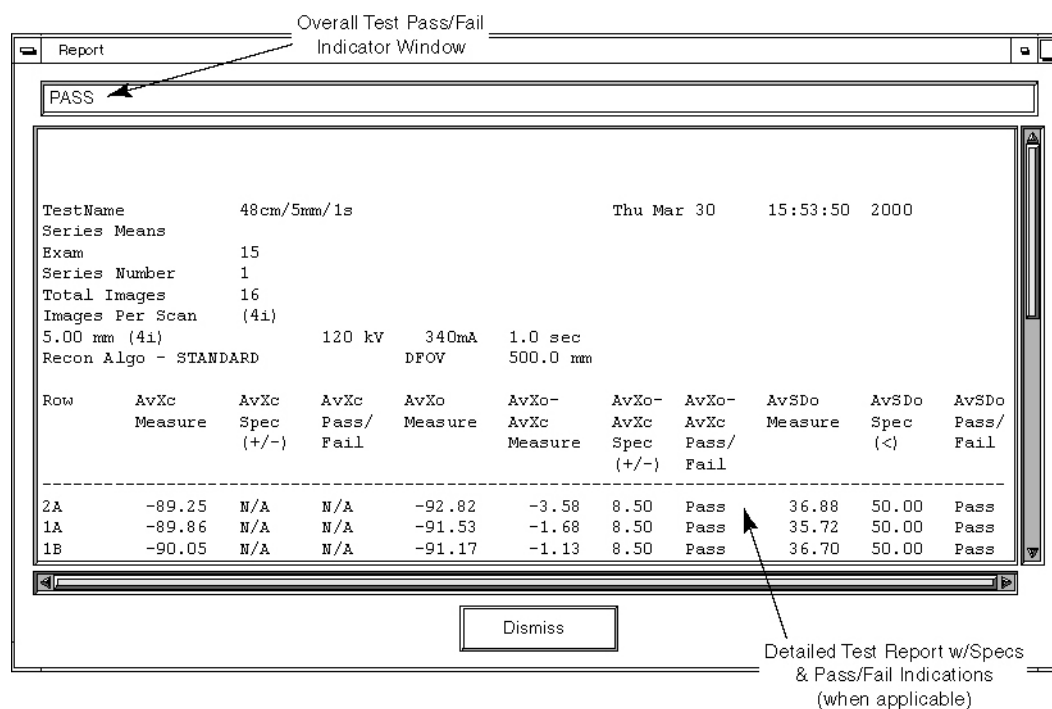


Figure 6-3 Image Analysis Tool User Interface - Test Results Report Window**6.4 48cm Phantom Image Series Image Performance Verification****6.4.1 Acquiring the 48cm Phantom Image Series**

- 1.) Mount the Phantom Holder on the head-end of the table.
- 2.) Mount the 48cm Phantom on the Phantom Holder.
- 3.) Align, level, & center the 48cm Phantom.
 - Align phantom using the internal laser lights.
 - Level phantom using bubble level and the Z Axis knob on the Phantom Holder.
 - Center phantom using the CENTER PHANTOM procedure in the left hand SCANNER UTILITIES selection and the X and Y Axis knobs on the Phantom Holder.
- 4.) Set up the system to scan a single, 64 image, 48cm Phantom series.

MANUAL SCAN PROTOCOL SETUP

Refer to [Table6-2](#) to manually set-up an Axial scan with the parameters shown.

Note: These parameters may differ from those found using the [Auto Scan Protocol Setup](#) below.

System	Series Description	Scan Type	Start Loc.	End Loc.	Total # of Images	Thick Speed	Interval (mm)	Gantry Tilt	SFOV	kV	mA	Total Exposure Time	DFOV (cm)	Recon Type
16 Slice	48 16x1.25/120kV /400mA/2s	Axial Full 2.0 sec	I9.375	S9.375	64	1.2516i	0.00	S0.0	Large	120	400	8.0 sec.	50.0	Std
4/8 Slice	48 Lg Series 1	Axial Full 1.0 sec.	I7.50	S7.50	16	5.0 4i	0.00	S0.0	Large	120	340	4.0 sec.	50.0	Std

Table 6-2 48cm Phantom Image Series Scan Parameters

AUTO SCAN PROTOCOL SETUP

- a.) On the Exam Rx desktop, select NEW PATIENT.
- b.) Type the following entries in the two listed Patient Information fields:
 - * Patient ID: **Service**
 - * Name: **48cm Phantom Image Series**
- c.) From the Protocol Selection display, click on the Service Tab under Anatomical Selector.
- d.) On the Service Protocol Selection window, click on MANUFACTURING.
- e.) On the Service Category Protocol List, click on IMAGE SERIES 48CM.
- f.) **16 Slice**
On the ExamRx protocol parameter display, skip to the 3rd Series.
(Series Description: 48 16x1.25/120kV/400mA/2s)
4/8 Slice:
On the ExamRx protocol parameter display, skip to the 2nd Series.
- g.) Set internal Landmark.
- 5.) Acquire a single, 64 image, 4 scan, image series of the 48 cm Phantom.

6.4.2 Image Performance Verification

6.4.2.1 Series Means (AvXo-AvXc and AvSDo) & Center Smudge

- 1.) Launch the Image Analysis Tool.
 - a.) From the Global Control Palette, click on the Service Desktop.
 - b.) From the Service Desktop upper navigational bar, click on the IMAGE QUALITY icon.
 - c.) In the left navigational area directory, click on IMAGE ANALYSIS.
- Note: During the Image Analysis process, you will need to switch between the Service Browser screen (to select the image series to analyze) and the Image Analysis screen (to run a particular Auto 1x analysis test). Use ALT/F1 to bring a window to the front. Use ALT/F3 to send a window to the back.
- 2.) Using the AUTO 1X Tool on the Image Analysis screen, generate a Series Means and Center Smudge report for the 1st series of the 48cm Phantom acquisition.
 - a.) From the Service Browser window, select the 48cm Phantom series acquired in Section 6.4.1.
 - b.) On the Image Analysis Tool window, click on the AUTO 1X diamond shaped button. (See [Figure 6-2, on page 234.](#))
 - c.) Click on the AUTO 1X rectangular button, select the IMGSER 48CM test from the pull-down menu, slide cursor to the right and select 16X1.25/120KV/400MA/2S (**16 Slice**) or 48CM/5MM/0.8S (**4/8 Slice**) from the test pull-down menu, and continue to slide cursor to the right to select AUTO.
 - d.) Click on the ACCEPT button.
 - 3.) Review the Image Analysis Report data and record the Series Means (AvXo-AvXc, and AvSDc) data and the Center Smudge data for each row in [Table 6-3](#) (**16 Slice**) or [Table 6-4](#) (**4/8 Slice**).
 - a.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS. (See [Table6-2.](#))
 - b.) Verify the scan parameter values displayed in the report match those in [Table6-2.](#)
 - c.) Record the Series Means (AvXo-AvXc, and AvSDc) data and the Center Smudge data for each row in [Table 6-3](#) (**16 Slice**), or [Table6-4](#) (**4/8 Slice**).

6.4.2.2 Band and Streak Artifact

- 1.) Review the 48cm Phantom images acquired in Section 6.4.1 for Bands and Streaks using the Image Works Desktop.
 - a.) From the Global Control Palette, click on the IMAGE WORKS Desktop.
 - b.) From the Image Works Browser window, select the 48cm Phantom series acquired in Section 6.4.1.
 - c.) From the Image Works Browser window, click on the VIEWER button.
 - d.) While paging through the 64 images in the series, analyze images for any Band or Streak artifacts. Adjust Window/Level using the center mouse button, as required.
 - e.) Note any image that appears to have a Band or Streak artifact.
- 2.) Verify any images noted above (step 1e) meet 48cm Phantom Band or Streak Image Performance requirements.
 - a.) From the Global Control Palette, click on the Service Desktop.
 - b.) From the Service Browser window, select an image of the 48cm Phantom series acquired in Section 6.4.1 that appeared to have a Band or Streak while reviewing the series in Image Works.
 - c.) On the Image Analysis Tool window, click on the AUTO 1X rectangular button, select the

- IMAGE SERIES 48CM test from the pull-down menu, slide cursor to the right and select 16X1.25/120KV/400MA/2S (**16 Slice**) or 48CM/5MM/0.8S (**4/8 Slice**) from the test pull-down menu, and continue to slide cursor to the right to select BAND or STREAK.
- d.) Click on the ACCEPT button.
- Note: If required, adjust Window/Level using the center mouse button.
- e.) Position and size the Band or Streak ROI using the left cursor button.
- Note: To ensure specifications are properly applied, do not adjust the Reference ROI's for the Band and Streak artifact tests. The Reference ROI's are adjustable after the 1st Accept Modification click.
- f.) Click on ACCEPT MODIFICATION twice, to generate a report.
- g.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS.
- h.) Repeat steps b through g, for each image noted in Step 1e.

Box Size = 1600 mm² to 2500 mm²

45 mm (+/- 5 mm) x 45 mm (+/- 5 mm)

45 (+/- 4 pixels) x 45 (+/- 4 pixels)

Box Positions: Box 1 = 0 mm x 0 mm
 Box 2 = 0 mm x -190 mm
 Box 3 = 190 mm x 0 mm
 Box 4 = 0 mm x 190 mm
 Box 5 = -190 mm x 0 mm

Row	Images	Brightness Uniformity (AvXo - AvXc)	Noise (AvSDo)	Center Smudge Row Values	Comments
8A	1, 17, 33, 49				
7A	2, 18, 34, 50				
6A	3, 19, 35, 51				
5A	4, 20, 36, 52				
4A	5, 21, 37, 53				
3A	6, 22, 38, 54				
2A	7, 23, 39, 55				
1A	8, 24, 40, 56				
1B	9, 25, 41, 57				
2B	10, 26, 42, 58				
3B	11, 27, 43, 59				
4B	12, 28, 44, 60				
5B	13, 29, 45, 61				
6B	14, 30, 46, 62				
7B	15, 31, 47, 63				
8B	16, 32, 48, 64				
Specifications		< +/- 8.5	< 50	< +/- 14	

Table 6-3 48cm Phantom CT# Brightness Uniformity, Noise & Center Smudge Row Performance Data Sheet (16 Slice**)**

Row	Images	Brightness Uniformity (AvXo - AvXc)	Noise (AvSDo)	Center Smudge Row Values	Comments
2A	1, 5, 9, 13				
1A	2, 6, 10, 14				
1B	3, 7, 11, 15				
2B	4, 8, 12, 16				
Specifications		< +/- 8.5	< 70	< +/- 14	

Table 6-4 48cm Phantom CT# Brightness Uniformity, Noise & Center Smudge Row Performance Data Sheet (4/8 Slice)

6.4.3 Failure Recommended Actions

6.4.3.1 Series Means (AvXo - AvXc or AvSDo) Failure Recovery

Specifications

Each Row of the series must pass 48cm Series Means (for the first series scan parameters) specifications:

- AvXo - AvXc: < +/- 8.5
- AvSDo: < 50.0 (16 Slice)
< 70.0 (4/8 Slice)

Recommended Recovery

- 1.) Repeat Detailed Cal for the 48cm Phantom.
- 2.) Repeat Sections 6.4.1 & 6.4.2, to verify Image Performance.

6.4.3.2 Center Smudge Failure Recovery

Specifications

80% the Rows must pass Center Smudge specifications when re-scanning:

Center Smudge:Smudge Factor < +/- 14.0

Recommended Recovery

- 1.) If one image of one Row fails (Row fails once):
 - Re-scan the 48cm Phantom acquiring two series of 64 images using the same protocol settings as listed in [Table6-2](#).
 - Analyze the two 64 re-scanned image series using the Image Analysis IMAGE SERIES 48CM -> 16X1.25/120KV/400MA (16 Slice) or 48CM/5MM/0.8S (4/8 Slice) -> AUTO tool.
 - Verify that all the 128 re-scanned images do not fail Series Means (AvXo - AvXc and AvSDo) or Center Smudge.
- 2.) If more than one image of one Row fails (Row fails more than once) or multiple rows are failing:
 - Repeat Detailed Cal for the 48cm Phantom.

- Re-scan the 48cm Phantom acquiring three series of 64 images using the same protocol settings as listed in [Table 6-2](#).
- Analyze the three re-scanned image series using the Image Analysis IMAGE SERIES 48CM -> 16X1.25/120KV/400MA (16 Slice) or 48CM/5MM/0.8S (4/8 Slice) -> AUTO tool.
- Verify that all the 192 re-scanned images do not fail Series Means (AvXo - AvXc and AvSDo) or Center Smudge.

6.4.3.3 Band or Streak Artifact Failure Recovery

Specifications

- Band Artifact: Band Factor< +/- 8.0
- Streak Artifact: Streak Factor< +/- 4.0

Recommended Recovery

- 1.) Troubleshoot Band or Streak artifact failures, as described in the *System Service Manual*.
- 2.) Repeat Detailed Cal for the 48cm Phantom.
- 3.) Repeat Sections 6.4.1 & 6.4.2, to verify Image Performance.

6.5 20cm QA Phantom Image Series Image Performance Verification

6.5.1 Acquiring the 20cm QA Phantom Image Series

- 1.) Mount the Phantom Holder on the head-end of the table.
- 2.) Mount the 20cm QA Phantom on the Phantom Holder.
- 3.) Align, level, & center the 20cm QA Phantom.
 - Align black line on phantom using the internal laser lights.
 - Level phantom using bubble level and the Z Axis knob on the Phantom Holder.
 - Center phantom using the CENTER PHANTOM procedure in the left head SCANNER UTILITIES selection and the X and Y Axis knobs on the Phantom Holder.
- 4.) Set up the system to scan three 20cm QA Phantom image series with a Recon of the 1st series.
 - a.) On the Exam Rx desktop, select NEW PATIENT.
 - b.) Type the following entries in the listed Patient Information following fields:
 - * Patient ID: **Service**
 - * Name: **20cm QA Phantom Image Series**
 - c.) From the Protocol Selection display, click on the Service Tab under Anatomical Selector.
 - d.) On the Service Protocol Selection window, click on MANUFACTURING.
 - e.) On the Service Category Protocol List, click on the appropriate Image Series selection as follows:

System	Protocol - First	Protocol - Second
BrightSpeed	Image Series QA	(none)

- f.) On the ExamRx protocol parameter display, select the 1st Series.
- g.) Set internal Landmark.

- 5.) Acquire the 1st 20cm QA Phantom image series and one Recon series (MTF and Contrast, Visible Lines).
- 6.) Acquire the 3rd 20cm QA Phantom image series.
- 7.) Acquire the last 20cm QA Phantom image series.

6.5.2 20cm QA Phantom Image Series Image Performance Verification

1st QA Phantom Image Series (4 Image MTF Average and Contrast Scale) Image Performance Verification

- 1.) Launch the Image Analysis Tool.
 - a.) From the Global Control Palette, click on the Service Desktop.
 - b.) From the Service Desktop upper navigational bar, click on IMAGE QUALITY.
 - c.) In the left navigational area directory, click on IMAGE ANALYSIS.

Note: During the Image Analysis process, you will need to switch between the Service Browser screen (to select the image series to analyze) and the Image Analysis screen (to run a particular Auto 1x analysis test). Use ALT/F1 to bring a window to the front. Use ALT/F3 to send a window to the back.

- 2.) Using the Auto 1x Tool on the Image Analysis screen, generate a QA#1 MTF and QA#1 Contrast Scale report for the 1st series of the 20cm QA Phantom acquisition.
 - a.) From the Service Browser window, select the 1st series of the 20cm QA Phantom series acquired in section 6.5.1.
 - b.) On the Image Analysis Tool window, click on the AUTO 1X diamond shaped button. (See [Figure 6-2, on page 234](#).)
 - c.) Click on the AUTO 1X rectangular button, select the IMGSER 20QA test from the pull-down menu, slide cursor to the right to select QA#1 from the test pull-down menu.
 - d.) Click on the ACCEPT button.
- 3.) Review the Image Analysis Report data and record the Per Image MTF, 4 Image MTF Average, and Per Image Contrast Scale data for each row in Table 10-7.
 - a.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS. (See [Figure 6-3, on page 235](#).)
 - b.) Record Per Image MTF and Per Image Contrast Scale data for each image in [Table 6-6](#).
 - c.) Record the 4 Image MTF Average in [Table 6-6](#).

Image	MTF	MTF 4-slice average	Contrast Scale	Comments
1		N.A.		
2		N.A.		
3		N.A.		
4		N.A.		
Specifications	N.A.	0.65 to 1.0	110.0 to 130.0	N.A.

Table 6-5 20cm QA#1 Phantom High Contrast Spatial Resolution Image Performance (MTF and Contrast Scale)

2nd QA Phantom Image Series (4 Bone Retro Image Visible Lines) Image Perf. Verification

- 1.) Using the Auto 1x Tool on the Image Analysis screen, generate a Per Image QA#1 Lines report for the 2nd series of the 20cm QA Phantom acquisition.
 - a.) From the Service Browser window, select the 1st image of the 2nd series of the 20cm QA Phantom series acquired in section 6.5.1 (on [page 239](#)).

- Note:
- b.) Click on the AUTO 1X rectangular button, select the IMGSER 20QA test from the pull-down menu, and slide cursor to the right to select QA#1/LINES/0.5S.
 - c.) Click on the ACCEPT button. The tool displays a Visible Lines pop-up window.
 - d.) On the Visible Lines pop-up window, click on the VISIBLE LINE button.
 - e.) From the Visible Line pull-down menu, select the letter (A through F) that matches the smallest line pair pattern that you can discern in the Image View Port.
 - f.) On the Visible Lines pop-up window, click on the OK button.
- 2.) Review the Image Analysis Report data and record the Per Image QA#1 Visible Lines data for the 1st image in [Table 6-6](#).
 - a.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS. (Refer to [Figure 6-3, on page 235](#).)
 - b.) Record the Per Image QA#1 Visible Lines data for the 1st image in [Table 6-6](#).
 - 3.) Repeat Steps 1 and 2, for each of the remaining three images in the 2nd 20cm QA Image Series.

Image	Line Patterns Visible	Comments
1		
2		
3		
4		
Specifications	B, C, D, E, F	N.A.

Table 6-6 20cm QA#1 Phantom High Contrast Spatial Resolution Image Performance (Visible Lines)

3rd QA Phantom Image Series (4 Image Visible Holes) Image Performance Verification

- Note:
- 1.) Using the Auto 1x Tool on the Image Analysis screen, generate a Per Image QA#2 Holes report for the 3rd series of the 20cm QA Phantom acquisition.
 - a.) From the Service Browser window, select the 1st image of the 3rd series of the 20cm QA Phantom series acquired in section 6.5.1.
 - b.) Click on the AUTO 1X rectangular button, select IMGSER 20QA from the pull-down menu, and slide cursor to the right to select QA#2 HOLES.
 - c.) Click on the ACCEPT button. The tool displays a Visible Hole pop-up window.
 - d.) On the Visible Hole pop-up window, click on the VISIBLE HOLE button.
 - e.) From the Visible Hole pull-down menu, select the number (1 through 5) that matches the smallest hole that you can discern in the pattern centered in the image displayed in the Image View Port.
 - f.) On the Visible Hole pop-up window, click on the OK button.
 - 2.) Review the Image Analysis Report data and record the Per Image QA#2 Visible Holes data for the 1st image in [Table 6-7](#).
 - a.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS. (Refer to [Figure 6-3, on page 235](#).)
 - b.) Record the Per Image QA#2 Visible Holes data for the 1st image in [Table 6-7](#).

- 3.) Repeat Steps 1 and 2 for the 3rd, 5th, and 7th images in the 3rd 20cm QA Image Series.

Image	Visible Holes Viewable at Window 20	Contrast Factor	Comments
1			
3			
5			
7			
Specifications	See Table 6-9	2.0 to 12.0	N.A.

Table 6-7 20cm QA#2 Phantom Low Contrast Detectability Image Performance (Visible Holes)

Contrast Factor Range (Box 1 Means - Box 2 Means)	Visible Number of Holes		Smallest Visible Hole Size
	Lower Limit *	Upper Limit *	
2.00 to 3.99	2	5	7.5 mm
4.00 to 7.99	3	5	5.0 mm
8.00 to 12.00	4	5	3.0 mm
* Required Number of Visible Holes depends on the Contrast Factor			

Table 6-8 20cm QA#2 Phantom Visible Hole Specifications

Last QA Phantom Image Series (QA#3 Small) Image Performance Verification

- 1.) Using the Auto 1x Tool on the Image Analysis screen, generate a QA#3 Small report for the last series of the 20cm QA Phantom acquisition.
 - a.) From the Service Browser window, select the last series of the 20cm QA Phantom series acquired in section 6.5.1 (on [page 239](#)).
 - b.) Click on the AUTO 1X rectangular button, select IMGSER 20QA from the pull-down menu, slide cursor to the right to select QA#3/SMALL/1.0S from the test pull-down menu, and continue to slide cursor to the right to select AUTO. (Refer to [Figure 6-3, on page 235](#).)
 - c.) Click on the ACCEPT button.
- 2.) Review the Image Analysis Report data and record the QA3 Small data for the last 20cm QA Image Series in [Table 6-9](#).
 - a.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS. (Refer to [Figure 6-3, on page 235](#).)
 - b.) Record the QA3 Small data (Row 2A1A and Row 2B1B Series Means AvXc and AvXo - AvXc), QA3 (AvSDc), and the Center Smudge Row data in [Table 6-9](#).

Box Size = 196 mm² to 256 mm²
 15 mm (+/- 1 mm) x 15 mm (+/- 1 mm)
 31 (+/- 2 pixels) x 31 (+/- 2 pixels)

Box Positions: Box 1 = 0 mm x 0 mm
 Box 2 = 0 mm x -80 mm
 Box 3 = 80 mm x 0 mm
 Box 4 = 0 mm x 80 mm
 Box 5 = -80 mm x 0 mm

Row	Images	AvXc	AvXo	AvXo - AvXc	AvSDo	AvSDc	Center Smudge Row Value	Comments
2A1A	1, 3, 5, 7				N.A.			
1B2B	2, 4, 6, 8				N.A.			
Specifications		+3.0 to -3.0	N.A.	+3.0 to -3.0	N.A.	+ 2.9 to +3.5	+2.2 to -2.2	

Table 6-9 20cm QA#3 Phantom CT#, Brightness Uniformity, Center Noise & Center Smudge Row Performance Data Sheet

- 3.) Review the last 20cm QA Phantom images acquired in Section 6.5.1 for Rings, Bands, and Streaks using the Image Works Desktop.
 - a.) From the Global Control Palette, click on the IMAGE WORKS DESKTOP.
 - b.) From the Image Works Browser window, select the last 20cm QA Phantom series acquired in section 6.5.1.
 - c.) From the Image Works Browser window, click on the VIEWER button.
 - d.) While paging through the eight images in the series, analyze images for any Ring, Band, or Streak artifacts. Adjust Window/Level using the center mouse button, as required.
 - e.) Note any image that appears to have a Ring, Band, or Streak artifact.
- 4.) Verify any images noted above (step 3e) meet 20cm QA Phantom Ring, Band, or Streak Image Performance requirements.
 - a.) From the Global Control Palette, click on the Service Desktop.
 - b.) From the Service Browser window, select an image of the 4th 20cm QA Phantom series acquired in section 6.5.1 that appeared to have a Ring, Band, or Streak while reviewing the series in Image Works.
 - c.) On the Image Analysis Tool window, click on the AUTO 1X rectangular button, select the IMGSER 20QA test from the pull-down menu, slide cursor to the right to select QA#3/SMALL/1.0S from the test pull-down menu, and continue to slide cursor to the right to select RING, BAND, or STREAK.
 - d.) Click on the ACCEPT button.

Note: If required, adjust Window/Level using the center mouse button.

 - e.) Position and size the Ring, Band, or Streak ROI, using the left cursor button.

Note: To ensure specifications are properly applied, do not adjust the Reference ROIs for the Ring, Band, and Streak artifact tests. The Reference ROIs are adjustable after the 1st Accept Modification click.

 - f.) Click on ACCEPT MODIFICATION twice to generate a report.
 - g.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS.
 - h.) Repeat steps b through g for each image noted in Step 3e.

6.5.3 Failure Recommended Actions

6.5.3.1 1st Image Series (4 Image MTF Average) Failure Recovery

Specifications

The 4 Image MTF Average must pass specifications:

4 Image MTF Average: 0.65 to 1.00

Recommended Recovery

- 1.) Repeat Detailed Cal for the 20cm QA Phantom.
- 2.) Repeat Auto CT# Adjust for the 20cm QA Phantom.
- 3.) Repeat Sections 6.5.1 & 6.5.2, to verify Image Performance.

6.5.3.2 2nd Image Series (Visible Lines) Failure Recovery

Specifications

The Largest 5 line pair patterns (coded F, E, D, C, and B) must be visible for each of the four images in this series.

Recommended Recovery

Note: The most common failure for this test is that the phantom has air bubbles that are obscuring the line pair patterns.

- 1.) Carefully inspect the 20cm QA Phantom for air bubbles. If required, refill the phantom with water, to eliminate all air bubbles.
- 2.) Repeat Sections 6.5.1 & 6.5.2, to verify Image Performance.

6.5.3.3 3rd Image Series (Visible Holes) Failure Recovery

Specifications

The number of visible holes in this image series must pass specifications as follows:

- Contrast Factor = 2.0 to 3.99:
Minimum of 2 holes must be visible (10mm & 7.5mm)
 - Contrast Factor = 4.0 to 7.99:
Minimum of 3 holes must be visible (10mm, 7.5mm & 5mm)
 - Contrast Factor = 8.0 to 12.0:
Minimum of 4 holes must be visible (10mm, 7.5mm, 5mm, & 3mm)
-

Recommended Recovery

- 1.) Repeat Detailed Cal for the 20cm QA Phantom.
- 2.) Repeat Auto CT# Adjust for the 20cm QA Phantom.

- 3.) Repeat Sections 6.5.1 & 6.5.2, to verify Image Performance.

6.5.3.4 4th Image Series Failure Recovery

Series Means

SPECIFICATIONS

Both Rows (2A1A, 1B2B) of the series must pass QA#3 Series Means specifications:

- AvXc: < +/- 3.0
- AvXo - AvXc: < +/- 3.0

RECOMMENDED RECOVERY

- 1.) Repeat Detailed Cal for the 20cm QA Phantom.
- 2.) Repeat Auto CT# Adjust for the 20cm QA Phantom.
- 3.) Repeat Sections 6.5.1 & 6.5.2, to verify Image Performance.

QA#3 (AvSDc)

SPECIFICATIONS

Both Rows (2A1A, 1B2B) of the series must pass QA#3 AvSDc specifications:

- AvSDc (< 5K Scans):2.90 to 3.50
- AvSDc (> 5K Scans):2.80 to 3.60

RECOMMENDED RECOVERY

- 1.) Repeat Detailed Cal for the 20cm QA Phantom.
- 2.) Repeat Auto CT# Adjust for the 20cm QA Phantom.
- 3.) Repeat Sections 6.5.1 & 6.5.2, to verify Image Performance.

Center Artifact or Center Smudge

SPECIFICATIONS

All eight images of the series must pass 20cm QA#3 Center Artifact and Center Smudge specifications:

- Center Artifact Factor: +/- 3.5
- Center Smudge Factor: +/- 2.2

RECOMMENDED RECOVERY

- 1.) Repeat Detailed Cal for the 20cm QA Phantom.
- 2.) Repeat Auto CT# Adjust for the 20cm QA Phantom.
- 3.) Repeat Sections 6.5.1 & 6.5.2, to verify Image Performance.

Ring, Band, or Streak Artifact Failure Recovery

SPECIFICATIONS

- Ring Artifact: Ring Factor < +/- 4.8
- Band Artifact: Band Factor +/- 2.8 from 0cm to 8.5cm
< +/- 2.8 > 8.5cm
- Streak Artifact: Streak Factor < +/- 4.0

RECOMMENDED RECOVERY

- 1.) Troubleshoot Ring, Band or Streak artifact failures as described in the Service Manual.
- 2.) Repeat Detailed Cal for the 20cm QA Phantom.
- 3.) Repeat Sections 6.5.1 & 6.5.2, to verify Image Performance.

Section 7.0

System Functional Test

Use the system tests in the following sections to exercise all aspects of the system and to ensure system integrity before releasing to the customer. Although the means, standard deviation, and resolution specifications do not apply during system functional tests, treat any artifact or image anomaly as a failure.

If you encounter a failure during the system tests:

- Record any evidence of artifacts, such as rings, streaks, shading, cupping, noise, or center artifacts.
- Correct artifacts, system test, or image series failures when they occur. Any delay in repairs could increase the number of retests.

- 1.) Place the QA phantom on the cradle.
 - Drive the table to an elevation of 100.
 - Align the line on the phantom with the internal laser lights.

Never scan above 50mA without first placing a phantom in the field of view. Levels in excess of 50mA can cause temporary radiation damage to the detector that lasts several hours. If you acquire image series cals with a radiation damaged detector, the cals may cause artifacts in subsequent image series scans.

- 2.) Select the service protocol, SYSTEM SCAN/CUST QUAL RELI.
- 3.) Run each series.

Note: **Stop the service protocol after the second helical series. It is not necessary to proceed past that point in the protocol.**

- 4.) Using Diagnostic Data Analysis, review the MSD Plots for each AIR scan.
- 5.) Complete the scans.

Section 8.0

Save System State

Use the following commands to create the System State DVD.

- 1.) Load a DVD into the mod drive on the front of the console.

- 2.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon, .

- 3.) Click on the UTILITIES icon. .

- 4.) Select SYSTEM STATE to open the System State Save/Restore menu.

- 5.) Select ALL

- 6.) Select SAVE

- 7.) When the save operation completes, select FILE and QUIT from the pull down menu.

- 8.) Remove the DVD from the drive

Chapter 7

Customer Options Installation & Verification

Note: Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

CT Options

- ☐ Injector functional tests completed.
- ☐ AWW functional tests completed.
- ☐ Filming/Camera/DASM functional tests completed.
- ☐ UPS functional tests completed.
- ☐ Network items installed and functional tests completed.
- ☐ Verify that all customer software options are installed and functional.

Section 2.0

Install Options

Refer to:

- GE Prints and schematics for mechanical (physical) location of option
- FDO shipment for identification of items
- Installation Specialist for installation instructions if they differ from print

2.1 5 inch MOD

Refer to documentation shipped with 5" MOD.

- 5" MOD Drive Option 5162230

2.2 Camera (Filming Device)

Refer to documentation shipped with camera

- Chapter 2 of this manual
- DICOM 2210573 GE Document
- DICOM Print 2152913

2.3 Advantage Windows Workstation (AWW)

Refer to the directions provided:

- Pre-install 2111833
- Service 2111831

Section 3.0

DICOM Network Introduction

BrightSpeed systems support two basic Networking Protocols:

- Advantage NET (IC, Signa 4.X, CT-HLA, CT/I ...)
- DICOM (CT/I, CT Synergy, Advantage Workstations, ...)

DICOM NETWORK INTRODUCTION

DICOM networks basically operate on the **tasks** or services that various devices on the network use or provide. These services are labeled as Application Entity Titles (AE Titles). The CT scanner system is a **user** of six DICOM Network Services and is a **provider** of two DICOM Services:

AS A DICOM SERVICE USER:

- Send or **Push** images to another network device.
- Send or **Push** images to a DICOM Printer.
- Review image database on another device and retrieve or **Pull** selected images from that device (Query/Retrieve User).
- Send or **Push** images to a an image storage device and obtain confirmation that the images have been archived (Storage Commitment).
- Obtain Patient Worklist Information from the Hospital HIS/RIS System.
- Store images on DVD-RAM media.

AS A DICOM SERVICE PROVIDER:

- Receive **Pushed** images from another network device
- Allow another network device to review the image database and to retrieve or **Pull** selected images (Query/Retrieve Provider)

For each DICOM Service that the CT system will be a **User** (except for storing images on DVD-RAM media), you must **declare** this device on the CT system using three menu selections. For some devices, you must declare not only the device, but each service (AE Title) that the device provides. For example, you may be required to declare a PACS System twice on the CT system: once as a destination to **push** images and, second, as destination that provides storage commitment capability after images have been **pushed**.

For each DICOM Service that the CT system will be a **Provider**, you must **declare** the CT system on the network device that will be using these services.

Information required to complete configuring a hospital DICOM network is provided by the hospital network administrator (hostnames, IP Addresses) and the DICOM Conformance Statement document (AE Titles, Port Numbers) provided with each DICOM compatible network device on the network.

Section 4.0

Before You Start

4.1 Network Physical Requirements

Before setting up the CT scanner system on the hospital network, verify the following physical items are complete:

- Scanner console, monitor, keyboard, and mouse are installed and connected.
- CT system power is ON
- Hospital Ethernet network RJ45 Class IV twisted pair cable is connected to the scanner console network receptacle.
- Hospital network connection is operational and is running 10baseT or 100baseT.

4.2 Network Identity Information

To declare the CT system on the network, ensure the following network identity information is available from the Hospital Network Administrator:

- Hostname (No more than 16 Characters)
- Internet Protocol (IP) Address
- Subnet Net Mask IP Address (if applicable)
- Broadcast Address (if applicable)

4.3 Scanner to DICOM Remote Hosts Network Information

To declare DICOM remote hosts (PACS systems, archival devices, review workstations) on the CT system, ensure the following information is available for each remote host:

From the Hospital Network Administrator:

- Hostname
- Internet Protocol (IP) Address
- Network Protocol (DICOM for CT Systems)

From the Remote Host Device DICOM Conformance Statement Document:

- DICOM Application Entity Title or AE Title (DICOM service that remote host provides or uses)
- DICOM Listening Port Number

4.4 Scanner to DICOM HIS/RIS Interface Network Information

To declare DICOM Hospital HIS/RIS Interface devices (Mitra and others) on the CT System, ensure the following information is available:

From the Hospital Network Administrator: *Internet Protocol (IP) Address*

From the HIS/RIS Interface Device DICOM Conformance Statement Document:

- DICOM Application Entity Title or AE Title (DICOM Service that the HIS/RIS interface provides)
- DICOM Listening Port Number

4.5 Scanner to DICOM Printers Network Information

To declare DICOM on the CT System, ensure the following information is available for each printer:

From the Hospital Network Administrator:

- Hostname
- Internet Protocol (IP) Address

From the Printer DICOM Conformance Statement Document:

- DICOM Application Entity Title or AE Title (DICOM service that remote host provides or uses)
- DICOM Listening Port Number

Section 5.0

Declaring the System on the Hospital Network

5.1 Enter Configuration Routine

- 1.) On the OC, open a Unix shell window.
- 2.) Enter root as a superuser:
Type: **su -** **ENTER** at the prompt.
Type: **#bigguy** **ENTER** at the password prompt.
- 3.) Change directory to scripts:
Type: **cd /user/g/scripts** **ENTER** at the root prompt.
- 4.) Launch the Install Utility:
Type: **reconfig** **ENTER** at the prompt.
The OC displays the Install Utility Window as shown in [Figure 7-1](#).

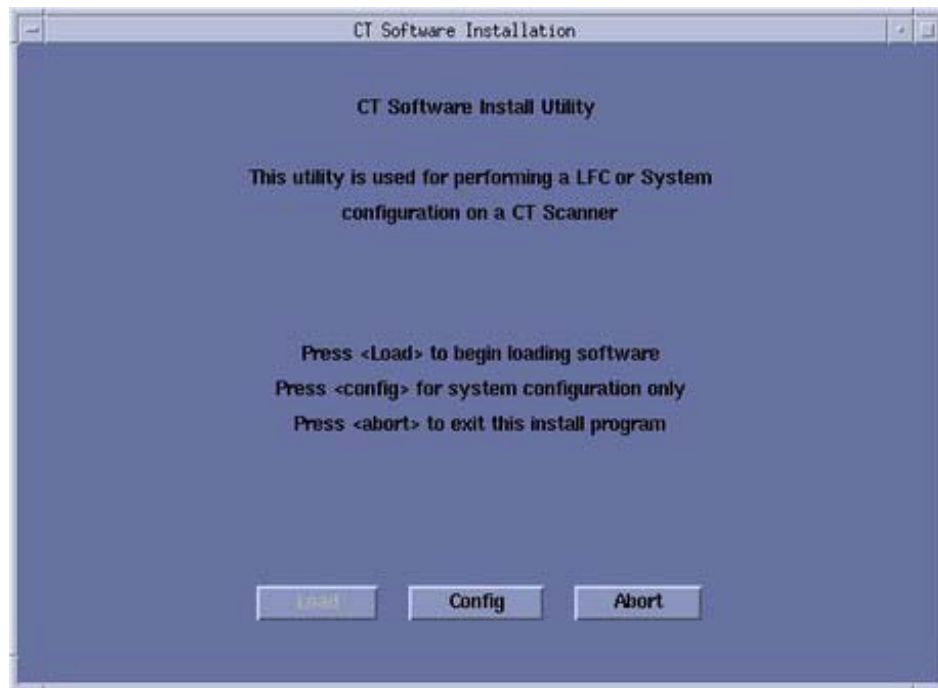


Figure 7-1 Install Utility Window

5.) Enter the Configuration Routine:

Using the mouse, click on the CONFIG button.

The OC displays the System Configuration - System Settings screen, as shown in [Figure 7-2](#).

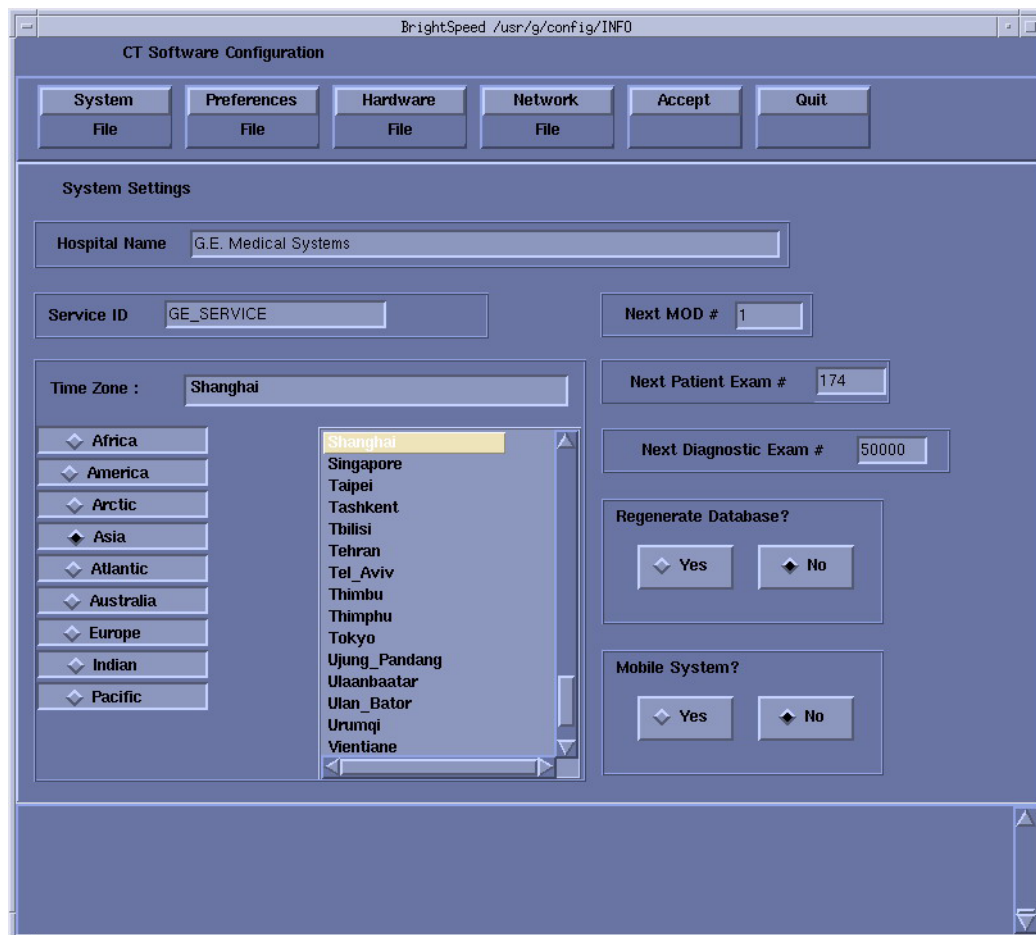


Figure 7-2 System Settings Screen

5.2 Configure Network Settings

This screen provides the ability to declare the CT system on a hospital network. Key information such as Host Name, IP Address, Net Mask (for CT systems on a subnet) must be obtained from the hospital network administrator.

- 1.) Select the NETWORK button to display the Network Settings screen as shown in [Figure 7-3](#).
- 2.) Enter the Suite Name.

The Suite Name is a means of identifying this particular CT system as a part of a group of CT Systems in a suite configuration. This Suite Name will appear on all image headers.

The Suite Name must start with a letter, followed by three alphanumeric characters (total MUST be four characters long). The name of the OC interface will be <Suite Name>_oc and the SBC interface will be <Suite Name>_sbc.

Figure 7-3 Networks Settings Screen

- 3.) Enter the hospital provided Host Name.
The Host Name identifies the network hostname and AE Title of the CT system.
The Host Name:
 - **MUST NOT** be <Suite Name>_oc or <SUITE NAME>_OC.
 - **MUST NOT** exceed 16 Characters.
 - **MUST** only contain the following characters: **A** through **Z**, **a** through **z**, **0** through **9**, **-** and **_**
- 4.) Enter the hospital provided IP Address.
- 5.) Enter the hospital provided Net Mask (if the CT system is on a subnet).
- 6.) Enter the Broadcast Address:
The Broadcast Address should be the same as the IP Address except for the bits of the host id portion (last digit group) set to 1s or 0s depending on the configuration of the network. The standard default is 1s but older SunOS machines used 0s.

Example: If the IP Address is 192.100.9.17, the Broadcast Address should be 192.100.9.255 if the network is configured to use 1's to specify the broadcast address.
If the network contains genesis based scanners or other SunOS 3.5 or 4.1 computers, the Broadcast Address should be 192.100.9.0.
- 7.) Enter the hospital provided Default Gateway IP Address in the Default Gateway field (if applicable). If the site network does not use a default gateway, leave the field blank.

- 8.) Select NIS (Yellow Pages database) Advanced Option only if requested by the hospital network administrator as follows:
 - a.) Select ADVANCED OPTIONS button on the Network Settings screen.
 - b.) Select Use NIS? button.
 - c.) Enter the hospital provided Domain Name.
- 9.) Record all the Network parameters in the *Software Installation Procedures* Document, or on the worksheet in 2.2 on page 199.

5.3 Initiate System Reconfiguration

- 1.) Select ACCEPT on the System Configuration Screen.

The system loads the application software, OS patches, and kernel changes, and configures the system on both the OC and the SBC.

This loading process takes approximately 15 minutes. While the load is going on, the results are displayed in a shell window, which closes when the loading process is complete. All the window output is logged to a file named:

`/var/adm/install.log.YYYYMMDDWWHHMMSS.`

(Where `YYYYMMDDWWHHMMSS` is the Date/Time that the loading process was started.)
- 2.) When the loading process and configuration changes are complete, the system displays a prompt to reboot. Click on YES.
- 3.) The system will automatically login as ctuser after the reboot. Select OK on the Autostart Disabled popup message.
- 4.) To startup Applications, in the console shell window, type **startup** ENTER.

Section 6.0

Declaring Remote Hosts on the CT System

6.1 Enter Remote Host Configuration Screen

- 1.) On the OC, select the IMAGE WORKS icon.
- 2.) Select NETWORK.



6.2 Declaring Advantage NET Remote Hosts on the Scanner

Use Advantage NET Protocol networks to communicate with older CT or MR Systems (MR Signa 4.x, CT-HLA, CT/I Systems, and Workstations that support the Advantage NET protocol). Advantage NET Protocol does not offer full compatibility with BrightSpeed DICOM formats.

Repeat the following procedure for each Advantage NET Remote Host device that the customer expects to have this CT system communicating with.

- 1.) Select REMOTE HOSTS from the pull down menu. The system displays the Remote Host Parameter Screen as shown in [Figure 7-4](#).

Figure 7-4 Advantage Net Network Protocol Parameter Settings

- 2.) Enter the hospital provided Host name.
- 3.) Enter the hospital provided Network Address (IP Address).

- 4.) Select ADVANTAGE NET as the Network Protocol.

The systems automatically de-highlights the remaining parameter fields on the Remote Host parameter selection screen. These are dedicated DICOM protocol parameters and do not apply to Advantage NET type devices.

- 5.) Select SAVE to store the parameter settings of the remote host.

6.3 Declaring DICOM Remote Hosts on the CT Scanner

Use DICOM protocol networks to communicate to DICOM devices such as CT/i, CT Synergy, DLX, MR Lx, and third party hosts.

Repeat the following procedure for each DICOM remote host device that the customer expects to have this CT system communicating with.

- 1.) Select REMOTE HOSTS from the pull down menu. The system displays the Remote Host Parameter screen as shown in [Figure 7-5](#).

Figure 7-5 DICOM Network Setting Protocol Parameter Settings

- 2.) Enter the hospital provided Host name.
- 3.) Enter the hospital provided Network Address (IP Address).
- 4.) Select DICOM as the Network Protocol.
The system automatically highlights the remaining parameter fields on the Remote Host parameter selection screen. These are dedicated DICOM Protocol parameters.
- 5.) Enter the TCP/IP Listening Port Number from the DICOM Conformance Statement provided with the device.

- 6.) Enter the AE Title from the DICOM Conformance Statement provided with the device.
Application Entity Titles (also known as ACR-Nema or Dicom Name) refer to the DICOM Network Services that a device provides to the CT System. For most devices, the AE Title is the same as the hostname (CT systems are equipped with this feature).
However, some devices such as PACS systems may have separate AE Titles and port numbers for each of the services that the PACS system provides. In these cases, you must enter a separate remote host (same hostname and IP Address) for each of the independent AE Title Services that the host provides (one host as an image **push-to** destination, another host as a **query/retrieve** provider, and another host as a **storage/commitment** provider).
Be sure to review the DICOM Conformance Statement for each device that will provide a remote host network service for the CT system (image **push-to** or store destination, Query/Retrieve, and Storage Commitment) to ensure that each service is correctly configured.
- 7.) Select the correct Archive Node choice for the device. The Archive Node selection field defines the ability of the remote host to act as a DICOM Storage/Commitment provider and indicate to the operator that a study/series/image was archived. Select:
 - AUTO to have the CT system automatically check to see if the designated remote host is a DICOM Storage/Commitment Provider.
 - YES if the device is the hospital designated DICOM Storage/Commitment Provider. During an Application Study Archive process, the local browser screen will indicate Archive Status = Y to the operator.
 - NO if the device is not a DICOM Storage/Commitment Provider.
- 8.) Select the correct Access to local host: settings. These two selections allow you to selectively block the remote host from using the BrightSpeed DICOM services as a provider (image **push-to** destination, and a **Query/Retrieve** provider).
 - Send Images? Set to YES if the customer wants the CT system to be able to have images **pushed** to the system from the applicable remote host. Set to NO if the customer wants to block an image **push** from the applicable remote host.
 - Query/retrieve images? Set to YES if the customer wants the remote host to be able to review the image database (query) and **pull** selected images from the database. Set to NO if the customer does not want the remote host to have this ability.
- 9.) Select the correct Custom search? setting. This selection allows the CT scanner to selectively search through the remote host's image database when the operator is using remote browser screen to **query** the remote host. The search parameters that the CT system allows the customer to use are: last name contains, patient ID, exam number, accession number, and exam date.
 - Select ON if the device supports custom searches as part of the devices **Query/Retrieve** DICOM Provider service.
 - Select OFF if the device does not support custom searches.
- 10.) Record all the remote host network parameters for each remote host in the *Software Installation Procedures* Document.
- 11.) Select SAVE to store the parameter settings of the remote host.

Section 7.0

Declaring the CT System on Remote Hosts

7.1 Declaring the Scanner on Advantage NET Protocol Devices/Systems

Refer to the appropriate service manual provided with the Advantage NET Protocol device or system to find instructions how to declare the CT System as an Advantage NET remote host.

7.2 Declaring the Scanner on DICOM Protocol Devices/Systems

Refer to the appropriate Service Manual provided with the DICOM protocol device or system to find instructions how to declare the CT System as a DICOM remote host.

The CT System provides two DICOM Services as a provider to remote hosts:

- A remote host can **push** images to the CT image database.
- A remote host can review the CT image database (query) and **pull** selected images (retrieve).

Use the following parameter information to configure the DICOM device/system to either **push** images to the CT scanner and/or perform a **Query/Retrieve** operation:

- **Hostname**: Provided by the Hospital Network Administrator. Exactly the same scanner assigned hostname entered in Network Configuration Screen.
- **Application Entity Title**: Exactly the same entry as the Hostname.
- **Network Address**: Provided by the Hospital Network Administrator. Exactly the same scanner assigned IP Address entered in Network Configuration Screen.
- **Network Protocol**: DICOM 3.0.
- **Port Number**: For all DICOM service that the CT System provides, use 4006.
- **Provider Type**: This field concerns the BrightSpeed DICOM **Query/Retrieve** provider capability. All CT systems are wstudy root systems, which allow queries at the exam, series, and image level.
- **Support Worklist**: This field concerns whether a DICOM **Query/Retrieve** provider capable device or system supports a filter search of the image database. All CT systems support a filtered search of the image database as part of the BrightSpeed DICOM **Query/Retrieve** provider capability.

Section 8.0 DICOM HIS/RIS Setup

8.1 Prerequisites

Most hospital HIS/RIS systems are not DICOM compatible and require a DICOM HIS/RIS Worklist Interface to provide patient scheduling information to the CT system. Contact your local HNS support engineer to determine exactly what DICOM HIS/RIS Interface is appropriate for the customer.

In addition, the CT system must have the ConnectPRO software option installed to utilize the DICOM Protocol Worklist capability.

8.2 Loading ConnectPRO Software Option on the CT System

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the CONFIGURATION icon.



- 3.) Click on INSTALL OPTIONS.

- 4.) Select INSTALL and click START. The console displays the Software Options window as shown in [Figure 7-6](#).

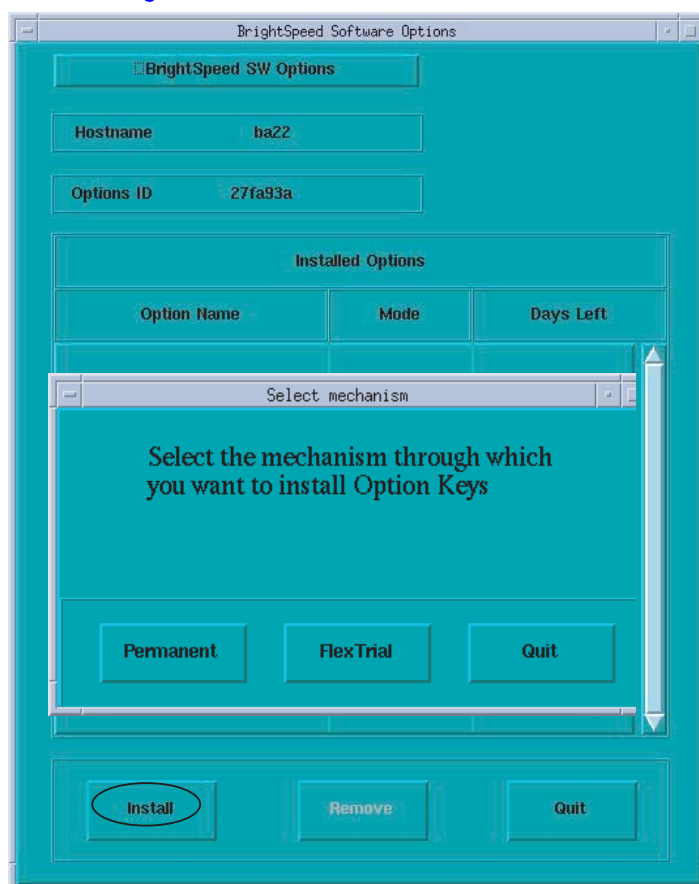


Figure 7-6 Options Window when First Selected

- 5.) Insert the ConnectPRO Options DVD into the DVD drive.
- 6.) Click on OK to continue.
- 7.) On the Software Option window, select the CONNECTPRO option in the Available Options window.
- 8.) Click on the INSTALL button.

The console may display a message box during the software loading operation. When the system has completed loading the software:

- the Installed Options window displays the ConnectPRO option, and
- the console displays the ConnectPro Setup window as shown in [Figure 7-7](#).

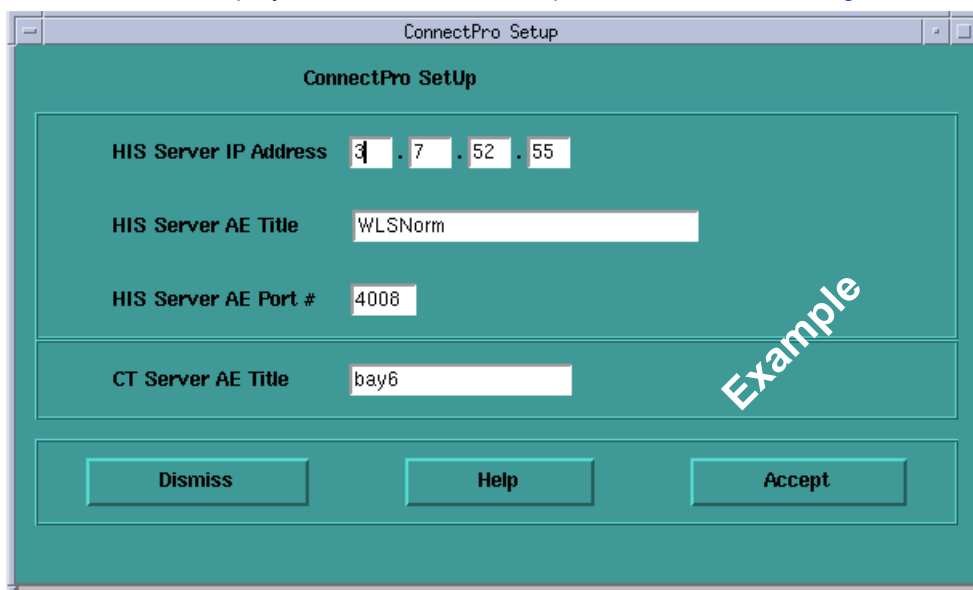


Figure 7-7 ConnectPro Setup Screen

- 9.) Enter the hospital provided HIS Server IP Address.
- 10.) Enter the HIS Server AE Title from the DICOM HIS/RIS Interface device DICOM Conformance Statement document.
- 11.) Enter the HIS Server AE Port # from the DICOM HIS/RIS Interface device DICOM Conformance Statement document.
- 12.) Record all the ConnectPRO HIS/RIS Network parameters.
The system will automatically load the BrightSpeed CT Server AE Title (Hostname of the system). CT Server AE Title will be used for PPS setup later.
- 13.) Click on the ACCEPT button.
- 14.) "Should PPS be enabled?" is popped up. Select Yes.
- 15.) The console displays the ConnectPro Setup window as shown in

PPS Server Setup

PPS Server (SCP) Setup

PPS Server IP Address 3 . 70 . 204 . 86

PPS Server AE Title DICOM_TEST_RMT

PPS Server AE Port # 4500

CT Server AE Title bay6

Dismiss Help Accept

Example

Figure 7-8 PPS Setup Screen

- 16.) Enter the hospital provided PPS Server IP Address.
- 17.) Enter the PPS Server AE Title from the DICOM HIS/RIS Interface device DICOM Conformance Statement document.
- 18.) Enter the PPS Server AE Port # from the DICOM HIS/RIS Interface device DICOM Conformance Statement document.
- 19.) Enter CT Server AE Title got from [Step 12](#)). Click [Accept].
- 20.) The system will check valid setup entries, execute a perl script that makes changes to the configuration files for HIS/RIS, and return to the Software Option window shown in [Figure 7-6](#).
- 21.) Click on the QUIT button and the subsequent OK button in the message window.
- 22.) Remove the DVD from the DVD drive, and write protect the side containing the ConnectPRO option.

8.3 PPS Setup

If PPS is not enabled during ConnectPro Option installation, PPS can also be setup when needed later.

From the tool chest located in upper right corner of either monitor, select Unix Shell.

Type '**hostname**'. The output is the CT scanner AE title that will be sent to the worklist server.

Type '**installhisris**'.

Then the ConnectPro Setup window as shown in [Figure 7-7](#). Follow the [Step 9\)](#) to [Step 19\)](#) in [Section 8.2 on page 262](#).

Section 9.0

DICOM Filming Devices Setup

9.1 Prerequisites

Before configuring DICOM filming devices (cameras, printers) on the CT System, ensure the following are complete:

- Filming Device Service Representative to assist in camera/printer setup for best image quality film presentation.
- Hospital DICOM network is operational.
- Filming device is connected to the DICOM network with the correct filmer DICOM interface.
- Filming device is DICOM protocol compatible.
- Filming device DICOM Conformance Statement document is available.

Note: **Filmer DICOM Application Entity Titles may be site specific. Make sure that you check with the Filming Device Service Representative and the hospital network administrator to ensure you are using the correct AE Title for the destination filming device.**

9.2 Declaring DICOM Filming Devices on the CT System



NOTICE
Potential For
Data Loss

Empty all filming queues before modifying camera parameters.

This section contains procedures for recording important Camera setup information. Use the table(s) at the end of this section to record information from the setup screens.



- 1.) Click on the SERVICE DESKTOP icon.
- 2.) On the Desktop Toolbar select CONFIGURATION -> INSTALL CAMERA. The Install Camera window appears, along with a warning message pop-up box, to remind you that all filming queues must be empty before you begin to update or delete a camera.
- 3.) The Graphical User Interface displayed shows a list of cameras installed (See Figure 7-10). First, you must click OK in the warning message box. See Figure 7-9.

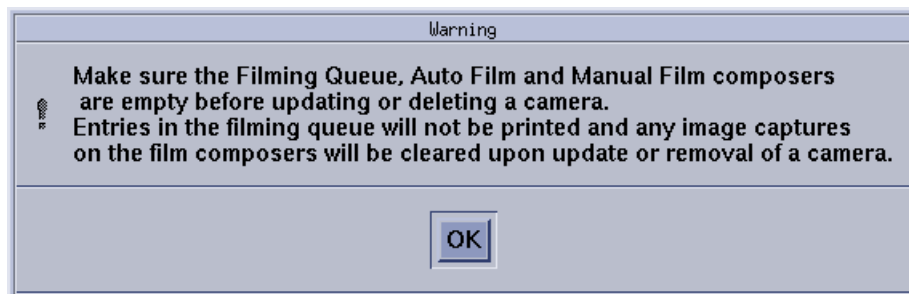


Figure 7-9 Warning Screen

- 4.) Now you are asked a series of questions.
- a.) To add a new camera, click the ADD button (See Figure 7-10).

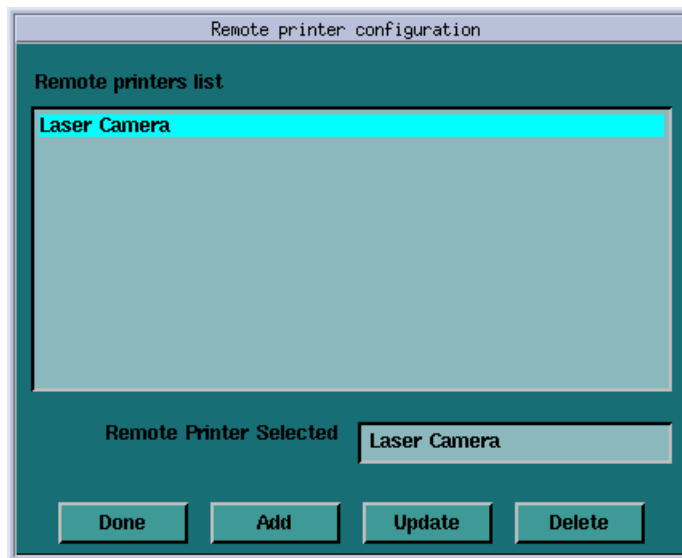


Figure 7-10 Printer Configuration GUI

- b.) A dialog window for the camera type (DASM/DICOM) appears. If no DASM is detected during the OC boot, the DASM button will be disabled (Figure 7-11). If a DASM is present and has not been detected, reboot the OC and run the camera configuration tool again.

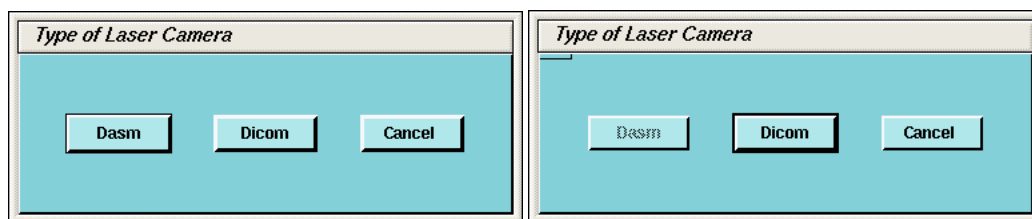


Figure 7-11 Dialog Box for camera Type

- 5.) To add a new laser camera, click DASM in the camera type dialog box. This brings up a list of available camera models. Select the appropriate model from the list and click SELECT (See Figure 7-12). Now you must configure it.

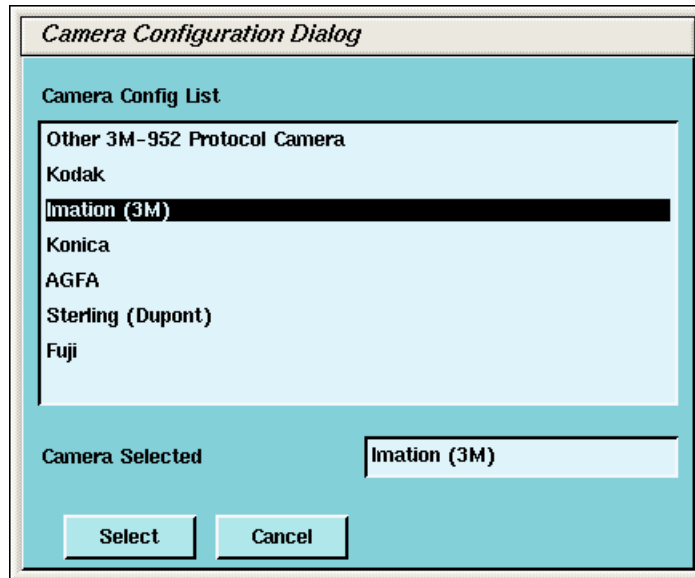


Figure 7-12 Camera Model Dialog (with DASM)

- DASM Interface is automatically detected as either Analogue or Digital
- Two Laser Options are available for laser cameras: SLIDES and ZOOM. Set this option only if the camera being installed supports slides and zoom. Setting the option allows it to be enabled or disabled at the application level.
- Camera manufactures provide two (2) Magnification Type options for cameras. The SMOOTH resolution blurs the image, while the SHARP resolution makes the image pixels more pronounced. The default is smooth.

Comment:

To film good images, and have them look like images filmed by other GE CT products, the following camera settings are suggested:

Kodak: SMOOTH

Dupont/Sterling: SMOOTH

3M/Imation (Laser Camera): SHARP

3M/Imation (Dry View): SMOOTH

Agfa: SMOOTH

- Select the appropriate File Format. Select ON from the drop down list boxes on the menu. Valid film formats are determined by the camera manufacture. IMATION for example, doesn't support 4x4, 2x4 or 1x2 and AGFA does not support 2x4) The DICOM print convention designates film formats by column and row (e.g. 12 on 1 film is 3x4).

When finished setting parameters, click on DONE and proceed to step 8.

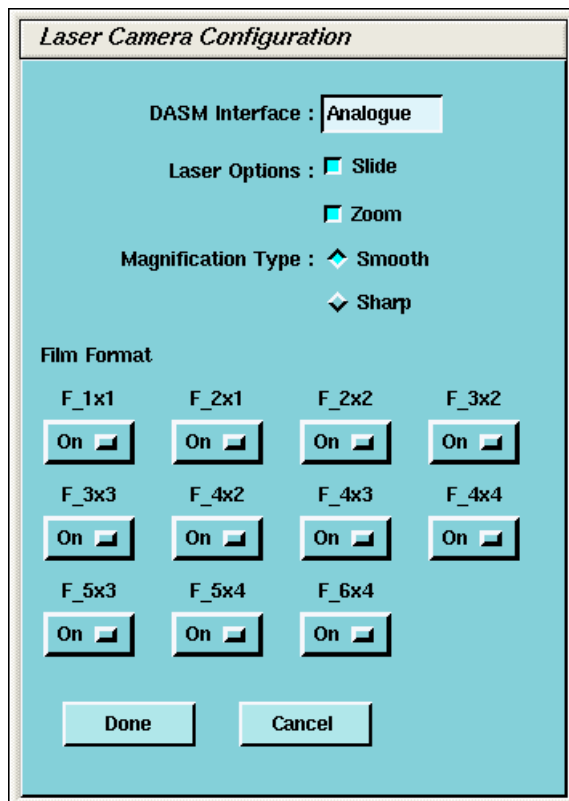


Figure 7-13 Laser Camera Configuration

- 6.) To add a new DICOM camera, click on ADD and then DICOM in the dialog box that appears.
 - a.) A list of camera models appears (See Figure 7-14). Select the appropriate model from the list and click SELECT. Clicking SELECT presets all the parameters to that models except the Network parameters.

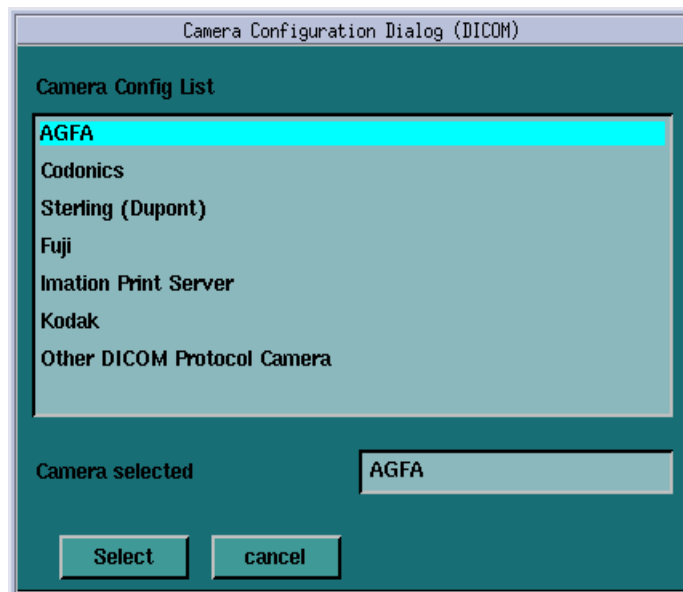


Figure 7-14 Camera Model Dialog (DICOM)

*Comment:
It's advised to
recheck the
preset
information with
the camera
manufacturer's
representative.*

- Selection of a different camera model clears the Image Quality parameters, because these are camera manufacture dependent.
- b.) Enter the Network Parameters (See Figure 7-15)
- >Device NameA unique name used to identify the camera.
 - >Host NameDICOM print server host name, as defined by the hospital.
 - >IP AddressDICOM print server IP address, as defined by the hospital.
 - >AE TitleDICOM print server application entity title, as defined by the print server. *You should consult the manufacturers DICOM Conformance Statement.*

Note: The Application Entity Title for the Camera may be site specific. Make sure that you check with the Camera Manufacturer's Representative and the hospital network administrator to ensure you are using the correct AE Title for the destination DICOM Print Camera.

- >TCP/IP Listen PortDICOM print server TCP/IP listen port, as defined by the server. *You should consult the manufacturers DICOM Conformance Statement.*
- >CommentsOptional comments used by the DICOM print server.

DICOM printer configuration parameters

Device Name:

Host Name:

IP address:

Application title:

TCP listen port:

Comments:

Medium Type:

Destination:

Film orientation:

Magnification type:

Standard film formats:

F_1x1 On	F_2x1 On	F_2x2 On	F_3x2 On
F_3x3 On	F_4x2 Off	F_4x3 On	F_4x4 On
F_5x3 On	F_5x4 On	F_6x4 On	

Figure 7-15 DICOM Camera Configuration

- c.) Medium Type specifies the type of film being used. Currently, only BLUE FILM and

CLEAR FILM are supported.

- d.) Set Destination to the final location for film output: either MAGAZINE or PROCESSOR.
- e.) Orientation selects film orientation. Only PORTRAIT is currently supported.
- f.) Set the Magnification Type. This parameter selects the algorithm used to interpolate pixels for proper film resolution. Set this parameter after consulting the camera manufacture to ensure optimal image quality. Choices are describe below:
 - >NoneNo interpolation. This option is not suported by all cameras.
 - >ReplicateAdjacent pixels are interpolated. This can result in "pixelized" images. *This algorithm is not normal preferred.*
 - >BilinearA 1st order interpolation of pixels. Results in images usuaully described as blurred. *This algorithm is not normal preferred.*
 - >CubicA 3rd order interpolation. Used with a large number of possible formulations. Camera manufactures define parameters called "smoothing type" to set coefficients used in this algorithm. The implementation of these "smoothing soefficients" is camera dependent.
- g.) Select the Standard Film Formats. Select the film format by choosing ON in the pull-down menu box located below each selection. See Figure 7-15. Valid film formats are set by the camera manufacture. IMATION for example, doesn't support 4x4, 2x4 or 1x2 and AGFA does not support 2x4) The DICOM print convention designates film formats by column and row (e.g. 12 on 1 film is 3x4).
- h.) After the camera has been configured, click the ADVANCED button. This creates the camera device file for you automatically and pops up the Advanced Parameters screen. See Figure 7-16.

Comment: For most Camera Manufacturers, the preferred selection is CUBIC.

The screenshot shows a dialog box titled "Camera Attribute Dialog". It contains several sections of controls:

- Image Quality Parameters** (Section Header)
- Configuration**: A text box containing "PERCEPTION_LUT=KANAMC".
- Maximum Density**: A text box containing "320".
- Minimum Density**: A text box containing "32".
- Smoothing Type**: A dropdown menu with "On" selected.
- Empty Density**: A dropdown menu with "Black" selected.
- Border Density**: A dropdown menu with "Black" selected.
- Film Size**: A dropdown menu with "<empty>" selected.
- Trim**: A dropdown menu with "Yes" selected.
- Priority**: A dropdown menu with "High" selected.
- Buttons**: "Done" and "cancel" buttons at the bottom.

Figure 7-16 Advanced Parameters (Camera Attribute Dialog)

- 7.) Advanced camera parameters control the image quality of films.

Note: For more information on the proper settings for these parameters, please see the Camera's DICOM Print Device Conformance Statement or the Camera Manufacturer Representative.

You may need to refer to a copy of the Conformance Statement as you are working with the Camera Manufacturer's Representative, to correctly set up the DICOM Print Camera I/Q and Time-out Settings.

- a.) **Configuration** - This parameter is camera manufacturer dependent as is typically used to specify the image contrast. The Configuration field may be up to 1024 characters long. The field will scroll automatically as text is entered. To review your entry, simply click and hold the middle mouse button, while the cursor is in the field, and drag the mouse towards the right (or left) as needed.

Note: Recommended Configuration Setting Values:

Agfa Drystar (MG3000) **PERCEPTION_LUT=KANAMORI (100)**
Imation Dryview (8700) **LUT=0,7**
Kodak Laser Printer 190 **CS434\CN0\PD1.20**

- b.) **Smoothing Type** - Set Smoothing Type to ON, and input the selected value. This parameter is used when the magnification type is CUBIC. It represents the coefficient for the image resolution algorithm. This parameter is camera manufacturer dependent, and should be re-verified with your radiology department.

Note: **Recommended Smoothing Type Starting Values and Ranges:**

Agfa DryStar (MG3000) Start Value: **140** Range: **137 - 150**
Imation Dryview (8700) Start Value: **3** Range: **3 - 13**
Kodak Laser Printer 190 Start Value: **Enhanced** Range: **Normal**

- c.) **Minimum and Maximum Density** - Used to set brightness of the images on film. The range of values is 0-4095, although the valid range for a specific camera is manufacture dependent. For **Maximum Density**, input the correct value into the text box. For **Minimum Density**, set it to ON and input the correct value in the text box.

Note: **Recommended Minimum and Maximum Density Starting Values:**

Agfa Drystar (MG3000) Min.: **20** or **23** Max: **300**
Imation Dryview (8700) Min.: **(Blank)** Max: **300**
Kodak Laser Printer 190 Min.: **20** Max: **300**

- d.) **Empty Image Density** - This parameter sets the density for empty film viewports. Typically, BLACK is used but WHITE is an available option. The minimum and maximum density values are used as the representation.
- e.) **Border Density** - This sets the density for the border used around the film viewports. Typically, BLACK is used but WHITE is an available option. The minimum and maximum density values are used as the representation.
- f.) **Film Size** - Allows the system to specify a particular film size, if selected.
- g.) **Trim** - YES produces a white (clear) box surrounding each image.
- h.) **Priority** - This sets the print priority.
- i.) If you have completed entry of advanced parameters, click DONE.

CAMERA DATA TABLES: TO LOCATE INSTALL CAMERA INFORMATION: CLICK ON THE SERVICE DESKTOP BUTTON. ON THE DESKTOP TOOLBAR SELECT UTILITIES -> INSTALL -> INSTALL CAMERA. THE INSTALL CAMERA WINDOW APPEARS. SELECT EACH OF THE CAMERAS THAT ARE INSTALLED FROM THE LIST OF INSTALLED CAMERAS, AND CLICK ON UPDATE TO VIEW THE CAMERA'S SETTINGS. RECORD THE VALUES USED TO SET UP EACH CAMERA IN THE TABLES THAT FOLLOW. EXTRA TABLES

ARE PROVIDED FOR MULTIPLE CAMERAS

- Note: You can determine this information by looking at the contents of the following files:
- For a DASM Camera: `/usr/g/ctuser/app-defaults/devices/camera.dev`
 - For a DICOM Print Camera: `/usr/g/ctuser/app-defaults/devices/name.cfg` where, **name.cfg** is the camera device name from the printer configuration GUI.
- Example: `more <filename from above> ENTER`

DASM CAMERA #1 SETUP

GUI SETTING	SELECTIONS	VALUE
Camera Type	Model Type of Camera	
DASM Type	Digital or Analog	☐ Digital ☐ Analog
Options	Slides or Zoom	☐ Slides ☐ Analog
Film	Smooth or Sharp	☐ Smooth ☐ Sharp
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Film Format Default	1x1, 2x1, 2x2, 3x2, etc.	

Table 7-1

DICOM CAMERA #1

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 7-2 DICOM Camera #1

GUI SETTING	SELECTIONS	VALUES
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 7-2 DICOM Camera #1

DICOM CAMERA #2

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 7-3 DICOM Camera #2

GUI SETTING	SELECTIONS	VALUES
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 7-3 DICOM Camera #2

DICOM CAMERA #3

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 7-4 DICOM Camera #3

9.3 Troubleshooting Tips

The following is a summary of troubleshooting information for DICOM print that was gathered during software testing and validation of the DICOM print feature.

There is also a significant amount of additional troubleshooting procedures, and the theory of the DICOM print feature in the System Service Manual. Should you have problems installing a DICOM print camera, first read the information in System Service Manual.

ERROR BRINGING UP THE CAMERA INSTALLATION/FILMING APPLICATION

- **Symptom:** After creating/modifying the DICOM print camera the install camera interface does not come up and the filming application indicates it cannot bring up the film composer.
Cause: The configuration field contains invalid information such as a backslash (\) as the final character in the field or brackets ({}).
- **Solution:** The camera.dev file in ~ctuser/app-defaults/devices must be manually edited to remove the offending characters in the set configuration line. Invalid characters include \{ }

CONFIGURATION INFORMATION FIELD

- **Symptom:** Cannot view the entire configuration field (> 25 characters)
- **Solution:** Hold down the middle mouse button and move the field contents

NEED TO SET DICOM PRINT ATTRIBUTES NOT SUPPORTED BY SOFTWARE

- **Symptom:** User wants the white border around each image box ON/OFF permanently for this system and it cannot be set as the default for the camera.
- **Solution:** Using your favorite editor, add the following line to the camera.dev file located in ~ctuser/app-defaults/devices after the DICOM print device has been otherwise configured.

For Trim Off - set	TRIM NO
For Trim On - set	TRIM YES
- **Symptom:** DICOM print camera supports multiple film sizes and the user only wants to print if the film size is correct for BrightSpeed (14x17). [Otherwise the camera will queue the films or return an error causing the queue to pause (based upon the DICOM print camera specifications).]
- **Solution:** Using your favorite editor, add the following line to the camera.dev file located in ~ctuser/app-defaults/devices after the DICOM print device has been otherwise configured.

To force a 14x17 film size - set	filmSize 14INX17IN
----------------------------------	--------------------

NEED TO PREVENT DICOM PRINT ATTRIBUTES FROM BEING SENT TO DICOM PRINT CAMERA

- **Symptom:** Some DICOM print attributes are optional, and may result in fatal errors. For example, the Fuji camera does not support the Empty Image Density parameter for the film box.
- **Solution:** Using your favorite editor, add the following line(s) to the camera.dev file located in ~ctuser/app-defaults/devices after the DICOM print device has been otherwise configured.

- To prevent sending the Smoothing Parameter set	FB_Smooth FALSE
--	-----------------

-
- | | |
|--|-----------------|
| - To prevent sending the Border Density set | FB_Border FALSE |
| - To prevent sending the Empty Image Density set | FB_EID FALSE |
| - To prevent sending the Minimum Density set | FB_MinD FALSE |
| - To prevent sending the Trim Parameter set | FB_Trim FALSE |

ERROR TRYING TO CONNECT TO THE DICOM PRINT CAMERA

- **Symptom:** DICOM print server can be reached (ping), but Application error indicates "Unable to start filming interface" and the help message talks about running the install.dasm (Association Error)
- **Solution:** The system is unable to complete the association. Check the AE Title and the Port number of the DICOM print server and correct them through the Install Camera procedure.

FILM COMPOSER ERROR NOT USABLE

- **Symptom:** Film composer error says "unrecognized status - code 0"
- **Solution:** Review the log file, the attention and status windows. These areas have the correct filming status (for example, **film jam** and **supply empty**).

DEBUGGING CONNECTION ISSUES DIFFICULT

- **Symptom:** The timeouts for the DICOM print are very long, which means one needs to wait a long time before you know the application is not working.
- **Solution:** The timeouts for the DICOM print were setup to ensure that the system would work regardless of whether the DICOM print camera was on a LAN or a WAN halfway around the world. The DICOM print timeouts for the association and DIMSE classes (for example, **N-GET**, **N-DELETE**) can be modified within the DICOM print camera installation. They can be reduced down to 90 seconds.

DICOM PRINT ERROR ON N-GET TIMEOUT CONFUSING

- **Symptom:** When the N-GET timeout goes off, the error message in the prslog file will be "Could not get printer status, invalid command sequence for N-GET".
- **Solution:** When the user sees the above error they may want to consider that the issue may be an inactivity timer on the N-GET DIMSE service.

DICOM PRINT CAMERA SLIDE SUPPORT

- **Symptom:** Current implementation of DICOM print does not allow selection of slide format.
- **Solution:** Feature not currently supported. Possibly in future releases.

CONFUSION ON FILM FORMAT NOTATION

- **Symptom:** GEMS Laser Camera and DICOM Print film format notations are opposite.
- **Solution:**
 - GEMS Laser Camera film format notation has always been row x col (for example, 12 on 1 = 4x3)
 - DICOM Print Standard film format notation is col x row (for example, 12 on 1 = 3x4)

Section 10.0 Network Connections

BROAD-BAND

Broad-band is considered the standard network connection for BrightSpeed. (A dial-up modem is optional.) Broad-band connections should use one of the following Category 5 patch cables:

CAT Num	GE Part Num	Length
K9000WB	2215028-10	20 m
K9000KP	2215028-5	10 m
K9000JR	2215028-4	5 m
K9000WA	2215028-9	3 m

The CT system is connected to the network through the Console.

- A patch cable (not to exceed 10 feet) should be provided by the customer, and it is used to connect the console to a wall box. (See Notes on [Figure 7-17](#))
- Some customer-site units may require cable duct-work or conduit to route connecting network cables to the workstation, camera and console.
- The run from the hospital switch to the CT wall outlet must not exceed 290 ft. (88m). Bandwidth performance is degraded when the length reaches 300 ft. (91m) or greater.
- For the optional modem: **Two phone lines should be provided by the facility.** One line is for use with a modem and must be an analog line. The second line is a voice only line.

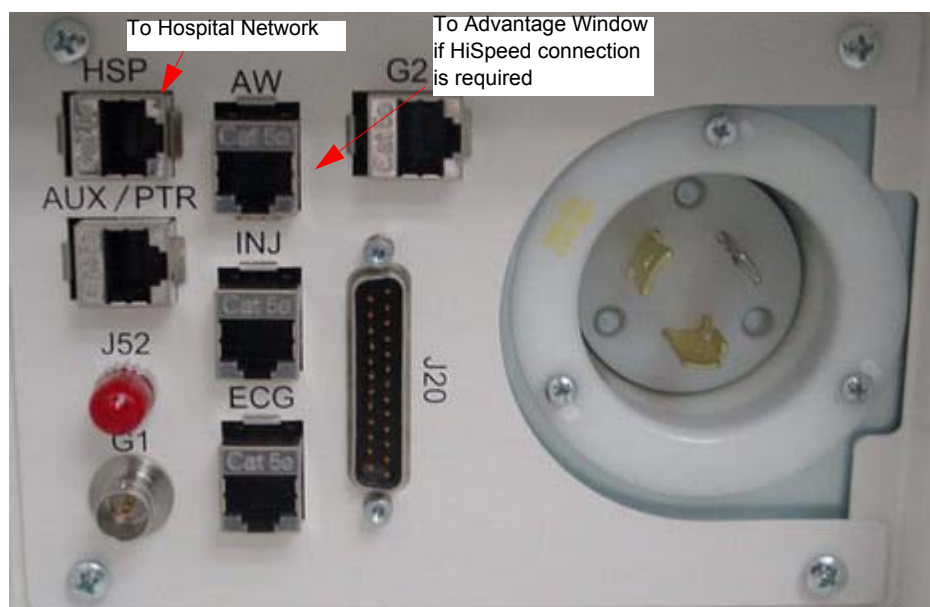


Figure 7-17 Console Rear Bulkhead

US PROCESS OVERVIEW

The United States network connectivity requirement for this product is broad-band. The US process relies on the Install Specialist to select a Customer Champion and identify an IT contact for the site. Together, those individuals then complete a site assessment to gauge what tasks are needed to fulfill the connection.

Anyone can contact the GE Connectivity team at 800.321.7937, Option #3, with questions.

CUSTOMER BROAD-BAND RESPONSIBILITIES

Provide GEMS Installation Specialist with an accurate site address, telephone number, contact name, and e-mail address for the:

- Customer Champion
 - Co-ordinate VPN activities between Radiology/Cardiology and the Information Technology (IT) departments
 - Act as a focal point in assuring site broad-band infrastructure meets GEMS requirements for connection as determined by a mutual assessment with the GEMS Connectivity team.
- IT Contact
 - Complete an equipment assessment with GEMS Connectivity team to determine site readiness for broad-band
 - Work with the Customer Champion to complete any identified infrastructure changes
 - Provide IP addresses for new CT equipment
 - Provide a VPN compatible appliance that will support the IPSec tunneling protocol and 3DES data encryption
 - To utilize an Internet Service Provider that supports static routing



GE Medical Systems

Remote Service Broadband - Customer Site Assessment

Site Name: _____ FE Name: _____
City, State: _____ FE Phone: _____
Date: _____ FE Email: _____

- | | Yes | No |
|---|---|--------------------------|
| 1. Does your site currently have a persistent (24x7) Internet connection? | ☐ | ☐ |
| 2. Is the GEMS Diagnostic Imaging equipment on the Local Area Network and will it be accessible to the Internet? | ☐ | ☐ |
| 3. Does your site have a VPN device today? | ☐ | ☐ |
| 4. Is the VPN device one of the models below? <i>If Yes, please select the model from the options below.</i> | ☐ | ☐ |
| ☐ a) Cisco Pix Firewalls
☐ b) Cisco Routers
☐ c) Cisco 3000 Series (Altiga acquisition)
☐ d) Checkpoint Firewalls Software Version 4.1 and higher
☐ e) Nortel Contivity Software Version 3.2 or higher
☐ f) Redcreek | ☐ g) Symantec (Raptor) firewalls
☐ h) Firebox
☐ i) Linux S/WAN
☐ j) Sidewinder
☐ k) Netscreen
☐ l) None
☐ m) Other _____ | |

**If No, the GEMS Connectivity Support Team can help determine device compatibility.*

5. Does your VPN device support "triple DES" Encryption? ☐ Yes ☐ No
6. Has approval been given to install this VPN connection? ☐ Yes ☐ No
- Site Approver's Name _____
7. Provide your VPN Installer information, this is the person who will be contacted to schedule the VPN install.

Customer Installer Name: _____
Installer Telephone Number: _____
Installer e-mail address: _____

Notes: _____

Field Engineer needs to provide compatible system information:

System ID	IP Address	Gateway Address
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____

[Additional System and IP Address Spaces Available on Page 2](#)

If you have questions or need assessment support contact your Zone Champ or:

Joe Gracz - HQ Support 1-262-524-5261
Joseph.Gracz@med.ge.com

Once you have completed both pages of this form, please send it to:

a) Judy Heyer judy.heyer@med.ge.com
b) Judy Heyer Fax# 414-918-4707

[Use the send button on page 2](#)



GE Medical Systems

System ID List & Compatibility Matrix

MR	Software Version	CT	Software Version
Include all Signa(LX, Ovation, OpenSpeed and CV/I) MR's System Ids and IP Addresses in assessment.		HiSpeed CT/e	Minimum 6.0
		HiSpeed CT/e dual	Minimum 2.06
		HiSpeedNX/I	Minimum 5.5
		LightSpeed QX/I	Minimum 1.3
Prior to completing an InSite checkout systems must be upgraded to Version 9.0 or 9.1 unless shown in the compatible list below.		LightSpeed Plus	Minimum 2.1
		LightSpeed Ultra	Minimum 3
		HiSpeed X/I	
		CT/I	Minimum 6.2
		Lxi, Dxi, Fxi, Zxi	Minimum 6.01
Signa TwinSpeed	Release 9.0		
Signa	Release 9.1		
Signa	Release 10.x		
Signa CV/I	CNV4		
Signa 3T			
Signa Profile	Minimum 7.66		
Signa Contour	Minimum 7.66		
Magnet Monitor	Minimum 2.3 software		
Network Products	Software Version	Nuclear/Pet	Software Version
AW 4.0	Version 4.0 or Above	Advance	Minimum 5.1
AW 4.0P		eNTEGRA	Minimum 2.03

X-Ray

<u>Cardiac</u>	
INNOVA	All
<u>Mammography</u>	
Seno2000D	All
<u>Digital Radiology</u>	
Revolution XQ/i	Version 10.12.5 or above
Revolution XR/d	Version 18.0

Multi Vendor

MR	PSI	CT	PSI
Philips ACS 1.5T	MZP 400	Picker MX-TWIN	CZM 400
Philips NT 2000	MZP 401	Picker MX 8000	CZM 401
Philips NT 3000	MZP 402	Picker PQ	CZM 500
Philips NT ACS	MZP 403	Picker PQ 2000	CZM 501
Picker Outlook 0.23T	MZM 301	Picker PQ 5000	CZM 502
Picker Edge 1.5T	MZM 800	Picker PQ 6000	CZM 503
Picker Vista 1.0T	MZM 801	Picker PQS	CZM 504
Picker Eclipse 1.5T	MZM 900	Siemens AR STAR	CZS 100
Picker Polaris 1.0T	MZM 901	Siemens AR.C	CZS 101
Siemens Harmony 1.0T	MZS 100	Siemens AR.HP	CZS 102
Siemens Impact 1.0T	MZS 200	Siemens AR.SP	CZS 103
Siemens Impact Expert 1.0T	MZS 201	Siemens AR.T	CZS 104
Siemens SP 1.0T	MZS 400	Siemens PLUS-4 EXP	CZS 400
Siemens SP 1.5T	MZS 401	Siemens PLUS-4 POW	CZS 401
Siemens SP4000 1.5T	MZS 403	Siemens PLUS-4 VOL	CZS 402
Siemens Symphony 1.5T	MZS 500		
Siemens Vision 1.5T	MZS 600		

Field Engineer needs to provide compatible system information:

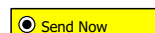
[illegible]

Once you have completed this form, please send it to:

Judy Heyer
Judy Heyer

email
Fax #

judy.heyer@med.ge.com
414-918-4707



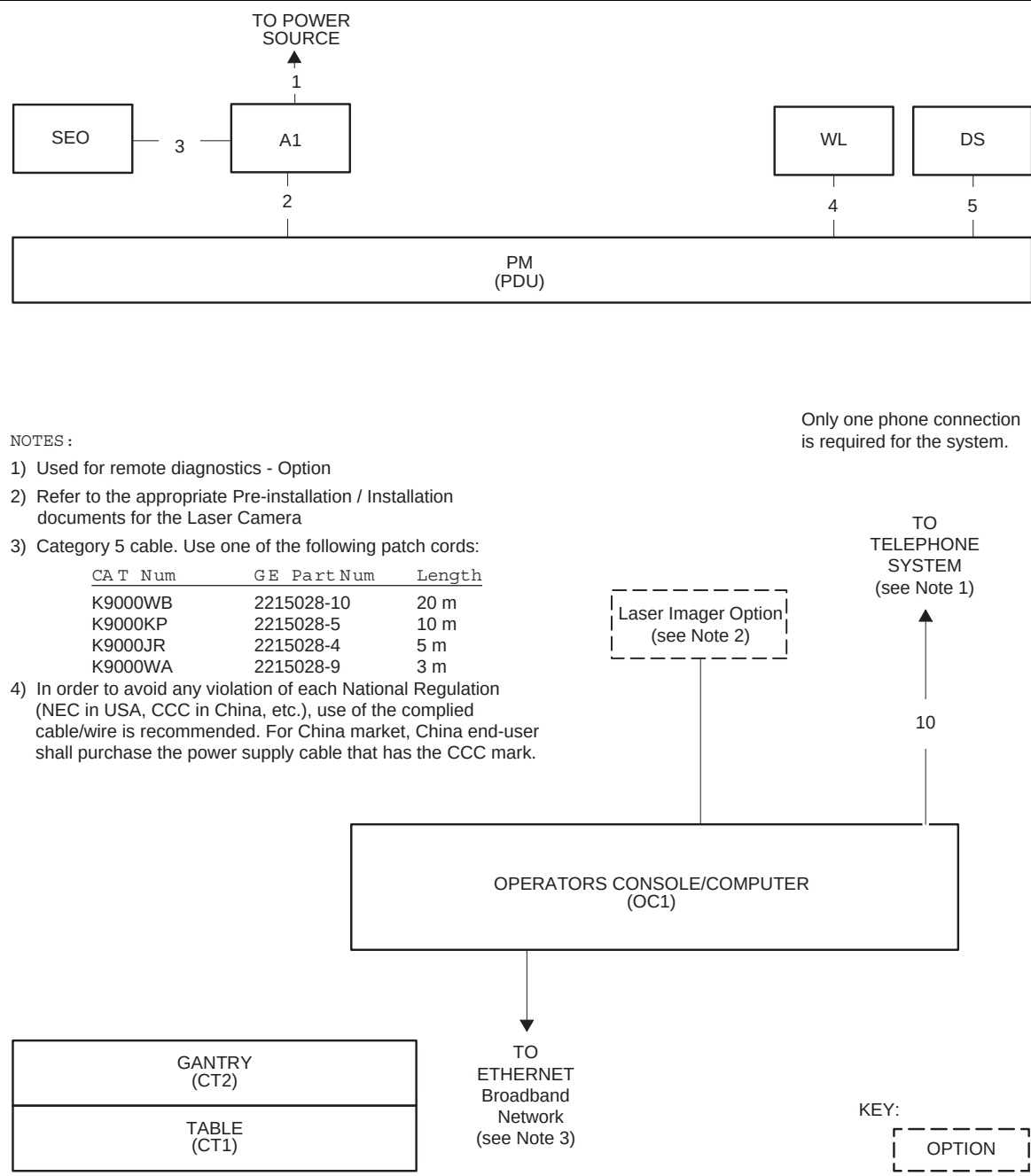


Figure 7-18 BrightSpeed System Interconnection Runs

Section 11.0

Modem Setup in IIP Configuration

If a phone Modem needs to be installed, follow below description to select correct modem type in "iipadmin config" GUI.



Figure 7-19 IIP Config GUI

For GOC4/LCGOC, a Series Port Modem will be used. At "CPU Series Port Name", Select /dev/ttyS0.

For All-In-One Console, see below:

If a USB Port Modem is used. at "CPU Series Port Name", Select /dev/ttyACM0.

If a Series Port Modem and USB Series converter cable is used. at "CPU Series Port Name", Select /dev/ttyUSB1.

Chapter 8

System-Level Safety Tests

Section 1.0

Patient Touch Leakage Test

1.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1 Engineer	10 mins	20 mins	N/A

1.2 Overview

This test should ensure that a patient undergoing a scan cannot touch or otherwise contact any conductive surfaces. Complete this procedure AFTER the installation of ALL options.

1.3 Preliminary Requirements

1.3.1 Tools and Test Equipment

Item	Qty
Standard service tool kit	1
Dale 600 / 601 Meter (from Tool Pool)*	1
Dale extended length leads*	1
*GE Healthcare performed validation of this procedure using the Dale 600/601 meter ONLY. Due to the unique nature of this meter, GE cannot guarantee the accuracy of this procedure if you use another meter.	

1.3.2 Safety



WARNING



POTENTIAL FOR SHOCK

GROUND WIRES WILL HAVE GROUND CURRENT PRESENT WITH POWER “ON”. FOLLOW APPROPRIATE SAFETY PROCEDURES FOR WORKING WITH AN ENERGIZED SYSTEM.



NOTICE

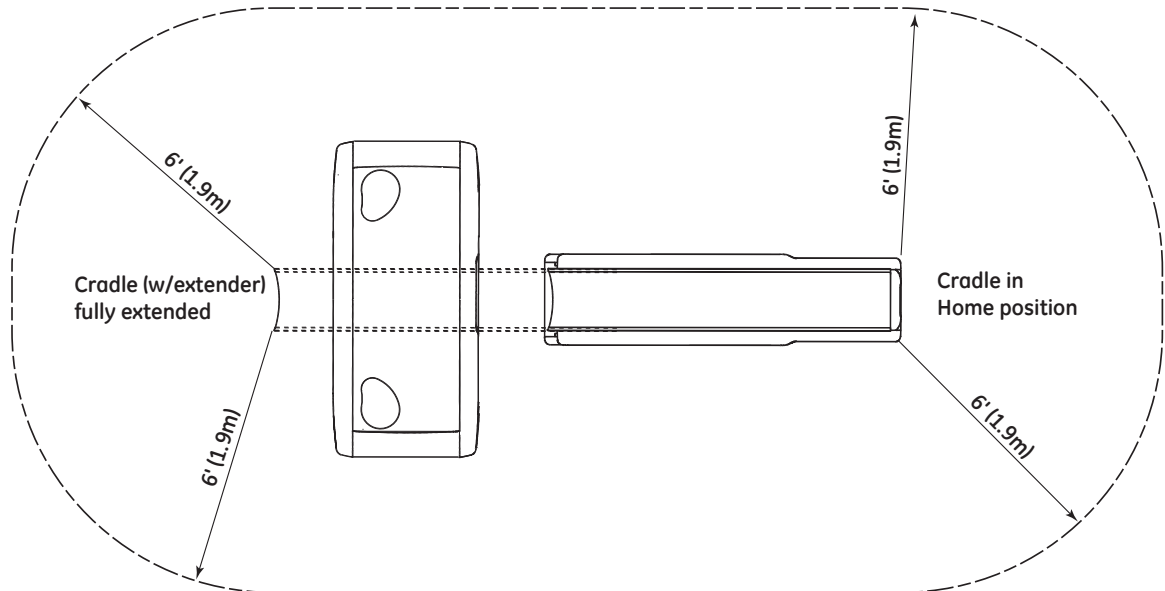
Follow ALL required safety and PPE procedures customary for your organization, when working on this product.

1.3.3 Required Conditions

- Only trained service personnel should service the GE Scanner.
- GE Healthcare performed validation of this procedure using the Dale 600/601 meter ONLY. GE cannot guarantee the accuracy of this procedure if you use another meter.
- You must remove the Footswitch cover.

1.3.4 Test Conditions

- Test with the Table at maximum elevation and again with the Table at minimum elevation.
- Test to cover ALL points in an envelope described by Table travel from minimum to maximum extension, including the table extender.
- Test from both the head and foot of the Table -- e.g., test assuming that a patient may lie either HEAD-FIRST or FEET-FIRST.
- Test to ensure that all conductive surfaces are 6 feet (1.9 m) below and 7 feet (2.1 m) above the Table surface -- e.g., test assuming that a patient may lie either FACE-DOWN or FACE-UP and can touch or otherwise contact conductive surfaces both above and below the table.
- Test to ensure that a protective envelope, free of ANY and ALL conductive surfaces, extends at least 6 feet (1.9 m) around the patient. Surfaces of concern include:
 - I.V. poles and tray assemblies
 - Smart step monitors and stands
 - Table bearing rails, if accessible
 - The A1 Disconnect
 - The PDU
 - All other conductive surfaces



1.4 Procedure



NOTICE GE Healthcare performed validation of this procedure with the Dale 600/601 meter **ONLY**. Due to the unique nature of this meter, GE cannot guarantee the accuracy of these procedures when performed with any meter other than the Dale 600/601.

1.4.1 Cover Removal

- 1.) Move the Table to ISO elevation.
- 2.) Remove the Footswitch covers and the Gantry left side cover. Refer to [Appendix A: Removal & Installation of Covers](#) for additional information.

1.4.2 Leakage Current Testing

Note: Refer to the Dale 601 Operator's Manual for instructions on the use of the Dale 600/601 Meter for measuring leakage current or refer to [Figure 8-1](#) for a quick overview.

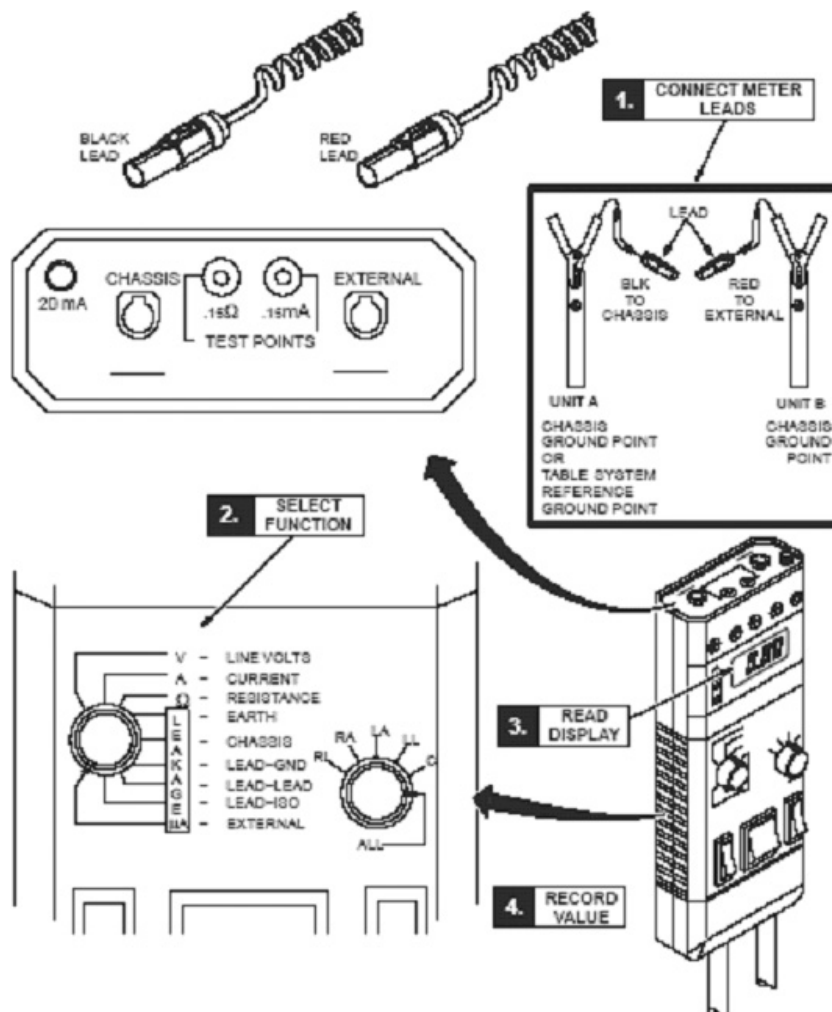


Figure 8-1 Using the Dale 600/601 Meter to Measure Leakage Current

- 1.) Plug the Dale 600/601 Meter into the outlet on the Gantry left side and confirm that the outlet is wired correctly by observing the three LED indicators on the meter.
- Note: Your meter must display a valid calibration sticker.
- 2.) Connect the meter leads to the meter as follows:
 - a.) Connect one end of the shorter black lead to the chassis plug and the other to the Table ground bus.
 - b.) Connect the red or longer black test lead to the external plug on top of the Dale 600/601 meter.
- Note: Some meters include a black lead and a red test lead, while other meters include two black leads. If your meter includes two black leads, the longer black lead is the test lead.
- 3.) Set the function switch on the Dale Meter to EXTERNAL and test that the meter is operational by touching the external lead to the meter's test terminal.
 - 4.) Keep the shorter black lead connected to the Table base ground bus and connect the red lead or longer black lead to the devices (components) under test, being sure to:
 - a.) Test with the power ON.
 - a.) Test between the system reference ground point (Table base) and the unit reference ground points (i.e. Gantry and Table, see [Figure 8-2](#)).
 - b.) Test all conductive surfaces and components within patient reach - within 6 feet (1.9 m) of the Table, and within 7 feet (2.1 m) above the surface of the Table, measuring at Table maximum travels. Refer to [Figure 8-2](#) for additional guidance.
- Note: At some sites, sinks and wall outlet cover plates may create areas of concern.
- c.) Test ALL optional components, such as in-room monitors, injectors, overhead monitor suspensions, and Table options.



NOTICE Beware of static discharge from the scan window, keypads, display, touch pads, and other plastic surfaces.

- 5.) Record the results of your tests in [Table 8-1](#), making certain that no leakage current exceeds the following limits:
 - * Critical care areas (invasive) - 10µa
 - * **General care areas - 20µa (Use this test criterion.)**
 - * Not intended for patient area - 50µa
- 6.) Complete the appropriate section of GE form e4879 to confirm the successful completion of this test. For additional information about this form, see Chapter 4, [Section 8.0 - GE and Regulatory Forms](#)

Table 8-1

Gnd Bus to	Install <20µA
Any Table ground points within the 6' range	
Any Gantry ground points within the 6' range	
Injector assembly metal surface	
Boom-in-Room metal surface	
Monitors or metal surface	
Sink or metal surface	
Installed Table accessories	
(other)	
(other)	
(other)	
(other)	

- 7.) Re-install the Footswitch cover and the Gantry left side cover. Refer to [Appendix A: Removal & Installation of Covers](#) for additional information.

PROCEDURE HINTS:

- Items with abnormally high or low measurements could indicate improper wiring, or loose or poor connections due to corrosion, painted surfaces, etc.
- High leakage could indicate a wiring error such as a neutral connected to the ground.
- Fluctuating ground currents could indicate a short, poor connection, or facilities ground problem causing leakage currents from other areas of the facility to flow through our system grounds.
- Refer to the illustrations below for visual depictions of the required measurements.

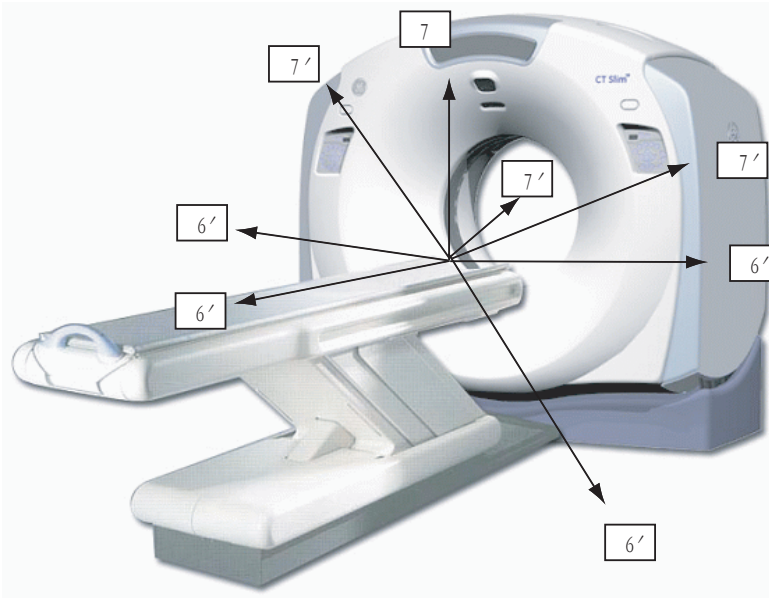


Figure 8-2 Measurement Points

1.5 Finalization

No finalization required.

Section 2.0

CT System Chassis Leakage Test

2.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1 Engineer	N/A	120 mins	N/A
1 Customer Electrician	N/A	120 mins	N/A

2.2 Overview

Unless required by your state, the CT System Chassis Leakage Test is an optional procedure.

2.2.1 Time

2 Hour labor on site, includes:

- Removal of covers
- Leakage test
- Re-installation of covers
- Recording of site data on the CT System Chassis Leakage Test Completion Form, if required by your state. Forward the completed form to your Project Manager of Installation.

2.3 Preliminary Requirements

2.3.1 Tools and Test Equipment

Item	Qty
Standard FE Service Tool Kit	1
GEH LOTO Kit	1
Dale 600 or 601 Leakage Current Meter	1

*GE Healthcare performed validation of this procedure using the Dale 600/601 meter ONLY. Due to the unique nature of this meter, GE cannot guarantee the accuracy of this procedure if you use another meter.

2.3.2 Safety



DANGER POTENTIAL FOR ELECTRIC SHOCK
GROUND WIRES WILL HAVE GROUND CURRENT PRESENT WITH POWER "ON".
THIS TEST IS PERFORMED WITH POWER "ON" - SYSTEM IN STAND-BY!



DANGER POTENTIAL FOR ELECTRIC SHOCK
SERVICING HARDWARE WITH POWER "ON."
EQUIPMENT SERVICE CAN ONLY BE PERFORMED SAFELY WITH THE MAIN
POWER "DISCONNECT" TAGGED AND LOCKED OUT.



WARNING Follow ALL required safety, PPE, and arc-flash procedures customary for your organization, when working on this product.

2.3.3 Required Documentation

Refer to Equipment Service section – LOTO PPE for detailed procedure.

2.3.4 Required Conditions

- Only trained service personnel should service the GE Scanner.
- Servicing hardware with power "On." Equipment service can only be performed safely with the main power "disconnect" Tagged- and Locked-Out.
- GE Healthcare performed validation of this procedure using the Dale 600/601 meter ONLY. GE cannot guarantee the accuracy of this procedure if you use another meter.
- You must remove the Footswitch cover and Gantry front, top, and side covers.

2.4 Procedure



NOTICE Follow LOTO and other safety procedures found in this manual before starting this procedure.

2.4.1 Cover Removal

- 1.) Remove the Gantry top and side covers. Guide rails on each side of the top cover will aid in removal. Refer to [Appendix A: Removal & Installation of Covers](#) for details.

Note: Fans located on the top cover make that cover heavier.

- 2.) Power down the console and follow the appropriate GEH Lockout/Tagout procedures.
- 3.) Remove the Gantry front cover, following the procedure in [Appendix A: Removal & Installation of Covers](#).

- 4.) Remove the Table footswitch top cover to gain access to the ground cables and ground bar.

Note: Do NOT disconnect ANY grounds at this time.

2.4.2 Electrical Connection Removal

- 1.) Confirm that all system grounds are securely attached to the system ground bus and NOT the Table base.
- 2.) Direct the electrician to remove all external electrical connections made during installation, including:
 - a.) Main system ground at PDU
 - b.) Power feeder flex connection at PDU
 - c.) Room door interlocks and room warning light connections
 - d.) Any and all other external ground connections to the system.
- 3.) Confirm the removal of all external gantry, table, console, and PDU connections, and have the electrician do the same.

Note: Some wires, such as the room warning light, may have external power and wire nuts, which should be installed to protect from arcing.

- 4.) Replace all covers removed in section [2.4.1](#) except for the Table footswitch cover.
- 5.) Follow the Lockout/Tagout procedure for re-energizing power and boot to application level. If not already done, remove the footswitch assembly top cover while the system boots.

2.4.3 System Ground Wire Testing

- 1.) Plug the Dale 600/601 Leakage Current Meter into one of the outlets on the gantry.
- 2.) Connect the meter leads to the meter as follows:
 - a.) Connect one end of the shorter black lead to the chassis plug and the other to the table ground bus.
 - b.) Connect the red or longer black test lead to the external plug on top of the Dale 600/601 meter.
- 3.) Set the function switch on the Dale 600/601 meter to EXTERNAL. Using the red or longer black external lead, touch the meter's test terminal to confirm that the meter is operational.

Note: For more information, refer to the Dale 600/601 Operator's Manual or see [Figure 8-1](#).

- 4.) With the system at application-level and all components functional, test the system ground wires as follows:
 - a.) Set-up the meter.
 - b.) Connect the shorter black lead to the system ground bus.
 - c.) Remove a system ground wire.
 - d.) Connect the red or longer black lead to the ground wire.
 - e.) Read the value on the meter.
 - f.) Replace the system ground wire.
 - g.) Record the results of your tests in [Table 8-2](#) for later use when filling-out the CT System Chassis Leakage Test Completion form.
 - h.) Repeat, testing all system ground wires ONE-AT-A-TIME. [Table 8-2](#) lists each system ground wire.

Note: The measured leakage current must not exceed 5 MA in any ground wire.

Table 8-2 System Ground Measurements

Components	Results	Date	Tester
PDU Leakage 1/0			
Gantry Leakage 1/0			
Console Leakage #2			
Table Leakage #10			
Option Grounds, if present			

2.4.4 System Power-Down

- 1.) After completing all tests, follow the Lockout/Tagout procedure to power down the system.
- 2.) Direct the electrician to re-install all electrical connections, conduits, cables, and wires removed in section 2.4.2 and to secure all connections per NEC code.
- 3.) Check to confirm that all connections are securely tightened.
- 4.) Re-install all removed system covers, except for the gantry right-side cover near the service switch panel.

2.4.5 System Power-Up and Test

Note: Be sure to complete this section of the procedure on-site.

- 1.) Follow the Lockout/Tagout procedure for re-energizing power.
- 2.) Turn-on the gantry service switches and power-up the console.
- 3.) Check that no cables remain in the gantry rotating path.
- 4.) Return cover dollies to storage areas.
- 5.) Check that the table controls and footswitch function properly.
- 6.) Re-test the system by completing a system functional scan. If installed, be sure to test the room warning light and the door interlock also at this time.
- 7.) Complete the CT System Chassis Leakage Test Completion Form, if required by your state, and forward the completed form to your Project Manager of Installation.

Note: Your product service information CD-ROM contains a copy of the CT System Chassis Leakage Test Completion form. For more information, see [GE and Regulatory Forms on page 189](#)

2.5 Finalization

No finalization required.

AppendixF

Additional Characterization Procedures

For information related to alignment, setup, and calibration procedures, please refer to the BrightSpeed System Service Methods, Direction 5193754-800.

AppendixG

System Configuration Data Sheets

REQUIREMENT

Record valuable system information in the data sheets that follow. Consult with your customer or network administrator to obtain the information. Understanding how the customer plans to use their CT scanner, and their network and filming expectation reduces the time required to reconfigure the system.

- [Table 8-3 Manual Film Composer Options, on page 297](#)
- [Table 8-4 System Network Configuration, on page 297](#)
- [Table 8-5 Networking Application \(Image transfer\) Configuration, on page 298](#)
- [Table 8-6 DASM Laser Camera Configuration, on page 298](#)
- [Table 8-7 DICOM Print Camera Configuration, on page 299](#)
- [Table 8-8 DICOM Print Camera Advanced Configuration, on page 299](#)

MANUAL FILM COMPOSER OPTIONS

MANUAL FILM COMPOSER OPTIONS

Slide Format (if available):

Greyscale:

Auto Printing:

Auto Clear Page:

Icon Labels:

Expose Order:

No. of Copies:

Table 8-3 Manual Film Composer Options

SYSTEM NETWORK CONFIGURATION

SYSTEM NETWORK CONFIGURATION

	FIELD NAME:	SETENV NAME:	FIELD VALUE:
System Settings:	Service ID	SERVICE_ID	
	Hospital Name	HOSPITAL_NAME	
	Exam Number *	* ask customer or checklog	
	DAS Type	DASTYPE	
	PDU Type	PDUTYPE	
Network Settings:	Gateway Host Name	GATEWAY_HOSTNAME	
	Gateway IP	GATEWAY_IP	
	Gateway Net Mask	GATEWAY_NETMASK	

Table 8-4 System Network Configuration

SYSTEM NETWORK CONFIGURATION			
	FIELD NAME:	SETENV NAME:	FIELD VALUE:
	Gateway Broadcast Mask	GATEWAY_BROADCAST	
	Suite Name	SUITEID	
Option	Network Printer IP Address		
Option	HIS Server IP Address		
Option	HIS Server AE Title		
Option	HIS server AE Port		
Option	CT Server AE Title		
Option	Connect Pro IP Address		

Table 8-4 System Network Configuration (Continued)

NETWORK APPLICATION (IMAGE TRANSFER) CONFIGURATION

Record the network application (image transfer) configuration.

NETWORKING APPLICATION (IMAGE TRANSFER) CONFIGURATION				
AE TITLE OR HOST NAME	NETWORK ADDRESS	NETWORK PROTOCOL	PORT NUMBER	COMMENTS

Table 8-5 Networking Application (Image transfer) Configuration

HOST ETHERNET ADDRESS

_____ : _____ : _____ : _____ : _____

CAMERA APPLICATION CONFIGURATION

Record the camera application configuration for the DASM or DICOM print camera.

DASM LASER CAMERA CONFIGURATION	
Camera Type:	

Table 8-6 DASM Laser Camera Configuration

DASM LASER CAMERA CONFIGURATION	
DASM Type:	
Film Smooth/Sharp Setting:	
Options:	
Valid Film Formats:	
Default Film Formats:	

Table 8-6 DASM Laser Camera Configuration (Continued)

DICOM PRINT CAMERA CONFIGURATION	
Camera Type:	
Host Name:	
IP Address:	
AE Title:	
TCP/IP Listen Port:	
Comments (Optional):	
Valid Film Formats:	
Default Film Formats:	
Destination:	
Orientation:	
Medium Type:	
Magnification Type:	

Table 8-7 DICOM Print Camera Configuration

DICOM PRINT CAMERA ADVANCE CONFIGURATION	
Smoothing Type:	

Table 8-8 DICOM Print Camera Advanced Configuration

DICOM PRINT CAMERA ADVANCE CONFIGURATION

Configuration:	
Minimum Density:	
Maximum Density:	
Empty Density:	
Border Density:	
Association Timeout:	
Session Timeout:	
N-Set Timeout:	
N-Action Timeout:	
N-Create Timeout:	
N-Delete Timeout:	
N-Get Timeout:	

Table 8-8 DICOM Print Camera Advanced Configuration (Continued)

CONFIGURATION

Note: Type the text shown in **boldface**, and press the ENTER key on the keyboard.

NEXT STEPS

Resume installation following instructions in [Chapter 5: Restore System State \(Section 2.3\)](#).

AppendixH

Symbols

SYMBOL	PUBLICATION	DESCRIPTION
	417-5032	Alternating Current
	335-1	Three-phase Alternating Current
	335-1	Three-phase Alternating Current with neutral conductor
		Direct Current
	417-5019	Protective Earth (Ground)
	348	Attention, consult ACCOMPANYING DOCUMENTS
	417-5008	OFF (Power: disconnection from the mains)
	417-5007	ON (Power: connection to the mains)
		Warning, HIGH VOLTAGE
		Emergency Stop
		Type B

Table 8-9 Symbols

SYMBOL	PUBLICATION	DESCRIPTION
	417-5339	X-ray Source Assembly Emitting
	417-5009	Standby
		Start
		Table Set
		Abort
		Intercom
		(on Operator Console) Power On: light on Standby: light off

Table 8-9 Symbols

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